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A Textbook of

CALCULUS I

(For all Engineering Programs of Pokhara University)

Dr. Narayan Prasad Pahari

Dr. Santosh Ghimire
Arun Kumar Bhandari
Prem Gurung

Dr. Jeevan Kafle
Madhav Prasad Poudel
Durgesh Ojha

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Unit 1

Limit, Continuity and Derivative

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- set, relation between sets, set operations, set of numbers (natural, integers, rational, irrational numbers)
- real number systems, different types of intervals
- relation from a set to another set
- function (domain, range, co-domain, image, pre-image)
- graph of the functions (polynomial, Exponential, Logarithmic and Trigonometric)
- basic skill on evaluation of limit, continuity and derivative.

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- limit, continuity and differentiability with their various properties.
- left hand and right hand derivatives.
- physical and geometrical interpretation of first, second and third order derivatives.
- differentials.
- indeterminate limits using L' Hospital's rule.
- higher order derivatives and their evaluation for product functions using Leibnitz method.

1.0 Introduction

In this unit, we will review on fundamental concepts of calculus. We mainly focus on concept of limit, continuity and differentiability together with their various properties and relations. For calculus, we need the concept of function. We first review functions and then proceed to limit.

Sir Isaac Newton
(1642–1727)

Calculus was developed independently by English physicist and mathematician Sir Isaac Newton (1642–1727) and German mathematician Gottfried Wilhelm Leibnitz (1646–1716) in the seventeenth century. He found that the mathematics of his time was not sufficient to solve the problems that he was interested in, so he introduced new concept in mathematics, the calculus. About the same time, another mathematician, Leibnitz developed the same idea as Newton with interest in calculating the velocity of an object at any instant. Important problems in physics, chemistry, engineering, economics, biology, and the other sciences involve finding the rate at which one quantity changes with respect to another, which is the main objective of the derivative.

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1.1.1 Functions

Let A and B be two sets. Then, a function $f: A \to B$ is a rule that assigns every element of A to an unique element of B.

In other words, let A and B be two sets, then the rule $f: A \to B$ is said to be a function if

i. every element of A has an image in B.

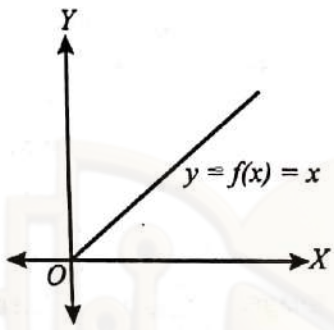
ii. the image in B must be unique.

The set A is called the domain and set B is called the co-domain of f. Also, $f(A) = \{y : y = f(x)$ for some $x \in A\}$ is called range of f. Clearly, image $f(A) \subseteq B$. We write $y = f(x)$ for the rule of the function.

1.1.2 Different types of functions

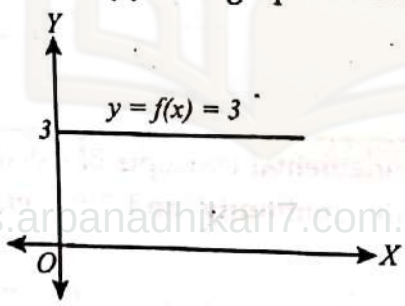
Identity function

A function $f: \mathbb{R} \to \mathbb{R}$ is said to be identity if $f(x) = x$, for all $x \in \mathbb{R}$. The graph is always a straight line and passes through the origin.



Constant function

A function $f: \mathbb{R} \to \mathbb{R}$ is said to be constant if $f(x) = c$, for all $x \in \mathbb{R}$. The domain of the function f is $\mathbb{R}$ and its range contains single element $\{c\}$. The graph is a straight line parallel to the x-axis.



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Polynomial function

A polynomial function is defined by $y = f(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$, where n is a non-negative integer and $a_0, a_1, a_2, \ldots, a_n \in \mathbb{R}, a_n \neq 0$. The highest power of x in the expression is called the degree of the polynomial function. Polynomial functions are further classified based on their degrees:

- **Constant function:** If the degree is zero, the polynomial function is a constant function.
- **Linear function:** The polynomial function with degree one is called linear function. The domain and range are both set of real numbers $\mathbb{R}$. The graph is always a straight line.

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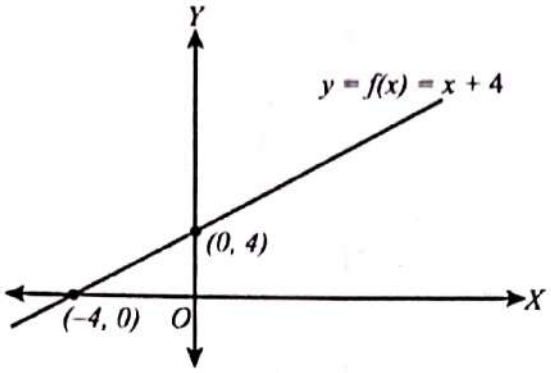
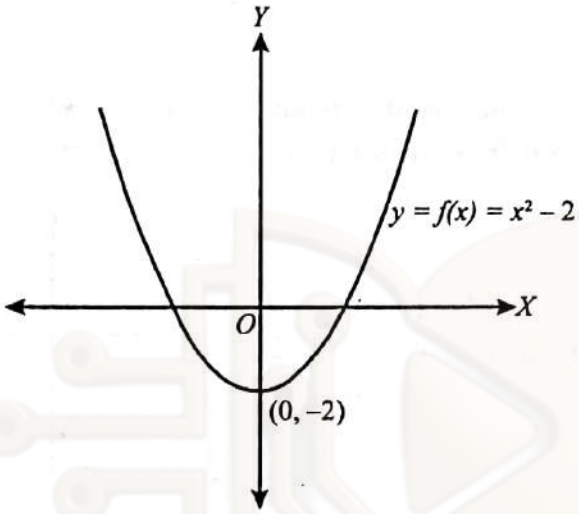
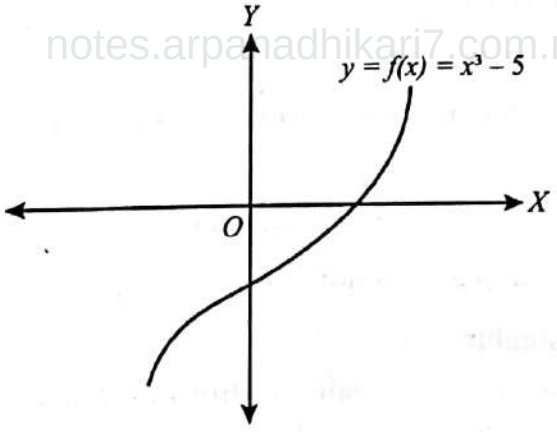


- **Quadratic function:** A polynomial function of degree two is called a quadratic function. It is expressed as $f(x) = ax^2 + bx + c$, where $a \neq 0$ and a, b, c are constants. The graphical representation of a quadratic function is a parabola. For example, the graph of $y = f(x) = x^2 - 2$ is a parabola which is shown in figure.



- **Cubic function:** A cubic function is a polynomial function of degree three and is denoted by $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$ and a, b, c and d are constants. The graph for $f(x) = ax^3 + bx^2 + cx + d$ is a semi-cubical parabola. For example, the graph of $y = f(x) = x^3 - 5$ is as shown in figure.



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Rational function

A rational function is any function which can be written as $f(x)/g(x)$ in which numerator $f(x)$ and denominator $g(x)$ are polynomial functions and $g(x) \neq 0$.

For example, $f(x) = \frac{x}{x-1}$ is a rational function.

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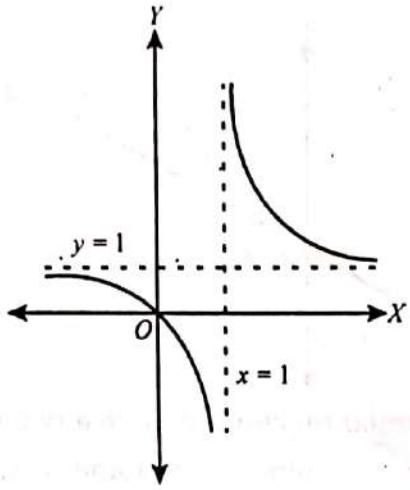


1.1.3 Absolute value of a real number

Let x be any real number. Then, the absolute value of x is denoted by $|x|$ and is defined by

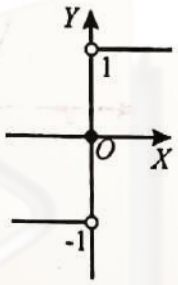
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Which is the measurement of distance of x from zero and is known as distance function in number line. We note that the positive numbers lie on the right and negative numbers lie on the left of zero.

1.1.4 Signum function

A function $f: \mathrm{R} \to \mathrm{R}$ defined by $f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$

is known as *signum function*.



1.1.5 Even function

A function $f(x)$ is said to be even function on the domain D if $x \in D \Rightarrow -x \in D$ and $f(-x) = f(x)$ for all $x \in D$.

Example: $f(x) = x^{2n}$, for all integers n is an even function.

The function $g(x) = \cos x$ is also an even function.

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1.1.6 Odd function

A function $f(x)$ is said to be odd function on the domain D if $x \in D \Rightarrow -x \in D$ and $f(-x) = -f(x)$ for all $x \in D$.

Example: $f(x) = x^{2n\pm1}$, for all integers n is an odd function.

The function $g(x) = \sin x$ is also an odd function.

1.1.7 Functional value and calculus

Let f be a real valued function. Then the value of function f at $x = a$ denoted by $f(a)$, is a real number given by function f at a. This real number $f(a)$ is called functional value. So when $f(a)$ is finite real number, then we say function is defined at a.

If value $f(a)$ is ∞ or $-\infty$, we say function f is not defined at a. If $f(a)$ takes an indeterminate form $\left(\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0^0, \infty^0, \text{etc.}\right)$, then we say functional value does not exist at $x = a$.

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We illustrate these with the help of following functions:

i. $f(x) = x^2 - 5x + 7$

The value of f at $x = 2$ is $f(2) = 2^2 - 5 \cdot 2 + 7 = 1$ which is a real number. So $f(2)$ is defined at $x = 2$. In fact $f(a)$ is defined for all real numbers a.

ii. $g(x) = \frac{1}{x-2}$

Here, at $x = 2$, $g(2) = \frac{1}{2-2} = \frac{1}{0}$, which is not defined, as division by zero is not defined.

iii. $h(x) = \frac{x^2-4}{x-2}$

Here, at $x = 2$, $h(2) = \frac{0}{0}$, which is an indeterminate and hence the function is not defined.

So, functional value of h at 2 does not exist. In such case h has no real value at 2. Is it possible to find the value of h at a nearest neighbour of 2? Answer to this question is 'yes'. Here we introduce the concept of limit which gives the desired value. In fact the concept of calculus begins from this point. We discuss all the concept of calculus in detail in this section. We begin with limit of a function.

1.2 Limit of a function

Let f be a function defined on an open interval containing a number a.

We say that L is the limit of $f(x)$ as x approaches to a, written as

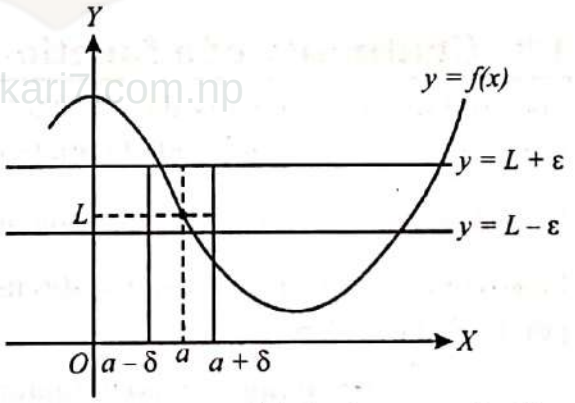
$$\lim_{x \to a} f(x) = L,$$

if for every $\varepsilon > 0$ there exists a number $\delta > 0$ such that

$0 < |x - a| < \delta$ whenever $|f(x) - L| < \varepsilon$.

We should note the function $f(x)$ may not be defined at $x = a$.

This definition says that the distance between $f(x)$ and L can be made arbitrarily small by taking distance between x and a sufficiently small. We now illustrate this situation geometrically.



Geometrically, the definition of limit can be depicted in the figure. Here, for given $\varepsilon > 0$, we draw two horizontal line $y = L + \varepsilon$ and $y = L - \varepsilon$ and $\lim_{x \to a} f(a) = L$ means that we can find a number $\delta > 0$, that if we restrict our x in the interval $(a - \delta, a$, the corresponding curve $y = f(x)$ lies between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$.

Based on this idea, we now introduce the concept of left and right hand limits. We say L_1 is the left hand limit of $f(x)$ as x approaches to a from the left if we can make the values of $f(x)$ arbitrarily close to by taking x very close to a through the values of x less than a. We write

$$f(a^-) = \lim_{x \to a^-} f(x) = L_1, \text{ for left hand limit.}$$

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Similarly, L_2 is said to be the right hand limit of $f(x)$ as x approaches to a if the distance between $f(x)$ and L_2 can be made as close as we want by taking x very close to a through the values greater than a. We write

$$f(a^+) = \lim_{x \to a^+} f(x) = L_2.$$

The limit $\lim_{x \to a} f(x) = L$ exists, if and only if, $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = L$.

1.2.1 Properties of limit

Various properties of limit are listed as:

a. The limit of the sum or the difference of the functions $f(x)$ and $g(x)$ is the sum or difference of the limit of the functions, i.e.,

$$\lim_{x \to a} [f(x) \pm g(x)] = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x).$$

b. The limit of the product of the functions $f(x)$ and $g(x)$ is the product of the limit of the functions

$$\lim_{x \to a} [f(x) \cdot g(x)] = \left[\lim_{x \to a} f(x)\right] \left[\lim_{x \to a} g(x)\right].$$

c. The limit of the quotient of the function $f(x)$ and $g(x)$ is the quotient of the limits of the function, provided the limit of the denominator is not zero, i.e.,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}; \quad \lim_{x \to a} g(x) \neq 0.$$

d. The limit of the n^{th} root of a function $f(x)$ is the n^{th} root of the limit of the function

i.e., $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$.

1.3 Continuity of a function

The continuous process is the one that takes place gradually without interruption and these types of process are represented with the help of continuous functions.

Mathematically, $f(x)$ is said to be continuous at a point $x = a$ if $\lim_{x \to a} f(x) = f(a)$.

Most common functions that we discuss are continuous everywhere except some particular points. Some of them are:

1. Every constant function is continuous on $(-\infty, \infty)$.
2. Every polynomial function is continuous at each point of $(-\infty, \infty)$.
3. The sine and cosine functions are continuous on $(-\infty, \infty)$.
4. The exponential function is continuous at each point on $(-\infty, \infty)$.
5. The logarithmic function is continuous at each point on $(0, \infty)$.

The functions which are continuous in $(-\infty, \infty)$ are said to be everywhere continuous function.

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Example 1. Show that $f(x) = \begin{cases} 2x+1 & \text{for } 0 < x \le 2 \\ 3x-1 & \text{for } x > 2 \end{cases}$ is continuous at $x = 2$.

Solution:

The left hand limit of $f(x)$ at $x = 2$ is $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (2x + 1) = 5$.

The right hand limit of $f(x)$ at $x = 2$ is $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (3x - 1) = 2 \times 3 - 1 = 5$.

Also, functional value of $f(x)$ is $f(2) = 2 \times 2 + 1 = 5$.

Thus, $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(2) = 5$.

Hence, $f(x)$ is continuous at $x = 2$.

Example 2. Show that $f(x) = \begin{cases} 2x+1 & \text{for } 0 < x \le 3 \\ 4x-5 & \text{for } 3 < x \le 4 \\ x^2+3 & \text{for } x > 4 \end{cases}$
is continuous at $x = 3$ but is discontinuous at $x = 4$.

Solution:

i. For the continuity at $x = 3$.

The left hand limit of $f(x)$ at $x = 3$ is

$$\lim_{x \to 3^-} (2x + 1) = 2 \times 3 + 1 = 7 \text{ and}$$

Also, right hand limit of $f(x)$ at $x = 3$ is

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (4x - 5) = 4 \times 3 - 5 = 7.$$

and functional value of $f(x)$ at $x = 3$ is

$$f(3) = 2 \times 3 + 1 = 7.$$

Thus, $\lim_{x \to 3} f(x) = \lim_{x \to 3^-} f(x) = f(3)$.

Hence, $f(x)$ is continuous at $x = 3$.

ii. For the continuity at $x = 4$

The left hand limit of $f(x)$ at $x = 4$ is

$$\lim_{x \to 4^-} f(x) = \lim_{x \to 4} (4x - 5) = 4 \times 4 - 5 = 11$$

and the right hand limit of $f(x)$ at $x = 4$ is

$$\lim_{x \to 4^+} f(x) = \lim_{x \to 4^+} (x^2 + 3) = 4^2 + 3 = 19.$$

Thus, $\lim_{x \to 4} f(x) \neq \lim_{x \to 4^+} f(x)$.

Hence, $f(x)$ is discontinuous at $x = 4$.

We next discuss a special type of discontinuity called removal discontinuity.

1.3.1 Removable discontinuity of a function

Let $y = f(x)$ be a function. Then the function $f(x)$ is said have removal discontinuity at $x = a$ if $\lim_{x \to a} f(x)$ exists but it is different from $f(a)$. In such case $f(x)$ can be made continuous by redefining its functional value that is equal to the limiting value so that $\lim_{x \to a} f(x) = f(a)$. The new function defined so is called the continuous extension of $f(x)$. This will be clear from following example.

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Example 3. Show that the function $f(x) = \begin{cases} 3x^2 & \text{for } x < 1 \\ 2 & \text{for } x = 1 \\ x^2 + 2 & \text{for } x > 1 \end{cases}$ is discontinuous at $x = 1$. What type of discontinuous is it and how it can be redefined to make it continuous?

Solution:

The left hand limit of $f(x)$ at $x = 1$ is

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} 3x^2 = 3 \times 1^2 = 3$$

and the right hand limit of $f(x)$ is

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (x^2 + 2) = 1^2 + 2 = 3.$$

Also, the functional value of $f(x)$ at $x = 1$ is $f(1) = 2$.

Thus, $\lim_{x \to 1} f(x) = \lim_{x \to 1^+} f(x) \neq f(1)$.

Hence, $f(x)$ is discontinuous at $x = 1$. This is removal type of discontinuity. To make it a continuous function, it should be redefined as

$$g(x) = \begin{cases} 3x^2 & \text{for } x < 1 \\ 3 & \text{for } x = 1 \\ x^2 + 2 & \text{for } x > 1 \end{cases} \quad \text{at } x = 1.$$

The function $g(x)$ is continuous extension of the given function $f(x)$, which differs from $f(x)$ only for functional value.

Example 4. Test the continuity of the function $f(x) = \begin{cases} (1 + 3x)^{1/x} & \text{for } x \neq 0 \\ e^3 & \text{for } x = 0 \end{cases}$ at $x = 0$.

Solution:

The limiting value of $f(x)$ at $x = 0$ is

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} (1 + 3x)^{1/x} = \lim_{x \to 0} \left[(1 + 3x)^{\frac{1}{3x}}\right]^3 = e^3$$

Also, the functional value of $f(x)$ at $x = 0$ is $f(0) = e^3$.

Thus, $\lim_{x \to 0} f(x) = f(0)$.

Hence, $f(x)$ is continuous at $x = 0$.

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Exercise 1.1

1. Check if the limit exists for the given functions and evaluate if it exists.

i. $f(x) = \begin{cases} x + 2 & \text{for } x \geq 0 \\ 4x + 2 & \text{for } x < 0 \end{cases}$

ii. $f(x) = \begin{cases} 3x + 1 & \text{for } x \geq 2 \\ 2x^2 - 1 & \text{for } x < 2 \end{cases}$

iii. $f(x) = \begin{cases} 3x - 2 & \text{for } x \geq 2 \\ 2x^2 + 1 & \text{for } x < 2 \end{cases}$

2. Test the continuity of

i. $f(x) = \begin{cases} (1 + x)^{1/x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases}$ at $x = 0$.

ii. $f(x) = \begin{cases} x^2 & \text{for } x \neq 1 \\ 2 & \text{for } x = 0 \end{cases}$ at $x = 1$.

iii. $f(x) = \begin{cases} \frac{1}{2} & \text{for } 0 \leq x \leq 1 \\ 2^{-x} & \text{for } 1 < x \leq 2 \end{cases}$ at $x = 1$.

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3. Find the value of k for which the function

$$f(x) = \begin{cases} \frac{\sin 2x}{5x} & \text{for } x \neq 0 \\ k & \text{for } x = 0 \end{cases}$$ is continuous at $x = 0$.

4. Let a function be $f(x) = \begin{cases} -x & \text{for } x \leq 0 \\ x & \text{for } 0 < x < 1 \\ 2 - x & \text{for } x \geq 1 \end{cases}$.

Show that it is continuous at $x = 0$ and $x = 1$.

5. A function is defined as

$$f(x) = \begin{cases} 2 - x^2 & \text{for } x < 2 \\ 3 & \text{for } x = 2 \\ x - 4 & \text{for } x > 2 \end{cases}$$.

Is the function $f(x)$ continuous at $x = 2$? If not why? Restate to make it a continuous function.

Answers

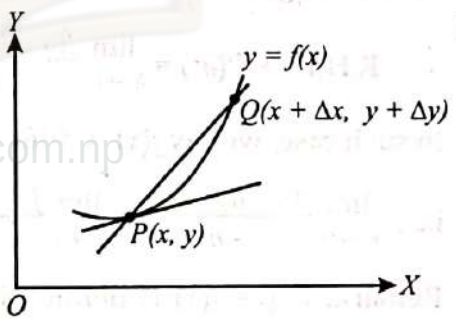
1. i. limit exists, 2 ii. limit exists, 7 iii. limit does not exist
2. i. discontinuous ii. discontinuous iii. continuous
3. $k = \frac{2}{5}$
5. discontinuous, $g(x) = \begin{cases} 2 - x^2 & \text{for } x < 2 \\ -2 & \text{for } x = 2 \\ x - 4 & \text{for } x > 2 \end{cases}$ is continuous extension of $f(x)$

1.4 Derivatives

Derivative of a function is the most fundamental concept that uses limit. The derivative is the limit of average rate of change of function with respect to independent variable and hence gives instantaneous rate of change of a function (one quantity) with respect to the variable (other quantity).

Definition of derivative

Let $y = f(x)$ be a function whose graph is shown in the figure. Let us take a point $P(x, y)$ on the curve of $f(x)$. Consider small change (increment) Δx in x and Δy in y, so that the point $Q(x + \Delta x, y + \Delta y)$ is very near to P.



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The secant line PQ will be reduced to a tangent at P if the point Q approaches to P.

i.e., as $Q \to P$, the secant line PQ becomes tangent at P.

Hence, slope of the tangent is approximated by the limiting slope of secant PQ as $Q \to P$.

Here, slope of secant $PQ = \frac{\Delta y}{\Delta x}$.

By actual change in y from point P to Q,

$$y + \Delta y = f(x + \Delta x)$$

or, $$\Delta y = f(x + \Delta x) - f(x)$$

$$\therefore \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Taking limit as $Q \to P$ i.e., $\Delta x \to 0$, we get

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad \ldots . \text{(i)}$$

This limit, if exists (i.e., finite) is called the derivative of $f(x)$ and is denoted by $\frac{dy}{dx}$ for $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ and $f'(x)$ for $\lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$, i.e., $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x)$.

Hence, the derivative is the limit rate of change of y with respect to x and gives instantaneous rate of change.

Moreover, the rate of change (i.e., derivative) at some particular point on the curve of $y = f(x)$, gives slope of tangent to the curve of $y = f(x)$ at that point.

1.4.1 Left hand derivatives and right hand derivatives

As we have seen that derivative is a limit of rate of change and gives slope of tangent at a given particular point on a curve of $y = f(x)$, we can evaluate the derivatives from left side of the point or right side of point as in the case of limit. In such case the derivative of $y = f(x)$ at $x = a$ is denoted by $\left(\frac{dy}{dx}\right)_{x=a}$ and exists if both left hand derivative (i.e. limit) exists at $x = a$ and right hand derivative (i.e., limit) exists at $x = a$ and are equal.

Left hand derivative L.H.D. of $y = f(x)$ at $x = a$ is defined by

$$\text{L.H.D.} = f'(a^-) = \lim_{h \to 0^-} \frac{f(a - h) - f(a)}{-h}$$, where $h > 0$ is a very small quantity that represents change in x (i.e., $\Delta x = h$).

Also, the right hand derivative, R.H.D. of $y = f(x)$ at $x = a$ is defined by

$$\text{R.H.D.} = f'(a^+) = \lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h}.$$

In such case, we say $f(x)$ is differentiable at $x = a$ if $f'(a^-) = f'(a^+)$.

i.e., $\lim_{h \to 0^-} \frac{f(a - h) - f(a)}{-h} = \lim_{h \to 0^+} \frac{f(a + h) - f(a)}{h}$.

Remark: If $y = f(x)$ is defined in an interval $[a, b]$ or (a, b) it is possible to define right hand derivative only at left end a and left hand derivative at right end b of the intervals.

In such case, we take right and left hand derivatives as derivatives of function.

Example 5. Check the differentiability of a function $f(x) = |x| + 1$ at $x = 0$.

Solution:

The given function is $f(x) = |x| + 1$ and given point is $a = 0$.

Now, L.H.D. $= \lim_{h \to 0} \frac{f(a - h) - f(a)}{-h}$

$= \lim_{h \to 0} \frac{f(0 - h) - f(0)}{-h}$

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$$= \lim_{h \to 0} \frac{(h+1)-(0+1)}{-h}$$

$$= \lim_{h \to 0} \frac{h+1-1}{-h}$$

$$= -1$$

$\therefore$ L.H.D. $= f'(0^-) = -1$.

Again, R.H.D. $= \lim_{h \to 0^+} \frac{f(a+h)-f(a)}{h}$

$$= \lim_{h \to 0^+} \frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h \to 0^+} \frac{(h+1)-(0+1)}{h}$$

$$= \lim_{h \to 0^+} \frac{h+1-1}{h}$$

$$= 1$$

$\therefore$ R.H.D. $= f'(0^+) = 1$.

Here, $f'(0^-) \neq f'(0^+)$, so the derivative of $f(x)$ does not exists.

Example 6. Check the differentiability of the function $f(x) = 2x^2$ at $x = 1$.

Solution:

The given function is $f(x) = 2x^2$ and given point is $a = 1$.

Now, L.H.D. $= \lim_{h \to 0^-} \frac{f(a-h)-f(a)}{-h}$

$$= \lim_{h \to 0} \frac{f(1-h)-f(1)}{-h}$$

$$= \lim_{h \to 0^-} \frac{2(1-h)^2-2.1}{-h}$$

$$= \lim_{h \to 0^-} \frac{2(1-2h+h^2)-2}{-h}$$

$$= \lim_{h \to 0^-} \frac{2-4h+2h^2-2}{-h}$$

$$= \lim_{h \to 0^-} \frac{2h(-2+h)}{-h}$$

$$= 4.$$

and R.H.D. $= \lim_{h \to 0^+} \frac{f(a+h)-f(a)}{h}$

$$= \lim_{h \to 0^+} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h \to 0^+} \frac{2(1+h)^2-f(1)}{h}$$

$$= \lim_{h \to 0^+} \frac{2+4h+2h^2-2}{h}$$

$$= \lim_{h \to 0^+} \frac{2h(2+h)}{h} = 4$$

Thus, L.H.D. = R.H.D. = 4. So $f(x)$ is differentiable at $x = 1$.

Relation between differentiability and continuity of a function at a point is given by the following theorem.

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Theorem 1.

If a function $f(x)$ is differentiable at a point $x = a$, it is necessarily continuous at that point, but the converse may not be true, i.e., every continuous function may not be differentiable.

Proof:

Suppose that the function $y = f(x)$ is differentiable at $x = a$, i.e.,

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h} \text{ exists. i.e., } f'(a) \text{ is a finite number.}$$

Now, $\lim_{h\to 0}[f(a+h)-f(a)] = \lim_{h\to 0}\left[\frac{f(a+h)-f(a)}{h}\right]\times h$

or, $\lim_{h\to 0}[f(a+h)-f(a)] = f'(a)\times \lim_{h\to 0} h$

or, $\lim_{h\to 0}[f(a+h)-f(a)] = \text{finite number} \times 0$

or, $\lim_{h\to 0}[f(a+h)-f(a)] = 0$

$\therefore \lim_{h\to 0} f(a+h) = f(a).$

i.e., limiting value and functional values are equal. Hence, the function $f(x)$ is continuous at $x = a$.

For the converse let us consider a function $f(x) = |x|$.

Then, by definition, $|x| = \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases}$.

For the continuity at $x = 0$.

$$\text{Left hand limit} = \lim_{x\to 0^-} f(x)$$
$$= \lim_{x\to 0^-} |x|$$
$$= \lim_{x\to 0^-} (-x)$$
$$= -\lim_{x\to 0^-} x$$
$$= 0.$$

Also, Right hand limit $= \lim_{x\to 0^+} f(x)$

$$= \lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0.$$

And $f(0) = 0$.

$\therefore \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0).$

Therefore, the function is continuous at $x = 0$.

Again, for differentiability at $x = 0$.

Left hand derivative at $x = 0$ is

$$\text{L.H.D.} = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{f(-h)-f(0)}{-h}$$

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$$= \lim_{h \to 0} \frac{|-h| - 0}{-h}$$

$$= \lim_{h \to 0} \frac{-(-h)}{-h} = -1.$$

and right hand derivative at $x = 0$ is

$$\text{R.H.D.} = \lim_{h \to 0^+} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0^+} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0^+} \frac{|h| - 0}{h}$$

$$= \lim_{h \to 0^+} \frac{h}{h} = 1.$$

So, left hand derivative ≠ right hand derivative.

Hence, the function $f(x) = |x|$ is not differentiable at $x = 0$.

This shows that every differentiable function is continuous but every continuous function may not be differentiable at the given point.

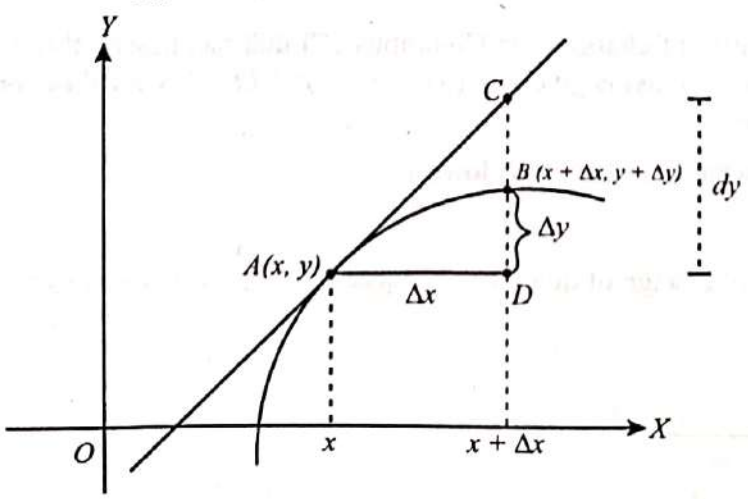
1.4.3 Differentials

The notion of differential is approximation to very small quantities. Let $y = f(x)$, be a function of independent variable x.

As we can choose independent variable at our convenience, we take Δx as small change in x, it is also represented by dx i.e., $dx = \Delta x$ is called differential in x i.e., small change in x is same as actual change.

Due to this small change $\Delta x = dx$ in x there is actual change Δy in y and is given by $\Delta y = f(x + \Delta x) - f(x)$ which is measured along the curve of $y = f(x)$ as shown in figure. This actual change in y along the curve can be approximated (differential) by the change in y along tangent and is denoted by dy and thus we can write

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x).$$



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Therefore, the approximate change dy in y (approximated by the change along tangential line for the change in y along curve) is given by

$$dy = f'(x)\,dx.$$

In the figure AD represents change Δx in x, which is same for both changes in y along curve Δy and along the tangent dy. We take $\Delta x = dx$, for independent variable.

But, BD represents Δy along the curve which is approximated by $CD = dy$ along the tangent. Note that $\Delta y \neq dy$ but given by $dy = f'(x)\,dx$.

1.4.2 Physical meaning of derivative

Here we discuss the derivative (limit of average change of rate) as instantaneous rate of change. Let $y = f(x)$ be a function and let x changes from x_1 to x_2 such that $\Delta x = x_2 - x_1$.

Then, corresponding change in y is $\Delta y = f(x_2) - f(x_1)$.

Then, $\frac{\Delta y}{\Delta x}$ gives the average rate of change of y with respect to x over the interval $[x_1, x_2]$. We now change this average rate of change into instantaneous rate of change by letting x_2 approaches to x_1 or by letting Δx approaches to 0. Consequently, $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ gives the instantaneous rate of change at $x = x_1$.

But, $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$ = derivative of $f(x)$ at $x = x_1$.

Hence, we can conclude that instantaneous rate of change of y with respect to x is measured by derivative of y with respect to x.

The rate of change of a quantity with respect to another quantity is regarded as the physical interpretation of derivative. It is needed everywhere to quantify (measure) the rate of change, since everything changes over the time. The rate of change of y (distance) with respect to x (time), i.e., $\left(\frac{dy}{dx}\right)$ gives the velocity of moving particles.

Rates of change occur in all the fields of knowledge from engineering to social sciences. In engineering an engineer needs to find the rate at which water flows into (or out) of a reservoir. Sometimes they need to calculate the rate of change of pressure with respect to height.

Example 7. The quantity of charge Q in Columbus (C) that has passed through a point in a wire up to time t (in seconds) is given by $Q(t) = t^3 - 2t^2 + 6t + 2$. Calculate the current when:

i. $t = 0.5$ ii. $t = 1$

iii. At what time current is lowest.

Solution:

We note that rate of change of quantity of charge Q is $\frac{dQ}{dt}$ which gives the current at time t.

$$\therefore \text{ Current } = \frac{dQ(t)}{dt}$$

$$= \frac{d(t^3 - 2t^2 + 6t + 2)}{dt}$$

$$= 3t^2 - 4t + 6.$$

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i. At $t = \frac{1}{2}$ second, Current $(I) = 3 \cdot \left(\frac{1}{2}\right)^2 - 4 \cdot \left(\frac{1}{2}\right) + 6$

$= \frac{3}{4} - 2 + 6$

$= \frac{3}{4} + 4$

$= \frac{19}{4}$ C/sec.

ii. At $t = 1$ second, Current $(I) = 3 \cdot (1)^2 - 4 \cdot 1 + 6$

$= 3 - 4 + 6 = 5$ C/sec.

iii. For the time of lowest current, we have

Current $I = \frac{dQ(t)}{dt}$

$= 3t^2 - 4t + 6.$

We now find minimum value of I. For this solve $\frac{dI}{dt} = 6t - 4 = 0$

$\therefore \quad t = \frac{2}{3}, \ \frac{d^2I}{dt^2} = 6 > 0.$

So by second derivative test, I is lowest at $t = \frac{2}{3}$ sec.

Example 8. Show that $f(x) = \begin{cases} x^2 + 3 & \text{for } x < 1 \\ 3x + 1 & \text{for } x \geq 1 \end{cases}$ is not differentiable at $x = 1$.

Solution:

The given function is $f(x) = \begin{cases} x^2 + 3 & \text{for } x < 1 \\ 3x + 1 & \text{for } x \geq 1 \end{cases}$ and given point is $a = 1$.

Now, left hand derivative at $x = 1$ is

$$\text{L.H.D.} = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

$$= \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \to 0} \frac{\{(1-h)^2 + 3\} - 4}{-h}$$

$$= \lim_{h \to 0} \frac{1 - 2h + h^2 + 3 - 4}{-h}$$

$$= \lim_{h \to 0} \frac{-h(2-h)}{-h}$$

$$= 2$$

$$\text{Again, R.H.D.} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{\{3(1+h) + 1\} - 4}{h}$$

$$= \lim_{h \to 0} \frac{3 + 3h + 1 - 4}{h}$$

$$= 3$$

Hence, L.H.D. ≠ R.H.D.

Therefore the function is not differentiable at $x = 1$.

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Example 9. The function is defined by $f(x) = \begin{cases} x^2 - 2 & \text{for } x \le 2 \\ x^2 - 4x + 6 & \text{for } x > 2 \end{cases}$

Show that $f(x)$ is continuous at $x = 2$ but not differentiable at $x = 2$.

Solution:

The given function is $f(x) = \begin{cases} x^2 - 2 & \text{for } x \le 2 \\ x^2 - 4x + 6 & \text{for } x > 2 \end{cases}$

For the continuity at $x = 2$

Left hand limit $= \lim_{x \to 2} f(x)$

$= \lim_{x \to 2} (x^2 - 2)$

$= 4 - 2$

$= 2$

Right hand limit $= \lim_{x \to 2^+} f(x)$

$= \lim_{x \to 2^+} (x^2 - 4x + 6)$

$= 4 - 8 + 6$

$= 2$

Also, $f(2) = 4 - 2 = 2$

Since L.H.L = R.H.L = $f(2)$, the function is continuous at $x = 2$.

Again for differentiability at $x = 2$,

L.H.D. $= \lim_{h \to 0} \dfrac{f(a-h) - f(a)}{-h}$

$= \lim_{h \to 0} \dfrac{f(2-h) - f(2)}{-h}$

$= \lim_{h \to 0} \dfrac{\{(2-h)^2 - 2\} - 2}{-h}$

$= \lim_{h \to 0} \dfrac{4 - 4h + h^2 - 4}{-h} = 4$

and R.H.D. $= \lim_{h \to 0^+} \dfrac{f(a+h) - f(a)}{h}$

$= \lim_{h \to 0^+} \dfrac{f(2+h) - f(2)}{h}$

$= \lim_{h \to 0^+} \dfrac{(2+h)^2 - 4(2+h) + 6 - 2}{h}$

$= \lim_{h \to 0^+} \dfrac{4 + 4h + h^2 - 8 - 4h + 4}{h}$

$= \lim_{h \to 0^+} \dfrac{h^2}{h} = 0.$

Hence, L.H.D $\neq$ R.H.D.

Therefore the given function is continuous at $x = 2$ but is not differentiable at that point.

Example 10. Examine the continuity and differentiability of the function at (i) $x = 0$ and (ii) $x = \dfrac{\pi}{2}$.

$$f(x) = \begin{cases} 1 & \text{when } x \in (-\infty, 0) \\ 1 + \sin x & \text{when } x \in \left[0, \dfrac{\pi}{2}\right) \\ 2 + \left(x - \dfrac{\pi}{2}\right)^2 & \text{when } x \in \left[\dfrac{\pi}{2}, \infty\right) \end{cases}$$

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Solution:

The function is $f(x) = \begin{cases} 1 & \text{when } x \in (-\infty, 0) \\ 1 + \sin x & \text{when } x \in \left[0, \frac{\pi}{2}\right) \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & \text{when } x \in \left[\frac{\pi}{2}, \infty\right) \end{cases}$

i. Fo

L.H.

$= 1.$

R.H.L. $= \lim_{x \to 0^+} f(x)$

$= \lim_{x \to 0^+} (1 + \sin x)$

$= 1.$

Functional value at $x = 0$ is $f(0) = 1 + \sin 0 = 1$.

Since L.H.L. = R.H.L. = functional value, the functio ''ous a'

Now for differentiability at $x = 0$

L.H.D. $= \lim_{h \to 0^-} \frac{f(a-h) - f(a)}{-h}$

$= \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$

$= \lim_{h \to 0^-} \frac{1-1}{-h} = 0$

and R.H.D. $= \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$

$= \lim_{h \to 0^+} \frac{f(0+h) - f(0)}{h}$

$= \lim_{h \to 0^+} \frac{1 + \sin(0+h) - (1 + \sin 0)}{h}$

$= \lim_{h \to 0^+} \frac{\sin h}{h} = 1$

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Since L.H.D. ≠ R.H.D., the function is not differentiable at $x = 0$.

ii. For continuity at $x = \frac{\pi}{2}$,

L.H.L. $= \lim_{x \to \frac{\pi}{2}^-} (1 + \sin x) = 2$

R.H.L. $= \lim_{x \to \frac{\pi}{2}^+} \left\{2 + \left(x - \frac{\pi}{2}\right)^2\right\} = 2$

Functional value at $x = \frac{\pi}{2}$ is $f\left(\frac{\pi}{2}\right) = 2$.

Since L.H.L. = R.H.L. = the functional value at $x = \frac{\pi}{2}$.

Hence, the function is continuous at $x = \frac{\pi}{2}$.

Now for differentiability at $x = \frac{\pi}{2}$

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$$\text{L.H.D.} = \lim_{h\to 0}\frac{f(a-h)-f(a)}{-h}$$

$$= \lim_{h\to 0}\frac{f\left(\frac{\pi}{2}-h\right)-f\left(\frac{\pi}{2}\right)}{-h}$$

$$= \lim_{h\to 0}\frac{1+\sin\left(\frac{\pi}{2}-h\right)-\left[2+\left(\frac{\pi}{2}-\frac{\pi}{2}\right)^2\right]}{-h}$$

$$= \lim_{h\to 0}\frac{1+\cos h-2}{-h}$$

$$= \lim_{h\to 0}\frac{\cos h-1}{-h}$$

$$= \lim_{h\to 0}\frac{2\sin^2\frac{h}{2}}{h}$$

$$= \lim_{h\to 0}\frac{2\sin^2\frac{h}{2}}{\left(\frac{h}{2}\right)^2}\times\frac{1}{2}h = 0$$

$$\text{R.H.D.} = \lim_{h\to 0}\frac{f\left(\frac{\pi}{2}+h\right)-f\left(\frac{\pi}{2}\right)}{h}$$

$$= \lim_{h\to 0}\frac{2+\left(\frac{\pi}{2}+h-\frac{\pi}{2}\right)^2-\left[2+\left(\frac{\pi}{2}-\frac{\pi}{2}\right)^2\right]}{h}$$

$$= \lim_{h\to 0}\frac{h^2}{h} = 0$$

Since L.H.D. = R.H.D., the function is differentiable at $x = \frac{\pi}{2}$.

Example 11. Show that the function $f(x) = |x| + |x-1|$ is continuous at $x = 1$ but is not differentiable at that point.

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Solution:

The given function is $f(x) = |x| + |x-1|$.

Now for continuity, the limiting value of $f(x)$ at $x = 1$ is

$\lim_{x\to 1} f(x) = \lim_{x\to 1} |x| + |x-1| = 1 + 1 - 1 = 1$

Also, functional value of $f(x)$ at $x = 1$ is $f(1) = |1| + |1-1| = 1$.

Since $\lim_{x\to 1} f(x) = f(1)$, the function is continuous at $x = 1$.

Now for differentiability

$$\text{L.H.D.} = \lim_{h\to 0}\frac{f(a-h)-f(a)}{-h}$$

$$= \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h}$$

$$= \lim_{h\to 0}\frac{|1-h|+|1-h-1|-\{|1|+|1-1|\}}{-h}$$

$$= \lim_{h\to 0}\frac{1-h+h-1}{-h} = 0$$

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and R.H.D. $= \lim_{h \to 0^+} \frac{f(a+h)-f(a)}{h}$

$= \lim_{h \to 0^+} \frac{f(1+h)-f(1)}{h}$

$= \lim_{h \to 0^+} \frac{|1+h|+|1+h-1|-\{|1|+|1-1|\}}{h}$

$= \lim_{h \to 0^+} \frac{1+h+h-1}{h}$

$= \lim_{h \to 0^+} \frac{2h}{h}$

$= 2.$

Thus, L.H.D. ≠ R.H.D. Therefore the function is not differentiable at $x = 1$.
Hence the function is continuous at $x = 1$ but is not differentiable at that point.

Exercise 1.2

1. Show that the given functions are differentiable at the given point. Find their values
   i. $f(x) = 3x^3$ at $x = 2$ ii. $f(x) = 4x^2 - x$ at $x = 1$
   iii. $f(x) = x^3 - 4x^2 - 1$ at $x = -1$

2. Discuss whether $f(x)$ is continuous and differentiable at the origin for $f(x) = \begin{cases} 2+x & \text{if } x \geq 0 \\ 2-x & \text{if } x < 0 \end{cases}$.

3. If $f(x) = \begin{cases} x & \text{for } x < 1 \\ 2-x & \text{for } 1 \leq x < 2 \\ -2+3x-x^2 & \text{for } x \geq 2 \end{cases}$
   Then check the continuity and differentiability at
   i. $x = 1$ ii. $x = 2$

4. A function $f(x)$ is defined as follows
   $f(x) = \begin{cases} 4-x^2 & \text{for } x < 2 \\ x-2 & \text{for } x \geq 2 \end{cases}$
   Show that $f(x)$ is continuous at $x = 2$ but is not differentiable at $x = 2$.

5. Let a function be defined as $f(x) = \begin{cases} 5x-4 & \text{for } 0 < x \leq 1 \\ 4x^2-3x & \text{for } 1 < x < 2 \\ 3x+4 & \text{for } x \geq 2 \end{cases}$
   Show that $f(x)$ is continuous as $x = 1$ and 2. Also prove that $f'(x)$ exists for $x = 1$ but not at $x = 2$.

6. If $f(x) = \begin{cases} 3+2x & \text{for } -\frac{3}{2} < x \leq 0 \\ 3-2x & \text{for } 0 < x < \frac{3}{2} \end{cases}$
   Show that $f(x)$ is continuous at $x = 0$ but is not differentiable at that point.

7. Show that $f(x) = \begin{cases} 2+x & \text{if } x \leq 0 \\ 2-x & \text{if } x > 0 \end{cases}$ is continuous but not differentiable at the origin.

8. Show that $f'\left(\frac{1}{2}\right)$ does not exist if
   $f(x) = \begin{cases} x & \text{for } 0 \leq x < \frac{1}{2} \\ 1-x & \text{for } \frac{1}{2} \leq x \leq 1 \end{cases}$

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9. If $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2 - x & \text{for } 1 \le x \le 2 \\ x - \frac{x^2}{2} & \text{for } x > 2 \end{cases}$

Show that

i. $f(x)$ is continuous at $x = 1$ and $x = 2$

ii. $f(x)$ is not differentiable at $x = 1$ but is differentiable at $x = 2$.

Answers

1. i. 36 ii. 7 iii. 11
2. continuous, not differentiable
3. i. continuous, not differentiable ii. continuous and differentiable

1.5 Indeterminate forms and their limits

If a function takes the limit in the following forms, we say that the limit is indeterminate.

i. Indeterminate q otients: $\lim_{x \to a} \frac{f(x)}{g(x)}$ takes the forms $\left(\frac{0}{0}\right)$ or $\left(\frac{\infty}{\infty}\right)$.

ii. Indeterminate products: $\lim_{x \to a} f(x)\, g(x)$ takes the form $(0 \cdot \infty)$.

rminate differences: $\lim_{x \to a} [f(x) - g(x)]$ takes the form $[\infty - \infty]$.

rminate powers: $\lim_{x \to a} [f(x)]\, g(x)$ takes the forms 0^0 or ∞^0 or 1^∞.

rivative to evaluate indeterminate limits

. Indeterm. e quo

L' Hospital R

Statemer .) are differentiable functions and $g'(x) \neq 0$ in an open interval that contains $x -$) = 0, im $_{\to a} g(x) = 0$, i.e., $\lim_{x \to a} \frac{f(x)}{g(x)} = \left(\frac{0}{0}\right)$, then

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$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

Proof: The functions $f(x)$ and $g(x)$ are differentiable at $x = a$, so $\lim_{x \to a} f'(x)$ and $\lim_{x \to a} g'(x)$ exists and

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}$$

$$= \lim_{x \to a} \frac{[f(x) - f(a)]/x - a}{[g(x) - g(a)]/x - a}$$

$$= \lim_{x \to a} \frac{f(x)}{g(x)} \text{ as } f(a)$$

$$= g(a) = 0.$$

$$\therefore \lim_{x \to a} \frac{f'(x)}{g'(x)} = \lim_{x \to a} \frac{f(x)}{g(x)}.$$

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Remark:

1. We can apply directly L' Hospital's rule if $\lim_{x \to a} \frac{f(x)}{g(x)} = \left(\frac{\infty}{\infty}\right)$ as $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{1/g(x)}{1/f(x)}$ is of the form $\left(\frac{0}{0}\right)$.

2. We can apply L' Hospital's rule repeatedly as long as we have $\left(\frac{0}{0}\right)$ or $\left(\frac{\infty}{\infty}\right)$ forms. We see that L' Hospital's rule is directly applicable for indeterminate quotients.

ii. Indeterminate products

Indeterminate product $(0 \cdot \infty)$ can be evaluated by using L' Hospital's rule by reducing into $\frac{0}{0}$ or $\frac{\infty}{\infty}$

i.e., $\lim_{x \to a} [f(x)\, g(x)]$ $\quad (0 \cdot \infty)$

$= \lim_{x \to a} \frac{f(x)}{1/g(x)}$ $\quad \left(\frac{0}{0}\right)$

For illustration see example 15.

iii. Indeterminate differences

Indeterminate difference $(\infty - \infty)$ can be evaluated by using L' Hospital's rule by reducing it into $\frac{0}{0}$

i.e., $\lim_{x \to a} [f(x) - g(x)]$ $\quad (\infty - \infty)$

$= \lim_{x \to a} \frac{\left[\frac{1}{g(x)} - \frac{1}{f(x)}\right]}{\frac{1}{f(x)\, g(x)}}$ $\quad \left(\frac{0}{0}\right)$

For illustration see example 16.

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iv. Indeterminate powers

Indeterminate powers 0^0, ∞^0, 1^∞ can be reduced into $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form by using logarithms.

i.e., $\lim_{x \to a} [f(x)]^{g(x)}$, $(0^0, \infty^0, 1^\infty)$

We use logarithm as follows.

Let, $y = [f(x)]^{g(x)}$

Taking logarithm on both sides, we get

$\log y = g(x) \log f(x)$

Taking limit $x \to a$ on both sides

$\lim_{x \to a} \log y = \lim_{x \to a} g(x) \log f(x)$

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We can take $g(x)$ in the denominator as $\frac{1}{g(x)}$ or $\log f(x)$ as $\frac{1}{\log f(x)}$ whichever makes the right hand side as $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Here, we use the fact that $\lim_{x \to a} \log y = \log \left(\lim_{x \to a} y \right)$.

For illustration see example 17.

Example 12. Evaluate: $\lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^{bx}$. $\left[\text{Recall that } e = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{x}\right]$.

Solution:

Given, $\lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^{bx} = \lim_{x \to \infty} \left[\left(1 + \frac{1}{x/a}\right)^{x \cdot a}\right]^{ab} = e^{ab}$

$\therefore \quad \lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^{bx} = e^{ab}$.

Example 13. Evaluate: $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$.

Solution:

Given, $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} \quad \left(\frac{0}{0}\right)$

$\therefore \quad \lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \to 0} \frac{e^x - 1}{2x} \quad \left(\frac{0}{0}\right)$

$= \lim_{x \to 0} \frac{e^x}{2} = \frac{1}{2}$

$\therefore \quad \lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}$.

Example 14. Evaluate: $\lim_{x \to \infty} \frac{\log x}{\sqrt{x}}$

Solution:

Given, $\lim_{x \to \infty} \frac{\log x}{\sqrt{x}} \quad \left(\frac{\infty}{\infty}\right)$

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$\therefore \quad \lim_{x \to \infty} \frac{\log x}{\sqrt{x}} = \lim_{x \to \infty} \frac{1/x}{1/2\sqrt{x}}$

$= \lim_{x \to \infty} \frac{2\sqrt{x}}{x}$

$= \lim_{x \to \infty} \frac{2}{\sqrt{x}}$

$= 0.$

$\therefore \quad \lim_{x \to \infty} \frac{\log x}{\sqrt{x}} = 0.$

Example 15. Evaluate: $\lim_{x \to \infty} x^3 e^{-x^2}$.

Solution:

Given, $\lim_{x \to \infty} x^3 e^{-x^2} \quad [\infty \cdot 0] \quad [\because e^{-\infty} = 0]$

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$$\therefore \lim_{x\to\infty} x^3e^{-x^2} = \lim_{x\to\infty} \frac{x^3}{e^{x^2}} \qquad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{3x^2}{2xe^{x^2}} \qquad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{6x}{2[e^{x^2} + 2x^2e^{x^2}]}$$

$$= \lim_{x\to\infty} \frac{6}{2[2xe^{x^2} + 4xe^{x^2} + 4x^3e^{x^2}]}$$

$$= \frac{6}{\infty} = 0$$

$$\therefore \lim_{x\to\infty} x^3e^{-x^2} = 0.$$

Example 16. Evaluate: $\lim_{x\to 1}\left[\frac{x}{x-1} - \frac{1}{\log x}\right]$.

Solution:

Given, $\lim_{x\to 1}\left[\frac{x}{x-1} - \frac{1}{\log x}\right]$ $[\infty - \infty]$

$$\therefore \lim_{x\to 1}\left[\frac{x}{x-1} - \frac{1}{\log x}\right] = \lim_{x\to 1}\left[\frac{x\log x - (x-1)}{(x-1)\log x}\right] \qquad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1}\left[\frac{\log x + \frac{x}{x} - 1}{\log x + \left(\frac{x-1}{x}\right)}\right] \qquad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1}\left[\frac{x\log x}{x\log x + (x-1)}\right] \qquad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{[\log x + x/x]}{[\log x + x/x + 1]}$$

$$= \lim_{x\to 1}\left(\frac{\log x + 1}{\log x + 2}\right) = \frac{1}{2}.$$

$$\therefore \lim_{x\to 1}\left[\frac{x}{x-1} - \frac{1}{\log x}\right] = \frac{1}{2}.$$

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Example 17. Evaluate: $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$.

Solution:

Given, $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$ (1^∞)

Let $y = \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$.

Taking logarithm on both sides

$$\log y = \frac{1}{x^2}\log\left(\frac{\sin x}{x}\right)$$

Taking limit as $x \to 0$, we get

$$\therefore \lim_{x\to 0} \log y = \lim_{x\to 0} \frac{\log\left(\frac{\sin x}{x}\right)}{x^2}$$

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$$= \lim_{x\to 0} \frac{\frac{x}{\sin x}\left(\frac{x\cos x - \sin x}{x^2}\right)}{2x}$$

$$= \lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

$$= \lim_{x\to 0} \frac{x\cos x - \sin x}{2x^3\left(\frac{\sin x}{x}\right)}$$

$$= \lim_{x\to 0} \frac{x\cos x - \sin x}{2x^3} \qquad \left(\frac{0}{0}\right) \quad \left[\because \lim_{x\to 0}\frac{\sin x}{x} = 1\right]$$

$$= \lim_{x\to 0} \frac{\cos x - x\sin x - \cos x}{6x^2} = -\frac{1}{6}\lim_{x\to 0}\frac{\sin x}{x}$$

$$\lim_{x\to 0} \log y = -\frac{1}{6}$$

or, $\log\left(\lim_{x\to 0} y\right) = -\frac{1}{6}$

or, $y = e^{-\frac{1}{6}}$

$\therefore \lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = e^{-\frac{1}{6}}$.

Example 18. Evaluate: $\lim_{x\to 0}(\tan 2x)^x$.

Solution:

Given, $\lim_{x\to 0}(\tan 2x)^x$ $[0^0]$

Let $y = (\tan 2x)^x$

Taking logarithm on both sides, we get

$\log y = x\log\tan 2x$

$\therefore \lim_{x\to 0}\log y = \lim_{x\to 0} x\log(\tan 2x)$ $[0\cdot\infty]$

$$= \lim_{x\to 0}\frac{\log(\tan 2x)}{1/x} \qquad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to 0}\frac{\frac{1}{\tan 2x}\sec^2 2x\,(2)}{\left(-\frac{1}{x^2}\right)}$$

$$= \lim_{x\to 0}\left[\frac{-2x^2}{\sin 2x\cos 2x}\right] \qquad \left(\frac{0}{0}\right)$$

$$= -4\lim_{x\to 0}\frac{x^2}{2\sin 2x\cos 2x}$$

$$= -4\lim_{x\to 0}\frac{x^2}{\sin 4x}$$

$$= -\lim_{x\to 0}\left(\frac{4x}{\sin 4x}\right)x = (-1)\cdot 0 = 0$$

Thus, $\log y = 0$

or, $y = e^0 = 1$

$\therefore \lim_{x\to 0}(\tan 2x)^x = 1$.

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Exercise 1.3

1. Evaluate the limit.

a. $\lim_{x\to 1}\frac{\log x}{x-1}$

b. $\lim_{x\to\infty}\frac{e^x}{x^2}$

c. $\lim_{x\to 0}\frac{\tan x - x}{x^3}$

d. $\lim_{x\to 0}\frac{\tan x - x}{x - \sin x}$

e. $\lim_{x\to 0}\frac{xe^x-(1+x)\log(1+x)}{x^2}$

f. $\lim_{t\to 0}\frac{e^{2t}-1}{\sin t}$

g. $\lim_{x\to\frac{\pi}{2}}\frac{\cos x}{1-\sin x}$

h. $\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)$

i. $\lim_{x\to 0}\log_{\tan x}(\tan 2x)$

j. $\lim_{x\to 0}\left(\frac{1}{x^2}-\cot^2 x\right)$

k. $\lim_{x\to 0}\left(\frac{1}{x^2}\right)^{\tan x}$

l. $\lim_{x\to\infty}x^{\frac{1}{x}}$

m. $\lim_{x\to 0}(\cot^2 x)^{\tan x}$

n. $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$

o. $\lim_{x\to 0}(\cos x)^{\cot^2 x}$

p. $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\tan x}$

q. $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}}$

Answers

1. a. 1 b. ∞ c. $\frac{1}{3}$ d. 2 e. $\frac{1}{2}$ f. 2
g. $-\infty$ h. $\frac{1}{2}$ i. 1 j. $\frac{2}{3}$ k. 1 l. 1
m. 1 n. 1 o. $e^{-\frac{1}{2}}$ p. 1 q. $e^{\frac{1}{3}}$

1.6 Higher order derivatives

The fundamental concept of the infinitesimal calculus, was used by Gottfried Wilhelm Leibniz and other mathematicians in the seventeenth century. The study was concerned with the conceptual and technical aspects of first-order and higher-order differentials on the development of the infinitesimal calculus.

Gottfried Wilhelm Leibniz
(1646-1716)

To understand the higher-order derivatives, first-order derivatives is to be understood which is already discussed in the earlier sections. The higher-order derivative is the extension of first order derivative. The process of higher order differentiation includes the successive differentiation of the given function with respect to the independent variable.

If $y = f(x)$ be a derivable function of x, its derivative denoted by $f'(x)$ (or $\frac{dy}{dx}$ or y' or y_1) and if $f'(x)$ is again differentiable, we denote the derivative of $f'(x)$ by $f''(x)$. Similarly, $f(x)$, $f'(x)$, $f''(x)$, . . . , $f^n(x)$, each of which is the derivative of the previous one. Proceeding in this way, we denote y_n or $\frac{d^n f(x)}{dx^n}$ or $f^n(x)$ to denote the n^{th} order derivative. We use notations

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$\frac{dy}{dx}$ for $f'(x)$, $\frac{d^2y}{dx^2}$ for $f''(x)$ ... $\frac{d^ny}{dx^n}$ for $f^n(x)$,

y_1 for $\frac{dy}{dx}$, y_2 for $\frac{d^2y}{dx^2}$ ... y_n for $\frac{d^ny}{dx^n}$

It is to be noted that $\frac{d^ny}{dx^n} = \frac{d}{dx}\left(\frac{d^{n-1}y}{dx^{n-1}}\right) = \frac{d^2}{dx^2}\left(\frac{d^{n-2}y}{dx^{n-2}}\right)$.

We prefer to use the notation $y_1, y_2, \ldots, y_n$ to denote the successive higher order derivative.

Remark:

1. Higher order derivatives are used for higher order approximations of a function.
2. Physically, the second order derivative gives the acceleration and third order derivative gives jerk (rate of change of acceleration with respect to time) of a body in motion.

1.6.1 Higher order derivatives of some special functions

Example 19. Find y_n, if $y = x^n$, where n is positive integer.

Solution:

Here, $y = x^n$.

Differentiating both sides with respect to x successively, we get

$y_1 = nx^{n-1}$ $\quad y_2 = n(n-1)x^{n-2}$

$y_3 = n(n-1)(n-2)x^{n-3}$

Proceeding in this way up to n steps, we get

$y_n = n(n-1)(n-2) \ldots [n-(n-1)]x^{n-n}$

$= n(n-1)(n-2) \ldots 1x^0$.

$\therefore$ $y_n = n!$

Hence, $y_n = D^n(x^n) = n!$

Note: $y_{n+1} = D^{n+1}(x^n) = 0$.

Example 20. Find y_n, if $y = (ax + b)^m$, where m is a positive integer greater than n.

Solution:

Here, $y = (ax + b)^m$.

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Differentiating successively with respect to x, we get

$y_1 = ma(ax + b)^{m-1}$ $\quad y_2 = m(m-1)a^2(ax + b)^{m-2}$

$y_3 = m(m-1)(m-2)a^3(ax + b)^{m-3}$

Proceeding in this way up to n steps, we get

$y_n = m(m-1)(m-2)\cdots(m-n+1)a^n(ax + b)^{m-n}$

$= \frac{1 \times 2 \times 3 \times \cdots \times (m-n)}{1 \times 2 \times 3 \times \cdots \times (m-n)} \times m(m-1)(m-2)\cdots(m-n+1)a^n(ax + b)^{m-n}$

$= \frac{m!}{(m-n)!}a^n(ax + b)^{m-n}$,

where m is a positive integer greater than n.

Thus, $y_n = D^n(ax + b)^m = \frac{m!}{(m-n)!}a^n(ax + b)^{m-n}$.

Note: If $m < n$, then $y_n = D^n[(ax + b)^m] = 0$.

If $m = n$, then $y_n = D^n[(ax + b)^m] = n!\,a^n$.

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Example 21. Find y_n, if $y = e^{ax}$.

Solution:

Here, $y = e^{ax}$.

Differentiating successively with respect to x, we get

$y_1 = ae^{ax}, y_2 = a^2e^{ax}, y_3 = a^3e^{ax}, \ldots, y_n = a^ne^{ax}$.

Thus, $y_n = D^n(e^{ax}) = a^ne^{ax}$

Note: i. If $y = e^x$, then $y_n = D^n(e^x) = e^x$.

ii. If $y = a^x$, then $y = e^{x \log_e a}$

$\therefore \quad y_n = D^n(a^x) = (\log_e a)^n a^x$.

Example 22. Find y_n, if $y = \frac{1}{x+a}$.

Solution:

Here, $y = (x + a)^{-1}$

Differentiating successively up to n steps, we get

$y_1 = -1(x + a)^{-2}$

$y_2 = (-1)(-2)(x + a)^{-3} = (-1)^2 2! (x + a)^{-3}$

$y_3 = (-1)(-2)(-3)(x + a)^{-4} = (-1)^3 3! (x + a)^{-4}$

Proceeding in this way up to n steps, we get

$y_n = (-1)^n n! (x + a)^{-(n+1)} = \frac{(-1)^n n!}{(x+a)^{n+1}}$.

$\therefore \quad y_n = D^n\left(\frac{1}{x+a}\right) = \frac{(-1)^n n!}{(x+a)^{n+1}}$.

Note: If $y = \frac{1}{(ax+b)^m}$, then $y_n = \frac{(-1)^n a^n (m+n-1)!}{(m-1)! (ax+b)^{m+n}}$.

Example 23. Find y_n if $y = \sin x$.

Solution:

Given, $y = \sin x$ .....(i)

Differentiating successively with respect to x, we get

$y_1 = \cos x = \sin\left(\frac{\pi}{2} + x\right)$

$y_2 = \cos\left(\frac{\pi}{2} + x\right) = \sin\left(\frac{2\pi}{2} + x\right)$

$y_3 = \cos\left(\frac{2\pi}{2} + x\right) = \sin\left(\frac{3\pi}{2} + x\right)$

$\vdots$

$y_n = \sin\left(\frac{n\pi}{2} + x\right)$

$\therefore \quad y_n = \sin\left(\frac{n\pi}{2} + x\right)$

Note: If $y = \cos x$, then $y_n = \cos\left(\frac{n\pi}{2} + x\right)$.

Example 24. Find y_n, if $y = \log(x + a)$.

Solution:

Here, $y = \log(x + a)$

Differentiating (i) successively with respect to x, we get

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$y_1 = \frac{1}{x+a} = (x+a)^{-1}$,

$y_2 = (-1)(x+a)^{-2}$

$y_3 = (-1)(-2)(x+a)^{-3} = \frac{(-1)^2 2!}{(x+a)^3}$

Proceeding in this way up to n steps, we get

$y_n = \frac{(-1)^{n-1}(n-1)!}{(x+a)^n}$

Note: If $y = \log(ax+b)$, then $y_n = \frac{(-1)^{n-1}(n-1)!\, a^n}{(ax+b)^n}$.

Example 25. Find y_n, if $y = \sin(ax+b)$.

Solution:

Here, $y = \sin(ax+b)$

Differentiating successively with respect to x, we get

$y_1 = a\cos(ax+b) = a\sin\left(\frac{\pi}{2} + (ax+b)\right)$

$y_2 = a^2\cos\left(\frac{\pi}{2} + (ax+b)\right) = a^2\sin\left(\frac{2\pi}{2} + (ax+b)\right)$

$y_3 = a^3\cos\left(\frac{2\pi}{2} + (ax+b)\right) = a^3\sin\left(\frac{3\pi}{2} + (ax+b)\right)$

Proceeding in this way up to n steps, we get

$y_n = a^n\sin\left(\frac{n\pi}{2} + (ax+b)\right)$.

$\therefore \quad y_n = D^n[\sin(ax+b)] = a^n\sin\left(\frac{n\pi}{2} + ax + b\right)$

Note: If $y = \cos(ax+b)$, then

$y_n = D^n[\cos(ax+b)] = a^n\cos\left(\frac{n\pi}{2} + ax + b\right)$.

Example 26. Find y_n, if $y = e^{ax}\sin bx$.

Solution:

Here, $y = e^{ax}\sin bx$

Differentiating successively with respect to x, we get

$y_1 = ae^{ax}\cdot\sin bx + be^{ax}\cos bx = e^{ax}(\sin bx \cdot a + \cos bx \cdot b)$

Put $a = r\cos\theta$ and $b = r\sin\theta$. So that $r = \sqrt{a^2+b^2}$ and $\theta = \tan^{-1}\left(\frac{b}{a}\right)$.

$\therefore \quad y_1 = e^{ax}[r\sin(bx+\theta)]$

or, $y_1 = re^{ax}\sin(bx+\theta)$.

Then, $y_2 = re^{ax}[\sin(bx+\theta)\cdot a + \cos(bx+\theta)\cdot b]$

Again, putting $a = r\cos\theta$ and $b = r\sin\theta$

$\therefore \quad y_2 = re^{ax}[\sin(bx+\theta)\cdot r\cos\theta + \cos(bx+\theta)\cdot r\sin\theta]$

$= r^2e^{ax}\sin(bx+\theta+\theta)$

$= r^2e^{ax}\sin(bx+2\theta)$

Similarly, $y_3 = r^3e^{ax}\sin(bx+3\theta)$

Proceeding in this way up to n steps, we get

$y_n = r^ne^{ax}\sin(bx+n\theta) = (a^2+b^2)^{n/2}e^{ax}\sin\left(bx + n\tan^{-1}\left(\frac{b}{a}\right)\right)$

Thus, $y_n = D^n(e^{ax}\sin bx) = (a^2+b^2)^{n/2}e^{ax}\sin\left(bx + n\tan^{-1}\left(\frac{b}{a}\right)\right)$.

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Example 27. If $y = e^{ax} \cos bx$, find y_n.

Solution:

Let $y = e^{ax} \cos bx$.

Differentiating with respect to x, we get

$y_1 = ae^{ax} \cos bx - be^{ax} \sin bx$

$y_1 = e^{ax} (a \cos bx - b \sin bx)$ .... (i)

Put $a = r\cos\theta$, $b = r\sin\theta$. Then,

$r^2 = a^2 + b^2$ and $\tan\theta = \frac{b}{a}$

i.e., $r = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

So, $y_1 = e^{ax}(r \cos\theta \cos bx - r \sin\theta \sin bx)$

$= re^{ax} (\cos\theta \cos bx - \sin\theta \sin bx)$

$y_1 = re^{ax}\cos(bx + \theta)$ .... (ii)

Again, differentiating with respect to x, we get

$y_2 = r[ae^{ax} \cos(bx + \theta) - be^{ax} \sin(bx + \theta)]$

$= re^{ax}[a \cos(bx + \theta) - b \sin(bx + \theta)]$

$= re^{ax} [r\cos\theta \cos(bx + \theta) - r \sin\theta \sin(bx + \theta)]$

$= r^2e^{ax} [\cos\theta .\cos(bx + \theta) - \sin\theta \sin(bx + \theta)]$

$= r^2e^{ax} \cos(bx + \theta + \theta)$

$y_2 = r^2e^{ax} \cos(bx + 2\theta)$

In general, $y_n = r^n e^{ax}\cos(bx + n\theta)$.

or, $y_n = [(a^2 + b^2)^{1/2}]^n e^{ax}\cos\left(bx + n\tan^{-1}\left(\frac{b}{a}\right)\right)$

$\therefore \quad y_n = (a^2 + b^2)^{n/2}e^{ax} \cos\left(bx + n\tan^{-1}\left(\frac{b}{a}\right)\right)$.

Example 28. Find y_n, if $y = \frac{1}{x^2 + a^2}$.

Solution:

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Here, $y = \frac{1}{x^2 + a^2} = \frac{1}{x^2 - i^2a^2} = \frac{1}{x^2 - (ia)^2} = \frac{1}{(x - ia)(x + ia)}$

Resolving into partial fractions, we get

$y = \frac{1}{2ia}\left[\frac{1}{x - ia} - \frac{1}{x + ia}\right]$

Differentiating up to n steps, we get

$y_n = \frac{1}{2ia}\left[D^n\left(\frac{1}{x - ia}\right) - D^n\left(\frac{1}{x + ia}\right)\right]$

Applying the formula $D^n\left(\frac{1}{x + a}\right) = \frac{(-1)^n n!}{(x + a)^{n+1}}$, we get

$y_n = \frac{1}{2ia}\left[\frac{(-1)^n n!}{(x - ia)^{n+1}} - \frac{(-1)^n n!}{(x + ia)^{n+1}}\right]$

$\therefore \quad y_n = \frac{(-1)^n n!}{2ia}[(x - ia)^{-(n+1)} - (x + ia)^{-(n+1)}]$ .... (i)

Put $x = r\cos\theta$ and $a = r\sin\theta$.

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So that, $r = \sqrt{x^2 + a^2}$ and $\theta = \tan^{-1}\left(\frac{a}{x}\right)$.

Now, $(x - ia)^{-(n+1)} = (r\cos\theta - i\, r\sin\theta)^{-(n+1)}$

$= r^{-(n+1)}[\cos\theta - i\sin\theta]^{-(n+1)}$

$= r^{-(n+1)}[\cos(-\theta) + i\sin(-\theta)]^{-(n+1)}$

Applying De'Moivre's Theorem,

$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, we have

$(x - ia)^{-(n+1)} = r^{-(n+1)}[\cos(n+1)\theta + i\sin(n+1)\theta]$ .... (ii)

Similarly, $(x + ia)^{-(n+1)} = [r\cos\theta + i\, r\sin\theta]^{-(n+1)}$

$= r^{-(n+1)}[\cos\theta + i\sin\theta)]^{-(n+1)}$

$= r^{-(n+1)}[\cos(n+1)\theta - i\sin(n+1)\theta]$ .... (iii)

Substituting the values of (ii) and (iii) in equation (i), we get

$$y_n = \frac{(-1)^n n!}{2ia} r^{-(n+1)}[\cos(n+1)\theta + i\sin(n+1)\theta - \cos(n+1)\theta + i\sin(n+1)\theta]$$

$$= \frac{(-1)^n n!}{2ia} r^{-(n+1)} \cdot 2i\sin(n+1)\theta \quad \ldots\ldots \text{(iv)}$$

Since $r = \frac{a}{\sin\theta}$

$\therefore\quad r^{-(n+1)} = \frac{a^{-(n+1)}}{(\sin\theta)^{-(n+1)}} = \frac{\sin^{n+1}\theta}{a^{n+1}}$.

Hence, (iv) becomes $y_n = \frac{(-1)^n n!}{a} \cdot \frac{\sin^{n+1}\theta}{a^{n+1}}\sin(n+1)\theta$

$= \frac{(-1)^n n!}{a^{n+2}}\sin^{(n+1)}\theta \cdot \sin(n+1)\theta$, where $\theta = \tan^{-1}\left(\frac{a}{x}\right)$.

Thus, $y_n = D^n\left(\frac{1}{x^2 + a^2}\right) = \frac{(-1)^n n!}{a^{n+2}}\sin^{n+1}\theta \cdot \sin(n+1)\theta$,

where $\theta = \tan^{-1}\left(\frac{a}{x}\right)$, which is required result.

Note 1: If $y = \frac{1}{(x+b)^2 + a^2}$, then

$y_n = \frac{(-1)^n n!}{a^{n+2}}\sin^{n+1}\theta \cdot \sin(n+1)\theta$, where $\theta = \tan^{-1}\left(\frac{a}{x+b}\right)$.

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Note 2: If $y = \tan^{-1}x$ $\quad \therefore y_1 = \frac{1}{1+x^2} = \frac{1}{x^2 + 1^2}$

Differentiating up to $(n-1)$ steps, we get

$y_n = D^{n-1}\left(\frac{1}{x^2 + 1^2}\right)$ .... (i)

Using, $D^n\left(\frac{1}{x^2 + a^2}\right) = \frac{(-1)^n n!}{a^{n+2}}\sin^{n+1}\theta \cdot \sin(n+1)\theta$, where $\theta = \tan^{-1}\left(\frac{a}{x}\right)$ and replacing $n-1$ for n, 1 for a in (i), we get

$y_n = (-1)^{n-1}(n-1)!\sin^n\theta \cdot \sin n\theta$, where $\theta = \tan^{-1}\left(\frac{1}{x}\right)$.

Example 29. Find y_n in the following functions:

i. $y = \frac{1}{a-x}$ ii. $y = x^{2n}$

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Solution:

i. Here, $y = \frac{1}{a-x} = (a-x)^{-1}$.

Differentiating with respect to x successively, we get

$y_1 = (-1)(a-x)^{-2}(-1) = 1 \,.\, (a-x)^{-2}$

$y_2 = (-2)(a-x)^{-3}(-1) = 1 \,.\, 2\,(a-x)^{-3} = 2!\,(a-x)^{-3}$

$y_3 = 1 \,.\, 2 \,.\, 3\,(a-x)^{-4} = 3!\,(a-x)^{-4}$

Proceeding in this way up to n steps, we get

$y_n = 1 \,.\, 2 \,.\, 3 \ldots n\,(a-x)^{-(n+1)} = n! \,.\, (a-x)^{-(n+1)}$

$\therefore \quad y_n = \frac{n!}{(a-x)^{n+1}}$.

ii. Here, $y = x^{2n}$

Differentiating with respect to x successively, we get

$y_1 = 2nx^{2n-1}$

$y_2 = 2n(2n-1)\,x^{2n-2}$

$y_3 = 2n(2n-1)(2n-2)\,x^{2n-3}$

Proceeding in this way up to n steps, we get

$y_n = 2n(2n-1)(2n-2) \ldots (2n-n+1)\,x^{2n-n}$

$= 2n(2n-1)(2n-2) \ldots (n+1)\,x^n$

Multiplying numerator and denominator by

$n(n-1)(n-2) \ldots 3.2.1$, we get

$$y_n = 2n(2n-1)(2n-2) \ldots (n+1) \times \frac{n(n-1)(n-2) \ldots 3.2.1}{n(n-1)(n-2) \ldots 3.2.1} x^n$$

Separating even and odd brackets

$$= \frac{[\{2n(2n-2)(2n-4) \ldots 6.4.2\}\ \{(2n-1)(2n-3)(2n-5) \ldots 5.3.1\}]}{n!} x^n$$

$$= \frac{2^n\{n(n-1)(n-2) \ldots 3.2.1\}\ \{(2n-1)(2n-3)(2n-5) \ldots 5.3.1\}}{n!} x^n$$

$$= 2^n \cdot \frac{n!}{n!}(2n-1)(2n-3)(2n-5) \ldots 5.3.1\,x^n$$

$\therefore \quad y_n = 2^n \{1.3.5 \ldots (2n-3)(2n-1)\}\,x^n$.

Example 30. Find y_n in the following functions:

i. $y = \sqrt{x}$ ii. $y = \frac{x^n}{x-1}$

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Solution:

i. Here, $y = \sqrt{x} = x^{1/2}$.

Differentiating successively with respect to x, we get

$$y_1 = \frac{1}{2}x^{-1/2}$$

$$y_2 = \frac{1}{2}\left(-\frac{1}{2}\right)x^{\left(-\frac{1}{2}-1\right)}$$

$$y_3 = \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)x^{\left(-\frac{1}{2}-2\right)}$$

$$= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)x^{\left(-\frac{1}{2}-2\right)}$$

$$= \frac{1}{2^3}(-1)^2\,1.3\,x^{\left(-\frac{1}{2}-2\right)}$$

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Proceeding in this way up to n steps, we get

$$y_n = \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\ldots\left(-\frac{1}{2}-(n-1)\right)x^{\left(-\frac{1}{2}-(n-1)\right)}$$

$$= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\ldots\left(-\frac{1}{2}-n+2\right)x^{\left(-\frac{1}{2}-n+1\right)}$$

$$= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\ldots\left\{-\left(\frac{2n-3}{2}\right)\right\}x^{\left(\frac{1}{2}-n\right)}$$

$$= \frac{1}{2^n}(-1)^{n-1}\,[1.3.5\ldots(2n-3)]\,x^{\left(\frac{1}{2}-n\right)}$$

ii. Here, $y = \dfrac{x^n}{x-1}$.

By division algorithm, $x^n = (x-1)(x^{n-1}+x^{n-2}+\ldots+1)+1$

$$\therefore\quad y = \frac{x^n}{x-1} = \frac{(x-1)(x^{n-1}+x^{n-2}+\ldots+1)+1}{(x-1)}$$

$$= \frac{(x-1)(x^{n-1}+x^{n-2}+\ldots+1)}{(x-1)} + \frac{1}{x-1}$$

$$y = (x^{n-1}+x^{n-2}+\ldots+1) + \frac{1}{x-1} \qquad \ldots\ (i)$$

Differentiating with respect to x up to n steps

$$y_n = 0 + D^n\left(\frac{1}{x-1}\right) \qquad [\because D^m(x^n) = 0, \text{ if } m > n]$$

$$= \frac{(-1)^n .n!}{(x-1)^{n+1}} \qquad \left[\because D^n\left(\frac{1}{x+a}\right) = \frac{(-1)^n n!}{(x+a)^{n+1}}, \text{ and } a = -1\right]$$

Example 31. Find y_n of the function: $y = e^x \sin x.\sin 2x$.

Solution:

We have, $y = \frac{1}{2}e^x(2\sin 2x.\sin x)$

$$= \frac{1}{2}e^x[\cos(2x-x) - \cos(2x+x)]$$

$$= \frac{1}{2}e^x[\cos x - \cos 3x]$$

$$= \frac{1}{2}e^x\cos x - \frac{1}{2}e^x\cos 3x.$$

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Differentiating both sides with respect to x up to n steps, we have

$$y_n = \frac{1}{2}D^n(e^x\cos x) - \frac{1}{2}D^n(e^x\cos 3x) \qquad \ldots\ (i)$$

Applying $D^n(e^{ax}\cos bx) = (a^2+b^2)^{n/2}e^{ax}\cos\left(bx + n\tan^{-1}\left(\frac{b}{a}\right)\right)$

Then, equation (i) becomes

$$y_n = \frac{1}{2}(1^2+1^2)^{n/2}e^x\cos(x + n\tan^{-1}1) - \frac{1}{2}.(1^2+3^2)^{n/2}e^x\cos\left(3x + n\tan^{-1}\left(\frac{3}{1}\right)\right)$$

$$= \frac{e^x}{2}[2^{n/2}\cos(x + n\tan^{-1}1) - 10^{n/2}\cos(3x + n\tan^{-1}3)]$$

$$y_n = \frac{1}{2}e^x\left[2^{n/2}\cos\left(x + \frac{n\pi}{4}\right) - 10^{n/2}\cos(3x + n\tan^{-1}3)\right]$$

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Example 32. Find y_n of the function $y = \frac{x}{x^2 + a^2}$.

Solution:

Here, $y = \frac{x}{x^2 + a^2} = \frac{x}{(x - ai)(x + ai)} = \frac{1}{2}\left[\frac{1}{x + ai} + \frac{1}{x - ai}\right]$

Differentiating up to n steps, we get

$$y_n = \frac{1}{2}\left[D^n\left(\frac{1}{x + ai}\right) + D^n\left(\frac{1}{x - ai}\right)\right] \quad \ldots\ (i)$$

Applying the formula, $D^n\left(\frac{1}{x + a}\right) = \frac{(-1)^n n!}{(x + a)^{n+1}}$. Then (i) becomes

$$y_n = \frac{1}{2}\left[\frac{(-1)^n n!}{(x + ai)^{n+1}} + \frac{(-1)^n n!}{(x - ai)^{n+1}}\right]$$

$$= \frac{1}{2}(-1)^n n! \left[(x + ai)^{-(n+1)} + (x - ai)^{-(n+1)}\right]$$

Putting $x = r\cos\theta$ and $a = r\sin\theta$, so that $\tan\theta = \frac{a}{x}$ and $r = \frac{a}{\sin\theta}$, then

$$\therefore\ y_n = \frac{(-1)^n n!}{2}\left[(r\cos\theta + ir\sin\theta)^{-(n+1)} + (r\cos\theta - ir\sin\theta)^{-(n+1)}\right]$$

$$= \frac{(-1)^n n!}{2} r^{-(n+1)}\left[(\cos\theta + i\sin\theta)^{-(n+1)} + (\cos\theta - i\sin\theta)^{-(n+1)}\right]$$

$$= \frac{(-1)^n n!}{2r^{n+1}}\left[\cos(n+1)\theta - i\sin(n+1)\theta + \cos(n+1)\theta + i\sin(n+1)\theta\right]$$

$$= \frac{(-1)^n n!}{2r^{n+1}}\ 2\cos(n+1)\theta$$

But $r = \frac{a}{\sin\theta}$, then $\frac{1}{r^{n+1}} = \frac{\sin^{n+1}\theta}{a^{n+1}}$.

$$\therefore\ y_n = (-1)^n n! \times \frac{\sin^{n+1}\theta}{a^{n+1}} \cdot \cos(n+1)\theta, \text{ where } \theta = \tan^{-1}\left(\frac{a}{x}\right)$$

$$\text{or,}\quad y_n = \frac{(-1)^n n!}{a^{n+1}} \sin^{n+1}\theta \cdot \cos(n+1)\theta$$

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Example 33. Find y_n of the function $y = \tan^{-1}\left(\frac{1 + x}{1 - x}\right)$.

Solution:

We know, $\tan^{-1}\left(\frac{A + B}{1 - AB}\right) = \tan^{-1}A + \tan^{-1}B$.

$$\therefore\ y = \tan^{-1}\left(\frac{1 + x}{1 - x}\right) = \tan^{-1}1 + \tan^{-1}x = \frac{\pi}{4} + \tan^{-1}x$$

Differentiating with respect to x, we get

$$y_1 = 0 + \frac{1}{1 + x^2} = \frac{1}{x^2 + 1^2}.$$

Differentiating up to $(n - 1)$ steps, we get

$$y_n = D^{n-1}\left(\frac{1}{x^2 + 1^2}\right) \quad \ldots\ (i)$$

Applying $D^n\left(\frac{1}{x^2 + a^2}\right) = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1}\theta \cdot \sin(n + 1)\theta$, $\theta = \tan^{-1}\left(\frac{a}{x}\right)$.

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$\therefore \quad D^{n-1}\left(\frac{1}{x^2+a^2}\right) = \frac{(-1)^{n-1}(n-1)!}{a^{n+1}} \sin^n\theta . \sin n\theta$, where $\theta = \tan^{-1}\left(\frac{a}{x}\right)$.

Then (i) gives, $y_n = \frac{(-1)^{n-1}(n-1)!}{1^{n+1}} \sin^n\theta . \sin n\theta$, where $\theta = \tan^{-1}\left(\frac{1}{x}\right)$.

$\therefore \quad y_n = (-1)^{n-1}(n-1)! \sin^n\theta . \sin n\theta$, where $\theta = \tan^{-1}\left(\frac{1}{x}\right)$.

Example 34. If $y = \frac{1}{x^x}$, show that $y_2(1) = 0$.

Solution:

Here, $y = \frac{1}{x^x} = x^{-x}$.

Taking log on both sides, $\log y = -x \log x$.

Differentiating with respect to x, we get

$$\frac{1}{y} \cdot y_1 = -\left(x \cdot \frac{1}{x} + \log x \cdot 1\right)$$

$\therefore \quad y_1 = -y(1 + \log x)$ .... (i)

When $x = 1$,

$y_1(1) = -y(1)(1 + \log 1) = -1(1 + 0) = -1$.

Again, differentiating equation (i), we get

$$y_2 = -y_1(1 + \log x) - \frac{y}{x}.$$

When $x = 1$, $y_2(1) = -y_1(1) \cdot (1 + \log 1) - \frac{y(1)}{1}$ .... (ii)

But $y(1) = 1$ and $y_1(1) = -1$, so (ii) gives,

$y_2(1) = -(-1)(1 + 0) - 1 = 1(1) - 1 = 0$.

$\therefore \quad y_2(1) = 0$.

We next discuss a standard result on higher order derivatives popularly known as Leibnitz's theorem, which gives a rule to find the n^{th} derivative of product of two functions.

1.6.2 Leibnitz's theorem

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If u and v be two functions of x, then the n^{th} derivative of their product, i.e., $(uv)_n$ is given by the formula $(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots + {}^nC_r u_{n-r} v_r + \ldots + uv_n$ and ${}^nC_r = \frac{n!}{r!(n-r!)}$ where, the suffices of u and v denote the order of the differentiation of u and v with respect to x.

Proof:

We use the method of induction to prove this theorem.

Let $y = uv$.

By actual differentiation, we get

$$y_1 = u_1 v + u v_1$$
$$= {}^1C_0 u_1 v + {}^1C_1 u v_1$$
$$y_2 = u_2 v + u_1 v_1 + u_1 v_1 + u v_2$$
$$= u_2 v + 2u_1 v_1 + u v_2$$
$$= {}^2C_0 u_2 v + {}^2C_1 u_1 v_1 + {}^2C_2 u v_2$$

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So the theorem is true for $n = 2$.

Thus we see that the theorem is true for $n = 1, 2$. So let us assume that the theorem is true for some positive integer $k < n$. We shall prove that this theorem is also true for $n = k + 1$. So by assumption,

$$y_k = u_k v + {}^kC_1 u_{k-1} v_1 + {}^kC_2 u_{k-2} v_2 + \ldots + {}^kC_r u_{k-r} v_r + \ldots + u v_k.$$

Differentiating both sides with respect to x, we get

$$y_{k+1} = u_{k+1} v + u_k v_1 + {}^kC_1 u_k v_1 + {}^kC_1 u_{k-1} v_2 + {}^kC_2 u_{k-1} v_2 + \ldots + {}^kC_{r-1} u_{k-r+1} v_r + {}^kC_r u_{k-r+1} v_r + \ldots + u v_{k+1}$$

$$= u_{k+1} v + (1 + {}^kC_1) u_k v_1 + ({}^kC_1 + {}^kC_2) u_{k-1} v_2 + \ldots + ({}^kC_{r-1} + {}^kC_r) u_{k+1-r} v_r + \ldots + u v_{k+1}$$

But we know,

$$1 + {}^kC_1 = {}^{k+1}C_1, \; {}^kC_1 + {}^kC_2 = {}^{k+1}C_2 \text{ and, } {}^kC_{r-1} + {}^kC_r = {}^{k+1}C_r$$

$$\therefore \quad y_{k+1} = u_{k+1} v + {}^{k+1}C_1 u_k v_1 + {}^{k+1}C_2 u_{k-1} v_2 + \ldots + {}^{k+1}C_r u_{k+1-r} v_r + \ldots + u v_{k+1}.$$

This shows that the given statement is true for $n = k + 1$ whenever it is true for $n = k$, but it is true for $n = 1$ and 2. So by induction method, the theorem is true for all positive integers. This completes the proof.

Example 35. Find y_n if $y = x^n e^x$.

Solution:

Let $y = x^n e^x$ .... (i)

Here, we take $u = e^x$ and $v = x^n$.

$u_1 = e^x, u_2 = e^x, \ldots, u_{n-1} = e^x, u_n = e^x$.

$v_1 = n x^{n-1}, v_2 = n(n-1)x^{n-2}, \ldots, v_n = n!$.

Differentiating (i) up to n steps by applying Leibnitz's theorem, we have

$$y_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots + {}^nC_n u v_n$$

$$= e^x x^n + n e^x . n x^{n-1} + \frac{n(n-1)}{2!} e^x . n(n-1) x^{n-2} + \ldots + 1 . e^x n!$$

$$= e^x \left[x^n + \frac{n^2}{1!} x^{n-1} + \frac{n^2(n-1)^2}{2!} x^{n-2} + \ldots + n! \right]$$

$$= e^x \left[x^n + \frac{n^2}{1!} x^{n-1} + \frac{n^2(n-1)^2}{2!} x^{n-2} + \ldots + \frac{(n!)^2}{n!} \right]$$

$$y_n = e^x \left[x^n + \frac{n^2}{1!} x^{n-1} + \frac{n^2(n-1)^2}{2!} x^{n-2} + \ldots + \frac{n^2 (n-1)^2 \ldots 1^2}{n!} \right]$$

Example 36. Find y_n of the function $y = e^x \log x$.

Solution:

Let $y = e^x \log x$ .... (i)

Let us take $u = e^x$ and $v = \log x$. Then,

$$u_1 = e^x, \quad v_1 = \frac{1}{x}$$

$$u_2 = e^x, \quad v_2 = (-1) x^{-2} = -\frac{1}{x^2}$$

$$u_3 = e^x, \quad v_3 = (-1)(-2) x^{-3}$$

$$u_n = e^x, \quad v_n = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

Differentiating (i) up to n steps by applying Leibnitz's theorem, we have

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$$y_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \cdots + {}^nC_n u v_n$$

$$= e^x \log x + {}^nC_1 e^x \cdot \frac{1}{x} + {}^nC_2 e^x \left(\frac{-1}{x^2}\right) + {}^nC_3 e^x \cdot \left(\frac{2}{x^3}\right) + \cdots + {}^nC_n e^x \cdot \frac{(-1)^{n-1}(n-1)!}{x^n}$$

$$= e^x \left[\log x + \frac{{}^nC_1}{x} - \frac{{}^nC_2}{x^2} + \frac{{}^nC_3}{x^3} 2! + \cdots + \frac{(-1)^{n-1} {}^nC_n (n-1)!}{x^n}\right]$$

Example 37. If $y = a \cos(\log x) + b \sin(\log x)$, then prove that:
a. $x^2 y_2 + x y_1 + y = 0$ b. $x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0$.

Solution:

a. Here, $y = a \cos(\log x) + b \sin(\log x)$ .... (i)

Differentiating both sides with respect to x, we get

$$y_1 = -a \sin(\log x) \cdot \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x}$$

or, $x y_1 = -a \sin(\log x) + b \cos(\log x)$

Again, differentiating with respect to x, we get

$$x y_2 + y_1 = -a \cos(\log x) \cdot \frac{1}{x} - b \sin(\log x) \cdot \frac{1}{x}$$

or, $x y_2 + y_1 = -\frac{1}{x} [a \cos(\log x) + b \sin(\log x)]$

or, $x y_2 + y_1 = -\frac{1}{x} \cdot y$ [$\because$ by using (i)]

$\therefore$ $x^2 y_2 + x y_1 + y = 0$, ... (ii)

Differentiating each terms of (ii) up to n steps by using Leibnitz's theorem, we get

$$D^n(x^2 y_2) + D^n(x y_1) + D^n(y) = 0$$

$$x^2 y_{n+2} + {}^nC_1 y_{n+1} (2x) + {}^nC_2 y_n (2) + x y_{n+1} + {}^nC_1 y_n (1) + y_n = 0$$

or, $x^2 y_{n+2} + (2 {}^nC_1 + 1) x y_{n+1} + (2 {}^nC_2 + {}^nC_1 + 1) y_n = 0$

But ${}^nC_1 = n$, and ${}^nC_2 = \frac{n(n-1)}{2}$.

$\therefore$ $x^2 y_{n+2} + (2n+1) x y_{n+1} + [n(n-1) + n + 1] y_n = 0$,

or, $x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0$.

Example 38. If $y^{1/m} + y^{-1/m} = 2x$, show that $(x^2-1) y_{n+2} + (2n+1) x y_{n+1} + (n^2 - m^2) y_n = 0$.

Solution:

We have, $y^{1/m} + y^{-1/m} = 2x$.

Differentiating with respect to x, we get

$$\frac{1}{m} y^{1/m-1} \cdot y_1 - \frac{1}{m} y^{-1/m-1} \cdot y_1 = 2$$

or, $\frac{y_1}{my} y^{1/m} - \frac{y_1}{my} y^{-1/m} = 2$

or, $\frac{y_1}{my} (y^{1/m} - y^{-1/m}) = 2$

or, $y_1 (y^{1/m} - y^{-1/m}) = 2my$

Squaring both sides, we get

$$y_1^2 (y^{1/m} - y^{-1/m})^2 = 4m^2 y^2$$

or, $y_1^2 [(y^{1/m} + y^{-1/m})^2 - 4 y^{1/m} \cdot y^{-1/m}] = 4m^2 y^2$

or, $y_1^2 [(2x)^2 - 4] = 4m^2 y^2$

or, $y_1^2 (x^2 - 1) = m^2 y^2$

or, $(x^2-1) y_1^2 = m^2 y^2$

Again, differentiating with respect to x, we get

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$(x^2-1).2y_1y_2+y^2_1(2x) = 2m^2yy_1$

or, $(x^2-1)y_2+xy_1-m^2y = 0$

Now, differentiating each terms with respect to x up to n times by using Leibnitz's rule, we get

$(x^2-1)y_{n+2}+{}^nC_1y_{n+1}(2x)+{}^nC_2y_n(2)+xy_{n+1}+{}^nC_1y_n-m^2y_n = 0$

or, $(x^2-1)y_{n+2}+(2n+1)xy_{n+1}+[n(n-1)+n-m^2]y_n = 0$

$\therefore$ $(x^2-1)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n = 0.$

Example 39. If $\log y = \tan^{-1}x$, then show that

a. $(1+x^2)y_2+(2x-1)y_1 = 0$

b. $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+n(n+1)y_n = 0.$

Solution:

a. We have,

$\log y = \tan^{-1}x$

$\therefore$ $y = e^{\tan^{-1}x}$ .... (i)

Differentiating with respect to x, we get

$y_1 = e^{\tan^{-1}x}\frac{1}{1+x^2} = \frac{y}{1+x^2}$ [$\because$ using equation (i)]

$\therefore$ $y_1(1+x^2)-y = 0$

Again, differentiating with respect to x, we get

$y_2(1+x^2)+y_1(2x)-y_1 = 0$

or, $(1+x^2)y_2+(2x-1)y_1 = 0$ .... (ii)

b. Now, differentiating equation (ii) with respect to x up to n times by using Leibnitz's rule, we get

$(1+x^2)y_{n+2}+{}^nC_1y_{n+1}(2x)+{}^nC_2y_n(2)+(2x-1)y_{n+1}+{}^nC_1y_n(2) = 0$

or, $(1+x^2)y_{n+2}+2nxy_{n+1}+\frac{n(n-1)}{2}(2y_n)+(2x-1)y_{n+1}+2ny_n = 0$

or, $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+[n(n-1)+2n]y_n = 0.$

$\therefore$ $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+n(n+1)y_n = 0.$

Example 40. If $y = \left(x+\sqrt{1+x^2}\right)^m$, show that

a. $(1+x^2)y_2+xy_1-m^2y = 0$

b. $(1+x^2)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n = 0.$

Solution:

a. Here, $y = \left(x+\sqrt{1+x^2}\right)^m$.

Differentiating with respect to x, gives

$y_1 = m\left(x+\sqrt{1+x^2}\right)^{m-1}\times\left(1+\frac{1}{2}\cdot\frac{2x}{\sqrt{1+x^2}}\right)$

$\therefore$ $y_n = \frac{m\left(x+\sqrt{1+x^2}\right)^m}{\sqrt{1+x^2}} = \frac{my}{\sqrt{1+x^2}}$

Now, squaring gives, $(1+x^2)y_1^2-m^2y^2 = 0.$

Again differentiating with respect to x, we get

$(1+x^2)2y_1y_2+y_1^2.2x-m^2.2yy_1 = 0$

$\therefore$ $(1+x^2)y_2+xy_1-m^2y = 0$ .... (i)

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b. Differentiating (i) with respect to x up to n steps by using Leibnitz's rule, we get

$$(1+x^2)y_{n+2} + {}^nC_1 y_{n+1}(2x) + {}^nC_2 y_n(2) + xy_{n+1} + {}^nC_1 y_n(1) - m^2 y_n = 0$$

or, $$(1+x^2)y_{n+2} + n.2xy_{n+1} + \frac{n(n-1)}{2}.2y_n + xy_{n+1} + ny_n - m^2 y_n = 0$$

or, $$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + [n(n-1) + n - m^2]y_n = 0$$

$\therefore$ $$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

Example 41. If $y = (\sin^{-1}x)^2$, prove that

a. $(1-x^2)y_2 - xy_1 - 2 = 0$

b. $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0.$

Solution:

a. We have, $y = (\sin^{-1}x)^2$ ... (i)

Differentiating equation (i) with respect to x, we get

$$y_1 = 2\sin^{-1}x \frac{1}{\sqrt{1-x^2}}$$

or, $(\sqrt{1-x^2})\, y_1 = 2\sin^{-1}x$

Squaring,

$(1-x^2)y_1^2 = 4(\sin^{-1}x)^2$

or, $(1-x^2)y_1^2 - 4y = 0$ [$\because$ using equation (i)]

Again, differentiating, we get

$(1-x^2)\, 2y_1y_2 - 2xy_1^2 - 4y_1 = 0$

or, $(1-x^2)y_2 - xy_1 - 2 = 0$ ... (ii) [$\because y_1 \neq 0$]

b. Now, differentiating equation (ii) with respect to x up to n steps by using Leibnitz's rule, we get

$$(1-x^2)y_{n+2} + {}^nC_1 y_{n+1}(-2x) + {}^nC_2 y_n(-2) - xy_{n+1} - {}^nC_1 y_n(1) - 0 = 0$$

or, $$(1-x^2)\, y_{n+2} + ny_{n+1}(-2x) + \frac{n(n-1)}{2} y_n(-2) - xy_{n+1} - ny_n = 0$$

or, $$(1-x^2)y_{n+2} - 2nx\, y_{n+1} - (n^2-n)\, y_n - xy_{n+1} - ny_n = 0$$

or, $$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - n + n)\, y_n = 0$$

$\therefore$ $$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0.$$

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Example 42. If $y = \sin(m\sin^{-1}x)$, prove that :

a. $(1-x^2)\, y_2 - xy_1 + m^2 y = 0$

b. $(1-x^2)y_{n+2} - (2n+1)\, xy_{n+1} + (m^2 - n^2)y_n = 0.$

Solution:

a. Here, $y = \sin(m\sin^{-1}x)$

Differentiating with respect to x, we get

$$y_1 = \cos(m\sin^{-1}x).\, m \frac{1}{\sqrt{1-x^2}}$$

or, $\sqrt{1-x^2}\, y_1 = m\cos(m\sin^{-1}x)$

Squaring gives,

$(1-x^2)\, y_1^2 = m^2\cos^2(m\sin^{-1}x)$

or, $(1-x^2)\, y_1^2 = m^2[1 - \sin^2(m\sin^{-1}x)]$

or, $(1-x^2)\, y_1^2 = m^2(1-y^2)$

or, $y_1^2(1-x^2) - m^2 + m^2y^2 = 0.$

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Again, differentiating with respect to x, we get

$(1-x^2)\,2y_1y_2 + y_1^2(-2x) - 0 + m^2 \cdot 2yy_1 = 0$

or, $2y_1[(1-x^2)y_2 - xy_1 + m^2y] = 0$

$\therefore$ $(1-x^2)y_2 - xy_1 + m^2y = 0$ .... (i)

b. Differentiating (i) up to n steps by using Leibnitz's rule, we get

$(1-x^2)y_{n+2} + {}^nC_1y_{n+1}(-2x) + {}^nC_2y_n(-2) - xy_{n+1} - {}^nC_1y_n + m^2y_n = 0$

or, $(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2} \cdot 2y_n - xy_{n+1} - ny_n + m^2y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - [n(n-1) + n - m^2]y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0.$

Example 43. If $y = \sin^{-1}x$, show that

a. $(1-x^2)y_2 - xy_1 = 0$

b. $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

c. $(y_{n+2})_0 = n^2(y_n)_0$. Find also the value of $(y_n)_0$.

Solution:

a. We have, $y = \sin^{-1}x$,

Differentiating with respect to x, we get

$$y_1 = \frac{1}{\sqrt{1-x^2}}$$

or, $y_1\left(\sqrt{1-x^2}\right) = 1$

Squaring both sides,

$y_1^2(1-x^2) = 1$

or, $y_1^2 - y_1^2x^2 = 1$ .... (i)

Differentiating equation (i) with respect to x, we get

$2y_1y_2 - y_1^2(2x) - x^2\,2y_1y_2 = 0$

or, $2y_1(y_2 - xy_1 - x^2y_2) = 0$

Since, $2y_1 \neq 0$,

So, $y_2 - xy_1 - x^2y_2 = 0$

or, $(1-x^2)y_2 - xy_1 = 0$ .... (ii)

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b. Now differentiating up to n steps by using Leibnitz theorem, we get

$y_{n+2}(1-x^2) + {}^nC_1y_{n+1}(-2x) + {}^nC_2y_n.(-2) - \{xy_{n+1} + {}^nC_1y_n.1\} = 0$

or, $y_{n+2}(1-x^2) + ny_{n+1}(-2x) + \frac{n(n-1)}{2!}y_n(-2) - xy_{n+1} - ny_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ .... (iii)

c. Put $x = 0$ in (iii)

$(1-0)(y_{n+2})_0 - (2n+1).0(y_{n+1})_0 - n^2(y_n)_0 = 0$

or, $(y_{n+2})_0 = n^2(y_n)_0$ .... (iv)

From (i) $(y_1)_0 = 1$, From (ii) $(y_2)_0 = 0$

Putting $n = 1, 3, 5, \ldots$ in (iv)

$(y_3)_0 = 1^2(y_1)_0 = 1^2 \cdot 1$

$(y_5)_0 = 3^2(y_3)_0 = 3^2 \cdot 1^2 \cdot 1$

$(y_7)_0 = 5^2(y_5)_0 = 5^2.3^2.1^2.1$

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Hence, in general, when n is odd

$(y_n)_0 = (n-2)^2 (n-4)^2 (n-6)^2 \ldots 3^2.1^2.1$

Again, putting $n = 2, 4, 6, \ldots$ in (iv),

$(y_4)_0 = 2^2 (y_2)_0 = 2^2.0 = 0$

$(y_6)_0 = 4^2 (y_4)_0 = 4^2.0 = 0$

$\vdots \quad \vdots \quad \vdots \quad \vdots$

So in general, when n is even $(y_n)_0 = 0$

$$(y_n)_0 = \begin{cases} (n-2)^2 (n-4)^2 (n-6)^2 \ldots 3^2.1^2 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Example 44. If $y = e^{a\sin^{-1}x}$, show that

a. $(1-x^2)y_2 - xy_1 - a^2y = 0$

b. $(1-x^2)\, y_{n+2} - (2n+1)\, xy_{n+1} - (n^2 + a^2)\, y_n = 0.$

Also, find $y_n(0)$ if $y = e^{a\sin^{-1}x}$.

Solution:

a. Here, $y = e^{a\sin^{-1}x}$

$\therefore$ $y(0) = e^0 = 1.$

Differentiating with respect to x, we get

$$y_1 = e^{a\sin^{-1}x} \cdot \frac{1}{\sqrt{1-x^2}} \cdot a$$

or, $$y = \frac{ay}{\sqrt{1-x^2}} \qquad \ldots\ (i)$$

$\therefore$ $(1-x^2)\, y_1^2 - a^2y^2 = 0$

Again, differentiating with respect to x, we get

$(1-x^2)\, 2y_1y_2 + y_1^2\,(-2x) - 2a^2yy_1 = 0$

$\therefore$ $(1-x^2)y_2 - xy_1 - a^2y = 0 \qquad \ldots\ (ii)$

Putting $x = 0$ in (i), $y_1(0) = ay(0) = a$

Putting $x = 0$ in (ii), we get

$y_2(0) - 0 - a^2y(0) = 0$

or, $y_2(0) - a^2 = 0$

$\therefore$ $y_2(0) = a^2.$

b. Differentiating up to n steps by applying Leibnitz's rule, we get

$(1-x^2)\, y_{n+2} + {}^nC_1 y_{n+1}(-2x) + {}^nC_2 y_n . (-2) - (xy_{n+1} + {}^nC_1 y_n . 1) - a^2y_n = 0$

or, $(1-x^2)\, y_{n+2} - 2nxy_{n+1} - n(n-1)\, y_n - xy_{n+1} - ny_n - a^2y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - [n(n-1) + n + a^2]y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)\, xy_{n+1} - (n^2 + a^2)y_n = 0 \qquad \ldots\ (iii)$

Putting $x = 0$ in (iii), we get

$y_{n+2}(0) - (2n+1) \cdot 0 - (n^2 + a^2)\, y_n(0) = 0$

or, $y_{n+2}(0) = (n^2 + a^2)\, y_n(0) \qquad \ldots\ iv)$

Also, we have $y_1(0) = a$ and $y_2(0) = a^2$

Case I: Putting $n = 2, 4, 6, \ldots$ in (iv), we get

$y_4(0) = (2^2 + a^2) \cdot y_2(0) = (2^2 + a^2)\, a^2$

$y_6(0) = (4^2 + a^2) \cdot y_4(0) = (4^2 + a^2)(2^2 + a^2)\, a^2$

$y_8(0) = (6^2 + a^2) \cdot y_6(0) = (6^2 + a^2)(4^2 + a^2)(2^2 + a^2)\, a^2$

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Thus, if n is even, then

$y_n(0) = [(n-2)^2 + a^2]\,[(n-4)^2 + a^2]\,[(n-6)^2 + a^2] \ldots (4^2 + a^2)(2^2 + a^2)\,a^2$ .... (A)

Case II: Putting $n = 1, 3, 5, \ldots$ in (iv), we get

$y_3(0) = (1^2 + a^2)a$

$y_5(0) = (3^2 + a^2)y_3(0) = (3^2 + a^2)(1^2 + a^2)a$

$y_7(0) = (5^2 + a^2)\,y_5(0) = (5^2 + a^2)(3^2 + a^2)(1^2 + a^2)\,.\,a$

Thus, if n is odd, then

$y_n(0) = [(n-2)^2 + a^2]\,[(n-4)^2 + a^2] \,..\, (3^2 + a^2)(1^2 + a^2)a.$ .... (B)

$\therefore$ From (A) and (B), we get

$$y_n(0) = \begin{cases} [(n-2)^2 + a^2]\,[(n-4)^2 + a^2] \ldots (4^2 + a^2)(2^2 + a^2)\,a^2, & \text{if } n \text{ is even} \\ [(n-2)^2 + a^2]\,[(n-4)^2 + a^2] \ldots (3^2 + a^2)(1^2 + a^2)a, & \text{if } n \text{ is odd} \end{cases}$$

Exercise 1.4

1. Find y_n in the following:

   i. $y = (a - bx)^m$ ii. $y = \dfrac{1}{a-x}$ iii. $y = (2 - 3x)^n$

   iv. $y = \dfrac{1}{x^2 + 25}$ v. $x^2 e^{ax}$ vi. $x^3 \sin x$

2. i. If $ax^2 + 2hxy + by^2 = 1$, show that $\dfrac{d^2y}{dx^2} = \dfrac{h^2 - ab}{(hx + by)^3}$.

   ii. If $y = ae^{mx} + be^{-mx}$, show that $y_2 - m^2y = 0$.

   iii. If $y = e^{x^2}$, show that $y_{n+1} - 2ny_{n-1} = 0$.

   iv. If $y = (\tan^{-1}x)^2$, show that: $(x^2 + 1)^2y_2 + 2x(x^2 + 1)y_1 - 2 = 0$.

3. If $y = x^{n-1}\log x$, show that $xy_n = (n-1)!$

4. If $y = \log(x + \sqrt{a^2 + x^2})$, show that

   i. $(a^2 + x^2)y_2 + xy_1 = 0$ ii. $(a^2 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + n^2y_n = 0$

5. If $y = (x^2 - 1)^n$, show that

   i. $(x^2 - 1)y_n + 2(n-1)xy_1 - 2ny = 0$

   ii. $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$

6. If $y = e^{a\tan^{-1}x}$, prove that

   i. $(1 + x^2)y_2 + (2x - a)y_1 = 0$

   ii. $(1 + x^2)y_{n+2} + (2nx + 2x - a)y_{n+1} + n(n+1)y_n = 0$

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Answers

1. i. $(-1)^n\, m(m-1)(m-2) \ldots (m-n+1)\, b^n\, (a - bx)^{m-n}$

   ii. $\dfrac{n!}{(a-x)^{n+1}}$

   iii. $(-1)^n 3^n\, n!$

   iv. $\dfrac{(-1)^n\, n!}{5^{n+2}} \sin^{n+1}\theta \sin((n+1)\theta)$ where, $\theta = \tan^{-1}\left(\dfrac{5}{x}\right)$

   v. $a^{n-2}\, e^{ax}\,[a^2x^2 + 2nax + n(n-1)]$

   vi. $x^3 \sin\left(\dfrac{n\pi}{2} + x\right) + 3nx^2 \sin\left(\dfrac{(n-1)\pi}{2} + x\right) + 3n(n-1)x \sin\left(\dfrac{(n-2)\pi}{2} + x\right) + n(n-1)(n-2) \sin\left(\dfrac{(n-3)\pi}{2} + x\right)$

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Multiple Choice Questions

1. The value of k which makes the function $f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), x \neq 0 \\ k, \text{ if } x = 0 \end{cases}$ continuous at $x = 0$ is
   a. 8 b. 2
   c. -1 d. none of these

2. $\lim_{n\to\infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$ equals to
   a. 3 b. 2
   c. 1 d. 0

3. If $f(x) = \frac{1 - \cos 2x}{x^2}$, $x \neq 0$ and $f(x) = k$ if $x = 0$ and f is continuous at $x = 0$, then $k =$
   a. 1 b. -1
   c. 0 d. 2

4. If $f(x) = \sin|x|$, then
   a. f is continuous for all $x \in R$ b. f is continuous for some $x \in R$
   c. f is differentiable for all $x \in R$ d. none of these

5. If $f'(x) = f(x)$ and $f(0) = 1$, then $\lim_{x\to 0} \frac{f(x) - 1}{x}$ is
   a. -1 b. 0
   c. 1 d. 2

6. If $\lim_{x\to 0} kx \cot x = \lim_{x\to 0} x \cot(kx)$, then the value of k is
   a. only 1 b. only -1
   c. ± 1 d. $\pm\frac{1}{2}$

7. For $a \neq b$, $\lim_{x\to 0} \frac{a^x - 1}{b^x - 1}$ equals to
   a. $\log_e ab$ b. $\log_e \frac{a}{b}$
   c. $\log_b a$ d. $\log_a b$

8. If $f(x) = 3^{ax}$, then $f'(x)$ is
   a. $\frac{3^{ax}}{\pi \log 3}$ b. $\frac{3^{ax}}{\pi}$
   c. $\frac{3^{ax}}{\log 3}$ d. $\pi \log 3\,(3^{ax})$

9. The function $f(x) = \cot x$ is discontinuous on the set
   a. $\{x = n\pi : n \in Z\}$ b. $\{x = 2x\pi : n \in Z\}$
   c. $\{x = (2n+1)\frac{\pi}{2} : n \in Z\}$ d. $\{x = \frac{n\pi}{2} : n \in Z\}$

10. If $f(x) = \begin{cases} mx + 1 \text{ if } x \leq \frac{\pi}{2} \\ \sin x + n \text{ if } x > \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then
    a. $m = 1, n = 0$ b. $m = \frac{n\pi}{2} + 1$
    c. $n = \frac{m\pi}{2}$ d. $m = \frac{\pi}{2}, n = \frac{\pi}{2}$

Answer

1.	d.	2.	a.	3.	d.	4.	a.	5.	c.
6.	c.	7.	c.	8.	d.	9.	a.	10.	c.

Additional Exercise

1. Test the continuity of
   i. $f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ at $x = 0$.
   ii. $f(x) = \begin{cases} \frac{x-1}{1 + e^{\frac{1}{(x-1)}}} & \text{for } x \neq 1 \\ 0 & \text{for } x = 1 \end{cases}$ at $x = 1$.
   iii. $f(x) = \begin{cases} (1+x)^{1/x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases}$ at $x = 0$.

2. Examine whether the given function are continuous or not at the given points.
   i. $f(x) = \begin{cases} -1 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$ at $x = 0$
   ii. $f(x) = \begin{cases} \frac{1}{2} - x & \text{when } 0 < x < \frac{1}{2} \\ \frac{1}{2} & \text{when } x = \frac{1}{2} \\ \frac{3}{2} - x & \text{when } \frac{1}{2} < x < 1 \end{cases}$ at $x = \frac{1}{2}$

3. Test the continuity of function $f(x)$ below at $x = 1$ and $x = 2$.
   $f(x) = \begin{cases} x^2 & \text{for } 0 < x < 1 \\ x & \text{for } 1 \leq x \leq 2 \\ \frac{x^3}{4} & \text{for } 2 < x \leq 3 \end{cases}$

4. Show that the function given below is continuous at $x = 0$ and discontinuous at $x = \frac{3}{2}$.
   $f(x) = \begin{cases} 3 + 2x & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x < \frac{3}{2} \\ -3 - 2x & \text{for } x \geq \frac{3}{2} \end{cases}$.

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5. Check the continuity of the function given below at $x = 0$ and $x = 1$.

$$f(x) = \begin{cases} 1 & \text{for } x < 0 \\ \sqrt{1-x^2} & \text{for } 0 \le x < 1 \\ x - 1 & \text{for } x \ge 1 \end{cases}.$$

6. Show that the function $f(x) = |x|$ is continuous at $x = 0$ but is not differentiable at that point.
7. Show that the function $f(x) = |x| + |x - 1|$ is continuous at $x = 0$ but is not differentiable at $x = 0$.
8. Prove $f'(0) = 0$ if

i. $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$

ii. $f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$

9. Show that $f(x) = \begin{cases} 7x - 1 & \text{for } x \le 1 \\ 6x & \text{for } x > 1 \end{cases}$ is continuous at $x = 1$ but is not differentiable at that point.
10. Show that the function $f(x) = \begin{cases} x^2 - 3x + 1 & \text{for } x \le 2 \\ 5x - 11 & \text{for } x > 2 \end{cases}$ is continuous but not differentiable at $x = 2$.
11. Find y_n in the following:

i. $y = e^x \sin^2 x$ ii. $y = \frac{x}{x^2 + 25}$

iii. $y = \frac{1}{x^2 + x + 1}$ iv. $y = \tan^{-1} \frac{\sqrt{1 + x^2} - 1}{x}$

v. $e^x \log x$

12. If $y = \sin mx + \cos mx$, prove that

i. $y_2 - m^2 y = 0$ ii. $y_n = m^n [1 + (-1)^n \sin 2mx]^{1/2}$

13. If $y = \sin^{-1} x$, show that

i. $(1 - x^2)y_2 - xy_1 = 0$

ii. $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2 y_n = 0$

iii. $(y_{n+2})_0 = n^2 (y_n)_0$, also find the value of $(y_n)_0$.

Answers

1. i. discontinuous ii. continuous iii. discontinuous
2. i. discontinuous ii. discontinuous
3. continuous at $x = 1$ and $x = 2$
5. continuous on $x = 0$ and $x = 1$
11. i. $\frac{1}{2} e^x [1 - 5^{n/2} \cos(2k + n \tan^{-1} 2)]$

ii. $\frac{(-1)^n n!}{5^{n+1}} \sin^{n+1}\theta \cos((n + 1)\theta)$ where, $\theta = \tan^{-1} \frac{5}{x}$

iii. $\frac{(-1)^n 2^{n+2} n!}{(\sqrt{3})^{n+2}} \sin^{n+1}\theta \sin((n + 1)\theta)$ where, $\theta = \tan^{-1} \frac{\sqrt{3}}{2x + 1}$

iv. $(-1)^{n-1} \frac{1}{2} (n - 1)! \sin^n\theta \sin n\theta$ where $\theta = \cot^{-1} x$

v. $e^x \left[\log x + \frac{{}^nC_1}{x} - \frac{{}^nC_2}{x^2} + \frac{{}^nC_3}{x^3} 2! + \ldots + \frac{(-1)^{n-1} (n - 1)!}{x^n}\right]$

13. iii. $y_n(0) = 0$ if n is even and $y_n(0) = (n - 2^2) \ldots 5^2 \cdot 3^2 \cdot 1^2$ if n is odd.

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Unit 2

Application of Derivative

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- idea of graph (curve) of a function.
- continuity and differentiability of a function (Polynomial, Exponential, Logarithmic and Trigonometric) at a point.
- higher order derivative.

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- mean value theorems (Rolle's and Lagrange's), their geometrical interpretations and applications.
- series expansion of various function by using higher order mean value theorems (Taylor's and Maclaurin's series expansion).
- asymptotes to curves and their determination.
- curve tracings.

2.0 Introduction

In this unit, we will focus on various applications of derivatives. We will see how derivative is used in various mean value theorems and in the representation (expansion) of continuous function by Taylor and Maclaurin's series. In addition to these, we will see how derivative is used in tracing (drawing graphs) curves of different functions and determining curvature and asymptotes to the curves of given function. We first begin with mean value theorem.

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2.1 Mean value theorem

We first discuss a case of issuing traffic ticket.

Suppose a person is driving on a straight highway on which the speed limit is 55 km/h. At 8 : 00 A.M. a traffic police records the velocity of the car as 45 km/h and few minutes later at 8 : 05 A.M., a second traffic police 5 km away from the first police, records the velocity of car as 50 km/h. In this situation, is the person eligible for over speeding ticket by traffic police? How can we answer this question? This question can be answered using a mathematical concept. Here, the person has travelled 5 kms in 5 minutes which is $\frac{1}{12}$ hour. So the person's average speed is 60 km/h. In mathematics, we have a standard theorem which says that an object's instantaneous velocity is the average velocity at least once in a given time frame. In other words, it says that

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$\left.\frac{dv}{dt}\right|_{t=c} = \frac{v(b) - v(a)}{b - a}$ for at least one value of c in (a, b).

So, we apply this theorem in the context of traffic ticketing and conclude that the average velocity of car, 60 km/h should be equal to velocity of car at some time in between 8:00 am to 8:05 am. Hence, the driver should be given a ticket for crossing the speed limit. This theorem which just used is popularly known as mean value theorem.

We need mean value theorem in various situations. We note that Rolle's theorem is a special case of mean value theorem. Mean value theorem can be used to approximate roots of many polynomial equations, to establish some standard inequalities and also to find limit. Moreover, mean value theorem is frequently used in numerical analysis and physics.

Michel Rolle (1652-1719)

Mean value theorem is also used to prove a number of other theorems in differential and integral calculus. In order to arrive at the mean value theorem, we need to begin with Rolle's theorem which was introduced by the French Mathematician Michel Rolle in a book entitled "Methode Dour Resoudre les Egalitez."

We now begin the discussion of Rolle's theorem.

2.1.1 Rolle's theorem

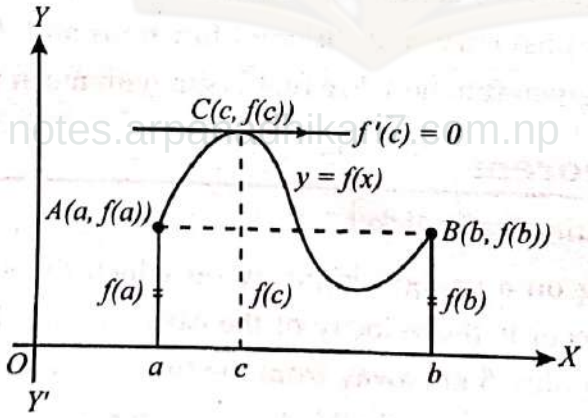
Let f be a function which satisfies:

i. f is continuous on the closed interval $[a, b]$

ii. f is differentiable on the open interval (a, b) and

iii. $f(a) = f(b)$.

Then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.



Proof:

Since $f(x)$ is continuous in $[a, b]$, so following cases arises:

Case I: $f(x) = k$, a constant. Then obviously $f'(x) = 0$ at every point in the interval and c can be taken any real number in (a, b).

Case II: If $f(x)$ is continuous and not constant in closed interval $[a, b]$, with $f(a) = f(b)$, so $f(x)$ must attain maximum or minimum value at some point $c \in (a, b)$.

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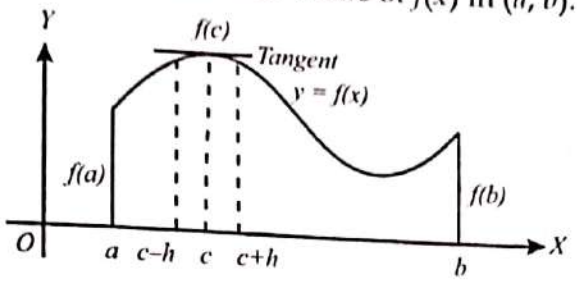
For definiteness, let $f(c)$ is a maximum value of $f(x)$ in (a, b).



Thus, $f(c + h) - f(c) \leq 0$ and $f(c - h) - f(c) \leq 0$ for all $h > 0$

$\therefore \quad \frac{f(c+h)-f(c)}{h} \leq 0$ and $\frac{f(c-h)-f(c)}{-h} \geq 0$ for all $h > 0$

Hence, $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h} \leq 0$ and $\lim_{h\to 0^-} \frac{f(c-h)-f(c)}{-h} \geq 0$.

This shows that right hand derivative $f'(c^+) \leq 0$ and left hand derivative $f'(c^-) \geq 0$.

Since $f'(x)$ exists at every value of x in (a, b) and so $f'(c)$ also exists. For this, above two limits must exist, and equal to zero.

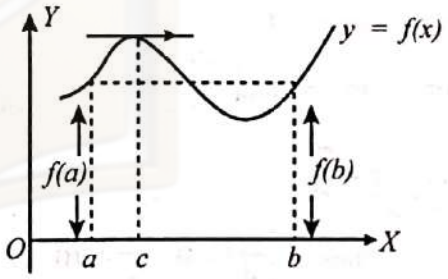
$\therefore \quad f'(c) = 0.$

Case III: If $f(x)$ attains minimum value we can similarly prove the theorem.

This completes the proof of the theorem.

2.1.2 Geometrical interpretation of Rolle's theorem

Let $y = f(x)$ be a continuous function on the interval $[a, b]$ and is differentiable at each point in (a, b) such that $f(a) = f(b)$. Then, there exists at least a point $c \in (a, b)$ where the tangent drawn to the curve is parallel to x-axis i.e., $f'(c) = 0$ as shown in figure.



This means that tangent is parallel to x-axis at the point $x = c$ in the interval (a, b).

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Remark:

1. Rolle's theorem guarantees existence of at least one point c such that $f'(c) = 0$. It does not restricts the existence of more than one points such that the tangent is parallel to x-axis.
2. If any of the conditions of the Rolle's theorem fails for given function, we cannot apply Rolle's theorem.

   For example, consider a function $f(x) = |x|$, $x \in [-1, 1]$.

   The function $f(x)$ is continuous on $[-1, 1]$ and $f(-1) = f(1) = 1$.

   But $f'(0)$ does not exist at $c = 0$ and $c \in (-1, 1)$. So, $f(x)$ is not differentiable at all points in $(-1, 1)$. Hence, we can not apply Rolle's theorem in this example.

We next verify Rolle's theorem taking some examples.

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Example 1. Verify Rolle's theorem for $f(x) = \frac{\sin x}{e^x}$ on $[0, \pi]$.

Solution:

We have, $f(x) = \frac{\sin x}{e^x}$ on $[0, \pi]$.

i. Since $f(x) = \frac{\sin x}{e^x}$ is continuous on $x \in [0, \pi)]$.

ii. Also, $f'(x) = \frac{e^x(\cos x - \sin x)}{(e^x)^2}$

$$f'(x) = \frac{\cos x - \sin x}{e^x} \text{ exists on } (0, \pi)$$

Hence, $f(x)$ is differentiable on $(0, \pi)$.

iii. $f(0) = \frac{\sin 0}{e^0} = 0$

$$f(\pi) = \frac{\sin \pi}{e^\pi} = \frac{0}{e^\pi} = 0$$

$\therefore$ $f(0) = f(\pi)$.

Thus all three conditions of Rolle's theorem are satisfied by f. To verify Rolle's theorem, we need to find $c \in (0, \pi)$ such that $f'(c) = 0$.

Using $f'(c) = 0$

or, $\frac{\cos c - \sin c}{e^c} = 0$

or, $\cos c - \sin c = 0$

or, $\cos c = \sin c$

or, $\tan c = 1$

or, $\tan c = \tan\left(\frac{\pi}{4}\right)$

$\therefore$ $c = \frac{\pi}{4}$

Thus $f'\left(\frac{\pi}{4}\right) = 0$, $\frac{\pi}{4} \in (0, \pi)$.

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Hence, Rolle's theorem is verified.

Example 2. Verify Rolle's theorem for the function $f(x) = (x-2)(x-3)(x-4)$ on $[2, 4]$.

Solution:

We have,

$$\begin{aligned} f(x) &= (x-2)(x-3)(x-4) \\ &= (x^2 - 3x - 2x + 6)(x-4) \\ &= (x^2 - 5x + 6)(x-4) \\ &= x^3 - 5x^2 + 6x - 4x^2 + 20x - 24 \\ &= x^3 - 9x^2 + 26x - 24 \end{aligned}$$

i. Since $f(x) = x^3 - 9x^2 + 26x - 24$ is continuous on $[2, 4]$.

ii. Here, $f'(x) = 3x^2 - 18x + 26$ and so $f'(x)$ exists on $(2, 4)$.

iii. Also, $f(2) = (2-2)(2-3)(2-4) = 0$

$f(4) = (4-2)(4-3)(4-4) = 0$

$\therefore$ $f(2) = f(4)$.

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Hence, $f(x)$ satisfies all the conditions of Rolle's theorem. To find c, we solve

$f'(c) = 0.$

or, $3c^2 - 18c + 26 = 0$

or, $c = \frac{-(-18) \pm \sqrt{(-18)^2 - 4.3.26}}{2.3}$

$= \frac{18 \pm \sqrt{324 - 312}}{6}$

$= \frac{18 \pm \sqrt{12}}{6}$

$= \frac{18 \pm 2\sqrt{3}}{6}$

Thus $c = \frac{9 \pm \sqrt{3}}{3} \in (2, 4)$ and $f'(c) = 0.$

Hence, Rolle's theorem is verified.

Example 3. Verify that the function $y = f(x) = x^2 - 4x + 3$ satisfies the Rolle's theorem in the interval $1 \leq x \leq 3$ and hence find the number c such that $f'(c) = -1$.

Solution:

We have $y = f(x) = x^2 - 4x + 3.$

i. Here, $f(x) = x^2 - 4x + 3$ is a polynomial and so it is continuous on $[1, 3]$.

ii. Again, $f'(x) = 2x - 4$ which exist for all $x \in (1, 3)$.
So $f(x)$ is differentiable on $(1, 3)$.

iii. Also, $f(1) = 1^2 - 4 \times 1 + 3 = 0$
$f(3) = 3^2 - 4 \times 3 + 3 = 0.$
$f(1) = f(3).$

Hence, all three conditions of Rolle's theorem is satisfied by f. Now,

$f'(c) = 0$

or, $2c - 4 = 0$

$2c = 4$

$\therefore \quad c = 2$

Thus, $2 \in (1, 3)$ and $f'(2) = 0.$

Hence, Rolle's theorem is verified.

For, $f'(c) = -1$

or, $2c - 4 = -1$

or, $2c = -1 + 4$

or, $2c = 3$

$\therefore \quad c = \frac{3}{2} \in (1, 3).$

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2.1.3 Lagrange's mean value theorem

Lagrange's mean value theorem is also popularly known as 'mean value theorem.'

Statement

Let f be a function which satisfies:

i. f is continuous on closed interval $[a, b]$.

ii. f is differentiable on open interval (a, b).

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Then there exists at least a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof:

We first define, $\phi(x) = f(x) + kx$

where k is a constant to be determined so that $\phi(a) = \phi(b)$.

i.e., $f(a) + ka = f(b) + kb$

or, $k(a-b) = f(b) - f(a)$

or, $k = \frac{f(b) - f(a)}{a - b}$

$\therefore \quad k = -\frac{f(b) - f(a)}{b - a}$ .... (i)

Thus, the function $\phi(x)$ is such that

i. Continuous on $[a, b]$.

As $f(x)$ and x are continuous functions on $[a, b]$,

ii. Again, $\phi'(x) = f'(x) + k$ .... (ii)

Since $f'(x)$ exists in (a, b). So, $\phi'(x)$ exists in (a, b).

Moreover, $\phi(a) = \phi(b)$.

$\therefore$, ϕ satisfies all the condition of Rolle's theorem. Hence by Rolle's theorem, there exists a least a point $c \in (a, b)$ such that

$\phi'(c) = 0$

or, $f'(c) + k = 0$ [by equation (ii)]

or, $f'(c) = -k$

or, $f'(c) = -\left(-\frac{f(b) - f(a)}{b - a}\right)$ [by equation (i)]

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$\therefore \quad f'(c) = \frac{f(b) - f(a)}{b - a}$

which completes the proof of the theorem.

Corollary:

1. Since c lies between a and b. So c can be expressed in the form

$c = a + \theta(b - a)$, for $0 < \theta < 1$.

Putting $b - a = h$, then $b = a + h$ and $c = a + \theta h$.

Then above mean value theorem can be expressed as

$f(a + h) - f(a) = h f'(a + \theta h)$

i.e., $f(a + h) = f(a) + h f'(a + \theta h)$, where $0 < \theta < 1$

which shows we can approximate value of function in neighb h using its va at $x = a$.

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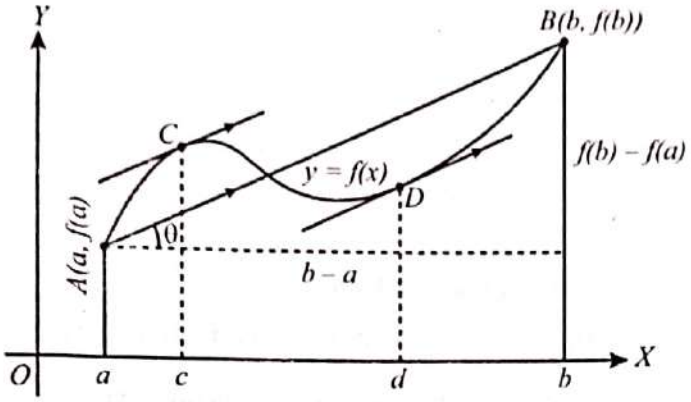
2. Let us put $a = x$, then

$f(x + h) = f(x) + h f'(x + \theta h)$ for $0 < \theta < 1$.

2.1.4 Geometrical interpretation of mean value theorem



Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b) as shown in figure. Then slope of the secant line joining A and B is

$$m = \tan\theta = \frac{\text{Rise}}{\text{Run}} = \frac{f(b) - f(a)}{b - a}.$$

Also, the slope of tangent at C is $f'(c)$. We see from the graph that tangent drawn at C is parallel to secant line joining end points A and B, i.e., $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Remark: There may be more than one point in (a, b) satisfying Lagrange's mean value theorem. This theorem guarantees existence of at least one point and does not restrict for more than one point.

We now see some applications of mean value theorem to establish some standard results.

Theorem 1.

If $f'(x) = 0$ for all x in (a, b) then $f(x)$ is constant on $[a, b]$.

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Proof:

Let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. Then by applying mean value theorem in $[x_1, x_2]$, we find $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = 0 \qquad [\because f'(x) = 0, \text{ for all } x \in (a, b)]$$

$$f(x_2) - f(x_1) = 0$$

$\therefore f(x_1) = f(x_2)$.

Hence, $f(x)$ is constant on $[a, b]$.

Theorem 2.

i. If $f'(x) > 0$ for all $x \in (a, b)$, then $f(x)$ is an increasing function.

ii. If $f'(x) < 0$, for all $x \in (a, b)$, then $f(x)$ is a decreasing function.

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Proof:

i. Let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. Then by applying mean value theorem, we find $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c)$$

$\therefore$ $f(x_2) - f(x_1) = (x_2 - x_1) f'(c)$

Here, $f'(c) > 0$ and $(x_2 - x_1) > 0$, so

$(x_2 - x_1) f'(c) > 0$

$\therefore$ $f(x_2) - f(x_1) > 0$

i.e., $f(x_2) > f(x_1)$. Hence, $f(x)$ is an increasing function.

Similarly, we can show that if $f'(x) < 0$, then $f(x)$ is decreasing function.

Example 4. Verify Lagrange's mean value theorem for the function $f(x) = (x-1)(x-2)(x-3)$ on $[1, 4]$.

Solution:

The given function is

$f(x) = (x-1)(x-2)(x-3)$

$\therefore$ $f(x) = x^3 - 6x^2 + 11x - 6$ .... (i)

i. since every polynomial function is continuous on $[1, 4]$, $f(x)$ is continuous on $[1, 4]$.

ii. here, $f'(x) = 3x^2 - 12x + 11$, which exists on $(1, 4)$.

Hence, $f(x)$ is differentiable on $(1, 4)$.

Thus, all the conditions of Lagrange mean value theorem are satisfied. To verify mean value theorem, we find $c \in (1, 4)$: $f'(c) = \frac{f(4) - f(1)}{4 - 1}$. So,

$$f'(c) = \frac{f(4) - f(1)}{4 - 1} = \frac{6 - 0}{3} = 2.$$

or, $3c^2 - 12c + 11 = 2$

or, $3c^2 - 12c + 9 = 0$

or, $c^2 - 4c + 3 = 0$

or, $(c-1)(c-3) = 0$

$\therefore$ $c = 1, 3$.

Clearly $3 \in (1, 4)$.

Hence, Lagrange's mean value theorem is verified.

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Example 5. Verify Lagrange's mean value theorem for the function $f(x) = x(x-1)(x-2)$ on $\left[0, \frac{1}{2}\right]$.

Solution:

We have, $f(x) = x(x-1)(x-2)$

$= x(x^2 - 3x + 2)$

$= x^3 - 3x^2 + 2x$.

i. Clearly $f(x) = x^3 - 3x^2 + 2x$ is continuous on $\left[0, \frac{1}{2}\right]$.

ii. Also $f'(x) = 3x^2 - 6x + 2$, exists on $\left(0, \frac{1}{2}\right)$. Thus, $f(x)$ is differentiable on $\left(0, \frac{1}{2}\right)$.

Hence, all conditions for Lagrange's mean value theorem are satisfied. Then,

$\frac{f(b) - f(a)}{b - a} = f'(c)$, for some $c \in \left(0, \frac{1}{2}\right)$

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or, $\dfrac{f\left(\frac{1}{2}\right) - f(0)}{\frac{1}{2} - 0} = 3c^2 - 6c + 2$

or, $\dfrac{\frac{3}{8} - 0}{\frac{1}{2} - 0} = 3c^2 - 6c + 2$

or, $\dfrac{3}{4} = 3c^2 - 6c + 2$

or, $12c^2 - 24c + 8 = 3$

or, $12c^2 - 24c + 5 = 0$

or, $c = \dfrac{-(-24) \pm \sqrt{(-24)^2 - 4 \times 12 \times 5}}{2 \times 12}$

$= \dfrac{24 \pm \sqrt{576 - 240}}{24}$

$= \dfrac{24 \pm \sqrt{336}}{24}$

$= \dfrac{24 \pm 4\sqrt{21}}{24}$

$\therefore \quad c = \dfrac{6 \pm \sqrt{21}}{6}$

Clearly the point $\dfrac{6 - \sqrt{21}}{6} \in \left(0, \dfrac{1}{2}\right)$.

Hence, mean value theorem is verified.

Example 6. Verify that mean value theorem for the function $f(x)$ defined by $f(x) = Ax^2 + Bx + C$, $A \neq 0$, on (a, b).

Solution:

We have, $f(x) = Ax^2 + Bx + C$

i. Since every polynomial function is continuous, $f(x)$ is continuous on $[a, b]$.

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ii. Also, $f'(x) = 2Ax + B$, which exists on (a, b). Thus, $f(x)$ is differentiable on (a, b). Then

$\dfrac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$

Here, $\dfrac{f(b) - f(a)}{b - a} = \dfrac{Ab^2 + Bb + C - Aa^2 - Ba - C}{b - a}$

$= \dfrac{A(b^2 - a^2) + B(b - a)}{(b - a)}$

$= \dfrac{A(b + a)(b - a) + B(b - a)}{(b - a)}$

$= A(b + a) + B$

Thus, $A(b + a) + B = 2Ac + B$

$2Ac = A(b + a)$

Clearly, $c = \dfrac{(b + a)}{2} \in (a, b)$.

Hence, Langrange's mean value theorem is verified.

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Exercise 2.1

1. Verify Rolle's theorem for

   i. $f(x) = x^2 - 8x + 15$ on $[0, 8]$

   ii. $f(x) = \frac{\cos x}{e^x}$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

   iii. $f(x) = x(x-8)e^{\left(-\frac{x}{2}\right)}$ on $[0, 8]$

   iv. $f(x) = e^x(\sin x - \cos x)$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

   v. $f(x) = \log\left(\frac{x^2 + ab}{(a+b)x}\right)$ on $[a, b]$, $a > 0$.

2. If $f(x) = (x-a)^m(x-b)^n$; m and n are positive integers, show that c in Rolle's theorem divides the segment $a \le x \le b$ in the ratio $m : n$.

3. Examine the validity of Rolle's theorem for the function
   $f(x) = |x - 2|$ on $[1, 3]$.

4. Discuss the applicability of Rolle's theorem for the function
   $f(x) = 1 - (x-1)^{\left(\frac{2}{3}\right)}$ on $[0, 2]$.

5. If $f(x) = \tan x$, $x \in [0, \pi]$. Is Rolle's theorem applicable to the function $f(x)$ on $[0, \pi]$?

6. Verify Lagrange's mean value theorem for

   a. $f(x) = x^2 - 2$ on $[1, 4]$

   b. $f(x) = e^x$ on $[0, 1]$

   c. $f(x) = \log x$ on $[1, e]$

   d. $f(x) = \sqrt{4 - x^2}$ on $[-2, 1]$

   e. $f(x) = x + \frac{1}{x}$ on $\left[-\frac{1}{2}, 2\right]$

7. Show that mean value theorem is not applicable for the function

$$f(x) = \begin{cases} x\cos\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases} \quad \text{in interval } (-1, 1)$$

8. For what value of a, m, b does the function

$$f(x) = \begin{cases} 3 & \text{for } x = 0 \\ -x^2 + 3x + a & \text{for } 0 < x < 1 \\ mx + b & \text{for } 1 \le x \le 2 \end{cases}$$

   satisfy the hypothesis of the mean value theorem on $[0, 2]$.

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9. Show that: $\frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}$ by using Lagrange's mean value theorem.

Answer

8. $a = 3, m = 1, b = 4$

2.2 Higher order mean value theorem

2.2.1 Taylor and Maclaurin's series

As we have observed in Lagrange's mean value theorem that it can be applied to approximate functional value at some neighbouring point $a + h$ of the point $x = a$ given by

$f(a + h) = f(a) + hf'(a + \theta h)$, where $a + \theta h \in (a, a + h)$, $0 < \theta < 1$.

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In this section, we generalize this approximation to some higher order of accuracy, using higher order derivatives.

Taylor's theorem

If $f(x)$ and its first n-derivatives $f', f'', \ldots, f^n$ are all such that:

i. are continuous in $[a, b]$.

ii. f^n is differentiable in (a, b), then there exists a number $c \in (a, b)$ such that

$$f(b) = f(a) + (b-a)f'(a) + (b-a)^2\frac{f''(a)}{2!} + \ldots + (b-a)^n\frac{f^n(a)}{n!} + (b-a)^{n+1}\frac{f^{n+1}(c)}{(n+1)!} \quad \ldots . \text{(i)}$$

Remark: If $b = a + h$ and $0 < \theta < 1$, then c can be written as $c = (a + \theta h)$ and statement (i) can be written as

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \ldots + \frac{h^n}{n!}f^n(a) + \frac{h^{n+1}}{(n+1)!}f^{n+1}(a + \theta h) \quad \ldots . \text{(ii)}$$

Equation (ii) can be applied to approximate value of $f(x)$ at $a + h$ by using values of $f(x)$ at a and $c = a + \theta h$ that belongs to $(a, a + h)$.

Either forms (i) and (ii) of Taylor's theorem are known as Taylor's series in finite forms.

Taylor's series (Taylor's theorem in infinite form)

The term $R_n = (b-a)^{n+1}\frac{f^{n+1}(c)}{(n+1)!} = \frac{h^{n+1}}{(n+1)!}f^{n+1}(a + \theta h)$

is known as remainder after n-terms. It can be shown that $R_n \to 0$ as $n \to \infty$ for all continuously differentiable functions (such as e^x, $\sin x$, $\cos x$, $\tan x$, etc.) then the Taylor's theorem in finite forms become Taylor's series in infinite form and is known as Taylor's series of $f(x)$.

The Taylor's theorem in the forms (i) or (ii) can be written in infinite form as:

$$f(b) = f(a) + (b-a)f'(a) + (b-a)^2\frac{f''(a)}{2!} + \ldots + (b-a)^n\frac{f^n(a)}{n!} + \ldots .$$

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or, $f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \ldots + \frac{h^n}{n!}f^n(a) + \ldots .$, where $h = b - a$

$$\therefore \quad f(a+h) = f(a) + f'(a)\,h + \frac{f''(a)}{2!}h^2 + \ldots + \frac{f^n(a)}{n!}h^n + \ldots .$$

which is Taylor's series in infinite form.

Maclaurin's series

The Taylor's series in infinite form is

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \ldots + \frac{h^n}{n!}f^n(a) + \ldots .$$

If we take $a = 0$ and $h = x$, then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \ldots + \frac{x^n}{n!}f^n(0) + \ldots .$$

$$\therefore \quad f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^n(0)}{n!}x^n + \ldots .$$

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Taylor's infinite series of $f(x)$ at $a = 0$ is known as Maclaurin's series representation of $f(x)$. This shows that every continuous function which is continuously differentiable can be represented by an infinite series around $a = 0$.

Method of undetermined coefficients

This method of undetermined coefficients is used to expand a function by Maclaurin's series.

We describe the method as follows:

For the given continuous function $f(x)$ we can find Maclaurin's series representation as

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots \quad \ldots . (i)$$

where $a_0, a_1, a_2, \ldots$ are constants to be determined known as undetermined coefficients.

To determine $a_0, a_1, a_2, \ldots$.

Differentiating (i) successively and taking $x = 0$ we get,

$$f'(0) = a_1$$

$$f''(0) = 2a_2, \qquad \therefore \ a_2 = \frac{f''(0)}{2!}$$

$$f'''(0) = 3 \cdot 2 \cdot 1\, a_3$$

$$\therefore \quad a_3 = \frac{f'''(0)}{3!}$$

and in general $a_n = \dfrac{f^n(0)}{n!}$

We note that $a_0 = f(0)$.

On substituting all $a_0, a_1, a_2 \ldots$ in (i), we get Maclaurin's series of $f(x)$.

Example 7. Expand $\log(1 + x)$ in ascending powers of x.

Solution:

Let $\log(1 + x) = a_0 + a_1x + a_2x^2 + \ldots + a_3x^3 + \ldots \quad \ldots . (i)$

Differentiating with respect to x, we get

$$\frac{1}{1+x} = a_1 + 2a_2x + 3a_3x^2 + \ldots \quad \ldots . (ii)$$

$$\therefore \quad 1 = (1 + x)(a_1 + 2a_2x + 3a_3x^2 + \ldots) \quad \ldots . (iii)$$

Equating the coefficients of x^n on both sides, we get

$$na_n + (n + 1)\, a_{n+1} = 0. \quad \ldots . (iv)$$

Putting $x = 0$ in (i) and (ii), we get

$a_0 = \log 1 = 0, a_1 = 1.$

Putting $n = 1, 2, 3, \ldots$ in (iv), we get

$a_2 = -\frac{1}{2}, \ a_3 = \frac{1}{3}, a_4 = -\frac{1}{4}$ and so on.

$$\therefore \quad \log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots .$$

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Example 8. Assuming the possibility of the expansion, show that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots .$$

Solutions:

Let $f(x) = e^x$ $\therefore$ $f^n(x) = e^x$.

Also, $f(0) = 1$, $f'(0) = 1$, $f''(0) = 1, f'''(0) = 1$

$f^{iv}(0) = 1$, $f^{v}(0) = 1$ $f^{vi}(0) = 1$ and so on.

Using Maclaurin's expansion of infinite series,

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^n(0)}{n!}x^n + \dots .$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots .$$

Example 9. Assuming the validity of expansion using Maclaurin's theorem, expand $(1 + x)^m$ in ascending powers of x.

Solution:

Let $f(x) = (1 + x)^m$ $\therefore f(0) = 1$

$f'(x) = m(1 + x)^{m-1}$ $\therefore f'(0) = m$

$f''(x) = m(m - 1)(1 + x)^{m-2}$ $\therefore f''(0) = m(m - 1)$

$f'''(x) = m(m - 1)(m - 2)(1 + x)^{m-3}$ $\therefore f'''(0) = m(m - 1)(m - 2)$ and so on.

Hence, using Maclaurin's expansion of infinite series, we have

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^n(0)}{n!}x^n + \dots .$$

or, $$(1 + x)^m = 1 + xm + \frac{x^2}{2!}m(m - 1) + \frac{x^3}{3!}m(m - 1)(m - 2) + \dots$$

$$= 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots$$

Example 10. Show that $\sqrt{x}$ can not be expanded by Maclaurin's infinite series.

Solution:

Since, Maclaurin's series extended to infinity is defined by

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^n(0)}{n!}x^n + \dots . \quad \dots . (i)$$

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Let $f(x) = \sqrt{x}, f'(x) = \frac{1}{2\sqrt{x}}$, then $f'(0)$ does not exist.

Similarly, $f''(0), f'''(0), \dots\dots, f^n(0)$ does not exist.

Hence, $\sqrt{x}$ can not be expanded by Maclaurin's series.

Example 11. Assuming the validity of the expansion, show that $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots .$

Solutions:

Let $f(x) = \log(1 + x)$.

Since, $f^n(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}$,

Here, Maclaurin's expansion in infinite form is valid for $-1 < x \le 1$.

Now, $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$,

$f'''(0) = 2!$, $f^{iv}(0) = -3!$, $f^{v}(0) = 4!$ and so on.

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By Maclaurin's series, we have

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^n(0)}{n!}x^n + \ldots .$$

$$\log(1+x) = 0 + x.1 + \frac{x^2}{2}(-1) + \frac{x^3}{6}.2 + \frac{x^4}{24}.(-6) + \ldots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots .$$

Example 12. Assuming the validity of the expansion, show that $e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{1}{8}x^4 + \ldots$.

Solution:

Let $f(x) = e^{\sin x}$ $\quad\therefore\quad f(0) = 1$

$f'(x) = \cos x.\, e^{\sin x}$ $\quad\therefore\quad f'(0) = 1$

$f''(x) = \cos^2 x\, e^{\sin x} - \sin x\, e^{\sin x}$ $\quad\therefore\quad f''(0) = 1$

$f'''(x) = \cos^3 x\, e^{\sin x} - 2\cos x \sin x\, e^{\sin x} - \cos x\, e^{\sin x} - \sin x \cos x\, e^{\sin x}$

$\therefore\quad f'''(0) = 1 - 0 - 1 - 0 = 0.$

Similarly, $f^{iv}(0) = -3.$

By Maclaurin's expansion,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \ldots$$

$$= 1 + x.1 + \frac{x^2}{2}.1 + \frac{x^3}{6}.0 - \frac{x^4}{24}.3 + \ldots$$

$$= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \ldots .$$

Example 13. Assuming the validity of the expansion expand $\sin x$ in powers of $x - \pi/2$.

Solutions:

Let $f(x) = \sin x.$

Then $f(x) = f[\pi/2 + (x - \pi/2)].$

By Taylor's theorem,

$$f(x) = f\left[\pi/2 + \left(x - \frac{\pi}{2}\right)\right]$$

$$= f\left(\frac{\pi}{2}\right) + \left(x - \frac{\pi}{2}\right)f'(\pi/2) + \frac{1}{2!}\left(x - \frac{\pi}{2}\right)^2 f''(\pi/2) + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 f'''(\pi/2) + \ldots . \qquad \ldots . \text{(i)}$$

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Now, $f(x) = \sin x$, $f(\pi/2) = 1$ and $f^n(x) = \sin\left(\frac{n\pi}{2} + x\right)$.

Putting $n = 1, 2, 3, \ldots$, we get

$f'(x) = \sin(\pi/2 + x) = \cos x \qquad \therefore f'(\pi/2) = \cos\frac{\pi}{2} = 0$

$f''(x) = \sin(\pi + x) = -\sin x, \qquad \therefore f''(\pi/2) = -\sin\frac{\pi}{2} = -1$

$f'''(x) = \sin\left(\frac{3\pi}{2} + x\right) = -\cos x, \qquad \therefore f'''(\pi/2) = -\cos\frac{\pi}{2} = 0.$

and so on. Substituting these values in (i), we get

$$f(x) = \sin x$$

$$= 1 + \left(x - \frac{\pi}{2}\right).0 + \frac{1}{2!}\left(x - \frac{\pi}{2}\right)^2.(-1) + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3.0 + \ldots$$

$$= 1 - \frac{1}{2!}\left(x - \frac{\pi}{2}\right)^2 + \ldots .$$

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Example 14. Assuming the validity of the expansion, expand e^x in powers of $x - 1$.

Solutions:

Let $f(x) = e^x$.

$\therefore \quad e^x = f(x) = f[1 + (x-1)]$

$$= f(1) + (x-1)f'(1) + \frac{1}{2!}(x-1)^2 f''(1) + \frac{1}{3!}(x-1)^3 . f'''(1) + \quad \ldots . \text{(i)}$$

(by Taylor's Theorem)

Now, $f(x) = e^x, \quad f(1) = e$

$f'(x) = e^x \quad f'(1) = e$

$f''(x) = e^x, \quad f''(1) = e$

$f'''(x) = e^x, \quad f'''(1) = e$ and so on.

Substituting these values in (i), we get

$$e^x = f(x) = e + (x-1)\,e + \frac{1}{2!}(x-1)^2 . e + \frac{1}{3!}(x-1)^3 e + \ldots$$

$$= e\left[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \ldots\right].$$

Example 15. Assuming the validity of expansion using Maclaurin's theorem, prove the following series :

$$e^x \sec x = 1 + x + x^2 + \frac{2}{3}x^3 + \ldots$$

Solution:

Let $y = f(x) = e^x \sec x$. $\quad \therefore \quad y(0) = 1.$

Now, $y_1(x) = e^x \sec x + e^x \sec x . \tan x.$

$= e^x \sec x\,(1 + \tan x)$

$= f(x)\,(1 + \tan x)$

$= y\,(1 + \tan x)$

Now, $y_1(0) = y(0)\,(1 + 0)$ $\quad \therefore \quad y_1(0) = 1.$

Again differentiating with respect to x gives,

$y_2 = y \sec^2 x + y_1\,(1 + \tan x)$

Now, $y_2(0) = y(0) . \sec^2 0 + y_1(0) . (1) = 1 \times 1 + 1 \times 1 = 2.$

$\therefore \quad y_2(0) = 2.$

Again, differentiating with respect to x, we get

$y_3 = y . 2 \sec^2 x . \tan x + y_1 \sec^2 x + y_1 \sec^2 x + y_2(1 + \tan x)$

$= 2y \sec^2 x . \tan x + 2y_1 . \sec^2 x + y_2(1 + \tan x)$

Now, $y_3(0) = 2\,y(0) \times 0 + 2\,y_1(0) . 1 + y_2(0)\,(1 + 0)$

$= 0 + 2 \times 1 + 2 \times 1 = 2 + 2 = 4$ and so on.

We note that $f^n(x) = y_n(x)$. The Maclaurin's series is given by,

$$y(x) = f(x) = y(0) + y_1(0)x + \frac{y_2(0)}{2!}x^2 + \frac{y_3(0)}{3!}x^3 + \frac{y_4(0)}{4!}x^4 + \frac{y_5(0)}{5!}x^5 + \frac{y_6(0)}{6!}x^6 + \ldots$$

$\therefore \quad e^x \sec x = 1 + x + x^2 + \frac{2}{3}x^3 + \ldots$

For some functions we can find their Maclaurin's series representation by termwise differentiation or integration of the series of other functions. We illustrate it by following examples.

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Example 16. Assuming the validity of expansion, find Maclaurin's series expansion of $f(x) = \log(\sec x)$. Find expansion of $\tan x$.

Solution:

We have given function,

$y = f(x) = \log(\sec x)$ ....(i)

$\therefore$ $y(0) = \log 1 = 0$

Differentiating (i) successively, we get

$y_1 = \frac{1}{\sec x} \sec x \tan x = \tan x$, and $y_1(0) = 0$

$y_2 = \sec^2 x$ and $y_2(0) = 1$

$y_3 = 2\sec x \cdot \sec x \tan x = 2y_2y_1$ and $y_3(0) = 2y_2(0)y_1(0) = 0$

$y_4 = 2y_3y_1 + 2y_2y_2 = 2y_1y_3 + 2y_2^2$ and $y_4(0) = 2(1) = 2$

$y_5 = 2y_1y_4 + 2y_2y_3 + 4y_2y_3$ and $y_5(0) = 0$

$y_6 = 2y_1y_5 + 2y_2y_4 + 6y_3^2 + 6y_2y_4$ and $y_6(0) = 2 \times 2 + 6 \times 2 = 16$

We note that $f^n(x) = y_n(x)$. The Maclaurin's series is given by,

$$y(x) = f(x) = y(0) + y_1(0)x + \frac{y_2(0)}{2!}x^2 + \frac{y_3(0)}{3!}x^3 + \frac{y_4(0)}{4!}x^4 + \frac{y_5(0)}{5!}x^5 + \frac{y_6(0)}{6!}x^6 + \ldots$$

$\therefore$ $\log(\sec x) = \frac{x^2}{2!} + \frac{2x^4}{4!} + \frac{16x^6}{6!} + \ldots$ ....(ii)

For the series of $\tan x$, differentiating each terms of (ii)

$\frac{d \log(\sec x)}{d(\sec x)} \times \frac{d \sec x}{dx} = \frac{2x}{2!} + \frac{8x^3}{4!} + \frac{16.6.x^5}{6!} + \ldots$

$\frac{1}{\sec x} \sec x \tan x = x + \frac{2}{3!}x^3 + \frac{16}{5!}x^5 + \ldots$

$\therefore$ $\tan x = x + \frac{2}{3!}x^3 + \frac{16}{5!}x^5 + \ldots$.

Example 17. Assuming the validity of expansion, find Maclaurian's representation of $y = f(x) = \frac{e^x}{e^x + 1}$ and hence determine the series of $\log(1 + e^x)$.

Solution:

The given function is,

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$y = f(x) = \frac{e^x}{e^x + 1}$ ....(i)

$y(0) = \frac{1}{2}$

Equation (i) can be written as, $(e^x + 1)y = e^x$ ....(ii)

Differentiating (ii) with respect to x, we get

$e^x y + (e^x + 1)y_1 = e^x$ ....(iii)

$\therefore$ $e^0 y(0) + (e^0 + 1)y_1(0) = e^0$

$\frac{1}{2} + 2y_1(0) = 1$

or, $2y_1(0) = \frac{1}{2}$

$\therefore$ $y_1(0) = \frac{1}{4}$

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Differentiating (iii) with respect to x, we get

$e^x y + e^x y_1 + e^x y_1 + (e^x + 1)\, y_2 = e^x$

$\therefore$ $e^x y + 2e^x y_1 + (e^x + 1) y_2 = e^x$ .... (iv)

and $e^0 y(0) + 2e^0 y_1(0) + (e^0 + 1)\, y_2(0) = e^0$

$\frac{1}{2} + 2\left(\frac{1}{4}\right) + 2y_2(0) = e^0$

$\therefore$ $y_2(0) = 0$

Differentiating (iv) with respect to x, we get

$e^x y + e^x y_1 + 2e^x y_1 + 2e^x y_2 + e^x y_2 + (e^x + 1) y_3 = e^x$

$\therefore$ $e^x y + 3e^x y_1 + 3e^x y_2 + (e^x + 1) y_3 = e^x$

$\therefore$ $e^0 y(0) + 3e^0 y_1(0) + 3e^0 y_2(0) + (e^0 + 1) y_3(0) = e^0$

$\frac{1}{2} + 3\left(\frac{1}{4}\right) + 3.0 + 2y_3(0) = 1$

$\therefore$ $2y_3(0) = 1 - \frac{1}{2} - \frac{3}{4} = \frac{4 - 2 - 3}{4}$

$\therefore$ $y_3(0) = -\frac{1}{8}$

$\therefore$ The Maclarin's series of $f(x) = \frac{e^x}{e^x + 1}$ is

$$y(x) = f(x) = y(0) + y_1(0)x + \frac{y_2(0)}{2!}x^2 + \frac{y_3(0)}{3!}x^3 + \frac{y_4(0)}{4!}x^4 + \frac{y_5(0)}{5!}x^5 + \frac{y_6(0)}{6!}x^6 + \ldots$$

or, $\frac{e^x}{e^x + 1} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{8(3!)} + \ldots$

$\therefore$ $\frac{e^x}{e^x + 1} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \ldots$

For the Maclaurin's series of $\log(e^x + 1)$ integrating each terms on both sides, we get

$$\int \frac{e^x dx}{e^x + 1} = \frac{1}{2}\int dx + \frac{1}{4}\int x dx - \frac{1}{48}\int x^3 dx + \ldots$$

$\therefore$ $\log(e^x + 1) = \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{48 \times 4} + \ldots + C$ .... (v)

For value of C, taking $x = 0$

$\log 2 = 0 + C$

$\therefore$ $C = \log 2$

$\therefore$ Required Maclaurin's series of $\log(1 + e^x)$ is

$\log(e^x + 1) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \ldots$.

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Exercise 2.2

1. Assuming the validity of the expansion, prove the following using Maclaurin's series:

i. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

ii. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$

iii. $\log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots$

iv. $\frac{1}{1 + x} = 1 - x + x^2 - x^3 + \ldots$

v. $e^{x/2} = 1 + \frac{x}{2} + \frac{1}{4}\frac{x^2}{2!} + \frac{1}{8}\frac{x^3}{3!} + \ldots$

vi. $e^{x \cos x} = 1 + x + \frac{x^2}{2} - \frac{x^3}{3} + \ldots$.

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2. Assuming the validity of the expansion, prove the following using Maclaurin's series.

i. $\log \sec x = \frac{x^2}{2!} + \frac{2x^4}{4!} + \frac{16x^6}{6!} + \ldots .$

ii. $\sin (e^x - 1) = x + \frac{x^2}{2} - \frac{5x^4}{24} + \ldots .$

iii. $\log (1 + \tan x) = x - \frac{x^2}{2!} + \frac{4}{3!} x^3 - \frac{14x^4}{4!} + \ldots .$

iv. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots .$

3. Prove the following expansions using Maclaurin's series.

i. $e^x \sec x = 1 + x + x^2 + \frac{2}{3} x^3 + \ldots .$

ii. $e^x \log (1 + x) = x + \frac{x^2}{2!} + \frac{2x^3}{3!} + \frac{9x^5}{5!} + \ldots .$

4. i. Expand tan x by Maclaurin's series in ascending powers of x and deduce the series for $\sec^2 x$.

ii. Expand $f(x) = \log (1 + e^x)$ as for as the term x^4 by Maclaurin's series and hence the expansion of $e^x / (1 + e^x)$.

Answers

4. i. $x + \frac{x^3}{3} + \frac{2}{15} x^5 + \ldots ; 1 + x^2 + \frac{2}{3} x^4 + \ldots$

ii. $\log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \ldots ; \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \ldots .$

2.3 Asymptotes

The word asymptote is derived from the Greek word *'asumptōtos'* which means "not falling together." The term was introduced by Apollonius of Perga in his work on conic sections. Apollonius of Perga was an Ancient Greek geometer and astronomer who is known for his work on conic sections. Beginning from the contributions of Euclid and Archimedes on the topic, he brought them to the state prior to the invention of analytic geometry.

Apollonius of Perga
(240 BCE – 190 BCE)

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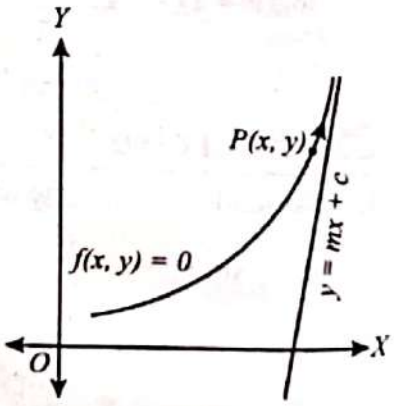
Definition

An asymptote to a curve is a straight line such that its perpendicular distance from a point on the curve approaches to zero as the point moves to infinity along the curve.

In other words, the line $y = mx + c$ is an asymptote to the curve $f(x, y) = 0$ if the perpendicular distance of $y = mx + c$ from a point in the curve is zero as the point $P(x, y)$ moves to infinity along the curve.

i.e. $\lim_{P(x,y) \to \infty} \left(\frac{y - mx - c}{\sqrt{1 + m^2}} \right) = 0$



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Remark:

1. If $y = y(x)$ is an equation of a curve then, without loss of generality, we can write it as $f(x, y) = 0$.
   For example, the parabola $y^2 = 4ax$, i.e., $y = y(x)$ can be written as $y^2 - 4ax = 0$, i.e., $f(x, y) = 0$.
2. From the definition of asymptote we see that an asymptote to a curve is a straight line which is tangent to the curve at infinity. (Tangent is a line that intersects a curve at two co-incident points.)
3. In the definition we have seen that a line is an asymptote to a curve if its perpendicular distance from a point on the curve is zero as the curve moves to infinity, it is obvious that the curve must extend up to infinity, hence the asymptote exists and defined for an unbounded (infinite) curves.
   For example, a parabola and hyperbola extend up to infinity so they have asymptote while a circle and ellipse which are finitely closed curves so they have no asymptote.
4. Finding an asymptote is just a task of determining equation of straight lines in relation to given curves. Note that there are generally three types of straight lines (i) lines parallel to x-axis, (ii) lines parallel to y-axis and (iii) lines neither parallel to x-axis nor parallel to y-axis. Such lines are called oblique lines and are given by $y = mx + c$.
5. It is worth to note that every straight line is the asymptote to itself.
6. For a given curve, how many asymptotes will be possible at maximum?
   We try to answer this question as follows:
   Consider the second degree equation in x and y as
   $$f(x, y) = x^2 + 3xy - 4y^2 = 0 \quad \ldots . (i)$$
   which is a second degree homogeneous equation and can be written as
   $$(x + 4y)(x - y) = 0$$
   $\therefore$ $x + 4y = 0$ and $x - y = 0$.
   Which are straight lines and hence asymptotes to themselves (or asymptotes to the given curve (i)).
   It is to be noted that not every second degree equations can be factored as above, but we can in general conclude that if an n-degree equation be factored into n linear factors then each factor will give rise to an asymptote, i.e., an n-degree curve will have n asymptotes at maximum. If we can find m asymptotes ($m < n$) then rest of ($n - m$) asymptotes are considered as imaginary asymptotes and the m-asymptotes are called real (possible) asymptotes.

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2.3.1 Types of asymptotes

1. Horizontal asymptotes (parallel to x-axis)

Let $y = f(x)$ be given function. Then the line $y = b$ becomes a horizontal asymptote to the curve if

$$\lim_{x \to \pm\infty} f(x) = b.$$

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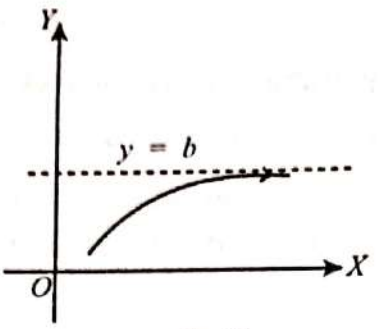

fig. (i)

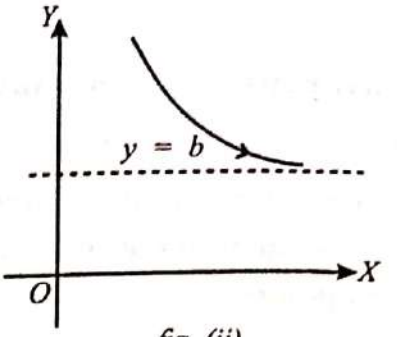

fig. (ii)

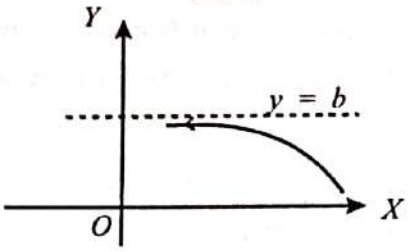

fig. (iii)

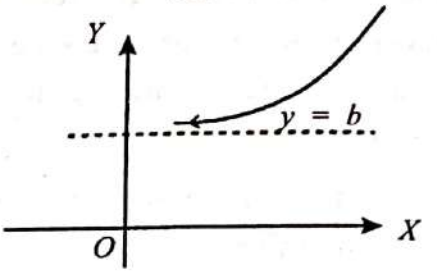

fig. (iv)

In fig. (i) and (ii), we see that $y \to b$ from below and above as $x \to +\infty$. So $y = b$ is an asymptote. Similarly, we see that $y \to b$ when $x \to -\infty$ in figure (iii) and (iv).

2. Vertical asymptotes (parallel to y–axis)

A line $x = a$ becomes a vertical asymptote to the curve $y = f(x)$ if $\lim_{x \to a} f(x) = \pm \infty$.

In the following figures, the dashed lines are used to illustrate the vertical asymptotes.

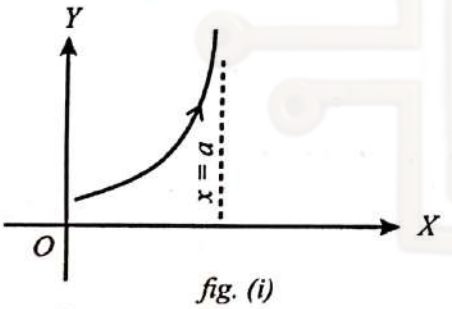

fig. (i)

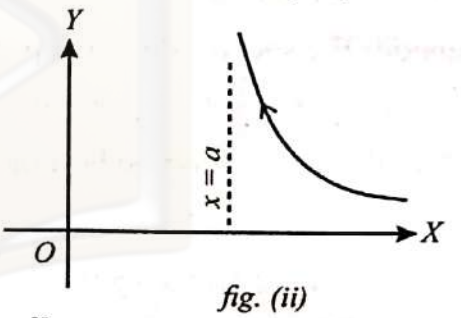

fig. (ii)

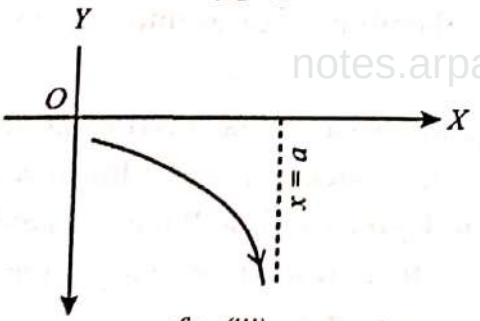

fig. (iii)

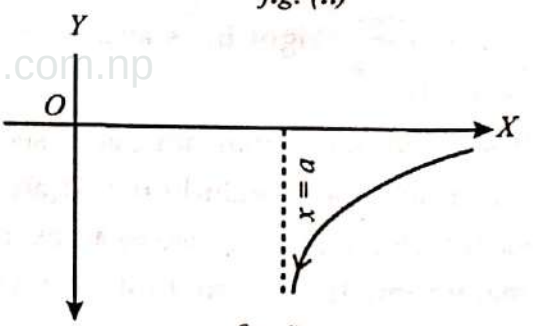

fig. (iv)

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The figure (i) has the asymptote $x = a$ in which $f(x) \to +\infty$ as $x \to a$ from the left. Similarly, the line $x = a$ represents an asymptote in which $f(x) \to +\infty$ as $x \to a$ from the right in fig. (ii) and so on.

Similarly, figure (iii) has the asymptote $x = a$ in which $f(x) \to -\infty$ as $x \to a$ from the left. Similarly, the line $x = a$ represents an asymptote in which $f(x) \to -\infty$ as $x \to a$ from the right in fig. (iv).

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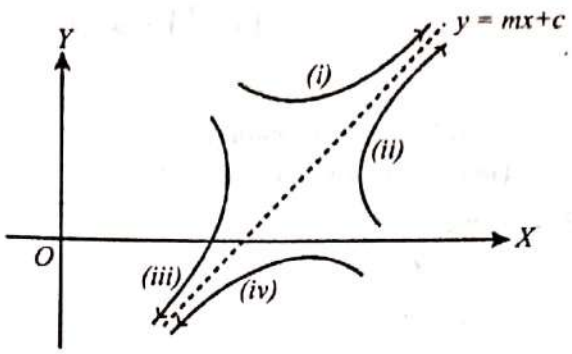
3. Oblique asymptotes

An asymptote to a curve is called *oblique* if it is an asymptote to the curve not parallel to either of the axes.

Thus, an oblique asymptote is of the form $y = mx + c$ with $m \neq 0$.

The various positions of an oblique asymptote to a curve is discussed as follows:



Case I. In (i), the curve approaches the asymptote from above to the right. In this case, as $x \to \infty$, $y - mx - c \to 0$ through values greater than 0.

Case II. In (ii), the curve approaches the asymptote from below to the right. In this case, as $x \to \infty$, $y - mx - c \to 0$ through values less than 0.

Case III. In (iii), the curve approaches the asymptote from above to the left. In this case, $y - mx - c \to 0$ through values greater than 0 as $x \to -\infty$.

Case IV. In (iv), the curve approaches the asymptote from below to the left and as $x \to -\infty$, $y - mx - c \to 0$ through values less than 0.

2.3.2 Determination asymptotes parallel to axes

1. If we can solve given equation for one variable in terms of other variable explicitly in the form of rational function, then the asymptotes parallel to axes (x-axis and y-axis) can be determined by the use of limit.

Example 18. Find the asymptotes of the curve $y = \dfrac{x^2}{(x-1)^2(x-2)}$ by using the limit.

Solution:

Here, $y = \dfrac{x^2}{(x-1)^2(x-2)}$ .... (i)

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Now, $\lim_{x \to 1} y = \lim_{x \to 1} \dfrac{x^2}{(x-1)^2(x-2)} = \infty$.

$\therefore \lim_{x \to 1} y = \infty$

So, $x = 1$ is an asymptote to (i).

Again, $\lim_{x \to 2} y = \lim_{x \to 2} \dfrac{x^2}{(x-1)^2(x-2)} = \infty$.

$\therefore \lim_{x \to 2} y = \infty$

So, $y = 2$ is an asymptote.

Also, $y \to 0$ as $x \to \infty$,

Since, $\lim_{x \to \infty} y = \lim_{x \to \infty} \dfrac{x^2}{(x-1)^2(x-2)}$

$= \lim_{x \to \infty} \dfrac{x^2}{x^2\left(1 - \frac{1}{x}\right)^2 (x-2)}$

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$$= \lim_{x \to \infty} \frac{1}{\left(1 - \frac{1}{x}\right)^2 (x - 2)}$$

$$= 0$$

$\therefore$ $y = 0$ is also an asymptote.

Hence, the asymptotes are $x = 1$, $x = 2$ and $y = 0$.

2. If we cannot solve for one variable in terms of other explicitly, the asymptote to the algebraic curve $f(x, y) = 0$ is determined as follows:

To find asymptote parallel to y-axis.

Suppose the general algebraic curve $f(x, y) = 0$ be of degree n. So it can be rewritten completely as

$$(a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \ldots + a_{n-1}xy^{n-1} + a_ny^n) + (b_1x^{n-1} + b_2x^{n-2}y + \ldots + b_{n-1}xy^{n-2} + b_ny^{n-1}) + \ldots + (c_2x^{n-2} + c_3x^{n-3}y + \ldots + c_ny^{n-2}) + \ldots = 0 \quad \ldots (i)$$

We can rearrange the terms in descending powers of y as,

$$a_ny^n + (a_{n-1}x + b_{n-1})y^{n-1} + (a_{n-2}x^2 + b_{n-2}x + c_{n-2})y^{n-2} + \ldots = 0 \quad \ldots (ii)$$

For the asymptotes parallel to y-axis of the form $x = a$, we have

$$\lim_{x \to \infty} y = \infty.$$

We must have $a_n = 0$, since second term equated to zero will give the line of the form $x = a$.

Therefore, the condition necessary for the curve to have asymptote parallel to y-axis is $a_n = 0$, i.e., $f(x, y)$ should not contain n degree term in y.

Thus, the asymptote parallel to y-axis is obtained from coefficient of second term in (ii).

i.e., $a_{n-1}x + b_{n-1} = 0$

$\therefore$ $x = -\frac{b_{n-1}}{a_{n-1}}$

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We note that the coefficient of second term is the coefficient of highest available power of y (less than degree n) in the equation $f(x, y) = 0$.

Similarly, we can determine asymptote parallel to x-axis that does not contain n degree term in x, by equating the coefficient of highest available power of x (less than degree n) in the equation $f(x, y) = 0$ to zero.

Example 19. Determine the asymptotes parallel to axes to the curve $(y - a)^2 (x^2 - a^2) = x^4 + a^4$.

Solution:

Given equation is $(y - a)^2 (x^2 - a^2) = x^4 + a^4$ ...

Equation (i) is of degree 4 and x^4 is present in the equation so it totes parallel to x-axis.

But, y^4 is absent in the equation, so it has asymptotes parallel to obtained by equating the coefficient of y^2 in (i) (coefficient of highest avail. y in the equation $(y^2 - 2ay + a^2)(x^2 - a^2) = x^4 + a^4$) to zero.

$\therefore$ $x^2 - a^2 = 0$

$x = \pm a$ are the asymptotes parallel to y-axis.

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2.3.3 Determination of oblique asymptotes

Let $y = mx + c$ .... (i)

be an oblique asymptote to the curve

$f(x, y) = 0$ .... (ii)

To determine above we need to find values of m and c

a. For value of m, by definition of asymptote the point on the curve moves to infinity, i.e., we have $x \to \infty$.

From (i) we can write,

$$\lim_{x\to\infty} \frac{(mx - y + c)}{\sqrt{m^2 + 1}} = 0$$

or, $$\lim_{x\to\infty} x\left(m - \frac{y}{x} + \frac{c}{x}\right) = 0$$

or, $$m = \lim_{x\to\infty}\left(\frac{y}{x} - \frac{c}{x}\right)$$

$$\therefore \quad m = \lim_{x\to\infty}\left(\frac{y}{x}\right)$$

b. For value of c, we can write

$$\lim_{x\to\infty} \frac{(mx - y + c)}{\sqrt{m^2 + 1}} = 0$$

$$\therefore \quad c = \lim_{x\to\infty} (y - mx)$$

On substituting value of m and c we get required oblique asymptotes.

Determination of values of m and c for an algebraic curve

An algebraic curve can, in general, be written as

$f(x, y) = 0$ .... (i)

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and let $y = mx + c$ .... (ii)

be its oblique asymptote.

If it is of degree n, then this can be expressed as

$$(a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \ldots + a_{n-1}xy^{n-1} + a_ny^n) + (b_0x^{n-1} + b_1x^{n-2}y + b_2x^{n-3}y^2 + \ldots + b_{n-2}xy^{n-2} + b_{n-1}y^{n-1}) + \ldots + (c_0x^{n-2} + c_1x^{n-3}y + c_2x^{n-4}y^2 + \ldots + c_{n-2}y^{n-2}) + \ldots = 0 \quad \ldots. \text{(iii)}$$

Where, $a_0, \ldots, a_n, b_0, \ldots, b_{n-1}, c_0, \ldots, c_{n-2}$ are real constants.

Equation (i) can be re-written as

$$x^n\left\{a_0 + a_1\left(\frac{y}{x}\right) + a_2\left(\frac{y}{x}\right)^2 + \ldots + a_n\left(\frac{y}{x}\right)^n\right\} + x^{n-1}\left\{b_0 + b_1\left(\frac{y}{x}\right) + b_2\left(\frac{y}{x}\right)^2 + \ldots + b_{n-1}\left(\frac{y}{x}\right)^{n-1}\right\}$$

$$+ x^{n-2}\left\{c_0 + c_1\left(\frac{y}{x}\right) + c_2\left(\frac{y}{x}\right)^2 + \ldots + c_{n-2}\left(\frac{y}{x}\right)^{n-2}\right\} + \ldots = 0$$

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$$\therefore \quad x^n \phi_n\left(\frac{y}{x}\right) + x^{n-1} \phi_{n-1}\left(\frac{y}{x}\right) + x^{n-2} \phi_{n-2}\left(\frac{y}{x}\right) + \ldots = 0 \qquad \ldots . \text{(iv)}$$

where, $\phi_n\left(\frac{y}{x}\right) = a_0 + a_1\left(\frac{y}{x}\right) + a_2\left(\frac{y}{x}\right)^2 + \ldots + a_n\left(\frac{y}{x}\right)^n$

$$\phi_{n-1}\left(\frac{y}{x}\right) = b_0 + b_1\left(\frac{y}{x}\right) + b_2\left(\frac{y}{x}\right)^2 + \ldots + b_{n-1}\left(\frac{y}{x}\right)^{n-1}$$

and $\phi_{n-2}\left(\frac{y}{x}\right) = c_0 + c_1\left(\frac{y}{x}\right) + c_2\left(\frac{y}{x}\right)^2 + \ldots + c_{n-2}\left(\frac{y}{x}\right)^{n-2}$

Since $y = mx + c$, equation (iv) becomes,

$$\therefore \quad x^n \phi_n\left(\frac{mx+c}{x}\right) + x^{n-1} \phi_{n-1}\left(\frac{mx+c}{x}\right) + x^{n-2} \phi_{n-2}\left(\frac{mx+c}{x}\right) + \ldots = 0$$

$$\therefore \quad x^n \phi_n\left(m + \frac{c}{x}\right) + x^{n-1} \phi_{n-1}\left(m + \frac{c}{x}\right) + x^{n-2} \phi_{n-2}\left(m + \frac{c}{x}\right) + \ldots = 0 \qquad \ldots . \text{(v)}$$

Using Taylor's infinite series expansion of each

$\phi_i\left(m + \frac{c}{x}\right)$ for $i = n, n-1, n-2$, we have

$$x^n\left[\phi_n(m) + \frac{c}{x}\phi'_n(m) + \frac{1}{2!}\frac{c^2}{x^2}\phi''_n(m) + \frac{1}{3!}\frac{c^3}{x^3}\phi'''_n(m) + \ldots\right]$$

$$+ x^{n-1}\left[\phi_{n-1}(m) + \frac{c}{x}\phi'_{n-1}(m) + \frac{1}{2!}\frac{c^2}{x^2}\phi''_{n-1}(m) + \frac{1}{3!}\frac{c^3}{x^3}\phi'''_{n-1}(m) + \ldots\right]$$

$$+ x^{n-2}\left[\phi_{n-2}(m) + \frac{c}{x}\phi'_{n-2}(m) + \frac{1}{2!}\frac{c^2}{x^2}\phi''_{n-2}(m) + \frac{1}{3!}\frac{c^3}{x^3}\phi'''_{n-2}(m) + \ldots\right] = 0$$

$$\therefore \quad x^n\phi_n(m) + x^{n-1}\{c\phi'_n(m) + \phi_{n-1}(m)\} + x^{n-2}\left\{\frac{c^2}{2!}\phi''_n(m) + c\,\phi'_{n-1}(m) + \phi_{n-2}(m)\right\}$$

$$+ x^{n-3}\left\{\frac{c^3}{3!}\phi'''_n(m) + \frac{c^2}{2!}\phi''_{n-1}(m) + \phi_{n-3}(m)\right\} + \ldots = 0 \qquad \ldots . \text{(vi)}$$

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Since $y = mx + c$ is asymptote to the curve of $f(x, y) = 0$.

Also we have, $m = \lim_{x \to \infty}\left(\frac{y}{x}\right)$ and $\lim_{x \to \infty}(y - mx - c) = 0$.

So for (vi) to be true we must have the coefficients of various powers of x equal to zero:

$$\therefore \quad \phi_n(m) = 0 \qquad \ldots . \text{(a)}$$

$$c\phi'_n(m) + \phi_{n-1}(m) = 0 \qquad \ldots . \text{(b)}$$

$$\frac{c^2}{2!}\phi''_n(m) + c\phi'_{n-1}(m) + \phi_{n-2} = 0 \qquad \ldots . \text{(c)}$$

$$\frac{c^3}{3!}\phi'''_n(m) + \frac{c^2}{2!}\phi''_{n-1}(m) + c\phi_{n-2}(m) + \phi_{n-3}(m) = 0 \qquad \ldots . \text{(d)}$$

Equation (a) gives n values of m and hence using these n values in (b) we get n values of c. On substituting these values of m and c in $y = mx + c$ we get required asymptotes to $f(x, y) = 0$.

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Remarks:

1. If $\phi_n(m) = 0$ gives twice repeated values of m i.e., repeated roots then (b) can not give values of c in such case we use (c) to determine values of c.
2. If $\phi_n(m) = 0$ gives thrice repeated values of m, we use (d) to determine values of c.
3. Since $m = \lim_{x \to \infty} \left(\frac{y}{x}\right)$, so to obtain the polynomials, $\phi_n(m)$, $\phi_{n-1}(m)$, $\phi_{n-2}(m)$, ..., we can simply take $y = m$ and $x = 1$.

This will be clear from the following example.

Example 20. Find oblique asymptotes to the curve $y^3 - x^2y + 2y^2 + 4y + x = 0$.

Solution:

We have given curve

$(y^3 - x^2y) + 2y^2 + (4y + x) = 0$ .... (i)

Let $y = mx + c$ .... (ii)

be oblique asymptote to curve (i).

Taking $y = m$ and $x = 1$ in (i)

$\phi_3(m) = m^3 - m$

$\phi_2(m) = 2m^2$

$\phi_1(m) = 4m + 1$

For the values of m,

$\phi_3(m) = 0$

$\therefore \quad m(m^2 - 1) = 0$

$\therefore \quad m = 0, m = 1, m = -1$

For values of c, $c\phi'_3(m) + \phi_2(m) = 0$

$c(3m - 1) + 2m^2 = 0 \Rightarrow c = -\frac{2m^2}{3m - 1}$

For $m = 0$, $c = 0$

For $m = 1 \Rightarrow c = -\frac{3}{3 - 1} = -1$

For $m = -1 \Rightarrow c = \frac{-2}{-4} = \frac{1}{2}$

Thus, after substituting the values of m and c in (ii), the oblique asymptotes are

$y = 0, y = x - 1, y = -x + \frac{1}{2}$.

Remark:

From above example, we see that sometimes while determining oblique asymptotes we may get $m = 0$ and $c = 0$ which gives $y = 0$ as an asymptote parallel to x-axis i.e., x-axis itself. The asymptotes parallel to axes are obtained by equating the coefficients of highest degree available to zero as discussed earlier.

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Example 21. Determine the asymptotes of the curve $y = \frac{x^2}{x^2+1}$.

Solution:

Here, $y = \frac{x^2}{x^2+1}$.

Now, $\lim_{x\to\infty} y = \lim_{x\to\infty} \frac{x^2}{x^2+1} = \lim_{x\to\infty} \frac{x^2}{x^2\left(1+\frac{1}{x^2}\right)} = \lim_{x\to\infty} \frac{1}{1+\frac{1}{x^2}} = 1.$

$\therefore$ $y = 1$ for $x \to \pm\infty$. Hence, $y = 1$ is an asymptote.

Again, we have $y - 1 = \frac{x^2}{x^2+1} - 1 = \frac{-1}{x^2+1} < 0.$

This shows that $y \to 1$ from the values less than 1 as $x \to +\infty$ or $x \to -\infty$.

So the curve approaches the asymptote $y = 1$ from the below to the right and from the below to the left.

Example 22. Determine the asymptotes of the curve $y = \frac{x^2+2x-1}{x}$.

Solution:

Now, $\lim_{x\to 0^-} y = \lim_{x\to 0^-} \left(\frac{x^2+2x-1}{x}\right) = \lim_{x\to 0^-} \left(x+2-\frac{1}{x}\right) = -\infty$ and

$\lim_{x\to 0} y = \lim_{x\to 0} \left(\frac{x^2+2x-1}{x}\right) = \lim_{x\to 0} \left(x+2-\frac{1}{x}\right) = \infty.$

Hence, $x = 0$ is an asymptote.

$\therefore$ The curve approaches the asymptote $x = 0$ downwards when $x \to 0^+$ and upwards when $x \to 0^-$.

Now we find oblique asymptote of the form $y = mx + c$,

$$m = \lim_{x\to\infty} \frac{y}{x}$$

$$= \lim_{x\to\infty} \left(\frac{x^2+2x-1}{x^2}\right) = 1 \text{ (finite)}$$

and $$c = \lim_{x\to\infty} (y - mx)$$

$$= \lim_{x\to\infty} (y - x)$$

$$= \lim_{x\to\infty} \left(\frac{x^2+2x-1}{x} - x\right)$$

$$= \lim_{x\to\infty} \left(\frac{2x-1}{x}\right) = 2.$$

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$\therefore$ $y = mx + c$ i.e., $y = x + 2$ is an oblique asymptote. Further,

$$\lim_{x\to\infty} (y - x - 2) = \lim_{x\to\infty} \left(\frac{x^2+2x-1}{x} - x - 2\right)$$

$$= \lim_{x\to\infty} \left(\frac{x^2+2x-1-x^2-2x}{x}\right)$$

$$= \lim_{x\to\infty} \left(-\frac{1}{x}\right) = 0.$$

Similarly, we can prove that $\lim_{x\to-\infty} (y - x - 2) = 0.$

Thus when $x \to +\infty$, $y - x - 2 \to 0$ through the values less then zero and negative value, and when $x \to -\infty$, $y - x - 2 \to 0$ through the values greater than zero.

The relative position of the curve and asymptotes are given below:

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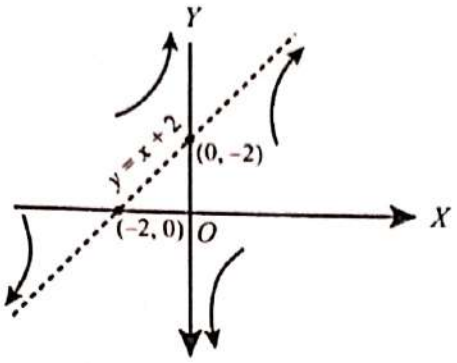

Example 23. Find all possible asymptotes for $y = \frac{2x-3}{x^2-3x+2}$.

Solution:

The given equation is $y = \frac{2x-3}{x^2-3x+2}$.

i.e., $x^2y - 3xy - 2x + 2y + 3 = 0$ which is of 3rd degree and the terms involving x^3 and y^3 are both absent. Hence there is no asymptote of the form $y = mx + c$.

To find the parallel asymptotes

Equating the coefficient of highest degree term of x to zero

i.e., coefficient of $x^2 = 0$ or, $y = 0$.

Again, equating the coefficient of highest degree term of y to zero

i.e., coefficient of $y = 0$ or, $x^2 - 3x + 2 = 0$

or, $(x-1)(x-2) = 0$ or, $x = 1$ and $x = 2$.

$\therefore$ Equation of asymptotes are $y = 0$, $x = 1$ and $x = 2$.

Example 24. Find all the asymptotes of the curve $(y-a)^2(x^2-a^2) = x^4 + a^4$

Solution:

The given equation can be written as

$(y^2 - 2ay + a^2)(x^2 - a^2) = x^4 + a^4$

or, $x^2y^2 - a^2y^2 - 2ax^2y + 2a^3y + a^2x^2 - a^4 = x^4 + a^4$

or, $(x^4 - x^2y^2) + 2ax^2y - a^2x^2 + a^2y^2 - 2a^3y + 2a^4 = 0$

This algebraic equation is of 4th degree. So there are at most four asymptotes. Since x^4 is present and y^4 is absent, hence there is no asymptote parallel to x-axis and has only asymptotes parallel to y-axis.

The equation of asymptotes parallel to y-axis can be obtained by equating the coefficient of y^2 to zero

i.e. $-x^2 + a^2 = 0$ $\therefore$ $x = \pm a$.

We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$, in 4th and 3rd degree terms, we get

$\phi_4(m) = 1 - m^2$ and $\phi_3(m) = 2am$.

Also, $\phi'_4(m) = -2m$.

Now, $\phi_4(m) = 0$ gives

$1 - m^2 = 0$ $\therefore$ $m = \pm 1$.

Now, $c = -\frac{\phi_3(m)}{\phi'_4(m)} = -\frac{2am}{-2m} = a$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the oblique asymptotes are

$y = \pm 1x + a$ i.e., $y \pm x = a$.

Hence, equations of asymptote are, $x = \pm a$ and $y \pm x = a$.

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Example 25. Find the asymptotes of the curve $y^2 = \frac{(a-x)^2}{a^2+x^2} \cdot x^2$

Solution:

Here, $y^2 = \frac{(a-x)^2}{a^2+x^2} \cdot x^2$

or, $x^4 - 2ax^3 - x^2y^2 + a^2x^2 - a^2y^2 = 0$ .... (i)

This equation is of 4th degree and the terms involving x^4 is present, hence there is no asymptote parallel to x-axis. Since, y^4 is absent, so the asymptotes parallel to y-axis can be obtained by equating the coefficient of y^2 to zero

i.e., $-x^2 - a^2 = 0$

or, $x^2 + a^2 = 0$ i.e. $x = \pm ia$, which are imaginary.

Hence, there is no real asymptote parallel to y-axis.

We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$ in 4th and 3rd degree terms, we get

$\phi_4(m) = 1 - m^2$ and $\phi_3(m) = -2a$.

Now, $\phi_4(m) = 0$ gives

$1 - m^2 = 0$

$\therefore$ $m = \pm 1$.

Also, $\phi'_4(m) = -2m$.

$\therefore$ $c = -\frac{f_3(m)}{f'_4(m)}$

$= -\frac{(-2a)}{-2m}$

$= -\frac{a}{m}$.

Now, when $m = 1$, $c = -a$ and when $m = -1$, $c = a$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the required asymptotes are

$y = 1x - a$, and $y = -1x + a$

or, $y - x + a = 0$, and $y + x - a = 0$

Example 26. Find the asymptotes of the following curve $(x^2 - y^2)(x + 2y + 1) + x + y + 1 = 0$.

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Solution:

The given curve can be written as

$x^3 + 2x^2y + x^2 - xy^2 - 2y^3 - y^2 + x + y + 1 = 0$

or, $x^3 + 2x^2y - xy^2 - 2y^3 + x^2 - y^2 + x + y + 1 = 0$.

This equation is of 3rd degree and the terms involving x^3 and y^3 are both present. Hence there are no asymptote parallel to x and y axis.

We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$ in 3rd and 2nd degree terms, we get

$\phi_3(m) = 1 + 2m - m^2 - 2m^3$ and $\phi_2(m) = 1 - m^2$.

Now, $\phi_3(m) = 0$ gives

$1 + 2m - m^2 - 2m^3 = 0$

or, $2m^3 + m^2 - 2m - 1 = 0$

or, $m^2(2m + 1) - 1(2m + 1) = 0$

or, $(2m+1)(m+1)(m-1) = 0$

$\therefore$ $m = 1, -1$ and $-\frac{1}{2}$.

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Also, $\phi'_3(m) = 2 - 2m - 6m^2$.

Now, $c = -\frac{\phi_2(m)}{\phi'_3(m)} = \frac{m^2 - 1}{2 - 2m - 6m^2}$.

If $m = 1$, then $c = 0$.

If $m = -1$, then $c = 0$.

If $m = -\frac{1}{2}$, then $c = -\frac{1}{2}$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the required asymptotes are

$y = 1x + 0$, $y = -1x + 0$ and $y = -\frac{1}{2}x - \frac{1}{2}$

or, $x - y = 0$, $x + y = 0$ and $x + 2y + 1 = 0$

Example 27. Find the asymptotes of the curve $x^2(x-y)^2 - a^2(x^2 + y^2) = 0$.

Solution:

Here, $x^2(x-y)^2 - a^2(x^2 + y^2) = 0$

or, $x^4 - 2x^3y + x^2y^2 - a^2x^2 - a^2y^2 = 0$.

This equation is of 4th degree and the terms involving x^4 is present, hence there is no asymptote parallel to x-axis.

To find asymptotes parallel to y-axis, equating the coefficient of highest degree term of y to zero. i.e., coefficient of $y^2 = 0$

or, $x^2 - a^2 = 0$ $\therefore$ $x = \pm a$.

We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$ in 4th and 3rd degree terms, we get

$\phi_4(m) = 1 - 2m + m^2$ and $\phi_3(m) = 0$.

Now, $\phi_4(m) = 0$ gives $m^2 - 2m + 1 = 0$

or, $(m-1)^2 = 0$

$\therefore$ $m = 1$ and 1.

Since the root $m = 1$ is repeated. So to find corresponding value of c, we shall use the formula

$$\frac{c^2}{2!}\phi''_4(m) + c\,\phi'_3(m) + \phi_2(m) = 0 \quad \ldots\ (i)$$

But $\phi_4(m) = 1 - 2m + m^2$ gives $\phi''_4(m) = 2$.

$\phi'_3(m) = 0$ and $\phi_2(m) = -a^2 - a^2m^2 = -2a^2$.

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Hence, equation (i) will become

$$\frac{c^2}{2} \cdot 2 + c \cdot 0 + (-a^2 - a^2) = 0$$

or, $c^2 - 2a^2 = 0$

$\therefore$ $c = \pm a\sqrt{2}$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the oblique asymptotes are

$y = x \pm a\sqrt{2}$

or, $x - y = \pm a\sqrt{2}$.

$\therefore$ Required asymptotes are $x = \pm a$ and $x - y = \pm a\sqrt{2}$.

Example 28. Find the asymptotes of the following curve $y^3 - xy^2 - x^2y + x^3 + x^2 - y^2 = 1$.

Solution:

Here, $y^3 - xy^2 - x^2y + x^3 + x^2 - y^2 - 1 = 0$.

This equation is of 3rd degree and the terms involving x^3 and y^3 are both present. Hence there are no asymptotes parallel to x and y axis.

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We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$ in 3rd and 2nd degree terms, we get

$\phi_3(m) = m^3 - m^2 - m + 1$ and $\phi_2(m) = 1 - m^2$.

Now, $\phi_3(m) = 0$ gives

$m^3 - m^2 - m + 1 = 0$

or, $m^2(m-1) - 1(m-1) = 0$

or, $(m-1)(m-1)(m+1) = 0$

$\therefore$ $m = 1, 1, -1$.

Also, $\phi'_3(m) = 3m^2 - 2m - 1$.

For $m = -1$, $c = -\dfrac{\phi_2(m)}{\phi'_3(m)}$

$= -\dfrac{(1-m^2)}{3m^2 - 2m - 1}$

$= -\dfrac{(1-1)}{3+2-1} = \dfrac{0}{4} = 0.$

Hence, $y = mx + c$ gives

$y = -1 \cdot x + 0$

i.e., $x + y = 0$.

But $m = 1$ is repeated root. So to find corresponding value of c, we shall apply the formula

$\dfrac{c^2}{2!}\phi''_3(m) + c\phi'_2(m) + \phi_1(m) = 0.$ .... (i)

But $\phi''_3(m) = 6m - 2$, $\phi'_2(m) = -2m$ and $\phi_1(m) = 0$.

Hence, from (i), we get

$\dfrac{c^2}{2!}(6m-2) + c(-2m) + 0 = 0$

or, $\dfrac{c^2}{2}(6.1-2) - 2c \cdot 1 = 0$

or, $2c^2 - 2c = 0$

or, $c^2 - c = 0$

$\therefore$ $c = 0$ and $c = 1$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the required asymptotes are

i.e. $y = 1 \cdot x + 0$ and $y = 1x + 1$

or, $y = x$ and $y = x + 1$.

$\therefore$ Required asymptotes are $x + y = 0$, $x - y = 0$ and $x - y + 1 = 0$

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Example 29. Find the asymptotes of the curve $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$.

Solution:

Here, $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$.

The equation is of 3rd degree and the terms involving x^3 and y^3 are both present. Hence there are no asymptotes parallel to y and x-axis.

We now find oblique asymptotes of the form $y = mx + c$. Putting $x = 1$ and $y = m$ in 3rd and 2nd degree term, we get

$\phi_3(m) = 1 + 3m - 4m^3$ and $\phi_2(m) = 0$.

Now, $\phi_3(m) = 0$ gives

$4m^3 - 3m - 1 = 0$

or, $4m^3 - 4m^2 + 4m^2 - 4m + m - 1 = 0$

or, $(m-1)(4m^2 + 4m + 1) = 0$

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or, $(m-1)(2m+1)^2 = 0$

or, $m = 1, m = -\frac{1}{2}, -\frac{1}{2}$

Also, $c = -\frac{\phi_2(m)}{\phi'_3(m)} = -\frac{0}{\phi'_3(m)} = 0.$

For $m = 1$, then $c = 0$

$\therefore$ $y = mx + c$ gives

$y = 1x + 0$

i.e., $y = x$ is an asymptote.

Again, for $m = \frac{-1}{2}$, a repeated root. To find corresponding value of c, we apply the formula

$$\frac{c^2}{2!}\phi''_3(m) + c\,\phi'_2(m) + \phi_1(m) = 0,$$

where, $\phi'_2(m) = 0$, $\phi_1(m) = -1 + m$, and $\phi''_3(m) = -24m$.

$\therefore$ $\frac{c^2}{2}.(-24m) + c.0 + (m-1) = 0$

or, $-12m\,c^2 + m - 1 = 0$ .... (i)

For $m = \frac{-1}{2}$, (i) gives

$$-12\left(\frac{-1}{2}\right)c^2 - \frac{1}{2} - 1 = 0$$

or, $6c^2 - \frac{3}{2} = 0$

$\therefore$ $c = \pm\frac{1}{2}$.

Substituting the value of m and corresponding value of c on $y = mx + c$, the asymptotes are

$y = -\frac{1}{2}x \pm \frac{1}{2}$

$\therefore$ $x + 2y = \pm 1$.

$\therefore$ Required asymptotes are $y = x$ and $x + 2y = \pm 1$.

Exercise 2.3

1. Obtain the horizontal and vertical asymptotes, if any, of the following curves:

i. $x^2y^2 = a^2(x^2 + y^2)$

ii. $x = y(x^2 - 1)$

iii. $x^2y^2 - 4(x-y)^2 + 2y - 3 = 0$

iv. $y = \frac{2x-3}{x^2-3x+2}$

v. $xy^2 - a^2(x-a) = 0$

vi. $x^2y^2 - x^2y - xy^2 + x + y + 1 = 0$

vii. $y = \frac{4x^2+4x-3}{x^2-4x+3}$

viii. $y = \frac{x^2-4x+5}{x^2+4x+3}$

ix. $y = \frac{5x}{x-3}$

2. Find the asymptotes, if any, of the following curves.

i. $y = \frac{x}{(x-1)^2(x-2)}$

ii. $xy^2 - x^2y = a^2(x+y) + b^2$

iii. $x^2y + xy^2 = a^3$

iv. $x^2y - xy^2 + xy + y^2 + x - y = 0$

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3. Find the asymptotes, if any, of the following curves.

i. $xy^2 - y^2 - x^3 = 0$ ii. $(x^2 - y^2)y - 4y^2 + 5x - 7 = 0$

iii. $y = \dfrac{x^2 + 2x - 1}{x}$ iv. $y = \dfrac{x^2 + 1}{1 + x}$

v. $(y - a)^2 (x^2 - a^2) = x^4 + a^4$ vi. $y(y - 1)^2 - x^2 y = 0$

vii. $x^3 + y^3 = 3axy$ viii. $y^2(x - 1) = x^3$

ix. $x^3 - 2y^3 + 2x^2y - xy^2 + xy - y^2 + 1 = 0$ x. $(x^2 - y^2)(x + 2y + 1) + x + y + 1 = 0$

xi. $(x - 1)(x - 2)(x + y) + x^2 + x + 1 = 0$ xii. $y^3 + 5x^2 = 1$

xiii. $x^2y^2 - 2x + 4 = 0$ xiv. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

4. Find the asymptotes of the following curves.

i. $x^3 + x^2y - xy^2 - y^3 + x^2 - y^2 - 2 = 0$ ii. $4x^3 - 3xy^2 - y^3 + 2x^2 - xy - y^2 - 1 = 0$

iii. $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$ iv. $y^3 + x^2y + 2xy^2 - y + 1 = 0$

v. $x^3 + x^2y - xy^2 - y^3 + 2xy + 2y^2 - 3x + y = 0$

vi. $y^3 - 5xy^2 + 8x^2y - 4x^3 - 3y^2 + 9xy - 6x^2 + 2y - 2x + 1 = 0$

vii. $x^2(x - y)^2 - a^2(x^2 + y^2) = 0$ viii. $(x^2 - y^2)^2 - 2(x^2 + y^2) + x - 1 = 0$

ix. $(y - a)^2 (x^2 - a^2) = x^4 + a^4$ x. $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$

Answer

1. i. $x = \pm a, y = \pm a$ ii. $x = \pm 1, y = 0$ iii. $x = \pm 2, y = \pm 2$ iv. $x = 1, x = 2, y = 0$
   v. $x = 0, y = \pm a$ vi. $x = 0, x = 1, y = 0, y = 1$ vii. $x = 1, x = 3, y = 4$
   viii. $x = -1, x = -3, y = 1$ ix. $x = 3, y = 5$
2. i. $x = 1, x = 2, y = 0$ ii. $x = 0, y = 0, x - y = 0$
   iii. $x = 0, y = 0, x + y = 0$ iv. $y = 0, x = 1, x - y + 2 = 0$
3. i. $x = 0, y = x + \dfrac{1}{2}, y = -x - \dfrac{1}{2}$ ii. $y = 0, y + x + 2 = 0, y - x + 2 = 0$
   iii. $x = 0, y = x + 2$ iv. $x + 1 = 0, y = x - 1$
   v. $x = \pm a, y + x = a, y - x = a$ vi. $y = 0, y + x = 1, y - x = 1$
   vii. $x + y + a = 0$ viii. $x = 1, 2y = 2x + 1, 2y + 2x + 1 = 0$
   ix. $x - y = 0, x + y + 1 = 0, x + 2y - 1 = 0$
   x. $x - y = 0, x + y = 0, x + 2y + 1 = 0$ xi. $x = 1, x = 2, x + y + 1 = 0$
   xii. No asymptote xiii. $x = 0, y = 0$ xiv. No asymptote
4. i. $y = x, y = -x, y = -x - 1$ ii. $y = x, y = -2x, y = -2x - 1$
   iii. $y = x, 2y + x - 1 = 0, 2y + x + 1 = 0$ iv. $y = 0, x + y - 1 = 0, x + y + 1 = 0$
   v. $x + y + 1 = 0, x + y - 2 = 0, x - y + 1 = 0$
   vi. $y = x, y = 2x + 2, y = 2x + 1$
   vii. $x = a, x = -a, x - y + a\sqrt{2} = 0, x - y - a\sqrt{2} = 0$
   viii. $y = x + 1, y = x - 1, y + x + 1 = 0, y + x - 1 = 0$
   ix. $y + x = a, y - x = a, x = a, x = -a$
   x. $x - y = 0, x + 2y - 1 = 0, x + 2y + 1 = 0$

2.4 Curve tracing

In this section, we apply all the ideas we learned in the previous sections to trace curves. While tracing a curve, we consider the following facts:

2.4.1 Symmetry of a curve

If a curve is symmetry then we only need to draw the half part of the graph and the rest is obtained by symmetry.

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1. Symmetry about x-axis

If the equation of the curve remains unchanged on replacing y by $-y$, then the curve is symmetrical about x-axis.

For example, the parabola $y^2 = 4ax$ is symmetrical about x-axis.

2. Symmetry about y-axis

If the equation of the curve remains unchanged on replacing x by $-x$, then the curve is symmetrical about the y-axis.

For example, the parabola $x^2 = 4ay$ is symmetrical about y-axis.

3. Symmetry about both axes

If the equation of the curve contains only even powers of both x and y, then the curve is symmetrical about both axes.

We note that the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$ are symmetrical about both axes.

4. Symmetry about opposite quadrants

If the equation of the curve remains unchanged when x is replaced by $-x$ and y by $-y$, then the curve is symmetrical in opposite quadrant.

For example, the hyperbola $xy = c^2$ is symmetrical in first and third quadrant.

5. Symmetry about the line y = x

If the equation of the curve remains unchanged when x and y are interchanged, then the curve is symmetrical about the line $y = x$.

For example, the folium $x^3 + y^3 = 3axy$ is symmetrical about the line $y = x$.

2.4.2 Origin

If the equation of the curve does not contain any constant term, then the curve passes through origin.

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For example, the parabola $y^2 = 4ax$ passes through origin. But the circle $x^2 + y^2 = a^2$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ do not pass through origin.

2.4.3 Equation of tangent at origin

If the curve passes through the origin, then the equation of the tangent at origin can be obtained by equating the lowest degree term of x and y to zero.

For example, if the curve is $x^2y^2 = a^2(x^2 - y^2)$, then

equation of tangent at origin is $x^2 - y^2 = 0$ i.e., $y = \pm x$.

Note: If $\frac{dy}{dx} = 0$, then the tangent is parallel to x-axis and if $\frac{dx}{dy} = 0$, then the tangent is parallel to y- axis.

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2.4.4 Noticeable points

The points of intersection of the curve and the coordinate axes or any special points of the curve are called noticeable points.

Putting $x = 0$ and $y = 0$ successively and obtaining the values of y and x, the points of intersection with the co-ordinate axes are found.

For example, the curve: $y^2 (a + x) = (3a^2 - x)$ meets x - axis at point $(3a^2, 0)$ and y-axis at points $\left(0, \pm\sqrt{3}\,a\right)$.

2.4.5 Increasing and decreasing nature of graph

We note that $f'(x)$ gives the slope of the curve $y = f(x)$ at $(x, f(x))$ so that one has to have the information about the derivative of $f(x)$. This is due to the fact that $f'(x)$ tells us the direction in which graph proceeds. In the tracing of a graph, we need to know about the increasing, decreasing and turning points of the graph. These nature of graph are obtained by derivative. We recall that if $f'(x) > 0$ for all x on an interval I, then f is of increasing nature whereas if $f'(x) < 0$ for all x on I, then f is of decreasing nature. We also find points x where $f'(x) = 0$ or $f'(x)$ does not exist. These x are called critical numbers which represent the turning points of graph. Once we know about the increasing and decreasing nature, we then also find the local maximum, local minimum values using the derivative test. We briefly recall the second derivative test. Suppose f'' is continuous near c. Then:

i. if $f'(c) = 0$ and $f''(c) > 0$, then f has local minimum at c.

ii. if $f'(c) = 0$ and $f''(c) < 0$, then f has local maximum at c.

We note that the points where $f'(x) = 0$ are called stationary points.

2.4.6 Stationary points, points of inflection and concavity

The stationary points are the points where the direction of curve changes and given by $\frac{dy}{dx} = 0$. A point P on a curve $y = f(x)$ is said to be a *point of inflection* if it is concave on one side and convex on the other side of P. At a point of inflection, the curve changes its direction of bending from concavity to convexity or vice versa. The two portions of the curve on the two sides of P lie on different sides of the tangent at P, i.e., the curve crosses the tangent at P. Thus a point, where the curve crosses the tangent is called a *point of inflection*.

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For a curve, there will be a point of inflection, if $\frac{d^2y}{dx^2} = 0$ but $\frac{d^3y}{dx^3} \neq 0$.

A curve is concave upward if $\frac{d^2y}{dx^2} > 0$ and convex upward if $\frac{d^2y}{dx^2} < 0$.

Applying all the ideas we discussed, we now trace some standard graphs as below.

Example 30. Trace the curve $x^2 + y^2 = 9$ (Circle)

Solution:

Here, $x^2 + y^2 = 9$. .... (i)

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So the curve (i) is symmetrical about both axes.

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II. Origin:Putting $x = 0$ in (i), then $y = \pm 3$ and putting $y = 0$, then $x = \pm 3$. Hence (i) passes through the points $(\pm 3, 0)$ and $(0, \pm 3)$.
So the curve doesn't pass through $(0, 0)$.

III. Equation of the tangents: Differentiating (i) with respect to x, we get

$$2x + 2y\frac{dy}{dx} = 0$$

or, $\frac{dy}{dx} = -\frac{x}{y} = -\frac{x}{\sqrt{9-x^2}}$.

$\therefore$ At $(\pm 3, 0)$: $\frac{dy}{dx} = \frac{-(\pm 3)}{0} = \pm\frac{3}{0}$

and at $(0, \pm 3)$, $\frac{dy}{dx} = -\frac{x}{y} = \frac{0}{\pm 3} = 0$.

Hence, equation of tangent at $(\pm 3, 0)$ are

$$y - 0 = \pm\frac{3}{0}(x \pm 3)$$

or, $x \pm 3 = 0$

$\therefore$ $x = \pm 3$.

Similarly, equations of tangent at $(0, \pm 3)$ are

$y \pm 3 = 0\ (x - 0)$

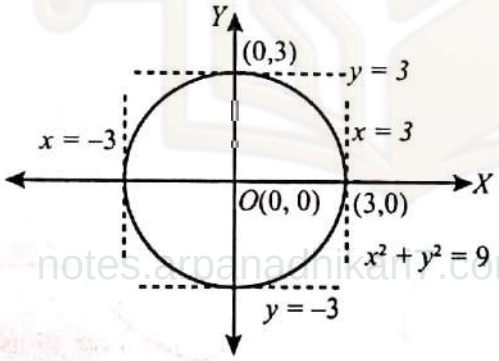
i.e., $y = \pm 3$.

IV. Nature of the graph: Since $\frac{dy}{dx} = -\frac{x}{y}$

For $-3 \le x \le 0$ and above the x-axis, $\frac{dy}{dx} = \frac{-x}{\sqrt{9-x^2}} > 0$.

So, it is an increasing function for $-3 \le x \le 0$ and $y > 0$.

V. Asymptote: This curve is bounded, so it has no asymptote. The figure is shown by the following circle.



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Example 31. Trace the curve: $y = (x^2 - x - 6)(x - 7)$.

Solution:

Here, $y = (x^2 - x - 6)(x - 7) = (x + 2)(x - 3)(x - 7)$ .... (i)

i. Symmetry: This curve is not symmetrical about the axes.

ii. Origin: It does not pass through origin. The curve passes through the points $(-2, 0)$, $(3, 0)$ and $(7, 0)$. Again, putting $x = 0$ then $y = 42$. So the curve cuts y - axis at point $(0, 42)$.

iii. Maximum and minimum points:

From (i), i.e., $y = x^3 - 8x^2 + x + 42$, we have

$\frac{dy}{dx} = 3x^2 - 16x + 1$ and $\frac{d^2y}{dx^2} = 6x - 16$.

For maximum and minimum points,

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$$\frac{dy}{dx} = 0$$

or, $3x^2 - 16x + 1 = 0$

or, $x = \frac{16 \pm \sqrt{256 - 12}}{6}$

$$= \frac{16 \pm \sqrt{244}}{6}$$

$$= \frac{16 \pm \sqrt{4 \times 61}}{6}$$

$\therefore$ $x = 5.27$ and $x = 0.13$.

These are stationary or turning points. Clearly $\frac{dy}{dx} > 0$ on $(-\infty, 0.13) \cup (5.27, \infty)$ and $\frac{dy}{dx} <$ on $(0.13, 5.27)$.

$\therefore$ $f(x)$ is increasing on $(-\infty, 0.13) \cup (5.27, \infty)$ and decreasing on $(0.13, 5.27)$.

At $x = 5.27$: $\frac{d^2y}{dx^2} > 0$ and at $x = 0.76$: $\frac{d^2x}{dy^2} < 0$.

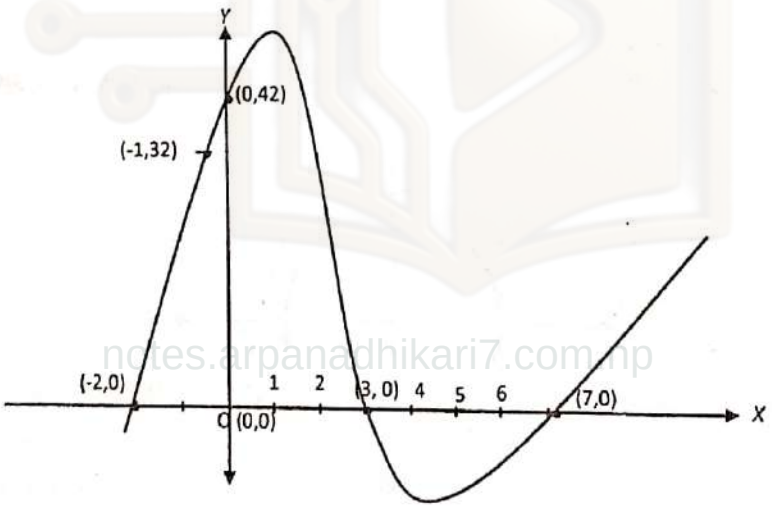
Hence, y has a minimum at $x = 5.27$ and maximum at $x = 0.76$

IV. Noticeable points on the curve:

The variable values of x and y on the curve are given below:

x:	$-\infty$	-2	-1	0	0.76	2	3	5.27	∞
y:	$-\infty$	0	32	42	Max.	20	0	Min.	∞

V. Asymptote: From the curve $x^3 - 8x^2 + x - y + 42 = 0$, we conclude that there is no asymptote. The sketch of the curve is given below:



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Example 32. Trace the curve $y = \frac{2x - 3}{x^2 - 3x + 2}$

Solution:

Here, $y = \frac{2x - 3}{x^2 - 3x + 2}$ .... (i)

I. Origin and points on axes:

Putting $x = 0$, then we get $y = -3/2$

and putting $y = 0$, then we get $2x - 3 = 0$ i.e., $x = 3/2$.

So the curve passes through $(0, -3/2)$ and $(3/2, 0)$.

Also, the curve does not pass through origin.

II. Differentiating, (i) with respect to x

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$$\frac{dy}{dx} = \frac{(x^2 - 3x + 2)\,2 - (2x - 3)(2x - 3)}{(x^2 - 3x + 2)^2}$$

$$= \frac{-2x^2 + 6x - 5}{(x^2 - 3x + 2)^2}$$

$$= -\frac{2\left[\left(x - \frac{3}{2}\right)^2 + \frac{1}{4}\right]}{(x-1)^2 (x-2)^2}.$$

Clearly, $\frac{dy}{dx} < 0$ for $-\infty < x < 1$, $1 < x < 2$ and $2 < x < \infty$.

This shows that the curve is strictly decreasing in each of the intervals $(-\infty, 1)$, $(1, 2)$ and $(2, +\infty)$.

III. Asymptotes:

From the curve (i) i.e., $y(x^2 - 3x + 2) - 2x + 3 = 0$.

The equation of asymptotes parallel to y-axis is given by

$x^2 - 3x + 2 = 0$ $\quad\therefore\quad$ $x = 1$ and $x = 2$.

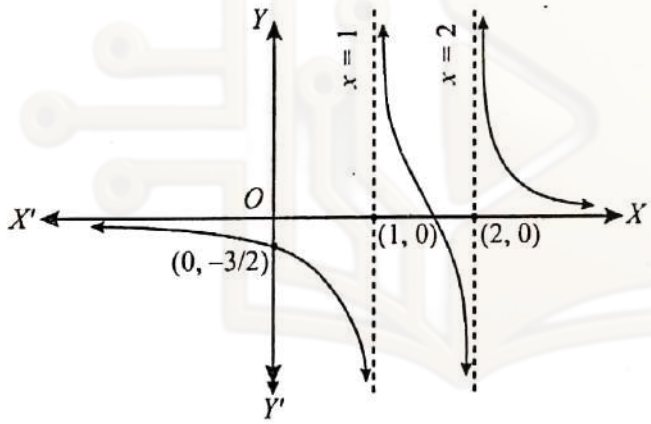
Thus, $x = 1$ and $x = 2$ are the asymptotes.

IV. Noticeable points on the curve:

The variation of x and y on the curve are given below:

x	$-\infty$	1	3/2	0	2	$+\infty$
y	0	∞	0	$-3/2$	∞	$-\infty$

The sketch of the curve is shown below:



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Example 33. Trace the curve $x^{2/3} + y^{2/3} = a^{2/3}$ (Hypocycloid).

Solution:

The given equation can be put in the form

$$\left(\frac{x^2}{a^2}\right)^{1/3} + \left(\frac{y^2}{a^2}\right)^{1/3} = 1. \qquad \ldots\ (i)$$

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So the curve is symmetrical about both axes.

II. Origin: If $y = 0$, we get $\left(\frac{x^2}{a^2}\right)^{1/3} = 1$ $\quad\therefore\ x^2 = a^2$ i.e., $x = \pm a$.

Hence, the curve (i) meets the x-axis at points $(a, 0)$ and $(-a, 0)$.

Similarly, if $x = 0$, we get $y = \pm b$. So the points on y-axis are $(0, b)$ and $(0, -b)$. Thus the curve (i) meets in the co-ordinate axes at points $(a, 0)$, $(-a, 0)$, $(0, b)$, $(0, -b)$. The curve does not pass through $(0, 0)$.

III. Equation of the tangents: Differentiating (i) with respect to x, we get

$\frac{dy}{dx} = -\frac{y^{1/3}}{x^{1/3}}$. At $(\pm a, 0)$: $\frac{dy}{dx} = 0$ and at $(0, \pm a)$: $\frac{dy}{dx} = \infty$.

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Hence, x-axis is the tangent at points $(\pm a, 0)$ and y-axis is tangent at points $(0, \pm a)$.

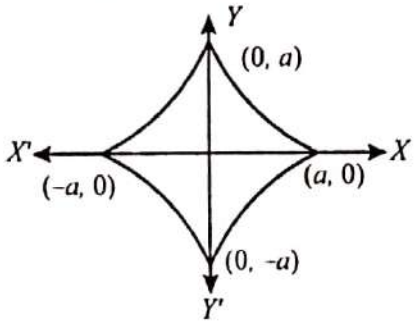
IV. Asymptote: This curve is bounded, so it has no real asymptote.

V. Nature of the graph:

From the curve (i), we have $\left(\frac{y^2}{a^2}\right)^{1/3} = 1 - \left(\frac{x^2}{a^2}\right)^{1/3}$.

If x is greater than a in magnitude, then $\frac{y^2}{a^2}$ is negative, and y is imaginary and hence no part of the curve goes beyond $x = \pm a$. Similarly, the curve does not go beyond $y = \pm a$.

The curve is shown in the following figure. This curve is popularly known as Hypocycloid.



Example 34. Trace the curve: $y^2 = (x - 2a)^3$.

Solution:

Here, $y^2 = (x - 2a)^3$ .... (i)

Symmetry: The equation of the curve remains unchanged when y is replaced by $-y$. So the curve is symmetrical about x-axis.

Origin: If $y = 0$, then (i) gives $(x - 2a)^3 = 0$ or, $x = 2a$. So, the curve passes through $(2a, 0)$. The curve does not pass through origin.

III. Nature of the graph: From (i), y becomes imaginary if $x = 2a$. So the curve can not extend to the left of $x = 2a$.

IV. Tangent: From the given curve (i), $\frac{dy}{dx} = \frac{3(x-2a)^2}{2y}$.

$\therefore$ $\frac{dy}{dx}$ at $(2a, 0) \to \infty$.

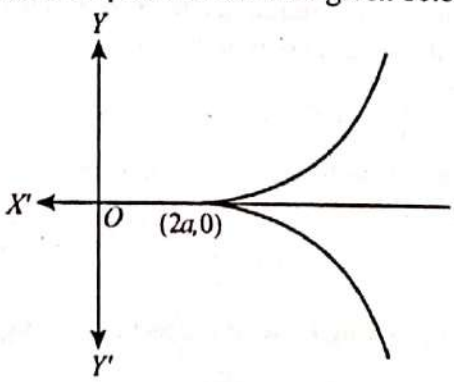
Hence, the tangent at $(2a, 0)$ is $x - 2a = 0$ i.e., $x = 2a$.

V. Increasing: If x increases from 0 to ∞, then y also increases from 0 to ∞.

VI. Noticeable points on the curve: The corresponding values of x and y on the curve are given below:

x:	$2a$	$3a$	$4a$	$6a$
y:	0	$a^{3/2}$	$2a\sqrt{2a}$	$8a\sqrt{a}$

With the help of above data, the shape of the curve is given below:



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Example 35. Trace the curve $y^2(a-x) = x^2(a+x)$.

Solution:

Here, $y^2(a-x) = x^2(a+x)$

$\therefore \quad y^2 = x^2\frac{a+x}{a-x}$ .... (i)

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So the curve is symmetrical about x-axis.

II. Origin: Putting $x = 0$, then we get $y = 0$ and putting $x = -a$, then $y = 0$. Hence the curve passes through the points $(0, 0)$ and $(-a, 0)$.

III. Tangents at origin: The tangents at the origin can be obtained by equating the lowest degree term with zero i.e., $ay^2 - ax^2 = 0$

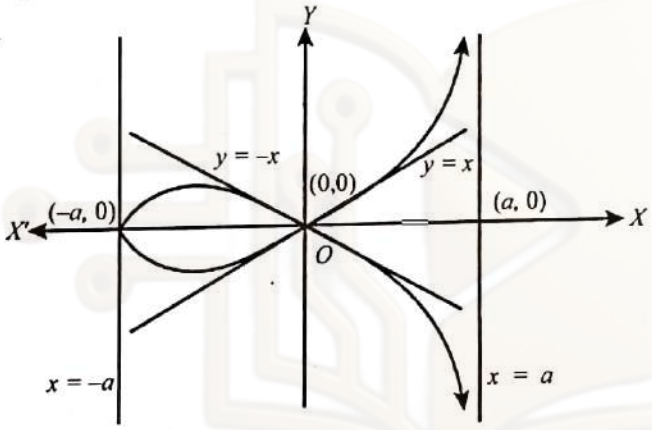
$\therefore \quad y = \pm x.$

IV. Nature of the curve: If $x > a$ or $x < -a$, then from (i), y becomes imaginary. So the curve cannot extends right of $x = a$ and left of $x = -a$.

Also from the curve (i), if $x \to a$, then $y \to \infty$. So the line $x = a$ is asymptote of (i).

V. Asymptote: Since y^3 is absent. Equating the coefficient of y^2 with zero, then we get, $a - x = 0$ i.e., $x = a$ is an asymptote.

VI. Point of inflection: No point of inflection. The shape of the curve is shown by the following figure.



Example 36. Trace the curve: $x^2(x^2+y^2) = a^2(x^2-y^2)$ (Lemniscate).

Solution:

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Here, $x^2(x^2+y^2) = a^2(x^2-y^2)$ .... (i)

or, $x^2y^2 + a^2y^2 = a^2x^2 - x^4$

or, $y^2 = \frac{x^2(a^2-x^2)}{x^2+a^2}$

$\therefore \quad y = \pm x\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ .... (ii)

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So it is symmetrical about both axes.

II. Origin: If $y = 0$, then from (ii), $x^2(a^2 - x^2) = 0$

$\therefore \quad x = 0$ and $x = \pm a$.

So, the curve passes through $(0,0)$, $(a,0)$ and $(-a,0)$.

III. Tangent at origin: The equation of tangent at origin can be obtained by equating the lowest degree terms with zero

i.e., $a^2(x^2 - y^2) = 0$

$\therefore \quad y = \pm x.$

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IV. Tangent at $(\pm a, 0)$: From the given curve, $y^2(x^2 + a^2) = a^2x^2 - x^4$.

Differentiating with respect to x

$$(x^2 + a^2) \cdot 2y\frac{dy}{dx} + 2xy^2 = 2xa^2 - 4x^3$$

or, $$\frac{dy}{dx} = \frac{2xa^2 - 4x^3 - 2xy^2}{2y(x^2 + a^2)}$$

$\therefore$ $$\frac{dy}{dx} \text{ at } (\pm a, 0) = \infty = \frac{1}{0}.$$

Hence, tangent at $(\pm a, 0)$ are

$$y - 0 = \frac{1}{0}(x \pm a) \text{ i.e., } x \pm a = 0$$

$\therefore$ $x = \pm a$.

So $x = \pm a$ are the tangents at $(\pm a, 0)$.

Since, y vanishes at $x = 0$, $x = \pm a$. So there are two loops formed between and $x = 0$, $x = a$ and $x = 0$, $x = -a$.

V. Asymptote: Since y^4 is absent, so, equating the coefficient of y^2 to zero in (i), we get

$x^2 + a^2 = 0$ i.e., $x = \pm ia$. i.e., imaginary values of x.

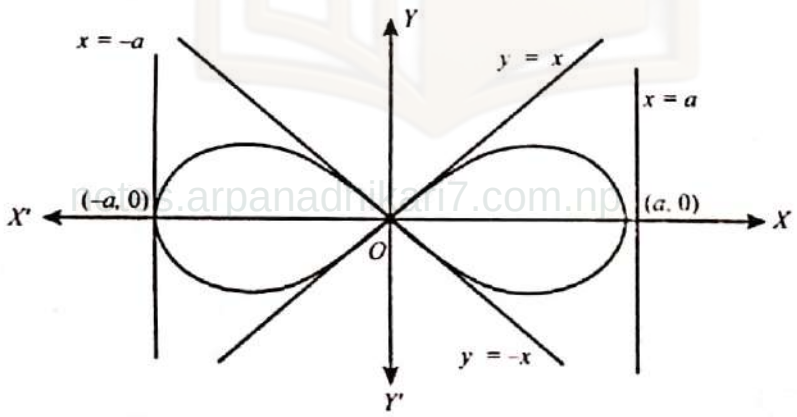
So the curve has no real asymptote.

VI. From (i) y is real when $-a \le x \le a$.

VII. Noticeable points on the curve: The noticeable values of x and y from (i) are given below:

x:	$-a$	$\frac{-a}{2}$	0	$\frac{a}{2}$	a
y:	0	$\pm\frac{a\sqrt{3}}{2\sqrt{5}}$	0	$\pm\frac{a\sqrt{3}}{2\sqrt{5}}$	0

The shape of the curve is given below and is called **Lemniscate.**



Example 37. Trace the curve: $a^2b^2 + x^2y^2 = a^2x^2$.

Solution:

From the given curve, we have

$$y^2 = \frac{a^2x^2 - a^2b^2}{x^2}$$

i.e., $$y^2 = \frac{a^2(x^2 - b^2)}{x^2} \quad \ldots\ (i)$$

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So the curve is symmetrical about both axes.

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II. Origin: Putting $y = 0$, then from (i), we get
$x^2 - b^2 = 0$

or, $x = \pm b$.

So the curve passes through $(-b, 0)$ and $(b, 0)$. Hence the curve does not pass through origin.

III. Asymptotes: The equation of asymptotes parallel to x-axis are obtained by equating the coefficient of x^2 with zero

i.e., $y^2 - a^2 = 0$

i.e., $y = \pm a$.

IV. Nature of the curve: If $x^2 < b^2$, then y becomes imaginary i.e., the curve does not exist in $-b < x < b$.

V. Equation of tangent: To find the equation of tangent. Differentiating w.r.t x, we get

$$2xy^2 + 2x^2y\frac{dy}{dx} = 2a^2x$$

or, $$\frac{dy}{dx} = \frac{2a^2x - 2xy^2}{2x^2y}$$

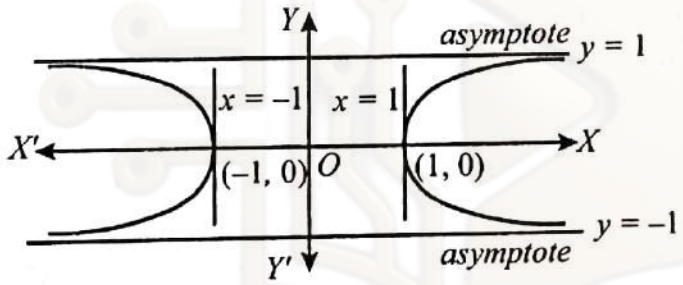
$\therefore$ $$\frac{dy}{dx} = \frac{a^2 - x^2}{xy} \Rightarrow \frac{a^2}{0} \text{ as } y \to 0.$$

$\therefore$ Equation of tangents at $(\pm b, 0)$ are

$y - 0 = \frac{a^2}{0}(x \pm b)$ or, $x \pm b = 0$ $\quad\therefore\quad x = \pm b.$

Hence, tangent at $(b, 0)$ and $(\pm b, 0)$ are $x = -b$ and $x = b$.

The shape of the curve is shown below:



Example 38. Trace the curve: $x^2(a^2 + y^2) = y^2(a^2 - y^2)$.

Solution:

Here, $x^2(a^2 + y^2) = y^2(a^2 - y^2)$ .... (i)

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I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So, the curve is symmetrical about both axes.

II. Tangent at origin: The curve (i) passes through (0, 0) i.e., origin. The equation of tangent at origin are obtained by equating the lowest degree terms of x and y to zero. i.e., $x^2 - y^2 = 0$

or, $y = \pm x$, which are real and distinct. So the origin is the node.

III. Intersection with co-ordinate axes: The curve meets x-axis only at the origin. Again, the curve meets y-axis in the points A $(0, a)$ and A'$(0, -a)$.

IV. Asymptote: The curve has no real asymptote.

V. Nature of the curve: From the curve (i), $x = \pm y\sqrt{\frac{a^2 - y^2}{a^2 + y^2}}$ .... (ii)

If x is real, when $a^2 - y^2 \geq 0 \Rightarrow y^2 \leq a^2$.

Thus the whole curve lie between $y = a$ and $y = -a$.

VI. Nature of the curve:

Differentiating (ii) with respect to y, we get

$$\frac{dx}{dy} = \pm\frac{a^4 - 2a^2y^2 - y^4}{(a^2 - y^2)^{1/2}(a^2 + y^2)^{3/2}} = \pm\frac{(a^2 - y^2)^{1/2}(a^2 + y^2)^{3/2}}{a^4 - 2a^2y^2 - y^4}.$$

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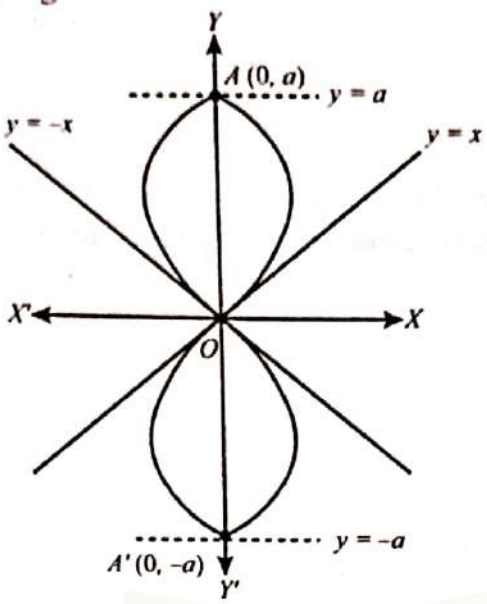
Now, $\frac{dy}{dx} = 0 \Rightarrow (a^2 - y^2)^{1/2} (a^2 + y^2)^{1/2} = 0$ i.e., $y = \pm a$ are only the real values.

When $y = \pm a$, then $x = 0$. Thus at $(0, \pm a)$, tangent are parallel to x-axis.

The rough sketch of the curve is given below:



Example 39. Trace the curve: $x(x^2 + y^2) = a(x^2 - y^2)$ stating the necessary techniques for tracing.

Solution:

Here, $x(x^2 + y^2) = a(x^2 - y^2)$. .... (i)

The necessary techniques for tracing the curve (i) are below:

I. Symmetry: The equation of the curve remains unchanged y is replaced by $-y$. So it is symmetrical about x-axis.

II. Origin: Clearly origin $(0, 0)$ is satisfied in the equation (i), so (i) passes through origin $(0, 0)$.

III. Noticeable points: Putting $x = 0$ on (i), then y becomes imaginary.

So the curve (i) never meet y-axis. Putting $y = 0$, then $x = a$. So the curve (i) meets the x-axis at $(a, 0)$.

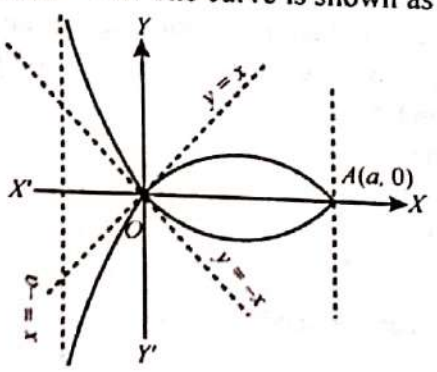
x	0	1	−1	2	−2	3
$y = x^2 - 12x - 6$	−6	−17	5	−22	10	−15
(x, y)	(0,−6)	(1, −17)	(−1,5)	(2,−22)	(−2,10)	(3, −15)

It passes through the points $(0, -6)$, $(1, -17)$, $(-1, 5)$, $(2, -22)$, $(-2, 10)$, $(3, -15)$.

IV. Tangents: The tangents at origin are $x^2 - y^2 = 0$ i.e., $y = \pm x$ and at point $(a, 0)$, tangent is $x = a$.

V. Region: Solving for y, we get $y^2 = \frac{(a-x)x^2}{(a+x)}$, y becomes real for $x \leq a$ and $x > -a$.

VI. Asymptotes: It has one asymptote $x = -a$. The curve is shown as in the figure below.



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Example 40. Trace the following curve: $x^2y^2 = a^2(y^2 - x^2)$.

Solution:

Here, $x^2y^2 = a^2(y^2 - x^2)$. .... (i)

I. Symmetry: The equation of the curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively. So, the curve is symmetrical about both axes.

II. Origin: If $x = 0$, then from (i), $y = 0$. So the curve passes through origin.

III. Nature of the graph: If $x > a$ numerically, then y becomes imaginary.
So the curve can't extend to the left of $x = -a$ and to the right of $x = a$.

IV. Tangent at origin: For the equation of tangent at origin, equating the lowest degree terms of x and y with zero

i.e., $y^2 - x^2 = 0$

or, $y = \pm x$.

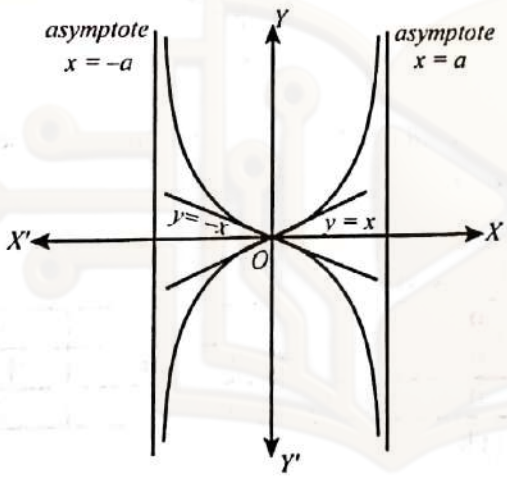
So, $y = x$ and $y = -x$ are the equations of tangent at (0, 0).

V. Asymptotes: Equating the coefficient of y^2 with zero, we get
$x^2 - a^2 = 0$

i.e., $x = \pm a$.

Hence, the equations of asymptote parallel to y axis are $x = \pm a$.

The shape of the curve is given below:



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2.4.7 Tracing of parametric curves

Whenever we need to trace the paths of moving particles, then these types of paths cannot be determined by the functions of type $y = f(x)$ or $x = g(y)$. The motion so formed can be represented by parametric curves. In parametric representation, both x and y are dependent on a parameter t and t varies on some closed interval $[a, b]$, i.e., $a \le t \le b$. We write $x = f(t)$, $y = g(t)$.

In parametric form we have:

- the set of points $(x, y) = (f(t), g(t))$ is the position of a particle at time t.
- if I is the closed interval $a \le t \le b$, the point $[f(a), g(a)]$ is the initial point and the point $[f(b), g(b)]$ is the terminal point.

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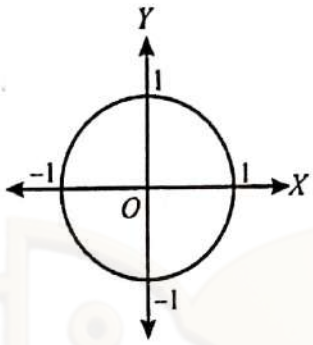
Example 41. Sketch the curve given by the parametric equations
$x = \cos t, y = \sin t$, for $0 \le t \le 2\pi$

Solution:

First we look at points on the curve for particular values of t as given in table.

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{3\pi}{2}$	2π
x	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	0	1
y	0	$\frac{1}{\sqrt{2}}$	1	$\frac{1}{\sqrt{2}}$	0	-1	0

Plotting these points we get curve



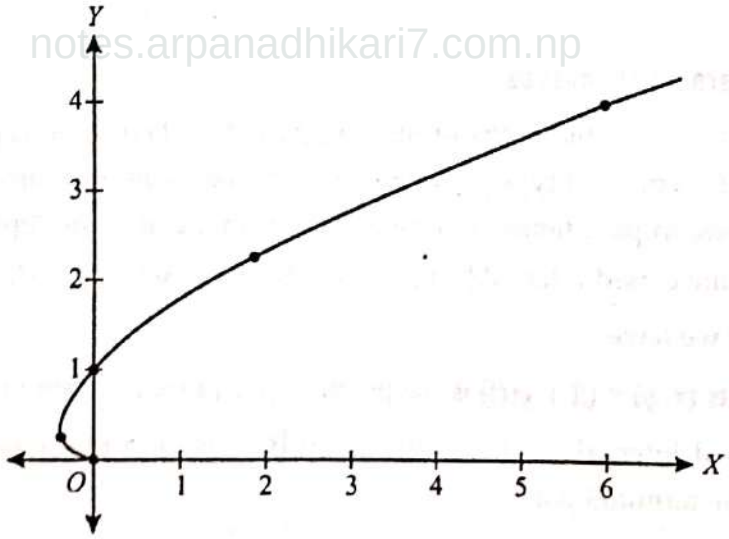
Example 42. Sketch the graph of *a* curve described by the following set of parametric equations
$x = t^3 - t,\ y = t^2$, for $0 \le t \le \infty$.

Solution:

Plotting the points t, x and y, we get

t	0	0.5	1	1.5	2
x	0	–0.375	0	1.875	6
y	0	0.25	1	2.25	4

Plotting the table in the graph



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Example 43. Sketch the curve given by the parametric equation
$x = t^2 = 2t, y = t + 1$ for $-2 \le t \le 4$.

Solution

The table for the given parametric curves is

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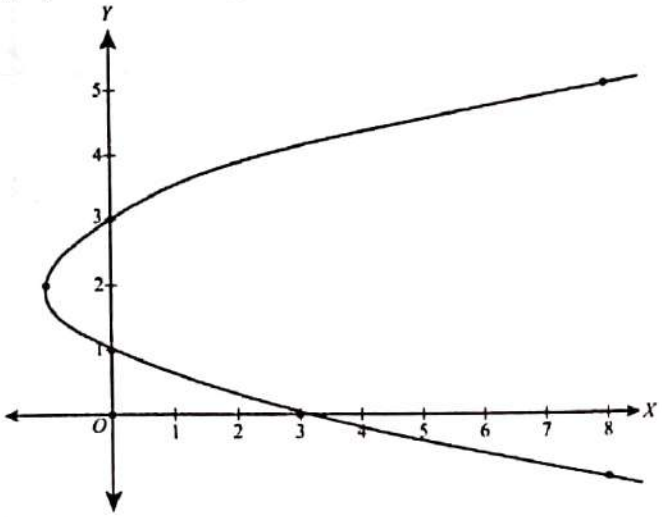
t	−2	−1	0	1	2	3	4
x	8	3	0	−1	0	3	8
y	−1	0	1	2	3	4	5

The plot of the above graph is as follows

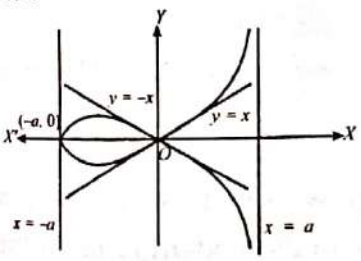


Exercise 2.4

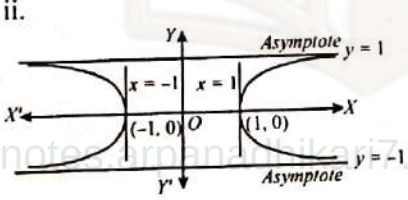
1. Trace the following curve:
   i. $y^2(a-x) = x^2(a+x)$
   ii. $x^2y^2 = x^2 - 1$
   iii. $x^2(a^2+y^2) = y(a^2-y^2)$
   iv. $y^2 = x^3$
   v. $x^3 + y^3 = 3axy$
2. Plot the following parametric equations.
   i. $x = \sin 2t$, $y = \cos 2t$ for $0 \le t \le 2\pi$
   ii. $x = t^2 - 2t$, $y = t + 1$ for $0 \le t \le 4$
   iii. $x = 2(t - \sin t)$, $y = 2(1 - \cos t)$ for $-4\pi \le t \le 4$
   iv. $x = t^2$, $y = t^3 - 3t$ for $-3 \le t \le t$

Answer

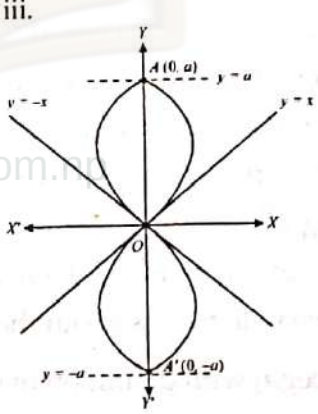
1. i.



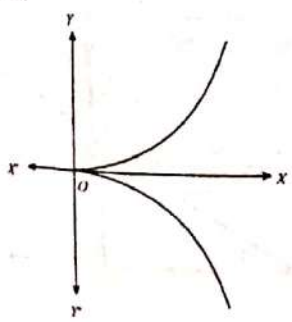
ii.



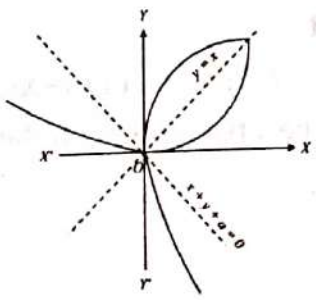
iii.



iv.



v.



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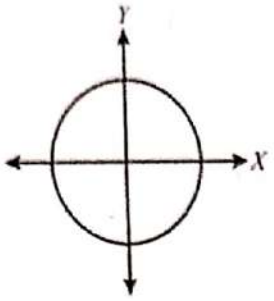
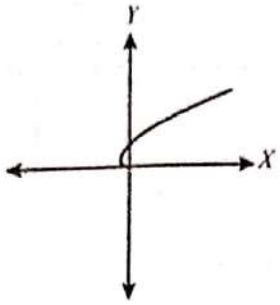
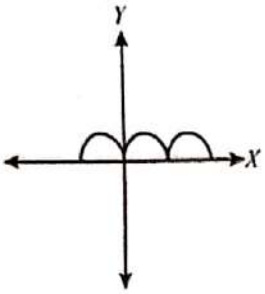
2. i.



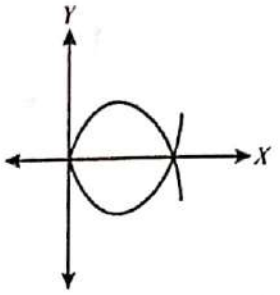
ii.



iii.



iv.



2.5 Curvature

2.5.0 Introduction

Nicolas Oresme(1320-1382)

The curvature of a curve is simply the rate at which the curve is turning. Since the tangent line shows the direction of the curve, this means that the curvature is the rate at which the tangent line is turning.

The first person to "see" curvature was Oresme (c. 1320-1382), Descartes's precursor in introducing the coordinates. He described it as a local measure of curve's bending, and named it with the Latin word "curvitas". Later he proposed that for circles, it can be quantified by the reciprocal of the radius in our modern convention. Kepler suggested the way to define a curvature for general curves by considering the "closest" circle at a point. Later it was named as osculating ("kissing") circle by Leibnitz in 1680. But it was Huygens, who first found a way to calculate curvature for general curves, and Newton gave the concept to its modern form. In this section, we will discuss about the determination of of curvature, their

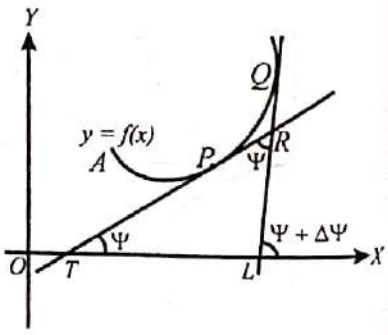
We first begin with definition of curvature at a point.

2.5.1 Curvature of a curve at a point

Let $P(x, y)$ be a point on the curve $y = f(x)$ and let $Q(x+\Delta x, y+\Delta y)$ be a point on the curve near P. Let A be a fixed point on the curve such that

Arc $AP = s$ and arc $AQ = s+\Delta s$.

Then, arc $PQ = \Delta s$



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Let PT and QL be tangents at P and Q to the curve respectively such that $PTX = \psi$ and $\angle QTX = \psi + \Delta\psi$.

Then $\angle TRL = \Delta\psi$.

The *curvature* at a point P to the curve $y = f(x)$ is the limiting value of $\frac{\Delta\psi}{\Delta s}$ when the point Q approaches to P along the curve i.e., when $\Delta s \to 0$. It is denoted by κ.

Thus, curvature at P, $\kappa = \lim_{\Delta s \to 0} \frac{\Delta\psi}{\Delta s} = \frac{d\psi}{ds}$.

Thus, the curvature is the rate of change of direction of the curve (and hence tangent) with respect to the arc length.

2.5.2 Radius of curvature at a point

The reciprocal of the curvature at any point P of a curve is called the *radius of curvature* at P. It is denoted by ρ.

Thus, radius of curvature at P is

$$\rho = \frac{1}{\kappa} = \frac{1}{\frac{d\psi}{ds}} = \frac{ds}{d\psi}.$$

We need to calculate the radius of curvature curves in different form. We first begin with the Cartesian curve.

2.5.3 Radius of curvature for a Cartesian curve $y = f(x)$

Let $y = f(x)$ be a curve.

Here, $\tan\psi = \frac{dy}{dx} = y_1$.

Differentiating both sides with respect to s, we get

$$\frac{d\tan\psi}{d\psi} \cdot \frac{d\psi}{ds} = \frac{d}{dx}\left(\frac{dy}{dx}\right)\frac{dx}{ds}.$$

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$$\sec^2\psi \cdot \frac{d\psi}{ds} = y_2 \cdot \frac{dx}{ds} \qquad \text{where } y_2 = \frac{d^2y}{dx^2}.$$

$$(1 + \tan^2\psi) \cdot \frac{d\psi}{ds} = y_2 \frac{dx}{ds}$$

$$\boxed{(1 + y_1^2)\frac{d\psi}{ds} = y_2 \frac{dx}{ds}} = y_2 \cos\psi$$

Now, $\rho = \frac{ds}{d\psi}$

$\therefore \frac{d\psi}{ds} = \frac{y_2 \cos\psi}{(1 + y_1^2)}$

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$$= \frac{1 + y_1^2}{y_2} \cdot \sec\psi \qquad \text{where, } \cos\psi = \frac{dx}{ds}$$

$$\therefore \quad \rho = \frac{(1 + y_1^2)\sqrt{1 + \tan^2\psi}}{y_2}$$

$$= \frac{(1 + y_1^2)\sqrt{1 + y_1^2}}{y_2}$$

$$\therefore \quad \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \text{ for } y_2 \neq 0.$$

The above formula of the radius of curvature fails to hold at the point where $y_1 = \infty$ (i.e., when tangent of the curve at the point is parallel to the y-axis). In such situation, the formula for the equation of the curve is expressed as $x = f(y)$ would be applicable.

Proceeding similarly, if the curve is of the form $x = f(y)$, then the radius of curvature is $\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}$, where $x_2 \neq 0$.

2.5.4 Radius of curvature for parametric curves

The equation of the curve be expressed parametric form as $x = \phi(t)$ and $y = \psi(t)$, where t is the parameters. Then,

$$y_1 = \frac{dy}{dx}$$

$$= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y'}{x'}$$

where the dashes denote derivatives with respect to t.

$$y_2 = \frac{d}{dt}\left(\frac{y'}{x'}\right) \cdot \frac{dt}{dx}$$

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$$= \frac{x' \frac{dy'}{dt} - y' \frac{dx'}{dt}}{x'^2} \cdot \frac{1}{x'} = \frac{x'y'' - y'x''}{x'^3}$$

Then, we have

$$\frac{(1 + y^2)^{3/2}}{y_2} = \frac{\left(1 + \left(\frac{y'}{x'}\right)^2\right)^{3/2}}{\frac{x'y'' - y'x''}{x'^3}} = \frac{\frac{(x'^2 + y'^2)^{3/2}}{x'^3}}{\frac{x'y'' - y'x''}{x'^3}}$$

$$\therefore \quad \rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''} \qquad \text{provided } x'y'' - y'x'' \neq 0.$$

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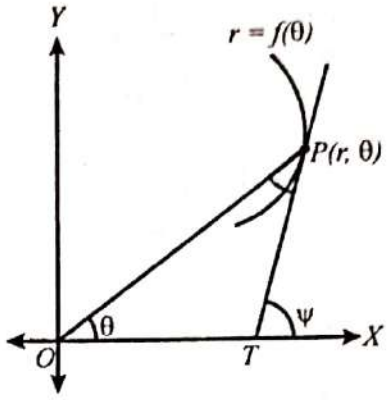
2.5.5 Radius of curvature for the polar curve $r = f(\theta)$

Let PT be tangent at point $P(r, \theta)$ on the curve $r = f(\theta)$.



Then, $OP = r$, $\angle POX = \theta$.

Suppose, $\angle OPT = \phi$, $\angle PTX = \psi$.

From figure,

$$\psi = \theta + \phi$$

$$\tan\psi = \tan(\theta + \phi)$$

$$\tan\psi = \frac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi} \quad \ldots.(i)$$

Also, we have, $\tan\psi = \frac{dy}{dx}$

Where $x = r\cos\theta$, $y = r\sin\theta$

$$\text{And, } \tan\psi = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta}(r\sin\theta)}{\frac{d}{d\theta}(r\cos\theta)}$$

$$= \frac{r\cos\theta + \sin\theta\frac{dr}{d\theta}}{\cos\theta\frac{dr}{d\theta} - r\sin\phi}$$

$$\therefore \tan\psi = \frac{\tan\theta + r\frac{d\theta}{dr}}{1 - \tan\theta \cdot r\frac{d\theta}{dr}} \quad \ldots.(ii)$$

Comparing (i) and (ii), we get

$$\tan\phi = r\frac{d\theta}{dr}$$

$$= \frac{r}{\frac{dr}{d\theta}} = \frac{r}{r_1}$$

or, $\phi = \tan^{-1}\left(\frac{r}{r_1}\right)$.

Also, we have

$$\psi = \theta + \phi$$

or, $\psi = \theta + \tan^{-1}\left(\frac{r}{r_1}\right)$.

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Differentiating with respect to θ, we get

$$\frac{dy}{d\theta} = 1 + \frac{1}{1 + \frac{r^2}{r_1^2}} \frac{d}{d\theta}\left(\frac{r}{r_1}\right)$$

$$= 1 + \frac{1}{\frac{r_1^2 + r^2}{r_1^2}} \cdot \frac{r_1 . r_1 - r . r_2}{r_1^2}$$

or, $$\frac{dy}{d\theta} = 1 + \frac{r_1^2}{r^2 + r_1^2} \cdot \frac{r_1^2 - rr_2}{r_1^2}$$

$$\therefore \quad \frac{dy}{d\theta} = 1 + \frac{r_1^2 - r\, r_2}{r^2 + r_1^2} \qquad \ldots . (i)$$

Also, we have

$$\frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$$

$$= \sqrt{r^2 + r_1^2} \qquad \ldots . (ii)$$

Then, $$\rho = \frac{ds}{d\psi} = \frac{\frac{ds}{d\theta}}{\frac{dy}{d\theta}}$$

$$= \frac{\sqrt{r^2 + r_1^2}}{1 + \frac{r_1^2 - r\, r_2}{r^2 + r_1^2}} \qquad [\because \text{From (i) and (ii)}]$$

$$= \frac{\sqrt{r^2 + r_1^2}\,(r^2 + r_1^2)}{r^2 + r_1^2 + r_1^2 - rr_2}$$

$$\therefore \quad \rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r\, r_2} \quad \text{provided } r^2 + 2r_1^2 - rr_2 \neq 0.$$

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2.5.3 Radius of curvature at origin

The radius of curvature ρ at the origin can be obtained by any one of the following methods:

1. Method of substitution
2. Method of expansion
3. Newtonian method.

1. Method of substitution

The radius of curvature ρ at the origin (0, 0) can be obtained by putting $x = 0, y = 0$ in the following formula

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \qquad \ldots . (i)$$

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But if y_1 at (0, 0) is ∞ (i.e., if the tangent at (0, 0) be y-axis or if the tangent at any point be parallel to y-axis) or y_1 at (0, 0) is of the form $\frac{0}{0}$ or y_2 at (0, 0) be zero, then the formula (i) fails to give the value of ρ. In such case, we find ρ by the following formula

$$\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}. \quad \ldots. \text{ (ii)}$$

Moreover, if the formula (ii) also fails to find the value of ρ at origin (0, 0), then the radius of curvature will be determined by the method of expansion.

2. Method of expansion

If (i) and (ii) in the method of expansion fail to find the value of ρ at origin, in such case, the radius of curvature at (0, 0) can be determined by the method of expansion with the help of Maclaurin's series.

Let the equation of the curve be $y = f(x)$. .... (i)

By using Maclaurin's series, we have

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \ldots \text{ to } \infty.$$

or, $y = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \ldots$ to ∞. .... (ii)

The curve passes through the origin (0, 0), therefore, $f(0) = 0$.

Then equation (ii) becomes

$$y = x f'(0) + \frac{x^2}{2!} f''(0) + \ldots \text{ to } \infty. \quad \ldots. \text{ (iii)}$$

From (i), $\frac{dy}{dx} = f'(x)$ .... (iv)

$\therefore$ At (0, 0), $\left(\frac{dy}{dx}\right)_{x=0} = f'(0) = p$, say. .... (v)

Again from (iv), $\frac{d^2y}{dx^2} = f''(x)$

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$\therefore$ At (0, 0), $\left(\frac{d^2y}{dx^2}\right)_{x=0} = f''(0) = q$, say. .... (vi)

Hence the equation (iii) becomes $y = f(x) = px + \frac{qx^2}{2!} + \ldots.$

Substituting the value of y in equation (i) and equating the coefficients of like powers of x, we get the value of p and q. Then ρ can be obtained by the following formula $\rho = \frac{(1 + p^2)^{3/2}}{q}$.

3. Newtonian method

This method of determining the radius of curvature at the origin is used when the curve passes through the origin and the tangent at the origin is either x-axis or y-axis.

If the curve passes through the origin, then the equation of the tangent at origin can be obtained by equating the lowest degree term of x and y to zero.

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Case I: If the curve passes through the origin, and the axis of x is the tangent at origin, then

$$\rho = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2}{2y}.$$

Case II: If the curve passes through the origin, and the axis of y is the tangent at origin, then

$$\rho = \lim_{\substack{y\to 0\\ x\to 0}} \frac{y^2}{2x}.$$

These two formulae are due to Newton.

Remark: If the equation of the curve be in polar form, then

$x = r\cos\theta,\ y = r\sin\theta.$

In this case, we use $\rho = \lim_{\theta\to 0}\left(\frac{r}{2\theta}\right)$.

We next discuss some examples:

Example 44. Find the radius of curvature at any point of the following curves.

i. $y = c\cosh\left(\frac{x}{c}\right)$ [cartesian form of curve]

ii. $x = a(t + \sin t), y = a(1 - \cos t)$ [parametric form of curve]

iii. $r = a(1 - \cos\theta)$ [polar curve]

Solution:

i. We have, $y = c\cosh\left(\frac{x}{c}\right)$ .... (i)

Differentiating (i) with respect to x, we get

$$\frac{dy}{dx} = c\sinh\left(\frac{x}{c}\right)\cdot\frac{1}{c} \qquad \left[\because \frac{d}{dx}(\cosh x) = \sinh x\right]$$

$$\therefore\ y_1 = \sinh\left(\frac{x}{c}\right) \qquad \text{.... (ii)}$$

Again, differentiating (ii) with respect to x, we get $y_2 = \frac{1}{c}\cosh\left(\frac{x}{c}\right)$

Now, radius of curvature ρ in cartesian form

$$= \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\left(1+\sinh^2\frac{x}{c}\right)^{3/2}}{\frac{1}{c}\cosh\frac{x}{c}}$$

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$$= \frac{c\left(\cosh^2\frac{x}{c}\right)^{3/2}}{\cosh\frac{x}{c}} \qquad [\because \cosh^2 x - \sinh^2 x = 1]$$

$$= \frac{c\cosh^3\frac{x}{c}}{\cosh\frac{x}{c}}$$

$$= c\cosh^2\frac{x}{c}$$

$$= c.\frac{y^2}{c^2} \qquad \left[\because y = c\cosh\frac{x}{c}\right]$$

$$\therefore\ \rho = \frac{y^2}{c}$$

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ii We have,

$x = a(t + \sin t)$ .... (i)

$y = a(1 - \cos t)$ .... (ii)

Then, $x' = a(1 + \cos t)$ $\quad x'' = -a\sin t$

$y' = a\sin t$ $\quad y'' = a\cos t$

Now, $\rho = \dfrac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}$

$= \dfrac{[a^2(1 + \cos t)^2 + a^2\sin^2 t]^{3/2}}{a(1 + \cos t).a\cos t - a\sin t.(-a\sin t)}$

$= \dfrac{[a^2(1 + 2\cos t + \cos^2 t) + a^2\sin^2 t]^{3/2}}{a^2\cos t + a^2\cos^2 t + a^2\sin^2 t}$

$= \dfrac{[a^2 + 2a^2\cos t + a^2\cos^2 t + a^2\sin^2 t]^{3/2}}{a^2\cos t + a^2(\cos^2 t + \sin^2 t)}$

$= \dfrac{[a^2 + 2a^2\cos t + a^2(\cos^2 t + \sin^2 t)^{3/2}}{a^2\cos t + a^2}$

$= \dfrac{(a^2 + 2a^2\cos t + a^2)^{3/2}}{a^2(1 + \cos t)}$

$= \dfrac{(2a^2 + 2a^2\cos t)^{3/2}}{a^2(1 + \cos t)}$

$= \dfrac{(2a^2)^{3/2} - (1 + \cos t)^{3/2}}{a^2(1 + \cos t)}$

$= \dfrac{2\sqrt{2}\,a^3(1 + \cos t)^{3/2}}{a^2(1 + \cos t)}$

$= 2\sqrt{2}\,a(1 + \cos t)^{1/2}$

$= 2\sqrt{2}\,a\left(2\cos^2\dfrac{t}{2}\right)^{\frac{1}{2}}$

$= 4a\cos\dfrac{t}{2}$.

iii. We have, $r = a(1 - \cos\theta)$

Then, $r_1 = a\sin\theta$, $r_2 = a\cos\theta$

Now, $\rho = \dfrac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$

$= \dfrac{[a^2(1 - \cos\theta)^2 + a^2\sin^2\theta]^{3/2}}{a^2(1 - \cos\theta)^2 + 2a^2\sin^2\theta - a(1 - \cos\theta).a\cos\theta}$

$= \dfrac{[a^2(1 - 2\cos\theta + \cos^2\theta) + a^2\sin^2\theta]^{3/2}}{a^2(1 - 2\cos\theta + \cos^2\theta) + 2a^2\sin^2\theta - a^2\cos\theta + a^2\cos^2\theta}$

$= \dfrac{[a^2 - 2a^2\cos\theta + a^2(\cos^2\theta + \sin^2\theta)]^{3/2}}{(a^2 - 2a^2\cos\theta + a^2\cos^2\theta + 2a^2\sin^2\theta - a^2\cos\theta + a^2\cos^2\theta)}$

$= \dfrac{(a^2 - 2a^2\cos\theta + a^2)^{3/2}}{a^2 - 3a^2\cos\theta + 2a^2(\cos^2\theta + \sin^2\theta)}$

$= \dfrac{(2a^2 - 2a^2\cos\theta)^{3/2}}{(3a^2 - 3a^2\cos\theta)} = \dfrac{2\sqrt{2}\,a^3(1 - \cos\theta)^{3/2}}{3a^2(1 - \cos\theta)}$

$= \dfrac{2\sqrt{2}}{3}a(1 - \cos\theta)^{1/2}$

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$$= \frac{2\sqrt{2}\,a}{3}\left(2\sin^2\frac{\theta}{2}\right)^{\frac{1}{2}}$$

$$= \frac{4a}{3}\sin\frac{\theta}{2}.$$

Example 45. Find the radius of curvature of the curve $y^2 = 4x$ at the vertex $(0, 0)$.

Solution:

Here, $y^2 = 4x$ .... (i)

Differentiating with respect to x, we get

$$2y\frac{dy}{dx} = 4$$

or, $$\frac{dy}{dx} = \frac{2}{y}$$

$\therefore$ $\frac{dy}{dx}$ at $(0, 0) = \infty$.

Thus $\frac{dy}{dx}$ at the vertex is infinite. So, the tangent at the vertex being y-axis and the formula $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$ is not applicable in this case.

Hence we apply $\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}$.

Now, from (i),

$$x = \frac{y^2}{4}$$

$\therefore$ $\frac{dx}{dy} = \frac{1}{2}y$ and

$$\frac{d^2x}{dy^2} = \frac{1}{2}.$$

$\therefore$ At the vertex $(0, 0)$, $x_1 = 0$ and $x_2 = \frac{1}{2}$.

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Hence $\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}$

$$= \frac{(1 + 0)^{3/2}}{\frac{1}{2}}$$

$$= \frac{(1)^{3/2}}{\frac{1}{2}}$$

$$= \frac{1}{\frac{1}{2}}$$

$$= 2.$$

$\therefore$ $\rho = 2$.

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Example 46. Find the radius of curvature of the curve $y = x^2(x - 3)$ at the points where the tangent is parallel to x-axis.

Solution:

Here, $y = x^2(x - 3) = x^3 - 3x^2$ .... (i)

$\therefore \quad \frac{dy}{dx} = 3x^2 - x.$

If the tangent is parallel to x-axis, then

$\frac{dy}{dx} = 0$

or, $3x^2 - 6x = 0$

$\therefore \quad x = 0$ and $x = 2$.

Then if $x = 0, y = 0$ and if $x = 2, y = -4$.

$\therefore$ Points are (0,0) and (2, –4).

Case I: At (0, 0): x-axis is tangent at origin, and hence

$$\rho = \lim_{x \to 0} \frac{x^2}{2y} \quad \text{.... (ii)}$$

But $y = x^2(x - 3)$, then (ii) gives

$$\rho = \lim_{x \to 0} \frac{x^2}{2x^2(x - 3)}$$

$$= \lim_{x \to 0} \frac{1}{2(x - 3)}$$

$$= -\frac{1}{6}.$$

$\therefore \quad \rho = -1/6.$

Case II: At (2, –4):

Here, $y = x^2(x-3) = x^3 - 3x^2$.

$y_1 = \frac{dy}{dx} = 3x^2 - 6x$

and $y_2 = \frac{d^2y}{dx^2} = 6x - 6.$

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At (2, –4): $y_1 = 3.4 - 12 = 12 - 12 = 0$ and $y_2 = 6.2 - 6 = 6.$

$$\therefore \quad \rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$= \frac{(1 + 0)^{3/2}}{6}$$

$$= \frac{1}{6}.$$

Hence, radius of curvatures are $\frac{1}{6}$ and $-\frac{1}{6}$.

Example 47. Show that the radius of curvature at any point of a circle is constant and its curvature at any point is also constant.

Solution:

Let the equation of circle in standard from is $x^2 + y^2 = a^2$ ....(i)

Differentiating with respect to x, we get

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$$2x + 2y\frac{dy}{dx} = 0$$

or, $\frac{dy}{dx} = -\frac{x}{y}$

$\therefore \quad y_1 = -\frac{x}{y}$

Again, differentiating with respect to x, we get

$$y_2 = -\frac{\frac{ydx}{dx} - x\frac{dy}{dx}}{y^2}$$

$$= -\frac{y.1 - x.\left(\frac{-x}{y}\right)}{y^2}$$

$$= -\frac{y^2 + x^2}{y^3}$$

$$= -\frac{a^2}{y^3} \qquad \text{[using (i)]}$$

Then, $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{\left(1 + \left(\frac{-x}{y}\right)^2\right)^{3/2}}{-\frac{a^2}{y^3}}$

$$= -\left(\frac{y^2 + x^2}{y^2}\right)^{3/2} \cdot \frac{y^3}{a^2}$$

$$= -\frac{(a^2)^{3/2}}{(y^2)^{3/2}} \cdot \frac{y^3}{a^2} \qquad \text{[using (i)]}$$

$$= -\frac{a^3}{y^3} \cdot \frac{y^3}{a^2}.$$

$\rho = -a$

$\therefore \quad \rho = a$, numerically which is constant

Also, curvature $(\kappa) = \frac{1}{r} = \frac{1}{a}$, which is also constant.

Example 48. Find the radius of curvature of the following curves at the origin:

i. $4x^2 - 3xy + y^2 - 3y = 0$ ii. $x^3 + y^3 = 3axy$

iii. $r = a\sin n\theta$ at $\theta = 0$

Solution:

i. We have, $4x^2 - 3xy + y^2 - 3y = 0$ .... (i)

The curve (i) passes through the origin.

Equating the lowest degree term to zero, we have $y = 0$

i.e. x-axis is the tangent to the curve at origin, so $\rho = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2}{2y}$

Dividing each term of the given equation by $2y$ and taking limit $\lim_{\substack{x\to 0\\ y\to 0}}$, we have

$$\lim_{\substack{x\to 0\\ y\to 0}} \left(4\frac{x^2}{2y} - \frac{3}{2}x - \frac{y}{2} - \frac{3}{2}\right) = 0$$

$\Rightarrow \quad 4\rho - 0 + 0 - \frac{3}{2} = 0$

$\Rightarrow \quad \rho = \frac{3}{8}$

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ii. We have, $x^3 + y^3 = 3axy$ .... (i)

The curve passes through the origin.

Equating lowest degree terms to zero, we get $3axy = 0$

i.e. $x = 0, y = 0$

When $y = 0$ (i.e., when the tangent is x-axis), then $\rho = \lim\limits_{\substack{x\to 0\\ y\to 0}} \frac{x^2}{2y}$

Dividing each term in (i) by $2xy$ and taking $\lim\limits_{\substack{x\to 0\\ y\to 0}}$, we have

$$\lim \left(\frac{x^2}{2y} + \frac{y^2}{2x}\right) = \frac{3a}{2}$$

$$\Rightarrow \lim \left(\frac{x^2}{2y} + \frac{xy}{4} \cdot \frac{1}{\frac{x^2}{2y}}\right) = \frac{3a}{2}$$

$$\Rightarrow \rho + \frac{1}{4} \cdot 0 \cdot 0 \cdot \frac{1}{\rho} = \frac{3a}{2}$$

$$\Rightarrow \rho = \frac{3a}{2}$$

When $x = 0$ (i.e., y-axis is the tangent), then $\rho = \lim\limits_{\substack{x\to 0\\ y\to 0}} \frac{y^2}{2x}$

Dividing each term of (i) by $2xy$ and taking $\lim\limits_{\substack{x\to 0\\ y\to 0}}$, we have

$$\lim \left(\frac{x^2}{2y} + \frac{y^2}{2x}\right) = \frac{3a}{2}$$

$$\Rightarrow \lim \left(\frac{x^2}{2y} + \frac{xy}{4} \cdot \frac{1}{\frac{x^2}{2y}}\right) = \frac{3a}{2}$$

$$\Rightarrow \frac{1}{4} \cdot 0 \cdot 0 \cdot \frac{1}{\rho} + \rho = \frac{3a}{2}$$

$$\therefore \rho = \frac{3a}{2}$$

iii. We have, $r = a\sin n\theta$

Then, $\rho = \lim\limits_{\theta\to 0} \left(\frac{r}{2\theta}\right) = \lim\limits_{\theta\to 0} \left(\frac{a\sin n\theta}{2\theta}\right) = \lim\limits_{\theta\to 0} \left(\frac{a}{2} \cdot \frac{\sin n\theta}{n\theta} \cdot n\right) = \frac{na}{2}$

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Example 49. If ρ_1 and ρ_2 be the radii of curvature at the ends of a focal chord of the parabola $y^2 = 4ax$, prove that $\rho_1^{-2/3} + \rho_2^{-2/3} = (2a)^{-2/3}$.

Solution:

The given equation of parabola is $y^2 = 4ax$ .... (i)

Any point on the parabola (i) is $x = at^2, y = 2at$

So, $x' = 2at, y' = 2a, \ x'' = 2a, \ y'' = 0$

Hence, $\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - x''y'} = \frac{(4a^2t^2 + 4a^2)^{3/2}}{2at.0 - 2a.2a} = \frac{(2a)^3 (1 + t^2)^{3/2}}{-(2a)^2} = -2a(1 + t^2)^{3/2}$

$\therefore \rho = 2a(1 + t^2)^{3/2}$ (Numerically)

Let the ends of a focal chord PQ of parabola be $P(at_1^2, 2at_1)$, $Q(at_2^2, 2at_2)$.

We know in case of parabola $t_1t_2 = -1$ .... (ii)

Then, at point P, $\rho_1 = 2a(1 + t_1^2)^{3/2}$

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and at point Q, $\rho_2 = 2a(1 + t_2^2)^{3/2}$

So, $\rho_1^{-2/3} + \rho_2^{-2/3} = [2a(1 + t_1^2)^{3/2}]^{-2/3} + [2a(1 + t_2^2)^{3/2}]^{-2/3}$

$= (2a)^{-2/3}(1 + t_1^2)^{-1} + (2a)^{-2/3}(1 + t_2^2)^{-2}$

$= (2a)^{-2/3}\left[\frac{1}{1 + t_1^2} + \frac{1}{1 + t_2^2}\right]$

$= (2a)^{-2/3}\left[\frac{1}{1 + t_1^2} + \frac{1}{1 + \left(-\frac{1}{t_1}\right)^2}\right]\left[\because t_1 t_2 = -1,\ t_2 = -\frac{1}{t_1}\right]$

$= (2a)^{-2/3}\left[\frac{1}{1 + t_1^2} + \frac{t_1^2}{t_1^2 + 1}\right]$

$= (2a)^{-2/3}\left[\frac{1 + t_1^2}{1 + t_1^2}\right]$

$= (2a)^{-2/3}$

$\therefore \quad \rho_1^{-2/3} + \rho_2^{-2/3} = (2a)^{-2/3}$.

Exercise 2.5

1. Find the radius of curvature at any point (x, y) of the cartesian curves.

   i. $y^2 = 4ax$ ii. $y = c\,\text{cosec}\,\frac{x}{a}$

   iii. $x^{2/3} + y^{2/3} = a^{2/3}$ iv. $xy = c^2$

   v. $y = a\cosh\left(\frac{x}{a}\right)$

2. Find the radius of the curvature for the parametric curves at the points indicated.

   i. $x = at^2, y = 2$ at (x, y)

   ii. $x = a\cos\theta, y = a\sin\theta$ at (x, y)

   iii. $x = a\cos\theta, y = b\sin\theta$ at (x, y)

   iv. $x = a(\theta + \sin\theta), y = a(1 - \cos\theta)$ at $\theta = 0$

   v. $x = a\cos^3\theta, y = a\sin^3\theta$ at $\theta = \frac{\pi}{4}$

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3. Find the radius of curvature for the polar curves at the points indicated:

   i. $r = ae^{\theta\cot\alpha}$ at (r, θ) ii. $r = a(1 - \cos\theta)$ at (r, θ)

   iii. $r^2\cos 2\theta = a^2$ at (r, θ) iv. $r^2 = a^2\cos 2\theta$ at $\theta = 0$

   v. $r = a(\theta + \sin\theta)$ at $\theta = 0$ vi. $r^m = a^m\cos m\theta$ at (r, θ)

4. Find the radius of curvature at the origin of the following curves.

   i. $y = x^4 - 4x^3 - 18x^2$ ii. $3x^2 + 4y^2 = 2x$

   iii. $3x^4 - 2y^4 + 5x^2y + 2xy - 2y^2 + 4x = 0$

   iv. $x^4 + y^2 = 6a(x + y)$

   v. $y^2 = \frac{x^2(a + x)}{a - x}$

5. Find the radius of curvature of the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point where it cuts the line $y = x$.

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Answer

1. i. $\frac{2(x+a)^{3/2}}{\sqrt{a}}$ ii. $a \sec \frac{x}{a}$ iii. $3(axy)^{1/3}$
   iv. $\frac{(x^2+y^2)^{3/2}}{2c^2}$ v. $\frac{y^2}{a}$
2. i. $2a(t^2+1)^{3/2}$ ii. a iii. $\frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{ab}$
   iv. $4a$ v. $\frac{3a}{2}$
3. i. $r\operatorname{cosec}\alpha$ ii. $\frac{2}{3}\sqrt{2ar}$ iii. $\frac{r^3}{a^2}$
   iv. $\frac{1}{3}a$ v. a vi. $\frac{a^m}{r^{m-1}(m+1)}$
4. i. $\frac{1}{36}$ ii. $\frac{1}{4}$ iii. 1
   iv. $a\sqrt{2}$ v. $6a\sqrt{2}$
5. $\frac{a}{\sqrt{2}}$

Multiple Choice Questions

1. For which of the function, Rolle's theorem is not applicable ?
   a. $f(x) = |x|$ b. $f(x) = x^2 - 4$
   c. $f(x) = \cos x$ d. $f(x) = x^4$
2. The equation of tangent to $y = \sin x$ at $\left(\frac{\pi}{2}, 1\right)$ is
   a. $x = 0$ b. $y = 0$
   c. $x = \frac{\pi}{2}$ d. $y = \frac{\pi}{2}$
3. The value of c given by mean value theorem for the function $f(x) = x^2 + 2x + 3$ on $[1, 3]$ is
   a. $\frac{5}{2}$ b. 2
   c. $\frac{3}{2}$ d. 0
4. For the curve $f(x) = (x-5)^2$ applying mean value theorem on $[3, 7]$, the tangent at which of the following point is parallel to the chord joining $A(3, 4)$ and B $(7, 4)$.
   a. $(0, 5)$ b. $(4, 1)$
   c. $(6, 1)$ d. $(5, 0)$
5. The value of c in Lagrange's mean value theorem for the function $|x|$ in the interval $[-1, 1]$ is
   a. 0 b. $\frac{1}{2}$
   c. $-\frac{1}{2}$ d. does not exist
6. For the function $f(x) = x + \frac{1}{x}$, $x \in [1, 3]$, the value of c for mean value Theorem is
   a. 1 b. $\sqrt{3}$
   c. 2 d. none of these

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7. For $a \neq 0, b \neq 0, y = ae^x$ and $y = be^{-x}$ intersect at right angle if

a. $a = \frac{1}{b}$ b. $a = b$

c. $a = -\frac{1}{b}$ d. $a + b = 0$

8. The curvature of a function $f(x)$ is zero. Which of the following functions could be $f(x)$?

a. $ax + b$ b. $ax^2 + bx + c$

c. $\sin x$ d. $\cos x$

9. The point of inflection of $f(x) = \log(x^2 + 4)$ occur at

a. 0 only b. $-2, 0, 2$

c. -1 and 1 d. -2 and 2

10. If $f(x) = 3x^4 - 4x^2 + 5$, then the interval for which $f(x)$ satisfies all the conditions of Rolle's theorem is

a. $[0, 2]$ b. $[-1, 1]$

c. $[-1, 0]$ d. $[1, 2]$

11. Tangent from origin at the parabola $y^2 + 4 = 4x$ are inclined at

a. 0 b. π

c. $\frac{\pi}{2}$ d. $\frac{\pi}{3}$

12. The curve $y = x^4$ is

a. concave up for all values of x.

b. concave down for all values of x.

c. concave up for any positive values of x.

d. concave up for only negative values of x.

13. Vertical asymptote to the curve $f(x) = \frac{x+5}{x^2 - 5x + 6}$ is

a. $x = 4$ b. $x = 3$

c. $x = 5$ d. $x = 6$

Answer

1. a. 2. d. 3. b. 4. d. 5. d. 6. b. 7. a.
8. a. 9. d. 10. b. 11. c. 12. a. 13. b.

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Unit 3

Integral Calculus

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- indefinite integrals, various techniques of integration
- evaluation of definite integral

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- definite integral, its physical interpretation and properties.
- fundamental theorem of integral calculus.
- improper integrals, comparison test.
- reduction formula, beta function and gamma function with their properties.

3.0 Introduction

In this chapter, we will focus on integral calculus. We begin with review of indefinite integration as reverse process of differentiation and the fundamental theorem of integral calculus. We then discuss definite integral along with various properties. We will also discuss improper integrals and beta function and gamma function.

Integration is the act of bringing smaller components into a single system. Newton was the first person to develop this idea about the concept of integration. Leibnitz also worked on integration independently. Newton neglected to publish his work until after Leibnitz. Leibnitz work was well known as compared to Newton's work.

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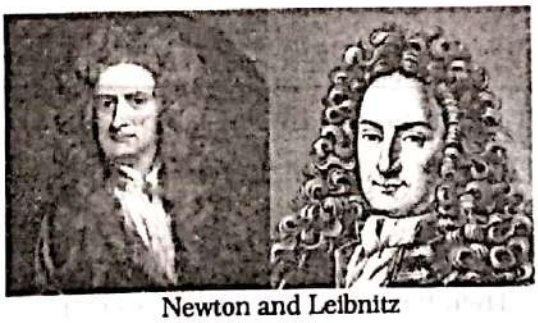

Newton and Leibnitz

After Newton and Leibniz brought the concept of integration through the fundamental theorem of calculus, it was seen as anti-differentiation, rather than its own concept. The notion of the integral as the limit of a summation process was certainly something that had been explored but it was treated more as a convenient way to treat difficult integrals, rather than as a fundamental support for integration theory.

We first discuss situations where we have to evaluate the total change from the rate of change of the quantity. We will apply this concept to measure the overall presence of a injected drug in the bloodstream in the course of a treatment. In medical language, the overall presence of a drug is called bioavailability.

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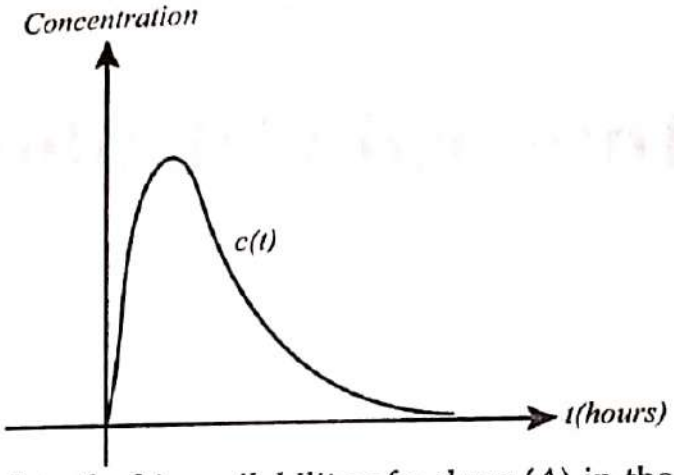
A unit bioavailability represents one unit concentration of the drug in the bloodstream for 1 hour. Normally, the concentration of a drug in the blood is not constant. In fact, the concentration in the blood increase as the drug is absorbed into the blood and then decreases as the drug is broken down and excreted. Figure below shows the concentration of drug in the blood as a function of t.



Suppose, we want to calculate the bioavailability of a drug (A) in the blood. Let $c(t)$ microgram per cm^3 be the concentration of it at a time t and we want to calculate the bioavailability for the time period $0 \le t \le T$. We note that during a small interval Δt, we have bioavailability $\approx c(t)\Delta t$ (concentration × time)

$\therefore$ Total bioavailability $= \Sigma c(t)\Delta t$.

Here, total bioavailability is obtained by adding all these quantities. This kind of phenomenon can be measured with the help of integrals. Hence, the total bioavailability on the interval $[0, T]$ is $\int_0^T c(t)\,dt$.

$$\therefore \quad \int_0^T c(t)\,dt = \Sigma\, c(t)\Delta t.$$

Thus, integral is a mathematical tool to calculate total quantity in various continuously changing phenomenon.

3.1 Review of indefinite integrals

Integration is the reverse process of differentiation. If the function $y = f(x)$ is differentiable, then we write,

$$\frac{d}{dx}(f(x)) = F(x)$$

Then the integral of $F(x)$ with respect to x is $f(x)$ and we write

$$\int F(x)\,dx = f(x)$$

Furthermore, $\frac{d}{dx}[f(x) + C] = \frac{d}{dx}[f(x)] + \frac{d}{dx}(C) = F(x) + 0 = F(x)$.

$$\therefore \quad \int F(x)\,dx = f(x) + C.$$

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Students are expected to have skills to evaluate indefinite integrals. Evaluation of various indefinite integrals is a fundamental skill in the chapters to come and in future courses too. In this chapter, we mainly focus on the techniques of evaluation of integrals. We basically review all the methods of integration that the students learned in their secondary level.

3.1.1 Methods of integration

The commonly used method of integration are illustrated as follows:

i. Substitution method
ii. Decomposition method
iii. Integration by parts method
iv. Partial fraction method

We illustrate these methods with examples.

Example 1. Evaluate: $\int \frac{1}{\sqrt{x}+x}\,dx$.

Solution:

We have $\int \frac{1}{\sqrt{x}+x}\,dx = \int \frac{1}{\sqrt{x}\,(1+\sqrt{x})}\,dx$

Put $t = (\sqrt{x}+1)$ then $dt = \frac{dx}{2\sqrt{x}} \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$

$\therefore \quad I = \int \frac{2}{t}\,dt = 2\log t + C = 2\log(1+\sqrt{x}) + C$

$\therefore \quad \int \frac{1}{\sqrt{x}+x}\,dx = 2\log(1+\sqrt{x}) + C.$

Example 2. Evaluate: $\int \frac{1+x^2}{\sqrt{1-x^2}}\,dx$.

Solution:

$I = \int \frac{1+x^2}{\sqrt{1-x^2}}\,dx$

Put $x = \sin\theta$ then $dx = \cos\theta\,d\theta$ and $\theta = \sin^{-1}x$

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$\therefore \quad I = \int \frac{1+\sin^2\theta}{\sqrt{1-\sin^2\theta}}\cos\theta\,d\theta$

$= \int \frac{1+\sin^2\theta}{\cos\theta}\cos\theta\,d\theta$

$= \int d\theta + \int \sin^2\theta\,d\theta$

$= \theta + \int\left(\frac{1-\cos 2\theta}{2}\right)d\theta$

$= \theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C = \frac{3\theta}{2} - \frac{\sin\theta\cos\theta}{2} + C$

$= \frac{3\sin^{-1}x}{2} - \frac{1}{2}x\sqrt{1-x^2} + C$

$\therefore \quad \int \frac{1+x^2}{\sqrt{1-x^2}}\,dx = \frac{3\sin^{-1}x}{2} - \frac{1}{2}x\sqrt{1-x^2} + C.$

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Example 3. Evaluate: $\int \frac{1}{x^2 - 8x + 20}\, dx.$

Solution:

Here, $I = \int \frac{1}{x^2 - 8x + 20}\, dx$

$$= \int \frac{1}{x^2 - 8x + 16 + 4}\, dx$$

$$= \int \frac{1}{(x-4)^2 + (2)^2}\, dx$$

$$= \frac{1}{2}\tan^{-1}\left(\frac{x-4}{2}\right) + C$$

$$\therefore \quad \int \frac{1}{x^2 - 8x + 20}\, dx = \frac{1}{2}\tan^{-1}\left(\frac{x-4}{2}\right) + C.$$

Example 4. Evaluate: $\int \frac{x}{x^2 + 6x + 40}\, dx.$

Solution:

Here, $I = \int \frac{x}{x^2 + 6x + 40}\, dx$

$$= \frac{1}{2}\int \frac{2(x+3) - 6}{x^2 + 6x + 13}\, dx$$

$$= \frac{1}{2}\int \frac{2(x+3)}{(x^2 + 6x + 13)}\, dx - \int \frac{6}{x^2 + 6x + 13}\, dx$$

$$= \frac{1}{2}\log(x^2 + 6x + 13) - 6\int \frac{1}{(x+3)^2 + (2)^2}\, dx$$

$$= \frac{1}{2}\log(x^2 + 6x + 13) - 6\,\frac{1}{2}\tan^{-1}\left(\frac{x+3}{2}\right) + C$$

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$$= \frac{1}{2}\log(x^2 + 6x + 13) - 3\tan^{-1}\left(\frac{x+3}{2}\right) + C.$$

Example 5. Evaluate: $\int \frac{2x^2 + 3x + 4}{x^2 + 6x + 10}\, dx.$

Solution:

By actual division

$$I = \int\left(2 - \frac{\frac{9}{2}(2x+6) + 11}{x^2 + 6x + 10}\right) dx$$

$$= 2\int dx - \frac{9}{2}\int \frac{2x+6}{x^2 + 6x + 10}\, dx + 11\int \frac{1}{(x+3)^2 + (1)^2}\, dx$$

$$= 2x - \frac{9}{2}\log|x^2 + 6x + 10| + 11\tan^{-1}(x+3) + C$$

$$\therefore \quad \int \frac{2x^2 + 3x + 4}{x^2 + 6x + 10}\, dx = 2x - \frac{9}{2}\log|x^2 + 6x + 10| + 11\tan^{-1}(x+3) + C$$

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Example 6. Evaluate: $\int \frac{1}{\sqrt{(x-\alpha)(x-\beta)}}\,dx$.

Solution:

Put $x - \alpha = t^2$ then $dx = 2t\,dt$

Then, $x - \beta = x - \alpha + \alpha - \beta = t^2 + (\alpha - \beta)$

$$\therefore \quad I = \int \frac{2t}{t.\sqrt{t^2 + (\alpha - \beta)}}\,dt$$

$$= \int \frac{2}{\sqrt{t^2 + (\sqrt{\alpha - \beta})^2}}\,dt$$

$$= 2\log\left(t + \sqrt{t^2 + \alpha - \beta}\right) + C$$

$$= 2\log\left(\sqrt{x-\alpha} + \sqrt{x-\beta}\right) + C$$

$$\therefore \quad \int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}} = 2\log\left(\sqrt{x-\alpha} + \sqrt{x-\beta}\right) + C.$$

Example 7. Evaluate: $\int \sin^2 x \cos^3 x\,dx$.

Solution:

$$I = \int \sin^2 x \cos^3 x\,dx$$

$$= \int \sin^2 x (1 - \sin^2 x) \cos x\,dx$$

$$= \int \sin^2 x \cos x\,dx - \int \sin^4 x \cos x\,dx$$

$$= \int \sin^2 x\,d(\sin x) - \int \sin^4 x\,d(\sin x)$$

$$= \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C.$$

$$\therefore \quad \int \sin^2 x \cos^3 x\,dx = \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C.$$

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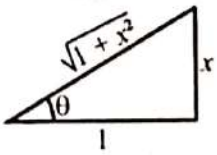
Example 8. Evaluate: $\int \frac{1}{x^2\sqrt{1+x^2}}\,dx$.

Solution:

Put $x = \tan\theta$ then $dx = \sec^2\theta\,d\theta$ and $1 + x^2 = 1 + \tan^2\theta = \sec^2\theta$.



$$\therefore \quad I = \int \frac{1}{x^2\sqrt{1+x^2}}\,dx$$

$$= \int \frac{\sec^2\theta}{\tan^2\theta \sec\theta}\,d\theta$$

$$= \int \frac{\sec\theta}{\tan^2\theta}\,d\theta$$

$$= \int \cot\theta \operatorname{cosec}\theta\,d\theta$$

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$= -\text{cosec}\theta + C$

$= -\frac{\sqrt{1+x^2}}{x} + C$

$\therefore \int \frac{1}{x^2\sqrt{1+x^2}}\,dx = -\frac{\sqrt{1+x^2}}{x} + C.$

Example 9. Evaluate: $\int \sqrt{\frac{a+x}{a-x}}\,dx.$

Solution:

Here, $I = \int \sqrt{\frac{a+x}{a-x}}\,dx$

$= \int \sqrt{\frac{(a+x)(a+x)}{(a-x)(a+x)}}\,dx$

$= \int \frac{a+x}{\sqrt{a^2-x^2}}\,dx$

Put $x = a\cos\theta$ then $dx = -a\sin\theta\,d\theta$

$\sqrt{a^2-x^2} = \sqrt{a^2-a^2\cos^2\theta} = a\sin\theta$

$\therefore I = \int \frac{a+x}{\sqrt{a^2-x^2}}\,dx$

$= \int \frac{a(1+\cos\theta)}{a\sin\theta} \cdot a\sin\theta\,d\theta$

$= -\int a\,d\theta - \int a\cos\theta\,d\theta$

$= -a\theta - a\sin\theta + C$

$= -a\cos^{-1}\frac{x}{a} - \sqrt{a^2-x^2} + C$

$\therefore \int \sqrt{\frac{a+x}{a-x}}\,dx = -a\cos^{-1}\left(\frac{x}{a}\right) - \sqrt{a^2-x^2} + C.$

Example 10. Evaluate: $\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}\,dx.$

Solution:

Here, $I = \int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}\,dx$

$= \int e^{\sin^{-1}x}\,d(\sin^{-1}x)$, since $d(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}\,dx.$

$= e^{\sin^{-1}x} + C$

$\therefore \int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}\,dx = e^{\sin^{-1}x} + C.$

Example 11. Evaluate: $\int \frac{1}{(x+3)\sqrt{x^2+6x+10}}\,dx.$

Solution:

Here, $I = \int \frac{1}{(x+3)\sqrt{x^2+6x+10}}\,dx = \int \frac{1}{(x+3)\sqrt{(x+3)^2+1}}\,dx$

Put $x + 3 = \frac{1}{y}$ then $dx = -\frac{1}{y^2}\,dy.$

Then, $I = \int \frac{1}{\frac{1}{y}\sqrt{\frac{1}{y^2}+1}}\left(-\frac{dy}{y^2}\right)$

$= -\int \frac{1}{\sqrt{y^2+1}}\,dy$

$= -\log|y + \sqrt{y^2+1}| + C$

$= -\log\left|\frac{1}{x+3} + \sqrt{\left(\frac{1}{x+3}\right)^2+1}\right| + C$

$= -\log\left|\frac{1}{x+3} + \frac{1}{x+3}\sqrt{1+(x+3)^2}\right| + C$

$= -\log\left|\frac{1+\sqrt{x^2+6x+10}}{x+3}\right| + C$

$\therefore \int \frac{1}{(x+3)\sqrt{x^2+6x+10}}\,dx = -\log\left|\frac{1+\sqrt{x^2+6x+10}}{x+3}\right| + C.$

Example 12. Evaluate: $\int \frac{1-x}{\sqrt{8+2x-x^2}}\,dx.$

Solution:

Here, $I = \int \frac{1-x}{\sqrt{8+2x-x^2}}\,dx$

Put $8 + 2x - x^2 = y^2$

or, $(2 - 2x)\,dx = 2y\,dy.$

or, $(1 - x)dx = y\,dy$

$\therefore \int \frac{(1-x)}{\sqrt{8+2x-x^2}} = \int \frac{y\,dy}{y}$

$= \int dy$

$= y + C$

$= (8 + 2x - x^2)^{\frac{1}{2}} + C$

$\therefore \int \frac{(1-x)}{\sqrt{8+2x-x^2}}\,dx = \sqrt{(8+2x-x^2)} + C$

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Example 13. Evaluate: $\int \frac{1}{4+5\sin x}\,dx$.

Solution:

Here, $I = \int \frac{1}{4+5\sin x}\,dx$

$$= \int \frac{1}{4\left(\sin^2\left(\frac{x}{2}\right)+\cos^2\left(\frac{x}{2}\right)\right)+5.2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)} \times \frac{\sec^2\left(\frac{x}{2}\right)}{\sec^2\left(\frac{x}{2}\right)}\,dx$$

$$= \int \frac{\sec^2\left(\frac{x}{2}\right)}{4\tan^2\left(\frac{x}{2}\right)+10\tan\left(\frac{x}{2}\right)+4}\,dx$$

Now, put $y = \tan\left(\frac{x}{2}\right)$

or, $\frac{dy}{dx} = \sec^2\left(\frac{x}{2}\right)\cdot\frac{1}{2}$

or, $\sec^2\left(\frac{x}{2}\right)\cdot dx = 2dy$

$\therefore I = \int \frac{2}{4y^2+10y+4}\,dy = \frac{2}{4}\int \frac{1}{y^2+\frac{5y}{2}+1}\,dy$

$$= \frac{1}{2}\frac{1}{y^2+2y\frac{5}{4}+\left(\frac{5}{4}\right)^2-\left(\frac{5}{4}\right)^2+1}\,dy$$

$$= = \frac{1}{2}\int \frac{1}{\left(y+\frac{5}{4}\right)^2-\frac{9}{16}}\,dy$$

$$= \frac{1}{2}\int \frac{1}{\left(y+\frac{5}{4}\right)^2-\left(\frac{3}{4}\right)^2}\,dy$$

$$= \frac{1}{2}\times\frac{1}{2\left(\frac{3}{4}\right)}\times\log\left[\frac{\left(y+\frac{5}{4}-\frac{3}{4}\right)}{\left(y+\frac{5}{4}+\frac{3}{4}\right)}\right]+C$$

$$= \frac{1}{3}\log\left(\frac{4y+2}{4y+8}\right)+C = \frac{1}{3}\log\left[\frac{2\tan\left(\frac{x}{2}\right)+1}{2\tan\left(\frac{x}{2}\right)+4}\right]+C$$

$\therefore \int \frac{1}{4+5\sin x}\,dx = \frac{1}{3}\log\left[\frac{2\tan\left(\frac{x}{2}\right)+1}{2\tan\left(\frac{x}{2}\right)+4}\right]+C.$

Example 14. Evaluate: $\int \frac{1}{4\sin x+3\cos x+13}\,dx$.

Solution:

Here, $I = \int \frac{1}{4\sin x+3\cos x+13}\,dx$

$$= \int \frac{1}{4.\left(2\sin\frac{x}{2}\cos\frac{x}{2}\right)+3\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)+13\left(\cos^2\frac{x}{2}+\sin^2\frac{x}{2}\right)} \times \frac{\sec^2\frac{x}{2}}{\sec^2\frac{x}{2}}\,dx$$

$$= \int \frac{\sec^2\frac{x}{2}}{8\tan\frac{x}{2}+10\tan^2\frac{x}{2}+16}\,dx$$

Now, put $y = \tan\frac{x}{2}$.

or, $2dy = \sec^2\frac{x}{2}\,dx$ then

$$\int \frac{1}{4\sin x+3\cos x+13}\,dx = \int \frac{2}{8y+10y^2+16}\,dy$$

$$= \frac{2}{10}\int \frac{1}{y^2+\frac{4y}{5}+\frac{8}{5}}\,dy$$

$$= \frac{1}{5}\frac{1}{y^2+2y\frac{2}{5}+\left(\frac{2}{5}\right)^2-\left(\frac{2}{5}\right)^2+\frac{8}{5}}\,dy$$

$$= \frac{1}{5}\int \frac{1}{\left(y+\frac{2}{5}\right)^2-\left(\frac{6}{5}\right)^2}\,dy$$

$$= \frac{1}{5}\left[\frac{1}{2.6}\times 5\log\left|\frac{y+\frac{2}{5}-\frac{6}{5}}{y+\frac{2}{5}+\frac{6}{5}}\right|\right]+C$$

$$= \frac{1}{12}\log\left|\frac{5y-4}{5y+8}\right|+C$$

$$= \frac{1}{12}\log\left|\frac{5\tan\left(\frac{x}{2}\right)-4}{5\tan\left(\frac{x}{2}\right)+8}\right|+C$$

$\therefore \int \frac{1}{4\sin x+3\cos x+13}\,dx = \frac{1}{12}\log\left|\frac{5\tan\left(\frac{x}{2}\right)-4}{5\tan\left(\frac{x}{2}\right)+8}\right|+C.$

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Example 15. Integrate $\int \frac{x + \sin x}{1 + \cos x} dx$.

Solution:

Let $I = \int \left(\frac{x + \sin x}{1 + \cos x}\right) dx$

$= \int \frac{x + \sin x}{2\cos^2\left(\frac{x}{2}\right)} dx$

$= \int \left(\frac{x}{2}\sec^2\left(\frac{x}{2}\right) + \tan\left(\frac{x}{2}\right)\right) dx$

$= \frac{1}{2}\int x \sec^2\left(\frac{x}{2}\right) dx + \int \tan\left(\frac{x}{2}\right) dx$

$= \frac{1}{2}\left[x \int \sec^2\left(\frac{x}{2}\right) dx - \int \left(\frac{dx}{dx}\int \sec^2\left(\frac{x}{2}\right) dx\right) dx\right] + \int \tan\left(\frac{x}{2}\right) dx$

$= \frac{1}{2} x\, 2\tan\left(\frac{x}{2}\right) - 2\int \tan\left(\frac{x}{2}\right) \cdot \left(\frac{1}{2}\right) dx + \int \tan\left(\frac{x}{2}\right) dx + C$

$= x \tan\left(\frac{x}{2}\right) + C$

$\therefore \int \frac{x + \sin x}{1 + \cos x} dx = x \tan\left(\frac{x}{2}\right) + C.$

Example 16. Evaluate: $\int \tan^{-1}\sqrt{\frac{1 - x}{1 + x}}\, dx$.

Solution:

Let, $I = \int \tan^{-1}\sqrt{\frac{1 - x}{1 + x}}\, dx$

Let $x = \cos\theta$

$\therefore dx = -\sin\theta\, d\theta$

Then, $I = \int \tan^{-1}\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} (-\sin\theta)\, d\theta$

$= \int \tan^{-1}\left[\sqrt{\frac{2\sin^2\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}}\right] (-\sin\theta)\, d\theta$

$= \int \tan^{-1}\left[\sqrt{\tan^2\frac{\theta}{2}}\right] (-\sin\theta)\, d\theta$

$= \int \tan^{-1}\tan\left(\frac{\theta}{2}\right) \times (-\sin\theta)\, d\theta$

$= \int -\frac{\theta}{2} \sin\theta\, d\theta$

$= -\frac{1}{2}\int \theta \sin\theta\, d\theta$

$= -\frac{1}{2}\left[\theta(-\cos\theta) - \int (-\cos\theta)\, d\theta\right] + C$

$= \frac{1}{2}\theta\cos\theta - \frac{1}{2}\sin\theta + C$

$= \frac{x\cos^{-1}x}{2} - \frac{1}{2}\sqrt{1 - x^2} + C$

$\therefore \int \tan^{-1}\sqrt{\frac{1 - x}{1 + x}}\, dx = \frac{x\cos^{-1}x}{2} - \frac{1}{2}\sqrt{1 - x^2} + C.$

Some standard integral formulae

A. i. $\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + C$

ii. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$

iii. $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$

iv. $\int \sqrt{x^2 - a^2}\, dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left(x + \sqrt{x^2 - a^2}\right) + C$

v. $\int \sqrt{x^2 + a^2}\, dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\log\left(x + \sqrt{x^2 + a^2}\right) + C$

B. **Partial Fraction Method:** The general rules for the partial fractions are as follows.

i. $\int \left(\frac{f(x)}{(ax + b)(cx + d)}\right) dx = \int \left[\frac{A}{(ax + b)} + \frac{B}{(cx + d)}\right] dx$

ii. $\int \left(\frac{f(x)}{(ax + b)(cx + d)^2}\right) dx = \int \left[\frac{A}{ax + b} + \frac{B}{cx + d} + \frac{C}{(cx + d)^2}\right] dx$

iii. $\int \left(\frac{f(x)}{(ax + b)(cx^2 + dx + e)}\right) dx = \int \left[\frac{A}{(ax + b)} + \frac{Cx + D}{(cx^2 + dx + e)}\right] dx$

C. If u is polynomial function of n degree and v is any integrable function, then we can write repeated integral as a single formula as,

$\int uv\, dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + u''''v_5 + \ldots$

where i, ii, iii, ... represent order of derivative and suffix 1, 2, 3, 4, 5 represent the order of integration.

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Example 17. Evaluate $\int e^x(\cos x - \sin x)\,dx$.

Solution:

Let, $I = \int e^x(\cos x - \sin x)\,dx$.

Here $f(x) = \cos x$ and $f'(x) = -\sin x$.

Applying $\int e^x[f(x) + f'(x)]\,dx = e^x f(x) + C$

$\therefore \quad I = e^x \cos x + C$.

Example 18. Evaluate $\int\sqrt{4x^2 - 4x + 5}\,dx$.

Solution:

Let, $I = \int\sqrt{4x^2 - 4x + 5}\,dx$

$$= \int\sqrt{4\left(x^2 - x + \frac{5}{4}\right)}\,dx$$

$$= 2\int\sqrt{\left(x^2 - x + \frac{1}{4} + \frac{5}{4} - \frac{1}{4}\right)}\,dx$$

$$= 2\int\sqrt{\left[\left(x - \frac{1}{2}\right)^2 + 1\right]}\,dx$$

$$= 2\left[\frac{\left(x - \frac{1}{2}\right)}{2}\sqrt{\left(x - \frac{1}{2}\right)^2 + 1} + \frac{1}{2}\log\left|\left(x - \frac{1}{2}\right) + \sqrt{\left(x - \frac{1}{2}\right)^2 + 1}\right|\right] + C$$

$$= \left(\frac{2x - 1}{2}\right)\sqrt{x^2 - x + \frac{5}{4}} + \log\left|\left(\frac{2x - 1}{2}\right) + \sqrt{x^2 - x + \frac{5}{4}}\right| + C$$

$$= \frac{1}{4}[(2x - 1)\sqrt{4x^2 - 4x + 5}] + \log\left|\frac{(2x - 1) + \sqrt{4x^2 - 4x + 5}}{2}\right| + C.$$

Example 19. Evaluate $\int\sqrt{2ax - x^2}\,dx$.

Solution:

Let, $I = \int\sqrt{2ax - x^2}\,dx$

$$= \int\sqrt{-(x^2 - 2ax)}\,dx$$

$$= \int\sqrt{-(x^2 - 2ax + a^2) + a^2}\,dx$$

$$= \int\sqrt{a^2 - (x - a)^2}\,dx$$

$$= \frac{x - a}{2}\sqrt{a^2 - (x - a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x - a}{a}\right) + C$$

$$= \frac{x - a}{2}\sqrt{2ax - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x - a}{a}\right) + C$$

$\therefore \quad \int\sqrt{ax - x^2}\,dx = \frac{x - a}{2}\sqrt{2ax - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x - a}{a}\right) + C.$

Example 20. Evaluate $\int x^5 \sin x\,dx$.

Solution:

Let, $I = \int x^5 \sin x\,dx$

$= -x^5\cos x + 5x^4\sin x + 20x^3\cos x - 60x^2\sin x - 120x\cos x + 120\sin x + C$ [successive integration]

$= \cos x(-x^5 + 20x^3 - 120x) + \sin x(5x^4 - 6x^2 + 120) + C$.

Example 21. Evaluate $\int\frac{x + 12}{x^2 - 13x + 42}\,dx$.

Solution:

Here, $\frac{x + 12}{x^2 - 13x + 42} = \frac{x + 12}{(x - 7)(x - 6)}$

Let $\frac{x + 12}{(x - 7)(x - 6)} = \frac{A}{x - 6} + \frac{B}{x - 7}$

or, $x + 12 = A(x - 7) + B(x - 6)$

or, $x + 12 = (A + B)x + (-7A - 6B)$

Equating the coefficients of x and the constant terms

$A + B = 1$ and $-7A - 6B = 12$

Solving we get $A = -18$, $B = 19$

$\therefore \quad \int\frac{x + 12}{x^2 - 13x + 42}\,dx = \int\left(\frac{-18}{x - 6} + \frac{19}{x - 7}\right)dx$

$= -18\log|x - 6| + 19\log|x - 7| + C$.

Example 22. Evaluate $\int\frac{x^2 - 4}{(x^2 + 1)(x^2 + 3)}\,dx$.

Solution:

Suppose, $x^2 = y$

We have, $\frac{x^2 - 4}{(x^2 + 1)(x^2 + 3)} = \frac{y - 4}{(y + 1)(y + 3)} = \frac{A}{y + 1} + \frac{B}{y + 3}$

$\therefore \quad y - 4 = A(y + 3) + B(y + 1)$ ....(i)

Taking $y = -3$ in equation (i)

$-7 = -2B$

$\therefore \quad B = \frac{7}{2}$

Again taking $y = -1$ in equation (i)

$-5 = 2A$

$\therefore \quad A = \frac{-5}{2}$

Then $\int\frac{x^2 - 4}{(x^2 + 1)(x^2 + 3)}\,dx = -\frac{5}{2}\int\frac{1}{x^2 + 1}\,dx + \frac{7}{2}\int\frac{1}{x^2 + 3}\,dx$

$= -\frac{5}{2}\tan^{-1}x + \frac{7}{2}\cdot\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3}} + C$

$\therefore \quad \int\frac{x^2 - 4}{(x^2 + 1)(x^2 + 3)}\,dx = -\frac{5}{2}\tan^{-1}x + \frac{7}{2}\cdot\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3}} + C.$

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Example 23. Evaluate $\int \frac{x^2+8}{x^2-5x+6}\,dx$.

Solution:

Here, $I = \int \frac{x^2+8}{x^2-5x+6}\,dx = \int \frac{x^2+8}{(x-2)(x-3)}\,dx$.

Since numerator and denominator have the same degree

By division, $I = \int\left(1 + \frac{5x+2}{(x-2)(x-3)}\right)dx$

Now, for $\frac{5x+2}{(x-2)(x-3)} = \frac{A}{(x-2)} + \frac{B}{(x-3)}$

or, $5x + 2 = A(x-3) + B(x-2)$

or, $5x + 2 = (A+B)x + (-3A-2B)$

Comparing the coefficients of x^1 and x^0

$5 = A + B$ and $-3A - 2B = 2$

Solving we get $A = -12$ and $B = 17$

$\therefore \quad I = \int dx + \int\left(\frac{-12}{x-2} + \frac{17}{x-3}\right)dx$

$= x - 12\log|x-2| + 17\log|x-3| + C.$

Example 24. Evaluate $\int \frac{2\sin x + 3\cos x}{3\sin x + 4\cos x}\,dx$.

Solution:

Here, $2\sin x + 3\cos x = A.d(3\sin x + 4\cos x) + B(3\sin x + 4\cos x)$

$= A(3\cos x - 4\sin x) + B(3\sin x + 4\cos x)$

$= (-4A + 3B)\sin x + (3A + 4B)\cos x$

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Comparing the coefficients of $\sin x$ and $\cos x$, we get

$-A + 3B = 2 \qquad 3A + 4B = 3$

On solving, $A = \frac{1}{25}$ and $B = \frac{18}{25}$

Then (i) becomes

$I = \frac{1}{25}\int \frac{d(3\sin x + 4\cos x)}{3\sin x + 4\cos x} + \frac{18}{25}\int \frac{2\sin x + 4\cos x}{3\sin x + 4\cos x}\,dx$

$= \frac{1}{25}\log(3\sin x + 4\cos x) + \frac{18}{25}\int dx$

$= \frac{1}{25}\log(3\sin x + 4\cos x) + \frac{18}{25}x + C.$

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Exercise 3.1

Integrate the following integrals

1. i. $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}}\, dx$ ii. $\int \frac{1}{\sqrt{2ax - x^2}}\, dx$

   iii. $\int \frac{x\, e^x}{(x+1)^2}\, dx$ iv. $\int \frac{x^2 + x + 1}{\sqrt{1 - x^2}}\, dx$

   v. $\int \frac{\cos x}{\sqrt{2\sin^2 x + 3\sin x + 4}}\, dx$ vi. $\int \frac{1}{x\,[10 + 7\log x + (\log x)^2]}\, dx$

   vii. $\int \frac{1}{\sqrt{x+a} + \sqrt{x+b}}\, dx$ viii. $\int \frac{3e^{2x} + 3e^{4x}}{e^x + e^{-x}}\, dx$

2. i. $\int \frac{1}{\sqrt{x^2 - 2x + 5}}\, dx$ ii. $\int \frac{\sqrt{x}}{1+x}\, dx$

   iii. $\int \frac{3x + 7}{2x^2 + 3x - 2}\, dx$ iv. $\int \frac{1}{(2x+1)\sqrt{x^2 + 2x + 2}}\, dx$

   v. $\int \frac{x}{\sqrt{x^2 + 4x + 5}}\, dx$

3. i. $\int \frac{1}{2 - 3\sin 2x}\, dx$ ii. $\int \frac{1}{3\sin x + 4\cos x}\, dx$

   iii. $\int \frac{1}{4\sin^2 x + 5\cos^2 x}\, dx$

4. i. $\int \frac{x\, e^x}{(x+1)^2}\, dx$ ii. $\int e^x (\tan x - \log \cos x)\, dx$

   iii. $\int x^5 e^x\, dx$

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5. i. $\int \frac{x}{(x-3)(x+1)}\, dx$ ii. $\int \frac{x+5}{(x+1)(x+2)^2}\, dx$

   iii. $\int \frac{x^3}{(x-2)(x-3)}\, dx$ iv. $\int \frac{2x}{(x^2+1)(x^2+3)}\, dx$

Answer

1. i. $\frac{1}{\sqrt{1+x^2}}(x - \tan^{-1} x) + C$ ii. $\sin^{-1}\left(\frac{x-a}{a}\right) + C$ iii. $\frac{e^x}{x+1} + C$

   iv. $\frac{3}{2}\sin^{-1} x - \frac{1}{2}(x+2)\sqrt{1-x^2} + C$ v. $\frac{1}{\sqrt{2}}\sinh^{-1}\left(\frac{4\sin x + 3}{\sqrt{23}}\right) + C$ vi. $\frac{1}{3}\log\left|\frac{\log x + 2}{\log x + 5}\right| + C$

   vii. $\frac{2}{3(a-b)}[(x+a)^{3/2} - (x+b)^{3/2}] + C$ viii. $e^{3x} + C$

2. i. $\log[(x-1) + \sqrt{x^2 - 2x + 5}] + C$ ii. $2[\sqrt{x} - \tan^{-1}\sqrt{x}] + C$

   iii. $\frac{3}{4}\log(2x^2 + 3x - 2) + \frac{19}{20}\log\left(\frac{2x-1}{2(x+2)}\right) + c$ iv. $-\frac{1}{\sqrt{5}}\log\left[\frac{(2x+6) + \sqrt{20x^2 + 40x + 40}}{2x+1}\right] + C$

   v. $\sqrt{x^2 + 4x + 5} - 2\log[(x+2) + \sqrt{x^2 + 4x + 5}] + C$

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3. i. $\frac{1}{2\sqrt{5}}\log\left[\frac{2\tan x-(3+\sqrt{5})}{2\tan x-(3-\sqrt{5})}\right]+C$ ii. $\frac{1}{5}\log\left[\tan\left(\frac{x+\tan^{-1}\frac{4}{3}}{2}\right)\right]+C$

iii. $\frac{1}{2\sqrt{5}}\tan^{-1}\left[\frac{2}{\sqrt{5}}\tan x\right]+C$

4. i. $\frac{e^x}{x+1}+C$ ii. $e^x\log\sec x+C$

iii. $(x^5-5x^4+20x^3-60x^2+120x-120)e^x+C$

5. i. $\frac{3}{4}\log|x-3|+\frac{1}{4}\log|x+1|+C$ ii. $4\log\left|\frac{x+1}{x+2}\right|+\frac{3}{x+2}+C$

iii. $\frac{x^2}{2}+5x-8\log|x-2|+27\log|x-3|+C$ iv. $\frac{1}{2}\log\left(\frac{x^2+1}{x^2+3}\right)+C$

3.2 Definite integral

While evaluating area under a curve, we need to calculate a limit of a sum. The limits of this type also arises in various other situations such as in finding length of curves, volumes of solids, centre of mass, force due to water pressure, work done as well as in various other contexts. This type of limit of sum is popularly known as the definite integral and is considered as a fundamental tool in various physical situations. In this section, we discuss about definite integrals together with its properties.

Joseph Fourier

3.2.1 Physical Interpretation of definite integral

Just as the notation $\frac{dy}{dx}$ reminds us that the derivative is the limit of a quotient of differences, the notation $\int_a^b f(x)\,dx$ reminds us that an integral is a limit of a sum. If $f(x)$ is a rate of change of a physical quantity, then the value of the integral $\int_a^b f(x)\,dx$ gives the total change in quantity between $x = a$ and $x = b$.

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We next discuss the fundamental theorem of calculus which makes it possible to compute areas without computing the integrals as a limit of sums.

3.2.2 Fundamental theorem of integral calculus

Suppose f is a continuous function defined on $[a, b]$.

i. If $F(x)=\int_a^x f(t)\,d(t)$, then $F'(x) = f(x)$.

ii. $\int_a^b f(x)\,dx = F(b) - F(a)$, where $F' = f$.

The fundamental theorem made the problems of determining area very easy. It also provides relation between differentiation and integration which are inverse process of each other. We also have the following result from fundamental theorem:

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$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

Which is the widely known net change theorem.

The definite integrals as a limit of a sum can be used for some instances as below:

i. Let $v(x)$ denote volume of water in a tank at a time x, then the derivative $v'(x)$ gives the rate at which water flows into the tank at a time x, and

$$\int_a^b x'(x)\,dx = v(b) - v(a)$$

gives the amount of water in the tank from time $x = a$ to $x = b$.

ii. Let $a(x)$ denote the acceleration of an object at time x, then

$$\int_{x_1}^{x_2} a(x)\,dx = \int_{x_1}^{x_2} v'(x)\,dx = v(x_2) - v(x_1)$$

gives the net change in velocity from time x_1 to time x_2.

Similarly, $\int_{x_1}^{x_2} v(x)\,dx = \int_{x_1}^{x_2} s'(x)\,dx = s(x_2) - s(x_1)$ gives the displacement of the particle from time x_1 to time x_2.

iii. Let P denotes a population. Then, $\frac{dP}{dx}$ gives the rate of growth of P. Then

$\int_{x_1}^{x_2} \frac{dP}{dx}\,dx = P(x_2) - P(x_1)$ is the net change in population from time x_1 to time x_2.

3.2.3 Properties of definite integral

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Various properties of definite integrals are listed as:

1. $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$
2. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
3. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \quad (a < c < b)$
4. $\int_0^a f(x)\,dx = \int_0^a f(a - x)\,dx$

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5. $\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{when } f(x) \text{ is even function} \\ 0 & \text{when } f(x) \text{ is odd function} \end{cases}$

Note that $f(x)$ is even if $f(-x) = f(x)$ and it is odd if $f(-x) = -f(x)$.

6. $\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{when } f(2a-x) = f(x) \\ 0 & \text{when } f(2a-x) = -f(x) \end{cases}$

7. $\int_0^{na} f(x)\,dx = n\int_0^a f(x)\,dx$ when $f(a+x) = f(x)$.

We now give the proof of some selected properties:

4. $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Proof:

R.H.S. $= \int_0^a f(a-x)\,dx$

Put $t = a - x$

Then, differentiating with respect to x, we get

$\frac{dt}{dx} = 0 - 1$

or, $dx = -dt$

When $x = 0$ then $t = a$

When $x = a$ then $t = 0$.

So, $\int_0^a f(a-x)\,dx = \int_a^0 f(t)(-dt) = \int_0^a f(t)\,dt = \int_0^a f(x)\,dx$ $\left[\because \int_a^b f(t)\,dt = \int_a^b f(x)\,dx\right]$

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Thus, $\int_0^a f(a-x)\,dx = \int_0^a f(x)\,dx$.

6. $\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{when } f(2a-x) = f(x) \\ 0 & \text{when } f(2a-x) = -f(x) \end{cases}$

Proof:

We have, $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_a^{2a} f(x)\,dx$ .... (i)

Put $x = 2a - t$ in the second integral.

Then, $dx = -dt$

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When $x = a$, then $t = a$ and when $x = 2a$, $t = 0$.

$$\therefore \int_a^{2a} f(x)\,dx = -\int_a^0 f(2a-t)\,dt$$

$$= \int_0^a f(2a-t)\,dt = \int_0^a f(2a-x)dx$$

Equation (i) becomes,

$$\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$$

$$= 2\int_0^a f(x)dx, \quad \text{if } f(2a-x) = f(x)$$

and $$= 0, \quad \text{if } f(2a-x) = -f(x)$$

Proofs of the rest of the properties are left as an exercise.

We now discuss some examples of application of properties in the evaluation of definite integrals.

Example 25. Show that $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx = \frac{\pi}{4}$.

Solution:

Let $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ .... (i)

By applying property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we have

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$$I = \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}dx$$

$$\therefore I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \text{.... (ii)}$$

Then, adding equation (i) and (ii),

$$I + I = \int_0^{\pi/2} \frac{\sqrt{\sin x}+\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$$

or, $$2I = \int_0^{\pi/2} dx$$

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or, $2I = [x]_0^{\pi/2}$

or, $2I = \frac{\pi}{2} - 0$

or, $I = \frac{\pi}{4}$

$\therefore \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$.

Example 26. Prove that $\int_0^{\pi/2} \frac{dx}{1 + \sqrt{\tan x}} = \frac{\pi}{4}$.

Solution:

Let $I = \int_0^{\pi/2} \frac{dx}{1 + \sqrt{\tan x}}$

$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$ .... (i)

By applying property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we have

$I = \int_0^{\pi/2} \frac{\sqrt{\cos\left(\frac{\pi}{2} - x\right)}}{\sqrt{\cos\left(\frac{\pi}{2} - x\right)} + \sqrt{\sin\left(\frac{\pi}{2} - x\right)}} dx$

$= \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ .... (ii)

Then, adding equation (i) and (ii)

$I + I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

or, $2I = \int_0^{\pi/2} dx$

or, $2I = [x]_0^{\pi/2}$

or, $2I = \frac{\pi}{2} - 0$

or, $I = \frac{\pi}{4}$

$\therefore \int_0^{\pi/2} \frac{dx}{1 + \sqrt{\tan x}} dx = \frac{\pi}{4}$.

Example 27. Evaluate $\int_0^{\pi/2} \frac{x\,dx}{\sin x + \cos x}$.

Solution:

Let $I = \int_0^{\pi/2} \frac{x\,dx}{\sin x + \cos x}$ .... (i)

By applying property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we have

$I = \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$

$= \int_0^{\pi/2} \frac{\frac{\pi}{2} - x}{\cos x + \sin x} dx$ .... (ii)

Adding (i) and (ii), we get

$I + I = \int_0^{\pi/2} \frac{x + \frac{\pi}{2} - x}{\sin x + \cos x} dx$

or, $2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{\sqrt{2} \cdot \frac{1}{\sqrt{2}} (\sin x + \cos x)} dx$

or, $I = \frac{\pi}{4\sqrt{2}} \int_0^{\pi/2} \frac{1}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x} dx$

$= \frac{\pi}{4\sqrt{2}} \int_0^{\pi/2} \frac{1}{\sin x \cdot \cos\frac{\pi}{4} + \cos x \cdot \sin\frac{\pi}{4}} dx$

$= \frac{\pi}{4\sqrt{2}} \int_0^{\pi/2} \frac{1}{\sin\left(\frac{\pi}{4} + x\right)} dx$

$= \frac{\pi}{4\sqrt{2}} \int_0^{\pi/2} \operatorname{cosec}\left(\frac{\pi}{4} + x\right) dx$

$= \frac{\pi}{4\sqrt{2}} \left[\log\left\{\operatorname{cosec}\left(\frac{\pi}{4} + x\right) - \cot\left(\frac{\pi}{4} + x\right)\right\} \right]_0^{\pi/2}$ $\left[\int \operatorname{cosec} x\,dx = \log(\operatorname{cosec} x - \cot x) + C \right]$

$= \frac{\pi}{4\sqrt{2}} \left[\log\left\{\operatorname{cosec}\left(\frac{\pi}{4} + \frac{\pi}{2}\right) - \cot\left(\frac{\pi}{4} + \frac{\pi}{2}\right)\right\} - \log\left\{\operatorname{cosec}\left(\frac{\pi}{4} + 0\right) - \cot\left(\frac{\pi}{4} + 0\right)\right\} \right]$

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$$= \frac{\pi}{4\sqrt{2}}\left[\log\left(\operatorname{cosec}\frac{3\pi}{4} - \cot\frac{3\pi}{4}\right) - \log\left(\operatorname{cosec}\frac{\pi}{4} - \cot\frac{\pi}{4}\right)\right]$$

$$= \frac{\pi}{4\sqrt{2}}\left\{\log\left(\sqrt{2} - (-1)\right) - \log\left(\sqrt{2} - 1\right)\right\}$$

$$= \frac{\pi}{4\sqrt{2}}\left\{\log\left(\sqrt{2}+1\right) - \log\left(\sqrt{2} - 1\right)\right\}$$

$$= \frac{\pi}{4\sqrt{2}}\log\frac{\sqrt{2}+1}{\sqrt{2}-1} \qquad \left[\because \log x - \log y = \log\frac{x}{y}\right]$$

$$= \frac{\pi}{4\sqrt{2}}\log\frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \frac{\pi}{4\sqrt{2}}\log\frac{(\sqrt{2}+1)^2}{2-1}$$

$$= \frac{\pi}{4\sqrt{2}}\log(\sqrt{2}+1)^2$$

$$= \frac{\pi}{4\sqrt{2}}2\log(\sqrt{2}+1) \qquad [\because \log x^p = p\log x]$$

$$= \frac{\pi}{2\sqrt{2}}\log(\sqrt{2}+1)$$

$$\therefore \int_0^{\pi/2} \frac{x}{\sin x + \cos x}\,dx = \frac{\pi}{2\sqrt{2}}\log(\sqrt{2}+1).$$

Example 28. Show that $\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\cos x\,dx = \frac{\pi}{2}\log\left(\frac{1}{2}\right)$.

Solution:

Let $I = \int_0^{\pi/2} \log\sin x\,dx$ ····(i)

By applying property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, we have

or, $I = \int_0^{\pi/2} \log\sin\left(\frac{\pi}{2} - x\right)dx$

$\therefore I = \int_0^{\pi/2} \log\cos x\,dx$ ····(ii)

Adding equation (i) and (ii), we get

$$2I = \int_0^{\pi/2} \log(\sin x\cos x)\,dx$$

$$= \int_0^{\pi/2} \log\left(\frac{2\sin x\cos x}{2}\right)dx$$

$$= \int_0^{\pi/2} \log\frac{\sin 2x}{2}\,dx$$

$$= \int_0^{\pi/2} (\log\sin 2x - \log 2)\,dx \qquad \left[\because \log\frac{a}{b} = \log a - \log b\right]$$

$$= \int_0^{\pi/2} \log(\sin 2x)\,dx - \int_0^{\pi/2} \log 2\,dx$$

Put $t = 2x$.

Then, $\frac{dt}{dx} = 2.1$

or, $dx = \frac{dt}{2}$

When $x = 0$, then $t = 0$ and

When $x = \frac{\pi}{2}$, then $t = \pi$

So, $2I = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt - \log 2\int_0^{\pi/2} dx$

or, $2I = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt - \log 2\,[x]_0^{\pi/2}$

or, $2I = \frac{1}{2}.2\int_0^{\pi/2} \log\sin t\,dt - \log 2\left[\frac{\pi}{2} - 0\right]$ $\left[\because \int_0^{2a} f(x)dx = n\int_0^a f(x)dx\right]$

or, $2I = \int_0^{\pi/2} \log\sin x\,dx - \frac{\pi}{2}\log 2$

or, $2I = I - \frac{\pi}{2}\log 2$

or, $I = -\frac{\pi}{2}\log 2$

or, $I = \frac{\pi}{2}\log 2^{-1}$

$\therefore I = \frac{\pi}{2}\log\left(\frac{1}{2}\right)$

Hence, $\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\cos x\,dx$

$= \frac{\pi}{2}\log\left(\frac{1}{2}\right)$.

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Example 29. Show that $\int_0^{\pi/2} \log(\tan x)\, dx = 0$.

Solution:

Let $I = \int_0^{\pi/2} \log(\tan x)\, dx$ .... (i)

By applying property $\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$, we get

$$I = \int_0^{\pi/2} \log\left[\tan\left(\frac{\pi}{2} - x\right)\right] dx$$

$$= \int_0^{\pi/2} \log\cot x\, dx$$

$$= \int_0^{\pi/2} \log\left(\frac{1}{\tan x}\right) dx$$

$$= \int_0^{\pi/2} [\log 1 - \log\tan x]\, dx$$

$$= \int_0^{\pi/2} [0 - \log\tan x]\, dx$$

$$I = -\int_0^{\pi/2} \log\tan x\, dx$$

or, $I = -I$ [$\because$ from (i)]

or, $I + I = 0$

or, $2I = 0$

$\therefore I = 0$

$\therefore \int_0^{\pi/2} \log(\tan x)\, dx = 0$.

Example 30. Show that $\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8}\log 2$.

Solution:

Let $I = \int_0^1 \frac{\log(1+x)}{1+x^2} dx$ .... (i)

Put $x = \tan\theta$.

Then, $dx = \sec^2\theta\, d\theta$

When $x = 0$ then $\tan\theta = 0$

$\therefore \theta = 0$

and when $x = 1$ then $\tan\theta = 1$

or, $\tan\frac{\pi}{4} = \tan\theta$

$\therefore \theta = \frac{\pi}{4}$

From (i)

Then, $I = \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{1+\tan^2\theta} \sec^2\theta\, d\theta$

$$= \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{\sec^2\theta} \sec^2\theta\, d\theta$$

$$= \int_0^{\pi/4} \log(1+\tan\theta)\, d\theta$$ .... (i)

Using property $\int_0^a f(x)\, dx = \int_0^a f(a-x) dx$, we get

$$I = \int_0^{\pi/4} \log\left(1 + \tan\left(\frac{\pi}{4} - \theta\right)\right) d\theta$$

$$= \int_0^{\pi/4} \log\left(1 + \frac{\tan\frac{\pi}{4} - \tan\theta}{1 + \tan\frac{\pi}{4}\tan\theta}\right) d\theta$$

$$= \int_0^{\pi/4} \log\left(1 + \frac{1-\tan\theta}{1+\tan\theta}\right) d\theta$$

$$= \int_0^{\pi/4} \log\left(\frac{1+\tan\theta+1-\tan\theta}{1+\tan\theta}\right) d\theta$$

$$= \int_0^{\pi/4} \log\frac{2}{1+\tan\theta}\, d\theta$$

$$= \int_0^{\pi/4} [\log 2 - \log(1+\tan\theta)]\, d\theta$$

$$= \int_0^{\pi/4} \log 2\, d\theta - \int_0^{\pi/4} \log(1+\tan\theta)\, d\theta$$

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[∵ from (i)]

$\therefore \quad I = \log 2 \int_0^{\pi/4} 1 \, d\theta - I$

or, $I = \log 2 \, [\theta]_0^{\pi/4} - I$.

or, $I + I = \log 2 \left(\frac{\pi}{4} - 0\right)$

or, $2I = \frac{\pi}{4} \log 2$

$\therefore \quad I = \frac{\pi}{8} \log 2$

Thus, $\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log 2$.

Example 31. Show that $\int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx = \frac{\pi^2}{4}$.

Solution:

Let $I = \int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx$ .... (i)

By applying property $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$, we have

$$I = \int_0^{\pi} \frac{(\pi - x) \tan(\pi - x)}{\sec(\pi - x) + \cos(\pi - x)} dx$$

or, $I = \int_0^{\pi} \frac{(\pi - x) \tan x}{\sec x + \cos x} dx$ .... (ii)

From equation (i) and (ii), we get

$$I + I = \int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx + \int_0^{\pi} \frac{(\pi - x) \tan x}{\sec x + \cos x} dx$$

or, $2I = \int_0^{\pi} \frac{\pi \tan x}{\sec x + \cos x} dx$

$$= \pi \int_0^{\pi} \frac{\tan x}{\sec x + \cos x} dx$$

$$= \pi \int_0^{\pi} \frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x} + \cos x} dx$$

$$= \pi \int_0^{\pi} \frac{\frac{\sin x}{\cos x}}{\frac{1 + \cos^2 x}{\cos x}} dx$$

$$= \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

Put $t = \cos x$.

Then $\frac{dt}{dx} = -\sin x$

or, $dt = -\sin x \, dx$

When $x = 0$ then $t = 1$

and when $x = \pi$ then $t = -1$

Then, $2I = \pi \int_1^{-1} \frac{-dt}{1+t^2}$

$$= -\pi \int_1^{-1} \frac{dt}{1+t^2}$$

$$= -\pi \, [\tan^{-1} t]_1^{-1}$$

$$= -\pi [\tan^{-1}(-1) - \tan^{-1}(1)]$$

$$= -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4}\right)$$

$$= -\pi \times \left(-\frac{2\pi}{4}\right)$$

or, $2I = \frac{\pi^2}{2}$

$\therefore \quad I = \frac{\pi^2}{4}$

Thus, $\int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx = \frac{\pi^2}{4}$.

Example 32. Prove that $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$.

Solution

Let $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$

$$= \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

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$$= \int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x}\, dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x}\, dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx$$

$$= \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\, dx - I$$

or, $2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\, dx$

Let $t = \cos x$

Differentiating with respect to x, we get

$\frac{dt}{dx} = -\sin x$

or, $-dt = \sin x dx$

When $x = 0$, then $t = \cos 0 = 1$

When $x = \pi$, then $t = \sin \pi = -1$

So, $2I = \pi \int_1^{-1} \frac{-1}{1+t^2}\, dt$

or, $2I = -\pi \int_1^{-1} \frac{1}{1+t^2}\, dt$

or, $2I = \pi \int_{-1}^{1} \frac{1}{1+t^2}\, dt$

or, $2I = \pi \left[\tan^{-1} t\right]_{-1}^{1}$

or, $2I = \pi \left[\tan^{-1} 1 - \tan^{-1}(-1)\right]$

or, $2I = \pi \left[\pi/4 - (-\pi/4)\right]$

or, $2I = \pi\left(\frac{\pi}{2}\right)$

or, $I = \frac{\pi^2}{4}$

$\therefore \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx = \frac{\pi^2}{4}$

Example 33. Show that: $\int_0^{\pi} \frac{x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2}\, dx = \frac{\pi^2 (a^2 + b^2)}{4a^3 b^3}$.

Solution:

Let $I = \int_0^{\pi} \frac{x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2}\, dx$

$$= \int_0^{\pi} \frac{(\pi - x)}{\{a^2 \sin^2 (\pi - x) + b^2 \cos^2 (\pi - x)\}^2}\, dx$$

$$= \int_0^{\pi} \frac{(\pi - x)}{(a^2 \sin^2 x + b^2 \cos^2 x)^2}\, dx$$

or, $I = \pi \int_0^{\pi} \frac{1}{(a^2 \sin^2 x + b^2 \cos^2 x)^2} dx - \int_0^{\pi} \frac{x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2}\, dx$

or, $2I = 2\pi \int_0^{\pi/2} \frac{1}{(a^2 \sin^2 x + b^2 \cos^2 x)^2}\, dx$ $\left[\because \int_0^{2a} f(x)\, dx = 2\int_0^{a} f(x)\, dx\right]$

or, $2I = 2\pi \int_0^{\pi/2} \frac{\sec^4 x}{(b^2 + a^2 \tan^2 x)^2}\, dx$ [$\because$ Dividing num. and deno. by $\cos^2 x$]

$$I = \pi \int_0^{\pi/2} \frac{\sec^2 x (1 + \tan^2 x)}{(b^2 + a^2 \tan^2 x)^2} dx$$

Put $a \tan x = b \tan \theta$.

Then, $a \sec^2 x dx = b \sec^2 \theta d\theta$

$$I = \pi \int_0^{\pi/2} \frac{\left(1 + \frac{b^2}{a^2} \tan^2 \theta\right) \frac{b}{a} \sec^2 \theta d\theta}{(b^2 + b^2 \tan^2 \theta)^2}$$

$$= \frac{\pi}{a^3 b^3} \int_0^{\pi/2} \frac{(a^2 + b^2 \tan^2 \theta) \sec^2 \theta d\theta}{\sec^4 \theta}$$

$$= \frac{\pi}{a^3 b^3} \int_0^{\pi/2} (a^2 \cos^2 \theta + b^2 \sin^2 \theta)\, d\theta$$

$$= \frac{\pi}{a^3 b^3} \left[\frac{a^2}{2} \int_0^{\pi/2} (1 + \cos 2\theta)\, d\theta + \frac{b^2}{2} \int_0^{\pi/2} (1 - \cos 2\theta)\, d\theta\right]$$

$$= \frac{\pi}{a^3 b^3} \left[\frac{a^2}{2}\left[0 + \frac{\sin 2\theta}{2}\right]_0^{\pi/2} + \frac{b^2}{2}\left[0 - \frac{\sin 2\theta}{2}\right]\right]_0^{\pi/2}$$

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$$= \frac{\pi}{a^2b^2}\left\{\frac{a^2}{2}\left(\frac{\pi}{2}+\frac{1}{2}, 0\right) + \frac{b^2}{2}\left(\frac{\pi}{2}-\frac{1}{2}, 0\right)\right\}$$

$$\therefore \quad I = \frac{\pi}{a^2b^2}\left(\frac{\pi}{4}\right)(a^2+b^2) = \frac{\pi^2(a^2+b^2)}{4a^2b^2}$$

Thus, $\int_0^{\pi} \frac{x\,dx}{(a^2\sin^2 x + b^2\cos^2 x)^2} = \frac{\pi(a^2+b^2)}{4a^2b^2}$.

Example 34. Show that $\int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx = \frac{\pi}{2}(\pi - 2)$

Solution:

Let $I = \int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx$

$$= \int_0^{\pi} \frac{x\sin x}{1+\sin x}\,dx \qquad \ldots\ldots (i)$$

Also, $I = \int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1+\sin(\pi - x)}\,dx = \int_0^{\pi} \frac{(\pi - x)\sin x}{1+\sin x}\,dx \qquad \ldots\ldots (ii)$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi} \frac{[x\sin x + (\pi - x)\sin x]}{(1+\sin x)}\,dx$$

$$= \int_0^{\pi} \frac{\pi\sin x}{1+\sin x}\,dx$$

$$\therefore \quad 2I = \pi\int_0^{\pi} \frac{\sin x + 1 - 1}{1+\sin x}\,dx$$

$$= \pi\int_0^{\pi} dx - \pi\int_0^{\pi} \frac{1}{1+\sin x}\,dx$$

$$= \pi[x]_0^{\pi} - \pi\int_0^{\pi} \frac{1}{\cos^2\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right) + 2\sin\left(\frac{x}{2}\right).\cos\left(\frac{x}{2}\right)}\,dx$$

$$= \pi^2 - \pi\int_0^{\pi} \frac{\sec^2 x/2}{(1+\tan x/2)^2}\,dx \qquad \ldots\ldots (iii)$$

Put $1 + \tan\left(\frac{x}{2}\right) = t$.

Then, $\sec^2\left(\frac{x}{2}\right)dx = 2dt$.

When $x = 0$, then $t = 1$.

When $x = \pi$, then $t = \infty$.

Hence, equation (iii) becomes

$$2I = \pi^2 - 2\pi\int_1^{\infty} \frac{1}{t^2}\,dt$$

$$= \pi^2 - 2\pi\left[-\frac{1}{t}\right]_1^{\infty}$$

$$= \pi^2 - 2\pi$$

or, $2I = \pi^2 - 2\pi$

$$\therefore \quad I = \frac{1}{2}(\pi^2 - 2\pi) = \frac{\pi}{2}(\pi - 2)$$

Thus, $\int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx = \frac{\pi}{2}(\pi - 2)$.

Example 35. Prove $\int_0^1 \cot^{-1}(1 - x + x^2)\,dx = \frac{\pi}{2} - \log 2$.

Solution:

Here, $I = \int_0^1 \cot^{-1}(1 - x + x^2)\,dx$

$$= \int_0^1 \tan^{-1}\left(\frac{1}{1 - x + x^2}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{1}{1 - x(1-x)}\right)dx$$

$$= \int_0^1 \tan^{-1}\frac{x + (1-x)}{1 - x(1-x)}\,dx$$

$$= \int_0^1 \tan^{-1} x\,dx + \int_0^1 \tan^{-1}(1-x)\,dx$$

$$= \int_0^1 \tan^{-1} x\,dx + \int_0^1 \tan^{-1}[(1-(1-x)]\,dx \quad \left[\because \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx\right]$$

$$= \int_0^1 \tan^{-1} x\,dx + \int_0^1 \tan^{-1} x\,dx$$

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$$= 2\int_0^1 \tan^{-1}x\,dx$$

$$= 2\left[\tan^{-1}x\int dx - \left(\frac{d\tan^{-1}x}{dx}\int dx\right)\right]_0^1 \quad \text{[integration by part]}$$

$$= 2\left[x\tan^{-1}x - \int\frac{1}{1+x^2}x\,dx\right]_0^1$$

$$= 2\left[x\tan^{-1}x - \frac{1}{2}\int\frac{2x}{1+x^2}dx\right]_0^1$$

$$= 2\left[x\tan^{-1}x - \frac{1}{2}\log(1+x^2)\right]_0^1$$

$$= 2\left\{\left[\tan^{-1}1 - \frac{1}{2}\log(1+1)\right] - \left[\tan^{-1}0 - \frac{1}{2}\log 1\right]\right\}$$

$$= 2\left[\tan^{-1}1 - \frac{1}{2}\log 2\right]$$

$$= 2\left[\frac{\pi}{4} - \frac{1}{2}\log 2\right]$$

$$= \frac{\pi}{2} - \log 2$$

$$\therefore \int_0^1 \cot^{-1}(1 - x + x^2)\,dx = \frac{\pi}{2} - \log 2.$$

Exercise 3.2

Prove the followings:

1. $\int_0^{\pi/2} \frac{\sin\theta}{\sin\theta + \cos\theta}\,d\theta = \frac{\pi}{4}$

2. $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{1 + \sqrt{\cot x}}\,dx = \frac{\pi}{4}$

3. $\int_0^a \frac{1}{x + \sqrt{a^2 - x^2}}\,dx = \frac{\pi}{4}$

4. $\int_0^a \frac{\sqrt{x}}{\sqrt{x} + \sqrt{a - x}}\,dx = \frac{a}{2}$

5. $\int_0^1 \frac{\log x}{\sqrt{1 - x^2}}\,dx = \frac{\pi}{2}\log\left(\frac{1}{2}\right)$

6. $\int_0^{\pi/4} \log(1 + \tan\theta)\,d\theta = \frac{\pi}{8}\log 2$

7. $\int_0^\infty \log\left(x + \frac{1}{x}\right)\frac{1}{1 + x^2}\,dx = \pi\log 2$

8. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x\cos x}\,dx = 0$

3.3 Improper integrals

In the definite integral $\int_a^b f(x)\,dx$ we consider that the function f is defined on the finite interval $[a, b]$. Moreover, we also assumed that f does not have an infinite discontinuity in $[a, b]$. In this section, we want to extend the notion of definite integral to the context when the interval $[a, b]$ is infinite and also we take into account the cases where $f(x)$ has infinite discontinuity at some points in $[a, b]$. The definite integrals so formed are called improper integrals. The most common improper integrals are:

i. $\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$ provided that $\int_a^t f(x)\,d(x)$ exists for all $t \geq a$.

ii. $\int_{-\infty}^a f(x)\,dx = \lim_{t\to-\infty}\int_t^a f(x)\,dx$, provided that $\int_t^a f(x)\,dx$ exists for all $t \leq a$.

iii. $\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx$

iv. If f is discontinuous at b, then $\int_a^b f(x)\,dx = \lim_{t\to b^-}\int_a^t f(x)\,dx$.

v. If f is discontinuous at a, then $\int_a^b f(x)\,dx = \lim_{t\to a^+}\int_t^b f(x)\,dx$.

vi. If f has discontinuity at some point $a < c < b$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \lim_{t\to c^-}\int_a^t f(x)\,dx + \lim_{t\to c^+}\int_t^b f(x)\,dx$$

In all of the above cases, the improper integrals are said to be convergent or divergent when the limit is finite and infinite.

We now discuss some examples:

Example 36. Evaluate $\int_2^\infty \frac{x}{x^2 - 1}\,dx$.

Solution:

$$\text{Here, } I = \int_2^\infty \frac{x}{x^2 - 1}\,dx$$

$$= \frac{1}{2}\int_2^\infty \frac{2x}{x^2 - 1}\,dx = \frac{1}{2}\lim_{t\to\infty}\int_2^t \frac{2x}{x^2 - 1}\,dx$$

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$$= \lim_{t\to\infty} \frac{1}{2}[\log(x^2-1)]_2^t$$

$$= \frac{1}{2}\lim_{t\to\infty}[\log(t^2-1) - \log 3].$$

Since $\lim_{t\to\infty} \log(t^2-3)$ does not exist, $\int_2^\infty \frac{x}{x^2-1}\,dx$ does not exist.

So the integral is divergent integral.

Example 37. Evaluate $\int_0^a \sqrt{\frac{a-x}{x}}\,dx$.

Solution:

Here, $I = \int_0^a \sqrt{\frac{a-x}{x}}\,dx$

Here the integrant $f(x)$ is undefined at $x = 0$

$$\therefore \quad I = \lim_{t\to 0^+} \int_t^a \sqrt{\frac{a-x}{x}}\,dx$$

Put $x = a\sin^2\theta$ then $dx = 2a\sin\theta\cos\theta d\theta$

If $x = t$ then $t = a\sin^2\theta$, or $\sin^2\theta = \frac{t}{a}$ i.e., $\theta = \sin^{-1}\sqrt{\frac{t}{a}}$ and

if $x = a$ then $a = a\sin^2\theta \Rightarrow \sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}$

$$\therefore \quad I = \lim_{t\to 0^+} \int_{\sin^{-1}\sqrt{\frac{t}{a}}}^{\pi/2} \sqrt{\frac{a-a\sin^2\theta}{a\sin^2\theta}}\,2a\sin\theta\cos\theta\,d\theta$$

$$= \lim_{t\to 0^+} \int_{\sin^{-1}\sqrt{\frac{t}{a}}}^{\pi/2} \frac{\cos\theta}{\sin\theta}\,2a\sin\theta\cos\theta\,d\theta$$

$$= \lim_{t\to 0^+} \int_{\sin^{-1}\sqrt{\frac{t}{a}}}^{\pi/2} 2a\cos^2\theta\,d\theta$$

$$= 2a\lim_{t\to 0^+} \int_{\sin^{-1}\sqrt{\frac{t}{a}}}^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= a\lim_{t\to 0^+}\left[\theta + \frac{\sin 2\theta}{2}\right]_{\sin^{-1}\sqrt{\frac{t}{a}}}^{\pi/2} = a\left[\frac{\pi}{2} + \frac{\sin 2\frac{\pi}{2}}{2}\right] = \frac{a\pi}{2}$$

$$\therefore \int_0^a \sqrt{\frac{a-x}{x}}\,dx = \frac{a\pi}{2}.$$

Example 38. Evaluate $\int_1^\infty \frac{\log x}{x^2}\,dx$.

Solution:

Here, $I = \int_1^\infty \frac{\log x}{x^2}\,dx$

Here by definition, $I = \lim_{b\to\infty} \int_1^b \frac{\log x}{x^2}\,dx$

Let $y = \log x$

Then, $\frac{dy}{dx} = \frac{1}{x}$

or, $dy = \frac{1}{x}\,dx$

When $x = 1$, $y = 0$

When $x = b$, $y = \log b$

Then, $I = \lim_{b\to\infty} \int_0^{\log b} \frac{y}{e^y}\,dy$

$$= \lim_{b\to\infty} \int_0^{\log b} ye^{-y}\,dy$$

$$= \lim_{b\to\infty} [-ye^{-y} - e^{-y}]_0^{\log b}$$

$$= \lim_{b\to\infty} [-\log b\, e^{-\log b} + e^{-\log b} + 1]$$

$$= 1$$

$$\therefore \int_1^\infty \frac{\log x}{x^2}\,dx = 1.$$

Example 39. Prove that $\int_0^\infty e^{-ax}\cos bx\,dx = \frac{a}{a^2+b^2}, a > 0$.

Solution:

Let, $I = \int e^{-ax}\cos bx\,dx = \frac{1}{b}e^{-ax}\sin bx + \frac{a}{b}\int e^{-ax}\cos bx\,dx$

$$= \frac{1}{b}e^{-ax}\sin bx - \frac{a}{b^2}e^{-ax}\cos bx - \frac{a^2}{b^2}\int e^{-ax}\cos bx\,dx$$

$$= \frac{1}{b}e^{-ax}\sin bx - \frac{a}{b^2}e^{-ax}\cos bx - \frac{a^2}{b^2}I$$

or, $\left(1 + \frac{a^2}{b^2}\right)I = e^{-ax}\left(\frac{\sin bx}{b} - \frac{a\cos bx}{b^2}\right)$

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or, $I = e^{-ax}\left(\frac{b\sin bx - a\cos bx}{a^2+b^2}\right)$

$\therefore \int e^{-ax}\cos bx\,dx = e^{-ax}\left(\frac{b\sin bx - a\cos bx}{a^2+b^2}\right)$

Now, $\int_0^\infty e^{-ax}\cos bx\,dx = \lim_{t\to\infty}\int_0^t e^{-ax}\cos bx\,dx$

$= \lim_{t\to\infty} e^{-ax}\left(\frac{b\sin bx - a\cos bx}{a^2+b^2}\right)_0^t$

$= \lim_{t\to\infty}\frac{[e^{-at}(b\sin bt - a\cos bt)] - (-a)}{a^2+b^2}$

Now, $\lim_{t\to\infty} e^{-at} = 0$ and $[b\sin bt - a\cos bt]$ is a finite value for all a and b.

$\therefore \lim_{t\to\infty} e^{-at}(b\sin bt - a\cos bt) = 0.$

Hence, $\int_0^\infty e^{-ax}\cos bx\,dx = \frac{a}{a^2+b^2}$, $a > 0$.

Exercise 3.3

Evaluate the following integrals, if exists

1. $\int_0^\infty \frac{1}{1+x^2}\,dx$
2. $\int_0^\infty \frac{x}{x^2+4}\,dx$
3. $\int_1^\infty \frac{1}{(1+x)^{3/2}}\,dx$
4. $\int_0^\infty xe^{-x^2}\,dx$
5. $\int_1^\infty \frac{x}{(1+x^2)^2}\,dx$
6. $\int_{-\infty}^{-1}\frac{1}{x^3}\,dx$
7. $\int_0^1 \log x\,dx$
8. $\int_0^\infty e^{-ax}\sin bx\,dx$
9. $\int_0^1 \frac{\sin^{-1}x}{\sqrt{1-x^2}}\,dx$
10. $\int_0^\pi \frac{1}{1+\cos x}\,dx$

Answer

1. $\frac{\pi}{2}$
2. does not exist
3. $\sqrt{2}$
4. $\frac{1}{2}$
5. $\frac{1}{4}$
6. 0
7. -1
8. $\frac{b}{a^2+b^2}$ for $a > 0$
9. $\frac{\pi^2}{8}$
10. does not exist

3.4 Comparison test for improper integrals

The comparison theorem for improper integrals depends on the comparison of functions without actually evaluating the integral itself. It states:

If $f(x) \geq g(x) \geq 0$ for all $x \geq a$ and are integrable functions over the interval $[a, \infty)$, then

i. if the integral $\int_a^\infty f(x)\,dx$ converges, then the integral $\int_a^\infty g(x)\,dx$ also converges.

ii. if the integral $\int_a^\infty g(x)\,dx$ diverges, then the integral $\int_a^\infty f(x)\,dx$ also diverges.

Considering in terms of area, the comparison test makes a lot of sense. If $f(x)$ is larger than $g(x)$ then the area under $f(x)$ must also be larger than the area under $g(x)$.

So, if $f(x) \geq g(x) \geq 0$, then the area under and $f(x)$ finite (*i.e.*, $\int_a^\infty f(x)\,dx$ converges) then the area under the $g(x)$ function must also be finite (*i.e.*, $\int_a^\infty g(x)\,dx$ converges). Similarly, if the area under the function $g(x)$ is infinite (*i.e.*, $\int_a^\infty g(x)\,dx$ diverges) then the area under the function $f(x)$ must also be infinite (i.e., $\int_a^\infty f(x)\,dx$ diverges).

Example 40. Determine whether the following integral are convergent or divergent

i. $\int_2^\infty \frac{\cos^2 x}{x^2}\,dx$ ii. $\int_{-\infty}^3 \frac{1}{x+e^3}\,dx$ iii. $\int_3^\infty \frac{1}{x-e^3}\,dx$

Solution

i. Since $0 \leq \cos^2 x \leq 1$ *for all* $x \geq 0$ and $f(x) = \frac{\cos^2 x}{x^2} \leq \frac{1}{x^2} = g(x)$. Moreover,

$\int_2^\infty f(x)\,dx = \int_2^\infty \frac{1}{x^2}\,dx$

$= \lim_{t\to\infty}\int_2^t x^{-2}\,dx$

$= \lim_{t\to\infty}\left[\frac{x^{-1}}{-1}\right]_2^t$

$= -\lim_{t\to\infty}[t^{-1} - 2^{-1}]$

$= -(0 - 2^{-1})$

$= \frac{1}{2}$.

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This shows that the integral $\int_2^\infty \frac{1}{x^2}\,dx$ converges. So by the Comparison Test, the integral $\int_2^\infty \frac{\cos^2 x}{x^2}\,dx$ must also converge.

iv. Since $\frac{1}{x+e^x} < \frac{1}{e^x} = e^{-x}$ for all $x \geq 3$ and

$$\int_3^\infty e^{-x}\,dx = \lim_{t\to\infty}\int_3^t e^{-x}\,dx$$

$$= \lim_{t\to\infty}\left[\frac{e^{-x}}{-1}\right]_3^t = -\lim_{t\to\infty}\left[e^{-t} - e^{-3}\right] = e^{-3},$$

which is finite. So, $\int_3^\infty e^{-x}\,dx$ is convergent. Therefore, by the Comparison test $\int_3^\infty \frac{1}{x+e^x}\,dx$ is also convergent.

iii. Since $\frac{1}{x-e^x} > \frac{1}{x}$ for all $x \geq 3$ and we know that $\int_3^\infty \frac{1}{x}\,dx$ diverges and so by the Comparison Test the integral $\int_3^\infty \frac{1}{x-e^x}\,dx$ must also diverge.

Example 41. Use the comparison of integrand, determine whether the following integral are convergent or divergent

i. $\int_2^\infty \frac{x^2}{\sqrt{x^5+1}}\,dx$ ii. $\int_2^\infty \frac{x^3}{\sqrt{x^7-1}}\,dx$

Proof:

i. Here, $f(x) = \frac{x^2}{\sqrt{x^5+1}} < \frac{x^2}{\sqrt{x^5}} = x^{-3/2} = g(x)$.

Clearly, both $f(x)$ and $g(x) = x^{-3/2}$ are integrable over $[2, \infty)$.

Moreover $\int_2^\infty g(x)\,dx = \int_2^\infty x^{-3/2}\,dx$

$$= \lim_{t\to\infty}\int_2^t x^{-3/2}\,dx$$

$$= \lim_{t\to\infty}\left[\frac{-2}{\sqrt{x}}\right]_2^t$$

$$= -2\lim_{t\to\infty}\left[\frac{1}{\sqrt{t}} - \frac{1}{\sqrt{2}}\right]$$

$$= -2\left[0 - \frac{1}{\sqrt{2}}\right]$$

$$= \sqrt{2} < \infty.$$

Therefore, $\int_2^\infty g(x)\,dx < \infty$. Hence by comparison test, the integral $\int_2^\infty \frac{x^2}{\sqrt{x^5+1}}\,dx$ also converges.

ii. Here, $g(x) = \frac{x^3}{\sqrt{x^7-1}} > \frac{x^3}{\sqrt{x^7}} = x^{-1/2} = f(x)$. Clearly, both $f(x)$ and $g(x)$ are integrable over $[2, \infty)$.

Moreover, $\int_2^\infty f(x)\,dx = \lim_{t\to\infty}\int_2^t x^{-1/2}\,dx = \lim_{t\to\infty} 2\left[\sqrt{x}\right]_2^t = 2(\infty - \sqrt{2}) = \infty$.

Therefore, the integral $\int_2^\infty f(x)\,dx$ diverges. Hence by comparison test, the integral $\int_2^\infty \frac{x^3}{\sqrt{x^7-1}}$ also diverges.

Exercise 3.4

1. Apply the comparison of the integrals, prove that following integrals are convergent.

i. $\int_a^\infty \frac{\sin^2 x}{x^2}\,dx\ (a > 0)$ ii. $\int_1^\infty \frac{1}{x\sqrt{1+x^2}}\,dx$

iii. $\int_a^\infty \frac{x}{x\sqrt{1+x^3}}\,dx\ (a > 0)$ iv. $\int_1^\infty \frac{1}{x^2+\sqrt{x}}\,dx$

Hints:

i. Let $I_1 = \int_a^\infty \frac{\sin^2 x}{x^2}\,dx\ (a > 0)$ and take $f(x) = \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2} = g(x)$. Since $I_2 = \int_a^\infty g(x)\,dx$ is known to be convergent, so does I_1.

ii. Let $f(x) = \frac{1}{x\sqrt{1+x^2}} < \frac{1}{x\sqrt{x^2}} = \frac{1}{x^2} = g(x)$. Since $\int_1^\infty \frac{1}{x^2}\,dx$ converges, so does $\int_1^\infty f(x)\,dx$.

iii. Let $f(x) = \frac{x}{(1+x)} < \frac{x}{x^3} = \frac{1}{x^2} = g(x)$ and $\int_a^\infty g(x)\,dx\ (a > 0)$ converges and hence so is $\int_a^\infty f(x)\,dx$.

iv. $f(x) = \frac{1}{x^2+\sqrt{x}} < \frac{1}{x^2} = g(x)$ and $\int_1^\infty \frac{1}{x^2}\,dx$ converges. Then proceed as (i).

3.5 Reduction formula

While evaluating certain types of integrals, we need to work with similar type of integrals. This type of method of evaluating integrals are generally termed as *Reduction Method*. We call this method as reduction method because the power n is reduced to some lower power of n.

This reduction method is generally used to evaluate the integrals of $\int x^n\,dx$, $\int e^{ax}x^n\,dx$, $\int \sin^n x\,dx$, $\int \cos^n x\,dx$, $\int \tan^n x\,dx$, etc.

We now discuss the reduction method by taking some examples.

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Example 42. Obtain the reduction formula for $\int x^n e^{-ax} dx$, $(n \neq -1)$ and hence find $\int x^3 e^{-ax} dx$.

Solution:

Let $I_n = \int x^n e^{-ax} dx$

$= x^n \int e^{-ax} dx - \int \left(\frac{d}{dx}(x^n) \int e^{-ax} dx\right) dx$

$= x^n \frac{e^{-ax}}{-a} - \int nx^{n-1} \frac{e^{-ax}}{-a} dx$

$= -\frac{x^n e^{-ax}}{a} + \frac{n}{a} \int x^{n-1} e^{-ax} dx$

$I_n = -\frac{x^n e^{-ax}}{a} + \frac{n}{a} I_{n-1}$ .... (i)

which is required reduction formula.

For next part:

Let $I_3 = \int x^3 e^{-ax} dx$

$= -\frac{x^3 e^{-ax}}{a} + \frac{3}{a} I_{3-1}$ [$\because$ by (i)]

$= \frac{-x^3 e^{-ax}}{a} + \frac{3}{a} I_2$

$= \frac{-x^3 e^{-ax}}{a} + \frac{3}{a}\left(-\frac{x^2 e^{-ax}}{a} + \frac{2}{a} I_1\right)$

$= \frac{-x^3 e^{-ax}}{a} - \frac{3}{a^2} x^2 e^{-ax} + \frac{6}{a^2} I_1$

$= \frac{-x^3 e^{-ax}}{a} - \frac{3}{a^2} x^2 e^{-ax} + \frac{6}{a^2}\left(-\frac{xe^{-ax}}{a} + \frac{1}{a} I_0\right)$

$= \frac{-x^3 e^{-ax}}{a} - \frac{3}{a^2} x^2 e^{-ax} - \frac{6x}{a^3} e^{-ax} + \frac{6}{a^3} I_0$

Since, $I_0 = \int x^0 e^{-ax} dx = -\frac{e^{-ax}}{a}$

So, $I_3 = \frac{-x^3 e^{-ax}}{a} - \frac{3}{a^2} x^2 e^{-ax} - \frac{6x}{a^3} e^{-ax} + \frac{6}{a^3}\left(-\frac{e^{-ax}}{a}\right)$

$\therefore I_3 = \frac{-x^3 e^{-ax}}{a} - \frac{3}{a^2} x^2 e^{-ax} - \frac{6x}{a^3} e^{-ax} - \frac{6}{a^4} e^{-ax}$.

Example 43. Obtain the reduction formula for $\int \tan^n x\, dx$ and hence find the value $\int \tan^5 x\, dx$.

Solution:

Let, $I_n = \int \tan^n x\, dx$

$= \int \tan^{n-2} x \tan^2 x\, dx$

$= \int \tan^{n-2} x (\sec^2 x - 1)\, dx$

$I_n = \int \tan^{n-2} x \sec^2 x\, dx - \int \tan^{n-2} x\, dx$.

Put $t = \tan x$.

Then, $dt = \sec^2 x\, dx$

So, $I_n = \int t^{n-2} dt - I_{n-2}$

$= \frac{t^{n-2+1}}{n-2+1} - I_{n-2}$

$I_n = \frac{t^{n-1}}{n-1} - I_{n-2}$

$\therefore \int \tan^n x\, dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$.

Taking $n = 5$ in above equation, we get

$\int \tan^5 x\, dx = \frac{\tan^4 x}{4} - I_2$

$= \frac{\tan^4 x}{4} - \tan x - I_0$

$\therefore \int \tan^5 x\, dx = \frac{\tan^4 x}{4} - \tan x - x$.

Example 44. Obtain the reduction formula for $\int \sin^n x\, dx$ and hence evaluate

i. $\int \sin^4 x\, dx$ ii. $\int_0^{\pi/2} \sin^n x\, dx$.

Solution:

Let $I_n = \sin^n x\, d = \int \sin^{n-1} x \sin x\, dx$.

Using integrating by parts, we get

$I_n = \sin^{n-1} x \int \sin x\, dx - \int \left(\frac{d}{dx}(\sin^{n-1} x) \int \sin x\, dx\right) dx$

$= \sin^{n-1} x (-\cos x) - \int (n-1) \sin^{n-2} x (\cos x)(-\cos x) dx$.

$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x\, dx$

$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x)\, dx$

$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x\, dx - (n-1) \int \sin^n x\, dx$

$I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n$

or, $I_n + (n-1) I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$

or, $nI_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$

$\therefore I_n = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$ ... (i)

which is required reduction formula for $\int \sin^n x\, dx$.

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Put $n = 4$ in (i), we get

$$I_4 = \int \sin^4 x\,dx$$

$$= -\frac{1}{4}\sin^{4-1}x\cos x + \frac{4-1}{4}I_{4-2}$$

$$= -\frac{\sin^3 x\cos x}{4} + \frac{3}{4}I_2$$

$$= -\frac{\sin^3 x\cos x}{4} + \frac{3}{4}\left[-\frac{1}{2}\sin^{2-1}x\cos x + \frac{2-1}{2}I_0\right]$$

$$= -\frac{\sin^3 x\cos x}{4} - \frac{3}{8}\sin x\cos x + \frac{3}{8}\int \sin^0 x\,dx$$

$$= -\frac{\sin^3 x\cos x}{4} - \frac{3}{8}\sin x\cos x + \frac{3}{8}x$$

$$\therefore \int \sin^4 x\,dx = -\frac{\sin^3 x\cos x}{4} - \frac{3}{8}\sin x\cos x + \frac{3}{8}x.$$

ii Let $J_n = \int_0^{\pi/2} \sin^n x\,dx$

Then, $J_n = \left[-\frac{1}{n}\sin^{n-1}x\cos x\right]_0^{\pi/2} + \frac{(n-1)}{n}J_{n-2}$ [using (i)]

or, $J_n = 0 + \frac{n-1}{n}J_{n-2}$

$\therefore J_n = \frac{n-1}{n}J_{n-2}$

which is required reduction formula for $\int_0^{\pi/2} \sin^n x\,dx$.

Example 45. If $I_n = \int_0^{\pi/4} \tan^n x\,dx$, show that $I_n + I_{n-2} = \frac{1}{n-1}$ and deduce the value of I_3.

Solution:

For first part:

Let, $I = \int_0^{\pi/4} \tan^n x\,dx$

$$= \int_0^{\pi/4} \tan^{n-2}x\tan^2 x\,dx$$

$$= \int_0^{\pi/4} \tan^{n-2}x(\sec^2 x - 1)\,dx$$

$$= \int_0^{\pi/4} \tan^{n-2}x\sec^2 x\,dx - \int_0^{\pi/4} \tan^{n-2}x\,dx.$$

or, $I_n = \int_0^{\pi/4} \tan^{n-2}x\sec^2 x\,dx - I_{n-2}$

Put $t = \tan x$.

or, $dt = \sec^2 x\,dx$

When $x = 0$, then $t = \tan 0 = 0$

When $x = \frac{\pi}{4}$, then $t = \tan\frac{\pi}{4} = 1$

So, $I_n = \int_0^1 t^{n-2}dt - I_{n-2}$

$$= \left[\frac{t^{n-2+1}}{n-2+1}\right]_0^1 - I_{n-2} = \left[\frac{t^{n-1}}{n-1}\right]_0^1 - I_{n-2} = \left[\frac{1^{n-1}}{n-1} - \frac{0}{n-1}\right] - I_{n-2}$$

or, $I_n = \frac{1}{n-1} - I_{n-2}$ ....(i)

$\therefore I_n + I_{n-2} = \frac{1}{n-1}$.

For last part:

$I_3 = \int_0^{\pi/4} \tan^3 x\,dx$

$= \frac{1}{3-1} - I_{3-2}$ [By equation (i)]

$= \frac{1}{2} - I_1$

$= \frac{1}{2} - \int_0^{\pi/4} \tan x\,dx$

$= \frac{1}{2} - [\log\sec x]_0^{\pi/4}$

$= \frac{1}{2} - \left\{\log\sec\frac{\pi}{4} - \log\sec 0\right\}$

$= \frac{1}{2} - \{\log\sqrt{2} - \log 1\}$

$= \frac{1}{2} - \log 2^{1/2}$

$= \frac{1}{2} - \frac{1}{2}\log 2$

$I_3 = \frac{1}{2}(1 - \log 2)$.

Example 46. Obtain the reduction formula for $\int \operatorname{cosec}^n x\,dx$ and find $\int \operatorname{cosec}^5 x\,dx$.

Solution:

Let, $I_n = \int \operatorname{cosec}^n x\,dx$

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$$= \int \text{cosec}^{n-2}x \, \text{cosec}^2x \, dx$$

$$= \text{cosec}^{n-2}x \int \text{cosec}^2x \, dx - \int\left(\frac{d}{dx}(\text{cosec}^{n-2}x)\int \text{cosec}^2x \, dx\right)dx$$

$$= \text{cosec}^{n-2}x\,(-\cot x) - \int(n-2)\,\text{cosec}^{n-3}x\,(-\text{cosec}x\cot x).(-\cot x)\,dx$$

$$= -\text{cosec}^{n-2}x\cot x - (n-2)\int \text{cosec}^{n-2}x\cot^2x\,dx$$

$$= -\text{cosec}^{n-2}x\cot x - (n-2)\int \text{cosec}^{n-2}x\,(\text{cosec}^2x - 1)dx$$

$$= -\text{cosec}^{n-2}x\cot x - (n-2)\int \text{cosec}^n x\,dx + (n-2)\int \text{cosec}^{n-2}x\,dx$$

or, $I_n = -\text{cosec}^{n-2}x\cot x - (n-2)\,I_n + (n-2)\,I_{n-2}$

or, $I_n + (n-2)\,I_n = -\text{cosec}^{n-2}x\cot x + (n-2)\,I_{n-2}$

or, $(1+n-2)\,I_n = -\text{cosec}^{n-2}x\cot x + (n-2)\,I_{n-2}$

or, $(n-1)\,I_n = -\text{cosec}^{n-2}x\cot x + (n-2)\,I_{n-2}$

$$\therefore \quad I_n = -\frac{\text{cosec}^{n-2}x\cot x}{n-1} + \frac{n-2}{n-1}I_{n-2} \qquad \ldots\ldots \text{(i)}$$

Putting $n = 5$ in (i), we get

Let $I_5 = \int \text{cosec}^5x\,dx$

$$= -\frac{\text{cosec}^{5-2}x\cot x}{5-1} + \frac{5-2}{5-1}I_{5-2} \qquad \text{[By using equation (i)]}$$

$$= -\frac{\text{cosec}^3x\cot x}{4} + \frac{3}{4}I_3$$

$$= -\frac{\text{cosec}^3x\cot x}{4} + \frac{3}{4}\left[-\frac{\text{cosec}^{3-2}x\cot x}{3-1} + \frac{3-2}{3-1}I_{3-2}\right]$$

$$= -\frac{\text{cosec}^3x\cot x}{4} - \frac{3}{8}\text{cosec}x\cot x + \frac{3}{8}I_1$$

$$= -\frac{\text{cosec}^3x\cot x}{4} - \frac{3}{8}\text{cosec}x\cot x + \frac{3}{8}\int \text{cosec}x\,dx$$

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$$\therefore \quad \int \text{cosec}^5x\,dx = -\frac{\text{cosec}^3x\cot x}{4} - \frac{3}{8}\text{cosec}x\cot x + \frac{3}{8}\log(\text{cosec}x - \cot x).$$

Exercise 3.5

1. Find the reduction formula for $\int x^n e^{ax}\,dx$.

2. Obtain the reduction formula for $\int \sec^n x\,dx$ and hence evaluate $\int \sec^7 x\,dx$.

3. Establish the reduction formula for $\int \cos^n x\,dx$ and hence evaluate

   i. $\int \cos^7 x\,dx$ ii. $\int_0^{\pi/2} \cos^7 x$

4. Find the reduction for $\int \cot^n x\,dx$ and hence evaluate $\int \cot^7 x\,dx$.

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Answer

1. $\frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$ 2. $\frac{\sec^{n-2}x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$ $\frac{\sec^5 x \tan x}{6} + \frac{5}{24}\sec^3 x \tan x + \frac{15}{24}\log(\sec x + \tan x) + C$

3. $\frac{\cos^{n-1}x \sin x}{n} + \left(\frac{n-1}{n}\right) I_{n-2}$ i. $\frac{\sin x \cos^6 x}{7} + \frac{6\sin x \cos^4 x}{35} + \frac{24}{105}\sin x \cos^2 x + \frac{48}{105}\sin x + C$ ii. $\frac{16}{35}$

4. $-\frac{\cot^{n-1}x}{n-1} - I_{n-2}$, $-\frac{\cot^6 x}{6} + \frac{\cot^4 x}{4} - \frac{\cot^2 x}{2} - \log(\sin x) + C$

3.6 Beta function and gamma function

Beta function and gamma function are special kind of integrals which arise in number of situations. In this section, we discuss their evaluation and various properties.

Beta function and gamma function were introduced by the Swiss mathematician Leonhard Euler in 18th century. Gamma function is the extension of factorial function to real numbers. Beta function (also known as Euler's integral of the first kind) is closely connected to gamma function; and is a generalization of the factorial function. Both beta function and gamma function are very important in calculus as complex integrals can be simplified by using beta and gamma functions.

Leonhard Euler
(1707–1783)

Beta function and gamma function are crucial in integral calculus and are considered tools to evaluate the integral of some specific functions. These are needed in the evaluation of trigonometric and exponential integrals. We begin with their definition.

3.6.1 Beta function

The integral $\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ $(m > 0,\ n > 0)$ is known as ***beta function*** (or first Eulerian integral) and is written as

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$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx,\ (m > 0, n > 0).$$

Properties of beta function

i. $\beta(m, n) = \beta(n, m)$

ii. $\beta(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}x \cos^{2n-1}x\,dx,$

To see the above result, we have by definition of beta function,

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx$$

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$$= \int_0^1 (1-x)^{m-1}(1-(1-x))^{n-1}dx \qquad \left[\because \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx\right]$$

$$= \int_0^1 (1-x)^{m-1}x^{n-1}dx$$

$$= \int_0^1 x^{n-1}(1-x)^{m-1}dx$$

$$= \beta(n, m)$$

$\therefore \quad \beta(m, n) = \beta(n, m).$

Again, $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx \qquad \ldots\ldots (i)$

Put $x = \sin^2 t$.

Then, $\frac{dx}{dt} = 2\sin t \cos t$.

$\therefore \quad dx = 2\sin t \cos t\, dt$

When $x = 0$, then $0 = \sin^2 t \Rightarrow t = 0$

When $x = 1$, then $1 = \sin^2 t \Rightarrow t = \frac{\pi}{2}$

Thus, $\beta(m, n) = \int_0^{\pi/2} (\sin^2 t)^{m-1}(1-\sin^2 t)^{n-1}\, 2\sin t \cos t\, dt$

$$= \int_0^{\pi/2} \sin^{2m-2}t \cdot \cos^{2n-2}t \cdot 2\sin t \cos t\, dt$$

$\therefore \quad \beta(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}t \cdot \cos^{2n-1}t\, dt.$

We next define gamma function.

3.6.2 Gamma function

The integral $\int_0^\infty e^{-x} x^{n-1}dx$, $(n > 0)$ is known as ***gamma function*** (or Second Eulerian integral) is written as

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}dx, \ (n > 0).$$

Properties of gamma function

i. $\Gamma(1) = 1$

We know, $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}dx$

Substituting $n = 1$, we get

$$\Gamma(1) = \int_0^\infty e^{-x} x^{1-1}dx$$

$$= \int_0^\infty e^{-x}\,dx = \left[\frac{e^{-x}}{-1}\right]_0^\infty = -[e^{-x}]_0^\infty = -[e^{-\infty} - e^{-0}] = -(0-1) = 1$$

$\therefore \quad \Gamma(1) = 1.$

ii. $\Gamma(n+1) = n\,\Gamma(n)$

We have, $\Gamma(n+1) = \int_0^\infty e^{-x}x^{(n+1)-1}\,dx = \int_0^\infty e^{-x}x^n\,dx.$

Integrating by parts,

$$= \left[x^n \frac{e^{-x}}{-1}\right]_0^\infty - \int_0^\infty (nx^{n-1})\frac{e^{-x}}{-1}dx = -[x^n e^{-x}]_0^\infty + n\int_0^\infty x^{n-1}e^{-x}dx = 0 + n\,\Gamma(n)$$

$\therefore \quad \Gamma(n+1) = n\Gamma(n).$

Further, $\Gamma(n+1) = n\,\Gamma(n)$

$= n(n-1)\,\Gamma(n-1)$

$= n(n-1)(n-2)\,\Gamma(n-2) = n(n-1)(n-1)\ldots n!$

$\therefore \quad \Gamma(n+1) = n!.$

Relation between beta function and gamma function

The relation between beta and gamma function is given by

$$\beta(m, n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$$

We prove this relation:

Here, $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

$$= \left[(1-x)^{n-1}\frac{x^m}{m}\right]_0^1 + (n-1)\int_0^1 (1-x)^{n-2}\frac{x^m}{m}\,dx$$

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$$= \frac{n-1}{m}\int_0^1 (1-x)^{n-2}\, x^m\, dx$$

Successive Integration yields

$$\beta(m,n) = \frac{(n-1)}{m}\frac{(n-2)}{m+1}\frac{(n-3)\ldots\ldots\ldots\ldots 2.1}{(m+2)\ldots(m+n-2)}\int_0^1 (1-x)^{n-n}\, x^{m+n-2}\, dx$$

$$= \frac{(n-1)!}{m(m+1)\ldots(m+n-2)}\left[\frac{x^{m+n-1}}{m+n-1}\right]_0^1$$

$$= \frac{(n-1)!}{m(m+1)\ldots(m+n-2)(m+n-1)}$$

$$= \frac{(n-1)!\,1.2.3\ldots\ldots(m-1)}{1.2.3\ldots\ldots(m-1)\,m(m+1)\ldots(m+n-1)}$$

$$= \frac{(n-1)!\,(m-1)!}{(m+n-1)!}$$

$$= \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$$

$$\therefore \beta(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}.$$

From above result, it can be deduced that

i. $\Gamma(m)\,\Gamma(1-m) = \frac{\pi}{\sin m\pi}$, $0 < m < 1$ ii. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

We now discuss the application of beta and gamma function in the evaluation of various standard integrals.

Example 47. Prove $\int_0^{\pi/2}\sin^n x\,dx = \int_0^{\pi/2}\cos^n x\,dx = \sqrt{\pi}\frac{\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{n+2}{2}\right)}$.

Solution:

We have, $\int_0^{\pi/2}\sin^m x\cos^n x\,dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$

Put $n = 0$ then $\int_0^{\pi/2}\sin^m x\,dx = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{m+1}{2}\right)}{2\Gamma\left(\frac{m+2}{2}\right)}$

Replacing m by n, we have $\int_0^{\pi/2}\sin^n x\,dx = \frac{\sqrt{\pi}\,\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{n+2}{2}\right)}$

Similarly $\int_0^{\pi/2}\cos^n x\,dx = \frac{\sqrt{\pi}\,\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{n+2}{2}\right)}$

$$\therefore \int_0^{\pi/2}\sin^n x\,dx = \int_0^{\pi/2}\cos^n x\,dx = \frac{\sqrt{\pi}\,\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{n+2}{2}\right)}.$$

Further properties of gamma function

i. $\int_0^{\pi/2}\sin^m x\cos^n x\,dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{m+n+2}{2}\right)}$.

ii. $\int_0^{\infty} e^{-x^2}dx = \frac{1}{2}\sqrt{\pi}$.

Solution:

i. Let $I = \int_0^{\pi/2}\sin^m x\cos^n x\,dx$

$$= \int_0^{\pi/2}(\sin^2 x)^{m/2}(\cos^2 x)^{n/2}dx$$

$$= \int_0^{\pi/2}(\sin^2 x)^{m/2}(1-\sin^2 x)^{n/2}dx$$

Let $t = \sin^2 x$, Then, $\frac{dt}{dx} = 2\sin x\cos x$

or, $dx = \frac{dt}{2\sin x\cos x}$

or, $dx = \frac{dt}{2\sqrt{(\sin^2 x)}\sqrt{(1-\sin^2 x)}}$

or, $dx = \frac{dt}{2\sqrt{(t)}\sqrt{(1-t)}}$

When $x = 0$, then $t = 0$

and when $x = \frac{\pi}{2}$, then $t = 1$

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So, $I = \int_0^1 t^{m/2}(1-t)^{n/2}\frac{dt}{2t^{1/2}(1-t)^{1/2}}$

$= \int_0^1 t^{m/2-1/2}(1-t)^{n/2-1/2}\frac{dt}{2}$

$= \frac{1}{2}\int_0^1 t^{(m+1)/2-1}(1-t)^{(n+1)/2-1}dt$

$= \frac{1}{2}\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$

$= \frac{1}{2}\frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{m+1}{2}+\frac{n+1}{2}\right)}$ $\left[\because \beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}\right]$

$= \frac{1}{2}\frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{m+n+2}{2}\right)}$

$\therefore \int_0^{\pi/2} \sin^m x \cos^n x\, dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$.

ii. Let $I = \int_0^\infty e^{-x^2} dx$.

Put $t = x^2$.

or, $dt = 2x\,dx$

or, $dx = \frac{dt}{2x} = \frac{dt}{2\sqrt{x^2}} = \frac{dt}{2\sqrt{t}}$

When $x = 0$ then $t = 0$.

and when $x = \infty$ then $t = \infty$.

So, $I = \int_0^\infty e^{-t}\frac{dt}{2\sqrt{t}}$

$= \frac{1}{2}\int_0^\infty e^{-t} t^{-1/2} dt$

$= \frac{1}{2}\int_0^\infty e^{-t} t^{1/2-1} dt$

$= \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$= \frac{1}{2}\sqrt{\pi}$

$\therefore \int_0^\infty e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$.

Example 48. Prove that: $\int_0^\infty e^{-x} x^n dx = n!$.

Solution:

L.H.S $= \int_0^\infty e^{-x} x^n dx = [x^n(-e^{-x})]_0^\infty + \int_0^\infty nx^{n-1}e^{-x} dx = \left[-\frac{x^n}{e^x}\right]_0^\infty + \int_0^\infty nx^{n-1}e^{-x} dx$

But $\lim_{x\to\infty} \frac{x^n}{e^x} = 0$, so $\left[-\frac{x^n}{e^x}\right]_0^\infty = 0$

$\therefore \int_0^\infty e^{-x}x^n\, dx = 0 + \int_0^\infty nx^{n-1}e^{-x} dx$

$= \int_0^\infty nx^{n-1}e^{-x}\, dx$ ....(i)

Now, if n is positive integer then

$\int_0^\infty nx^{n-1}e^{-x} dx = n\Gamma(n) = n(n-1)\Gamma(n-1)$

$= n(n-1)(n-2)\ldots 1\,\Gamma(1)$

$= n!$ $[\because \Gamma(1) = 1]$

$\therefore \int_0^\infty e^{-x}x^n dx = n!$.

Example 49. Show that $\int_0^{\pi/2} \sin^p\theta\cos^q\theta\, d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right), p, q > -1$.

Solution:

Let $I = \int_0^{\pi/2} \sin^p\theta\cos^q\theta\, d\theta,\ p > -1, q > -1$

$= \int_0^{\pi/2} (\sin^2\theta)^{p/2}(\cos^2\theta)^{q/2} d\theta = \int_0^{\pi/2} (\sin^2\theta)^{p/2}(1-\sin^2\theta)^{q/2} d\theta$

Put $t = \sin^2\theta$

Then $\frac{dt}{d\theta} = \frac{d}{d\theta}(\sin^2\theta)$

or, $dt = 2\sin\theta\cos\theta\, d\theta$

or, $d\theta = \frac{dt}{2\sin\theta\cos\theta}$

or, $d\theta = \frac{dt}{2\sqrt{\sin^2\theta}\sqrt{1-\sin^2\theta}}$

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or, $d\theta = \frac{1}{2\sqrt{t}\sqrt{1-t}}dt$

When $\theta = 0$, then $t = 0$.

When $\theta = \frac{\pi}{2}$, then $t = \sin^2\left(\frac{\pi}{2}\right) = 1$.

So, $I = \int_0^1 \frac{t^{p/2}(1-t)^{q/2}}{2\sqrt{t}\sqrt{1-t}}dt$

$= \int_0^1 \frac{t^{p/2}(1-t)^{q/2}}{2t^{1/2}(1-t)^{1/2}}dt$

$= \int_0^1 \frac{1}{2}t^{p/2-1/2}(1-t)^{q/2-1/2}dt$

$= \frac{1}{2}\int_0^1 t^{(p+1)/2-1}(1-t)^{(q+1)/2-1}dt$

$= \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$\therefore \int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$.

Example 50. Show that $\int_0^{\pi/2} \sin^3 x \cos^5 x dx = \frac{1}{24}$.

Solution:

Let $I = \int_0^{\pi/2} \sin^3 x \cos^5 x\, dx$

$= \frac{\Gamma\left(\frac{3+1}{2}\right)\Gamma\left(\frac{5+1}{2}\right)}{2\Gamma\left(\frac{3+5+2}{2}\right)}$

$= \frac{\Gamma(2)\Gamma(3)}{2\Gamma(5)}$

$= \frac{1!\,2!}{2 \times 4!}$ $[\because \Gamma(n) = (n-1)!]$

$= \frac{1 \times 1 \times 2}{2 \times 4 \times 3 \times 2}$

$= \frac{1}{24}$

$\therefore \int_0^{\pi/2} \sin^3 x \cos^5 x dx = \frac{1}{24}$.

Example 51. Show that $\beta(m, n) = \frac{(m-1)!\,(n-1)!}{(m+n-1)!}$, $m > 1$, $n > 1$, positive integers.

Solution:

We have, $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$, $m > 1$, $n > 1$

We know, $\Gamma(n) = (n-1)\Gamma(n-1)$

So, $\beta(m, n) = \frac{(m-1)\Gamma(m-1)(n-1)\Gamma(n-1)}{(m+n-1)\Gamma(m+n-1)}$

$= \frac{(m-1)(m-2)\Gamma(m-2)(n-1)(n-2)\Gamma(n-2)}{(m+n-1)(m+n-2)\Gamma(m+n-2)}$

Continuing this process

$\beta(m, n) = \frac{(m-1)(m-2)\cdots 2\cdot 1\cdot\Gamma(1)(n-1)(n-2)\cdots 2\cdot 1\,\Gamma(1)}{(m+n-1)(m+n-2)\cdots 2\cdot 1\,\Gamma(1)}$

$= \frac{\{(m-1)(m-2)\cdots 2\cdot 1\cdot\}\{(n-1)(n-2)\cdots 2\cdot 1\}}{(m+n-1)(m+n-2)\cdots 2\cdot 1}$ $[\because \Gamma(1) = 1]$

$= \frac{(m-1)!\,(n-1)!}{(m+n-1)!}$

$\therefore \beta(m, n) = \frac{(m-1)!\,(n-1)!}{(m+n-1)!}$

Example 52. Prove that $\int_0^1 \frac{1}{(1-x^6)^{1/6}}dx = \frac{\pi}{3}$.

Solution:

Let $I = \int_0^1 \frac{1}{(1-x^6)^{1/6}}dx$.

Put $x^6 = t$

Differentiating with respect to x, we get

or, $6x^5 dx = dt$

or, $6(x^6)^{5/6}dx = dt$

$\therefore dx = \frac{1}{6}t^{-5/6}dt$

If $x = 0$, then $t = 0$,

If $x = 1$, then $t = 1$

$\therefore I = \int_0^1 \frac{1}{(1-x^6)^{1/6}}dx$

$= \frac{1}{6}\int_0^1 \frac{t^{-5/6}}{(1-t)^{1/6}}dt$

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$$= \frac{1}{6}\int_0^1 t^{5/6}(1-t)^{-1/6}dt$$

$$= \frac{1}{6}\int_0^1 (t)^{1/6-1}(1-t)^{5/6-1}dt$$

$$= \frac{1}{6}\beta\left(\frac{1}{6}, \frac{5}{6}\right)$$

$$= \frac{1}{6}\frac{\Gamma(1/6)\,.\,\Gamma(5/6)}{\Gamma(1/6+5/6)}$$

$$= \frac{1}{6}\frac{\Gamma(1/6)\,.\,\Gamma(1-1/6)}{\Gamma(1)}$$

$$= \frac{1}{6}\frac{\pi}{\sin\frac{\pi}{6}}$$

$$= \frac{1}{6}\cdot\frac{\pi}{\frac{1}{2}} \qquad \left(\because \Gamma(m)\,\Gamma(1-m) = \frac{\pi}{\sin m\pi}\right)$$

$$= \frac{\pi}{3}$$

Example 53. Evaluate by using gamma function $\int_0^a x^3(a^2-x^2)^{5/2}dx$.

Solution:

Let $x = a\sin\theta$

Differentiating with respect to θ, we get

$$\frac{dx}{d\theta} = a\cos\theta$$

or, $dx = a\cos\theta d\theta$

when $x = 0$, then $0 = a\sin\theta$,

or, $\sin\theta = \sin 0$

or, $\theta = 0$

When $x = a$ then $a = a\sin\theta$

or, $\sin\theta = 1$

or, $\sin\theta = \sin\frac{\pi}{2}$

or, $\theta = \frac{\pi}{2}$

Then, $I = \int_0^a x^3(a^2-x^2)^{3/2}dx$

$$= \int_0^{\pi/2} (a\sin\theta)^3(a^2-a^2\sin^2\theta)^{3/2}a\cos\theta\, d\theta$$

$$= \int_0^{\pi/2} a^3\sin^3\theta.\ a^3\cos^3\theta.\ a\cos\theta d\theta$$

$$= a^6\int_0^{\pi/2}\sin^3\theta\cos^4\theta d\theta.$$

$$= a^6\frac{\Gamma\left(\frac{2+1}{2}\right)\Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{2+4+2}{2}\right)}$$

$$= a^6\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{5}{2}\right)}{2\Gamma(4)}$$

$$= a^6\frac{\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\cdot\left(\frac{3}{2}\right)\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{2\,.\,3!}$$

$$= a^6\frac{\frac{1}{2}\sqrt{\pi}\cdot\frac{3}{4}\cdot\sqrt{\pi}}{2\times3\times2\times1}$$

$$= \frac{a^6\pi}{32}$$

Exercise 3.6

1. Show that

   i. $\Gamma(7) = 720$

   ii. $\Gamma\left(\frac{11}{2}\right) = \frac{945}{32}\sqrt{\pi}$

   iii. $\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma(5)} = \frac{\pi}{64}$

2. By using gamma function, show that

   i. $\int_0^{\pi/6}\cos^4 3\theta\sin^2 6\theta\, d\theta = \frac{5\pi}{192}$

   ii. $\int_0^{\pi/2}\sin^4 x\cos^3 x\, dx = \frac{2}{35}$

   iii. $\int_0^{\pi/2}\sin^5 x\cos^3 x\, dx = \frac{1}{24}$

   iv. $\int_0^{\pi}\sin^6\frac{x}{2}\cos^4\frac{x}{2}dx = \frac{5\pi}{512}$

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3. Show that $\frac{\beta(m, n+1)}{n} = \frac{\beta(m+1, n)}{n} = \frac{\beta(m, n)}{m+n}$.

4. Show that $\int_0^1 \frac{x^{2n}}{\sqrt{1-x^2}}\,dx = \frac{(2n-1)(2n-3)\ldots\ldots 3.1}{2^n.n!}\frac{\pi}{2}$

5. Using gamma function, prove that

i. $\int_0^1 x^6\sqrt{1-x^2}\,dx = \frac{5\pi}{256}$

ii. $\int_0^a \frac{x^4}{\sqrt{a^2-x^2}}\,dx = \frac{3a^2\pi}{16}$

iii. $\int_0^1 \frac{x^6}{\sqrt{1-x^2}}\,dx = \frac{5\pi}{32}$

iv. $\int_0^{2a} x^{9/2}\sqrt{2a-x}\,dx = \frac{63}{8}\pi a^5$

v. $\int_0^1 x^2(1-x^2)^{3/2}\,dx = \frac{\pi}{32}$

vi. $\int_0^1 \frac{x^4}{\sqrt{1-x^2}}\,dx = \frac{3\pi}{16}$

6. Prove that $\int_0^{\infty}\sqrt{x}\,e^{-x^2}\,dx \times \int_0^{\pi}\frac{e^{-x^2}}{\sqrt{x}}\,dx = \frac{\pi}{2\sqrt{2}}$.

Multiple Choice Questions

1. $\int \frac{1}{1+\sin x}\,dx$ equals to

a. $\tan x + \sin x + C$
b. $\tan x - \sec x + C$
c. $\sec^2 x - \operatorname{cosec}^2 x + C$
d. $\sec x - \sec x \cdot \tan x + C$

2. Which of the following is not true?

a. $\Gamma(n) = n!$
b. $\Gamma(n) = \int_0^{\infty} x^{n-1}e^{-x}\,dx$
c. $\Gamma(n+1) = n\,\Gamma(n)$
d. $\Gamma(n+1) = n!$

3. The value of $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ is

a. 0
b. $\frac{\pi}{2}$
c. $\frac{\pi}{4}$
d. none of these

4. $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx = m\pi$, then the value of m is

a. 0
b. $\frac{1}{2}$
c. 1
d. 2

5. $\int_{-\pi}^{\pi} x^4\sin^5 x\,dx$ is equal to

a. 0
b. 1
c. -1
d. $\frac{1}{5}$

6. The value of the integral $\int_0^{\infty} \frac{1}{c^x}\,dx$ is

a. $\frac{1}{\log c}$
b. $\frac{2}{\log c}$
c. $\frac{\pi}{\log c}$
d. $\frac{1}{2\log c}$

7. $\int_0^{\frac{\pi}{2}} \frac{\sin\theta - \cos\theta}{\sqrt{\sin(2\theta)}}\,d\theta$ is equal to

a. 4
b. 2
c. 1
d. 0

8. The improper integral $\int_0^{\infty} \sin x\,dx$ equal to

a. 1
b. 0
c. diverges
d. does not exist

9. The value of $\int_0^{\frac{\pi}{2}} \sin^6 x\,dx$ equal to

a. 0
b. $\frac{\pi}{8}$
c. $\frac{\pi}{4}$
d. $\frac{15\pi}{96}$

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10. The value of $\int_0^\infty e^{-x^2} dx$ is equals to

a. $\sqrt{\pi}$ b. $\frac{\sqrt{\pi}}{2}$

c. $\frac{\sqrt{\pi}}{\sqrt{2}}$ d. $\frac{\pi}{2}$

11. The value of $\int_{-\infty}^{0} x^5 e^x dx$ equals to

a. 1 b. 199

c. $-5!$ d. $5!$

Answer

1. b	2. a	3. c	4. b	5. a	6. a
7. d	8. c	9. d	10. b	11. c	

❖❖❖

Unit 4

Application of Integral

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and on

- knowledge of graphs of various functions.
- evaluation Integrals of functions.

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

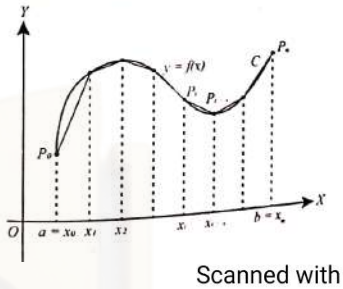
- area under curves and arc length of curves using integrals.
- surface area and volume of solid of revolution for Cartesian and parametric curves.

4.0 Introduction

In this unit, we focus on the applications of definite integrals. We will see how definite integrals can be used to evaluate arc length, area and volume of solid of revolution. We begin with the arc length.

4.1 Arc length

In this section, we discuss the derivation of arc length formula using definite integral. So, let C be a curve described by $y = f(x)$ on $[a, b]$. To find the length of C, we first find a polygonal approximation of C by diving $[a, b]$ into n number of sub intervals with the help of points say $x_0, x_1, \ldots, x_n$ and equal width Δx. Then corresponding points $P(x_i, y_i)$ lie on the curve C as shown below:



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Clearly length of curve $C = \lim_{n \to \infty} \sum_{i=0}^{n} |P_i P_{i+1}|$

Here, $|P_i P_{i+1}| = \Delta s_i = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}$

$= \sqrt{(\Delta x)^2 + (\Delta y_i)^2}$ .... (i)

Applying, mean value theorem on $[x_i, x_{i+1}]$, we get a point $x_i^* \in (x_i, x_{i+1})$ such that

$$f'(x_i^*) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \frac{\Delta y_{i+1}}{\Delta x}$$

$\therefore \quad \Delta y_{i+1} = f'(x_i^*)\, \Delta x.$

$$|P_i P_{i+1}| = \sqrt{(\Delta x)^2 + (f'(x_i^*)\, \Delta x)^2} = \Delta x \sqrt{1 + [f'(x_i^*)]^2}.$$

Then, length of curve $C = \lim_{n \to \infty} \sum_{i=0}^{n} \Delta s_i$

$$= \lim_{n \to \infty} \sum_{i=0}^{n} \Delta x \sqrt{1 + [f'(x_i^*)]^2}$$

$$= \lim_{n \to \infty} \sum_{i=0}^{n} \sqrt{1 + [f'(x_i^*)]^2}\, \Delta x$$

$$= \int_a^b \sqrt{1 + (f'(x))^2}\, dx \quad \text{(By definition of definite integral)}$$

So, length $= \int_a^b \sqrt{1 + (f'(x))^2}\, dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$

Hence, arc length of curve $y = f(x)$ in between two ordinates $x = a$ and $x = b$ is

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

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Remarks:

1. The length of the arc of the curve $x = f(y)$ included between two points $y = c$ and $y = d$ is

$$s = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy.$$

2. The length of the arc of the curve in polar form $r = f(\theta)$ included between two points whose vectorial angles are $\theta = \theta_1$ and $\theta = \theta_2$ is

$$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

3. If from (i) we can write $ds^2 = dx^2 + dy^2, dx = \Delta x$ and $dy = \Delta y$.

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We next discuss some examples on evaluation of arc length.

Example 1. Show that the perimeter of a circle $x^2 + y^2 = a^2$ is $2\pi a$.

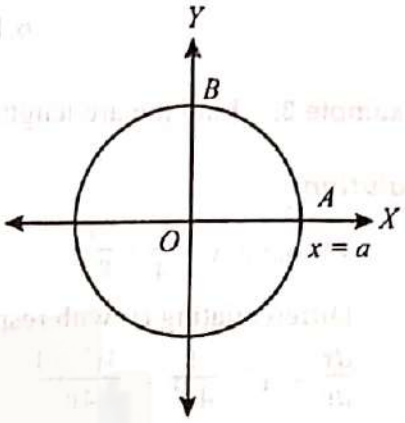
Solution:

We have, $y^2 = a^2 - x^2$

$2y\frac{dy}{dx} = -2x$

$\therefore \quad \frac{dy}{dx} = -\frac{x}{y}$

Now, length of perimeter AB in first quadrant where at A, x = a and at B, $x = 0$.



$$s = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= \int_0^a \sqrt{1 + \frac{x^2}{y^2}}\, dx$$

$$= \int_0^a \sqrt{\frac{x^2 + y^2}{y^2}}\, dx$$

$$= \int_0^a \sqrt{\frac{a^2}{a^2 - x^2}}\, dx$$

$$= \int_0^a \frac{a}{\sqrt{a^2 - x^2}}\, dx$$

$$= a\left[\sin^{-1}\frac{x}{a}\right]_0^a$$

$$= a\frac{\pi}{2} = \frac{\pi a}{2}$$

$\therefore$ Total perimeter $= 4 \times \frac{\pi a}{2} = 2\pi a$ units.

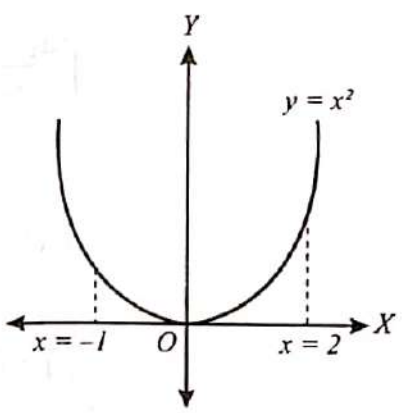
Example 2. Find the arc length of the curves $y = x^2$; $\quad -1 \le x \le 2$.

Solution:

We have, $y = x^2$ $\quad \ldots$ (i)

Differentiating (i) with respect to x

$\frac{dy}{dx} = 2x$



Now the arc length $s = \int_{-1}^{2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

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$$= \int_{-1}^{2} \sqrt{1+(2x)^2}\,dx = 2\int_{-1}^{2} \sqrt{\left(\frac{1}{2}\right)^2 + x^2}\,dx$$

$$= 2\left[\frac{x}{2}\sqrt{x^2+\frac{1}{4}} + \frac{1/4}{2}\log\left(x+\sqrt{x^2+\frac{1}{4}}\right)\right]_{-1}^{2}$$

$$= 2\left[\sqrt{\frac{17}{4}} + \frac{1}{8}\log\left(2+\sqrt{\frac{17}{4}}\right) + \frac{1}{2}\sqrt{\frac{5}{4}} - \frac{1}{8}\log\left(-1+\sqrt{\frac{5}{4}}\right)\right]$$

$$= \sqrt{17} + \frac{1}{2}\sqrt{5} + \frac{1}{8}\log\left(\frac{8+\sqrt{17}}{-2+\sqrt{5}}\right)$$

$$= 6.13 \text{ units.}$$

Example 3. Find the arc length of $x = \frac{y^4}{4} + \frac{1}{8y^2}$ from $y = 1$ to $y = 2$.

Solution:

We have, $x = \frac{y^4}{4} + \frac{1}{8y^2}$ .... (i) from $y = 1$ to $y = 2$

Differentiating (i) with respect to y

$$\frac{dx}{dy} = y^3 - \frac{1}{4y^3} = \frac{4y^6-1}{4y^3}.$$

Now arc length $s = \int_1^2 \left[1+\left(\frac{dx}{dy}\right)^2\right]^{1/2} dy$

$$= \int_1^2 \sqrt{\left[1+\left(\frac{4y^6-1}{4y^3}\right)^2\right]}\,dy$$

$$= \int_1^2 \sqrt{\frac{16y^6+16y^{12}-8y^6+1}{(4y^3)^2}}\,dy$$

$$= \int_1^2 \frac{\sqrt{(4y^6+1)^2}}{4y^3}\,dy$$

$$= \int_1^2 \frac{4y^6+1}{4y^3}\,dy$$

$$= \int_1^2 \left(y^3+\frac{1}{4y^3}\right)dy$$

$$= \left|\frac{y^4}{4} - \frac{1}{8y^2}\right|_1^2$$

$$= \left(\frac{16}{4} - \frac{1}{32}\right) - \left(\frac{1}{4} - \frac{1}{8}\right)$$

$$= \frac{128-1-8+4}{32}$$

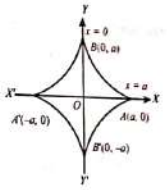
$$= \frac{123}{32} \text{ units.}$$

$\therefore$ Length of arc $= \frac{123}{32}$ units.

Example 4. Find perimeter of the asteroid $x^{2/3}+y^{2/3} = a^{2/3}$.

Solution:

The given equation of asteroid $x^{2/3}+y^{2/3} = a^{2/3}$ .... (i)



Differentiating with respect to x,

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$$

or, $\frac{dy}{dx} = -\frac{x^{-1/3}}{y^{-1/3}}$

or, $\frac{dy}{dx} = -\frac{y^{1/3}}{x^{1/3}}$

Using symmetry of curve in four quadrants perimeter of asteroid is given by

s = 4 × arc length AB, where at A, $x = a$ and at B, $x = 0$.

$$= 4 \times \int_0^a \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$= 4 \times \int_0^a \sqrt{1+\left(\frac{-y^{1/3}}{x^{1/3}}\right)^2}\,dx$$

$$= 4 \times \int_0^a \sqrt{\frac{x^{2/3}+y^{2/3}}{x^{2/3}}}\,dx$$

$$= 4 \times \int_0^a \sqrt{\frac{a^{2/3}}{x^{2/3}}}\,dx \qquad [\because \text{using equation (i)}]$$

$$= 4 \times \int_0^a \frac{a^{1/3}}{x^{1/3}}\,dx$$

$$= 4\int_0^a a^{1/3}x^{-1/3}\,dx$$

$$= 4a^{1/3}\int_0^a x^{-1/3}\,dx$$

$$= 4a^{1/3}\left[\frac{x^{-1/3+1}}{\frac{-1}{3}+1}\right]_0^a$$

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$$= 4a^{1/3}\left[\frac{x^{2/3}}{\frac{2}{3}}\right]_0^a$$

$$= \frac{3}{2} \times 4a^{1/3}[x^{2/3}]_0^a$$

$$= 6a^{1/3}[a^{2/3} - 0]$$

$$= 6a^{1/3}.a^{2/3}$$

$\therefore$ $s = 6a$ units.

Example 5. Find the perimeter of the cardioid $r = a(1 + \cos\theta)$ and show that the arc of the upper half is bisected by $\theta = \frac{\pi}{3}$.

Solution:

The given equation of cardioid is

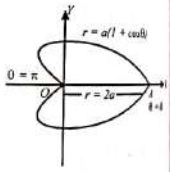
$r = a(1 + \cos\theta)$

Then $\frac{dr}{d\theta} = -a\sin\theta$.

$\therefore$ At A, $r = 2a$, $\theta = 0$ and at O, $r = 0$, $\theta = \pi$.

The curve is symmetrical about the initial line and length of upper half is given by $\theta = 0$ to $\theta = \pi$.

$\therefore$ Perimeter of cardioid $= 2 \times$ length of the arc in upper half



$$= 2\int_0^{\pi} ds$$

$$= 2.\int_0^{\pi} \frac{ds}{dr}\, dr.$$

$$= 2\int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$= 2\int_0^{\pi} \sqrt{(a(1+\cos\theta))^2 + (-a\sin\theta)^2}\, d\theta$$

$$= 2\int_0^{\pi} \sqrt{a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta)}\, d\theta$$

$$= 2a\int_0^{\pi} \sqrt{2 + 2\cos\theta}\, d\theta$$

$$= 2a\int_0^{\pi} \sqrt{2(1+\cos\theta)}\, d\theta$$

$$= 2a\int_0^{\pi} \sqrt{2.2\cos^2\frac{\theta}{2}}\, d\theta$$

$$= 4a\int_0^{\pi} \cos\frac{\theta}{2}\, d\theta$$

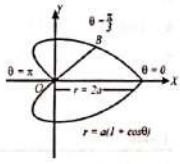
$$= 4a\left[\frac{\sin\frac{\theta}{2}}{\frac{1}{2}}\right]_0^{\pi}$$

$$= 8a\left[\sin\frac{\theta}{2}\right]_0^{\pi}$$

$$= 8a\left(\sin\frac{\pi}{2} - \sin\frac{0}{2}\right) = 8a(1-0) = 8a.$$

So, length of the arc of upper half $= \frac{8a}{2} = 4a$ units.

Also, the length of the arc, from $\theta = 0$ to $\theta = \frac{\pi}{3}$ is



$$s = \int_0^{\pi/3} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \qquad \left[\because \text{At } A,\ \theta = 0 \text{ and at } B,\ \theta = \frac{\pi}{3}\right]$$

$$= \int_0^{\pi/3} \sqrt{(a(1+\cos\theta))^2 + (-a\sin\theta)^2}\, d\theta$$

$$= \int_0^{\pi/3} a\sqrt{1 + 2\cos\theta + \cos^2\theta + \sin^2\theta}\, d\theta$$

$$= \int_0^{\pi/3} a\sqrt{2(1+\cos\theta)}\, d\theta$$

$$= 2a\int_0^{\pi/3} \cos\frac{\theta}{2}\, d\theta$$

$$= 2a\left[\frac{\sin\frac{\theta}{2}}{\frac{1}{2}}\right]_0^{\pi/3}$$

$$= 4a\left[\sin\frac{\theta}{2}\right]_0^{\pi/3}$$

$$= 4a\left[\sin\frac{\pi}{3} - \sin\frac{0}{2}\right]$$

$$= 4a\sin\frac{\pi}{6}$$

$$= 4a \times \frac{1}{2}$$

$$= 2a = \frac{1}{2}(4a) = \frac{1}{2}(\text{length of upper half}).$$

$\therefore$ The arc of the upper half is bisected by $\theta = \frac{\pi}{3}$ units.

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Exercise 4.1

1. Find the arc length of the following curves as indicated.
   i. $y^2 + 2y = 2x + 1, -1 \le x \le 7$
   ii. $x^{2/3} + y^{2/3} = a^{2/3}$
   iii. $y = \frac{1}{3}(x^2 + 2)^{3/2}, 0 \le x \le 3$
   iv. $y^2 = 4ax$ to the points cut off by line $3y = 8x$
   v. length of parabola $y^2 = 4x$ between the intersecting points by $y = 2x$
   vi $y = \frac{3}{4}x^{2/3} - \frac{3}{8}x^{2/3} + 5, -1 \le x \le 8$
2. Show that the length of the arc of the semi-cubical parabola $ay^2 = x^3$ from the vertex to the point (a, a) is $\frac{a}{27}(13\sqrt{13} - 8)$.
3. Show that the length of the curve $y = \log\frac{e^x - 1}{e^x + 1}$ from $x = 1$ to $x = 2$ is $\log\left(e + \frac{1}{e}\right)$.
4. Show that the length of the arc of the parabola $y^2 = 4ax$ from the vertex to one extremity of the latus rectum is $a(\sqrt{2} + \log(1 + \sqrt{2}))$.
5. Show that the length of the curve $y = \log\sec x$ from $x = 0$ to $x = \frac{\pi}{3}$ is $\log(2 + \sqrt{3})$.
6. Find the length of the curve $y^2 = (2x - 1)^3$ cut off by the line $x = 4$.
7. In the case of a catenary $y = a\cosh\frac{x}{a}$, show that the length of the arc measured from the vertex $(0, a)$ to any point (x, y) is $a\sinh\frac{x}{a}$ and hence prove that $s^2 = y^2 - a^2$.

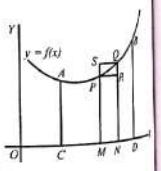
Answer

1. i. 9.29 units ii. $6a$ units iii. 12 units
   iv. $a\left(\log 2 + \frac{15}{16}\right)$ units v. $\log(1 + \sqrt{2}) + \sqrt{2}$ units vi $\frac{99}{8}$ units 6. $\frac{1022}{27}$

4.2 Area under a curve

Here we discuss the estimation of area bounded by curves.

Let $y = f(x)$ be the equation of the curve AB. The area $ACDB$ is bounded by the curve AB, the x-axis and the ordinates AC and BD at $x = a$ and $x = b$ respectively.



Let $P(x, y)$ be any point on the curve and $Q(x + \delta x, y + \delta y)$ a neighbouring point of P, so that δx and δy are increments in x and y. Let A be the area $APMC$ bounded by the curve, x-axis and fixed ordinate AC and variable ordinate PM then the elemental area δA = area $PMNQ$.

From the figure, we have,

Area of inner rectangle $MNRP \le$ area $MNQP \le$ Area of outer rectangle $MNQS$

i.e., $y\,\delta x \le \delta A \le (y + \delta y)\,\delta x$

or, $y \le \frac{\delta A}{\delta x} \le y + \delta y$

When $Q \to P$ then $\delta x \to 0$ and $\delta y \to 0$. Then, taking $\lim_{\delta x \to 0}$

$$y \le \lim_{\delta x \to 0}\frac{\delta A}{\delta x} \le \lim_{\delta y \to 0}(y + \delta y)$$

$$\left[\because \lim_{\delta x \to 0}\frac{\delta A}{\delta x} = \frac{dA}{dx}\right]$$

or, $y \le \frac{dA}{dx} \le y + 0$

or, $y \le \frac{dA}{dx} \le y$

So, $\frac{dA}{dx} = y = f(x)$

or, $dA = f(x)\,dx$.

Integrating, from $x = a$ to $x = b$,

$$A = \int_a^b f(x)\,dx.$$

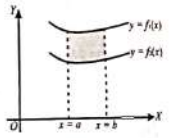
Thus, the area bounded by curve $y = f(x)$, ordinates $x = a$, $x = b$ and x-axis is $A = \int_a^b f(x)\,dx$.

Similarly, the area bounded by curve $x = f(y)$, between abscissa $y = c$, $y = d$ and y-axis is

$$A = \int_c^d f(y)\,dy.$$

4.2.1 Area between two curves

We now find the area between the curves $y = f_1(x)$ and $y = f_2(x)$ and the ordinate $x = a$ and $x = b$. Let $y = f_1(x)$ and $y = f_2(x)$ are two curves with graph as shown in figure.

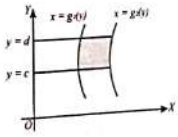


Then, the area bounded by curve $y = f_1(x)$, x-axis and ordinates $x = a$, $x = b$ is

$$A_1 = \int_a^b f_1(x)\,dx$$

and area bounded by curve $y = f_2(x)$, x-axis and ordinates $x = a$, $x = b$ is

$$A_2 = \int_a^b f_2(x)\,dx.$$



Hence, area between two curves $y = f_1(x)$ and $y = f_2(x)$ is

$$A = A_1 - A_2 = \int_a^b f_1(x)\,dx - \int_a^b f_2(x)\,dx = \int_a^b [f_1(x) - f_2(x)]\,dx.$$

Similarly, the area bounded by two curves $x = g_1(y)$ and $x = g_2(y)$ from abscissa $y = c$ to $y = d$.

$$A = \int_c^d g_1(y)\,dy - \int_c^d g_2(y)\,dy = \int_c^d [g_1(y) - g_2(y)]\,dy$$

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Example 6. Find the area enclosed by x-axis and the curve $y = 3x - 5x^2$.

Solution:

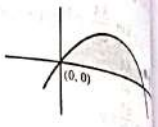
The given curve is $y = 3x - 5x^2$.

The curve intersect the x-axis at the point where $y = 0$

i.e. $3x - 5x^2 = 0$

or, $x(3 - 5x) = 0$

or, $x = 0$ and $\frac{3}{5}$.



The table for the graph is as follows:

x	0	0.33	0.66
y	0	0.45	0

$\therefore$ The curve intersects the x-axis at $(0, 0)$ and $\left(0, \frac{3}{5}\right)$. Now, the area enclosed by the curve and x-axis is given by

$$A = \int_0^{3/5} y\,dx$$

$$= \int_0^{3/5} (3x - 5x^2)\,dx$$

$$= \left[\frac{3x^2}{2} - \frac{5x^3}{3}\right]_0^{3/5}$$

$$= \frac{27}{50} - \frac{9}{25}$$

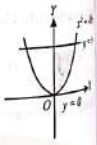
$$= \frac{9}{50} \text{ sq. units.}$$

$\therefore$ The required area is $\frac{9}{50}$ sq. units.

Example 7. Find the area bounded by the y-axis, the curve $x^2 = 4by$ and the line $y = b$.

Solution:

The given curve is $x^2 = 4by$. It is a parabola passing through origin and is symmetrical about y-axis. Now, the area is given by



$$A = \int_0^b x\,dy$$

$$= \int_0^b \sqrt{4by}\,dy$$

$$= 2\sqrt{b}\left[\frac{y^{3/2}}{3/2}\right]_0^b$$

$$= \frac{4\sqrt{b}}{3} b^{3/2} = \frac{4b^2}{3}$$

$\therefore$ The required area is $\frac{4b^2}{3}$ sq. units.

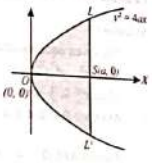
Example 8. Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum.

Solution:

The given equation of parabola is $y^2 = 4ax$ .... (i)

where $S(a, 0)$ be the focus and $O(0, 0)$ be the vertex of parabola (i)

Let LL' be the Latus rectum whose equation is $x = a$



Then the area of portion of upper half $OLSO$ $= \int_0^a y\,dx$ [$\because$ At O, $x = 0$ and at S, $x = a$]

$$= \int_0^a \sqrt{4ax}\,dx$$

$$= 2\sqrt{a}\int_0^a \sqrt{x}\,dx$$

$$= 2\sqrt{a}\left[\frac{x^{3/2}}{\frac{3}{2}}\right]_0^a$$

$$= \frac{4\sqrt{a}}{3}[x^{3/2}]_0^a$$

$$= \frac{4\sqrt{a}}{3}[a^{3/2} - 0] = \frac{4a^2}{3}$$

$\therefore$ Required area = Area of the portion $OLSL'O$

= 2 × area of portion $OLSO$

$= 2 \times \frac{4a^2}{3} = \frac{8a^2}{3}$ square units.

Example 9. Find the area bounded by the curve $y = x^3 - 12x$ and the x-axis.

Solution:

The given function is $y = x^3 - 12x$

For $y = 0$, $0 = x^3 - 12x$

or, $x(x^2 - 12) = 0$

or, $x = 0$ and $x = \pm\sqrt{12}$

$\therefore$ The table for the curve is as follows:

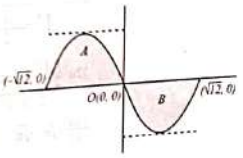
x	$-\sqrt{12}$	-1.732	0	1.732	$\sqrt{12}$
y	0	15.58	0	-15.58	0

Since the area of the region A and B are equal, by symmetry.

Total area $= 2\int_0^{\sqrt{12}} (x^3 - 12x)\,dx$

$$= 2\left[\frac{x^4}{4} - \frac{12x^2}{2}\right]_0^{\sqrt{12}}$$

$$= 2\left[\frac{x^4}{4} - 6x^2\right]_0^{\sqrt{12}}$$



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$$= 2\left[\frac{(\sqrt{12})^4}{4} - 6(\sqrt{12})^2\right]$$

$$= 2\left[\frac{144}{4} - 72\right]$$

$$= 2\left[\frac{144 - 288}{4}\right] = 72 \text{ sq. units.}$$

$\therefore$ The required area is 72 sq. units.

Example 10. Show that the area of the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ is $\frac{3}{8}\pi a^2$.

Solution:

The given equation of asteroid is $x^{2/3} + y^{2/3} = a^{2/3}$

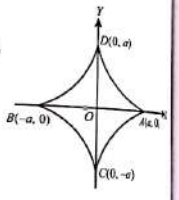
or $y = (a^{2/3} - x^{2/3})^{3/2}$

when $x = 0$, $y = \pm a$

$\therefore$ At O, $x = 0$ and at A, $x = a$.

$\therefore$ Area of ABCD $= 4 \times$ Area OAD $= 4\int_0^a y\,dx$

$$= 4\int_0^a (a^{2/3} - x^{2/3})^{3/2}\,dx \quad \ldots\ (i)$$



For integration, let $x = a\sin^3\theta$

Then, $dx = 3a\sin^2\theta\cos\theta\,d\theta$

If $x = 0$, $\theta = 0$

and when $x = a$, $\theta = \frac{\pi}{2}$

$\therefore$ Area of ABCD $= 4\int_0^{\pi/2} [(a^{2/3}) - (a\sin^3\theta)^{2/3}]^{3/2} \times 3a\sin^2\theta\cos\theta\,d\theta$

$$= 4a\int_0^{\pi/2} (1 - \sin^2\theta)^{3/2} \cdot 3a\sin^2\theta\cos\theta\,d\theta$$

$$= 12a^2\int_0^{\pi/2} \sin^2\theta\cos^4\theta\,d\theta$$

$$= \frac{12a^2\,\Gamma\left(\frac{2+1}{2}\right)\Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{2+4+2}{2}\right)}$$

$$= \frac{12a^2\,\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{2\,.\,3\,.\,2\,.\,1}$$

$$= \frac{12a^2}{32}\sqrt{\pi}\,.\,\sqrt{\pi} = \frac{3a^2\pi}{8} \text{ sq. units.}$$

Example 11. Find the area between the curves $x^2 = 4ay$ and $y^2 = 4ax$.

Solution:

The given equations of parabola

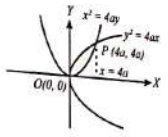
$x^2 = 4ay$

or, $y = \frac{x^2}{4a}$ .... (i)

and $y^2 = 4ax$ .... (ii)

or, $y = \sqrt{4ax}$

Solving equation (i) and (ii), we get



$$\left(\frac{x^2}{4a}\right)^2 = 4ax$$

or, $\frac{x^4}{16a^2} = 4ax$

or, $x^4 = 64a^3x$

or, $x^4 - 64a^3x = 0$

or, $x(x^3 - 64a^3) = 0$

or, $x = 0$ or $4a$

Thus, $x = 0$ at O and $x = 4a$ at P.

Then required area $= \int_0^{4a} (y_1 - y_2)\,dx$

$$= \int_0^{4a} \left(\sqrt{4ax} - \frac{x^2}{4a}\right) dx$$

$$= \int_0^{4a} 2\sqrt{a}\sqrt{x}\,dx - \int_0^{4a} \frac{x^2}{4a}\,dx$$

$$= 2\sqrt{a}\int_0^{4a} x^{1/2}dx - \frac{1}{4a}\int_0^{4a} x^2dx$$

$$= 2\sqrt{a}\left[\frac{x^{3/2}}{\frac{3}{2}}\right]_0^{4a} - \frac{1}{4a}\left[\frac{x^3}{3}\right]_0^{4a}$$

$$= \frac{2}{3}2\sqrt{a}\,(4a)^{3/2} - \frac{1}{4a}\left(\frac{(4a)^3}{3}\right)$$

$$= \frac{4}{3a}\sqrt{a}\,(2^2)^{3/2}.\,a^{3/2} - \frac{1}{4a}\cdot\frac{(4a)^3}{3}$$

$$= \frac{4}{3a}\sqrt{a}.\,2^3.\,a^{3/2} - \frac{(4a)^2}{3}$$

$$= \frac{32}{3}a^2 - \frac{16a^2}{3}$$

$$= \frac{32a^2 - 16a^2}{3}$$

$$= \frac{16a^2}{3} \text{ sq. units.}$$

$\therefore$ The required area is $\frac{16a^2}{3}$ sq. units.

Example 12. Find the area bounded by $x + y^2 = 0$ and $x + 3y^2 = 2$.

Solution:

The given functions are $x + y^2 = 0$ and $x + 3y^2 = 2$.

So both of them represent the parabola.

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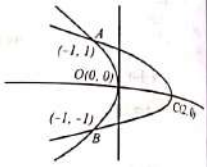
For $x + y^2 = 0$
or, $x = -y^2$
and for $x + 3y^2 = 2$
or, $x = -2 - 3y^2$

x	-1	0	-1
y	-1	0	1
x	-1	0	-1
y	-1	0	1

Clearly the intersection points are (-1, 1) and (-1, 1).



The required area of OACB $= \int_{-1}^{1} (x_1 - x_2)\, dy$

$= 2\int_{0}^{1} (-y^2 - 2 + 3y^2)\, dy$

$= 2\left[\frac{-y^3}{3} - 2y + \frac{3y^3}{3}\right]_0^1$

$= 2\left[-\frac{1}{3} - 2 + 1\right]$

$= -\frac{8}{3}$

$\therefore$ The required area is $\frac{8}{3}$ sq. units.

Example 13. Find the area bounded by the curves $y = x^4 - 2x^2$ and $y = 2x^2$.

Solution:

The given function is $y = y_1 = x^4 - 2x^2$ ....(i)

The table for the function is

x	-2	$-\sqrt{2}$	-1	0	1	2
y	8	0	-1	0	-1	8

and for $y = y_2 = 2x^2 \Rightarrow x^2 = \frac{1}{2}y$ ....(ii)

It is a parabola with vertex at (0, 0). Also

x	-2	-1	0	1	2
y	8	2	0	2	8

The intersection points of (i) and (ii) are $A(-2, 8)$, $O(0, 0)$ and $B(2, 8)$ and the figure is symmetrical with respect to y-axis.

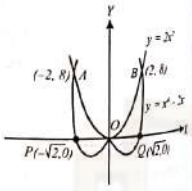
$\therefore$ A = 2 × Area of the right shaded part of y-axis.

$= 2 \times \int_0^2 (y_1 - y_2)\, dx$

$= 2x \int_0^2 [2x^2 - (x^4 - 2x^2)]\, dx$

$= 2\int_0^2 (4x^2 - x^4)\, dx$

$= 2\left[4\frac{x^3}{3} - \frac{x^5}{5}\right]_0^2$



$= 2\left[\left(\frac{32}{3} - \frac{32}{5}\right) - 0\right]$

$= 2 \times \frac{64}{15} = \frac{128}{15}$ sq. units

$\therefore$ Required area is $\frac{128}{15}$ sq. units.

Example 14. Find the area of the region and the circle $x^2 + y^2 = 4$ cut off by the line $x - 2y = -2$ in the first two quadrants.

Solution:

The given equation of circle is $x^2 + y^2 = 4$ ....(i)

So, $y = y_1 = \sqrt{4 - x^2}$.

It is a circle, centre at origin (0, 0) and radius 2.

The next equation is $x - 2y = -2$ ....(ii)

Which is a straight line.

x	-2	0
y	0	1

From (ii), we get

$x = 2y - 2 \Rightarrow y = \frac{x + 2}{2}$

Putting y in $x^2 + y^2 = 4$, we get

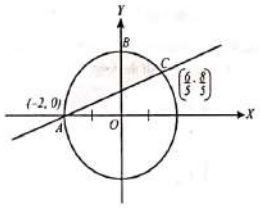
$(2y - 2)^2 + y^2 = 4$

$\Rightarrow 4y^2 - 8y + 4 + y^2 = 4$

$\Rightarrow 5y^2 - 8y = 0$

$\Rightarrow y(5y - 8) = 0$

$\therefore y = 0, \frac{8}{5}$



At $y = \frac{8}{5}$, $x = 2\left(\frac{8}{5}\right) - 2 = \frac{6}{5}$ and at $y = 0$, $x = -2$

$\therefore$ At point A, $x = -2$ and at point C, $x = \frac{6}{5}$.

Now, required area $= \int_a^b (y_1 - y_2)\, dx$

$= \int_{-2}^{6/5} \left(\sqrt{4 - x^2} - \frac{x + 2}{2}\right) dx$

$= \left[\frac{x\sqrt{4 - x^2}}{2} + \frac{4}{2}\sin^{-1}\frac{x}{2} - \frac{1}{2}\left(\frac{x^2}{2} + 2x\right)\right]_{-2}^{6/5}$

$= \left[\frac{6}{10}\left(4 - \frac{36}{25}\right)^{1/2} + 2\sin^{-1}\left(\frac{6}{10}\right) - \frac{1}{2}\left(\frac{36}{50} + \frac{12}{5}\right)\right] - \left[0 + 2\sin^{-1}(-1) - \frac{1}{2}(2 - 4)\right]$

$= \frac{24}{25} + 2\sin^{-1}\left(\frac{3}{5}\right) - 3.12 + \pi - 1$

$= 2\sin^{-1}(0.6) - 0.02$ sq. units

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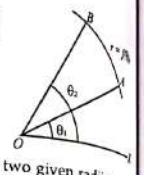
4.2.2 Area in polar coordinates

If $r = f(\theta)$, be a finite continuous and single valued function in the interval $\theta_1 < \theta < \theta_2$. Then the area A bounded by the curve $r = f(\theta)$ and two radius vectors from $\theta = \theta_1$ to $\theta = \theta_2$ is given by

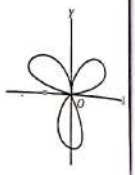


$$A = \frac{1}{2}\int_{\theta_1}^{\theta_2} r^2 d\theta = \frac{1}{2}\int_{\theta_1}^{\theta_2} [f(\theta)]^2 d\theta.$$

Note: The area bounded by the two curves $r_1 = f_1(\theta)$ and $r_2 = f_2(\theta)$, and two given radii vectors $\theta = \theta_1$ and $\theta = \theta_2$ is $\frac{1}{2}\int_{\theta_1}^{\theta_2} (r_1^2 - r_2^2)\, d\theta$.

Example 15. Find the area of loop of the curve $r = a\sin 3\theta$.

Solution:



We have, $r = a \sin 3\theta$(a three leaved role)

The curve consists of three loops with equal area. Also, for the loop in first quadrant

if $r = 0$, then $\sin 3\theta = 0 \Rightarrow \theta = 0$ and

if $r = a$, then $\sin 3\theta = 1 \Rightarrow \sin 3\theta = \sin\frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{6}$.

One loop is obtained when θ increases from 0° to 60°.

$\therefore$ A = area of half the loop

$$= \frac{1}{2}\int_0^{\pi/6} r^2 d\theta$$

$$= \frac{1}{2}\int_0^{\pi/6} a^2 \sin^2 3\theta d\theta$$

Put $t = 3\theta$

or, $\frac{dt}{d\theta} = 3$

or, $d\theta = \frac{dt}{3}$

when $\theta = 0$, $t = 0$ and when $\theta = \frac{\pi}{6}$, then, $t = \frac{\pi}{2}$

$\therefore$ $A = \frac{1}{2}\int_0^{\pi/2} \frac{1}{3} a^2 \sin^2 t\, dt$

$$= \frac{a^2}{6}\int_0^{\pi/2} \frac{1 - \cos 2t}{2} dt$$

$$= \frac{a^2}{12} \cdot \frac{\pi}{2} = \frac{\pi a^2}{24}$$

$\therefore$ Area of loop = $2 \cdot \frac{\pi a^2}{24} = \frac{\pi a^2}{12}$ sq. units.

Exercise 4.2

Find the area bounded by the given curves.

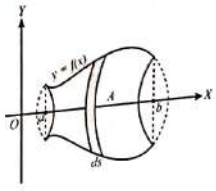
1. i. the circle $x^2 + y^2 = a^2$ ii. the elipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
2. i. the curve $y^2 = 4x$ and line $y = x$
   ii. the curve $y = \sin x$ and x-axis between $x = 0$ and $x = 2\pi$
   iii. the area bounded by $y = x^2$ between $x = 1$ and 3
   iv. the area bounded by x-axis and $y = 2x - x^2$
   v. the curve $y = 2 - x^2$ and $y = -x$
   vi. the curve $x = y^2$ and $x = -2y^2 + 3$
   vii. the curve $y^2 - 4x = 4$ and the line $4x - y = 16$
   viii. the curve $y = 5x - x^2 - 6$ and the x-axis
3. The equation of the curves are given below. Find the area between them.
   i. $x^2 = 4y$ and line $n = 4y - 2$ ii. $y = \sec^2 x$, $y = \sin x$ for $x = \left[0 \text{ to } \frac{\pi}{4}\right]$
   iii. $x = y^3$ and $x = y^2$ iv. $y = \sin\frac{\pi x}{2}$ and $y = x$
   v. $y = \sqrt{4 - x}$ in first quadrant vi. $x^2 = 4y$ and $y = |x|$
   vii. $x - y^3 = 0$ and $x - y = 0$ viii. $y = \sin x$, $y = \cos x$ in first quadrant
   ix. y-axis and $x = y^2 - y^3$ x. curve $y = x^2 + 1$ and line $y = -x + 3$
   xi. inside of $x^2 + y^2 = 1$ and outside of $y^2 = 1 - x$

Answer

1. i. πa^2 ii. πab
2. i. $\frac{8}{3}$ ii. 4 iii. $\frac{26}{3}$ sq. units iv. $\frac{4}{3}$
   v. $\frac{9}{2}$ vi. 4 vii. $\frac{243}{8}$ viii. $\frac{1}{6}$
3. i. $\frac{9}{8}$ iii. $\frac{1}{\sqrt{2}}$ iii. $\frac{1}{12}$ iv. $\frac{4 - \pi}{\pi}$
   v. $\frac{16}{3}$ vi. $\frac{16}{3}$ vii. $\frac{1}{2}$ viii. $\sqrt{2} - 1$
   ix. $\frac{1}{12}$ x. $\frac{9}{2}$ xi. $\frac{4}{3} + \frac{\pi}{2}$

4.3 Area of surface of revolution

When a curve is rotated about a line, then it forms a surface determined by $y = f(x)$. We now want to estimate these area of surface of revolution as shown in adjoining figure.



Using the similar concept as in the case of length of curve, the surface area of revolution A, formed by revolving $y = f(x)$, $a \le x \le b$ about x-axis is

$$A = \int_a^b 2\pi f(x)\, ds = \int_a^b 2\pi y \frac{ds}{dx} dx = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

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Similarly, if the curve $x = f(y)$, $c \le y \le d$ is revolved about y-axis, then

$$A = \int_c^d 2\pi f(y)\sqrt{1+[f'(y)]^2}\,dy = \int_c^d 2\pi x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy.$$

4.3.1 Area for parametric curves

Further, if the equation of the curve is in parametric form, the curved surface area S of the solid generated by the revolution about x-axis is given by

$$S = \int_{t_1}^{t_2} 2\pi y\frac{ds}{dt}dt = \int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt.$$

Similarly, if the arc is revolved around y-axis, then

$$S = \int_{t_1}^{t_2} 2\pi x\frac{ds}{dt}dt = \int_{t_1}^{t_2} 2\pi x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt.$$

For polar form

If the equation of the curve is in polar form i.e., $r = f(\theta)$ then the curved surface area S of the solid of revolution is give by

$$S = \int 2\pi y ds = \int_\alpha^\beta 2\pi y\frac{ds}{d\theta}d\theta = \int_\alpha^\beta 2\pi r\sin\theta\sqrt{r^2+\left(\frac{dr}{d\theta}\right)}\,d\theta$$

where, radii vectors $\theta = \alpha$ and $\theta = \beta$ about initial line.

Similarly, the area bounded by the curve $r = f(\theta)$, the radii of vector $\theta = \alpha$ and $\theta = \beta$ is revolved about the line $\theta = \frac{\pi}{2}$, the surface of revolution S is given by

$$S = \int 2\pi x ds$$

$$= \int_\alpha^\beta 2\pi x\frac{ds}{d\theta}d\theta$$

$$= \int_\alpha^\beta 2\pi r\cos\theta\sqrt{r^2+\left(\frac{dr}{d\theta}\right)}\,d\theta.$$

Example 16. Find the area of the surface generated by revolving the loop of the curve $3ay^2 = x(x-a)^2$ about x-axis.

Solution:

The given equation of curve is $3ay^2 = x(x-a)^2$ .... (i)

or, $6ay\frac{dy}{dx} = (x-a)^2 + 2x(x-a)$

$$\frac{dy}{dx} = \frac{(x-a)\{(x-a)+2x\}}{6ay}$$

or, $$\frac{dy}{dx} = \frac{(x-a)(3x-a)}{6ay}$$

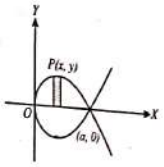
$\therefore$ Now the surface of revolution is given by

$$S = \int 2\pi y ds = 2\pi\int y\frac{ds}{dx}dx = 2\pi\int y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \quad \ldots.(ii)$$

Now, for $3ay^2 = x(x-a)$

If $x = 0, y = 0$

and if $x = a, y = 0$



$\therefore$ From (ii), $S = 2\pi\int_0^a y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

$$= 2\pi\int_0^a y\sqrt{1+\left(\frac{(x-a)(3x-a)}{6ay}\right)^2}\,dx$$

$$= 2\pi\int_0^a y\sqrt{1+\frac{(x-a)^2(3x-a)^2}{36a^2y^2}}\,dx$$

$$= 2\pi\int_0^a \sqrt{y^2+\frac{(x-a)^2(3x-a)^2}{36a^2}}\,dx$$

$$= 2\pi\int_0^a \sqrt{\frac{x(x-a)^2}{3a}+\frac{(x-a)^2(3x-a)^2}{36a^2}}\,dx \qquad \left[\because \text{From (i) } y^2 = \frac{x(x-a)^2}{3a}\right]$$

$$= 2\pi\int_0^a \frac{(x-a)}{6a}\sqrt{12ax+(3x-a)^2}\,dx$$

$$= \frac{2\pi}{6a}\int_0^a (x-a)(3x+a)dx$$

$$= \frac{\pi}{3a}\int_0^a (3x^2-2ax-a^2)\,dx$$

$$= \frac{\pi}{3a}(x^3-ax^2-a^2x)_0^a = -\frac{\pi a^2}{3}.$$

$\therefore$ Required surface area $= \frac{\pi a^2}{3}$ (Numerically).

Example 17. Find the area of surface generated by rotating the curve $y = e^x$, $0 \le x \le 1$ about x-axis.

Solution

The given curve is $y = e^x$.

$\therefore$ $\frac{dy}{dx} = e^x$

Now the surface of revolution is

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$$S = \int 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$= 2\pi \int_0^1 e^x \sqrt{1+e^{2x}}\,dx$$

Let, $u = e^x$, then $du = e^x\,dx$.

If $x = 0$ then $u = 1$ and if $x = 1$, then $u = e$.

$$\therefore \quad S = 2\pi \int_1^e \sqrt{1+u^2}\,du$$

Again, let $u = \tan\theta \Rightarrow du = \sec^2\theta\,d\theta$

If $u = 1$, $\theta = \frac{\pi}{4}$ and if $u = e$, $\theta = \tan^{-1}e = (\alpha$ say)

$$\therefore \quad S = 2\pi \int_{\pi/4}^{\alpha} \sec^3\theta\,d\theta$$

$$= 2\pi \cdot \frac{1}{2}[\sec\theta\tan\theta + \log(\sec\theta + \tan\theta)]_{\pi/4}^{\alpha}$$

$$= \pi[(\sec\alpha\tan\alpha + \log(\sec\alpha + \tan\alpha) - \sqrt{2} - \log(\sqrt{2}+1)]$$

Since $\tan\alpha = e$, $\sec^2\alpha = 1 + \tan^2\alpha = 1 + e^2$

$$\therefore \quad S = \pi[e\sqrt{1+e^2} + \log(e+\sqrt{1+e^2}) - \sqrt{2}\,\log(\sqrt{2}+1)].$$

Example 18. Find the area of the surface obtained by rotating the curve $y = \frac{x^2}{4} - \frac{\log x}{2}$, $1 \le x \le 2$ about x-axis.

Solution

The given curve is $y = \frac{x^2}{4} - \frac{\log x}{2} \Rightarrow \frac{dy}{dx} = \frac{x}{2} - \frac{1}{2x}$

Now, the surface of revolution of the curve is $1 \le x \le 2$ about y-axis.

$$S = \int_1^2 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$= \int_1^2 2\pi\left(\frac{x^2}{4} - \frac{\log x}{2}\right)\sqrt{1+\left(\frac{x}{2} - \frac{1}{2x}\right)^2}\,dx$$

$$= 2\pi \int_1^2 \left(\frac{x^2}{4} - \frac{\log x}{2}\right)\sqrt{\left(\frac{x}{2} + \frac{1}{2x}\right)^2}\,dx$$

$$= 2\pi \int_1^2 \left(\frac{x^2}{4} - \frac{\log x}{2}\right)\left(\frac{x}{2} + \frac{1}{2x}\right)dx$$

$$= \frac{\pi}{4}\int_1^2 (x^2 - 2\log x)\left(\frac{x^2+1}{x}\right)dx$$

$$= \frac{\pi}{4}\left[\int_1^2 x^2\left(\frac{x^2+1}{x}\right)dx\right] - \frac{\pi}{4}\left[\int_1^2 2\log x\left(\frac{x^2+1}{x}\right)dx\right]$$

$$= \frac{\pi}{4}\left[\int_1^2 x(x^2+1)\,dx\right] - \frac{\pi}{2}\left[\int_1^2 \log x\left(\frac{x^2+1}{x}\right)dx\right]$$

$$= \frac{\pi}{4}\left[\frac{x^4}{4} + \frac{x^2}{2}\right]_1^2 - \frac{\pi}{2}\left[\frac{x^2}{2}\left(\log x - \frac{1}{2}\right) + \frac{|\log x|^2}{2}\right]_1^2$$

$$= \frac{\pi}{4}\left[\frac{2^4}{4} - \frac{1^4}{4} + \frac{2^2}{2} - \frac{1^2}{2}\right] - \frac{\pi}{2}\left[\frac{2^2}{2}\left(\log 2 - \frac{1}{2}\right) + \frac{(\log 2)^2}{2} - \frac{1^2}{2}\left(\log 1 - \frac{1}{2}\right)\right]$$

$$= \pi\left(\frac{27}{16} - \log 2 - \frac{(\log 2)^2}{4}\right)$$ sq. units.

Example 19. Find the area of revolution obtained from the curve $x = \sin^2 t$, $y = \cos^2 t$ for $0 \le t \le \frac{\pi}{2}$ about y-axis.

Solution:

This curve represents a circle with centre at origin and radius 1.

We have, $x = \sin^2 t \Rightarrow \frac{dx}{dt} = 2\sin t\cos t$

and $y = \cos^2 t \Rightarrow \frac{dy}{dt} = -2\cos t\sin t$

Now the surface of revolution in parametric form is

$$S = \int_{t_1}^{t_2} 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$, where $t_1 = 0$ and $t_2 = \frac{\pi}{2}$.

$$= \int_0^{\pi/2} 2\pi \sin^2 t \sqrt{(2\sin t\cos t)^2 + (-2\cos t\sin t)^2}\,dt$$

$$= 2\pi \int_0^{\pi/2} \sin^2 t \sqrt{8\sin^2 t\cos^2 t}\,dt$$

$$= 4\sqrt{2}\,\pi \int_0^{\pi/2} \sin^3 t\cos t\,dt$$

$$= 4\sqrt{2}\,\pi \cdot \frac{1}{4}[\sin^4 t]_0^{\pi/2}$$

$$= \pi\sqrt{2}\,(1-0) = \sqrt{2}\pi$$ sq. units.

Exercise 4.3

1. Find the area of the surface obtained by rotating the curve about x-axis.
   i. $y = x^3$, $0 \le x \le 2$ ii. $y = \sqrt{x}$, $4 \le x \le 9$
   iii. $y = \cosh x$, $0 \le x \le 1$ iv. $y = \frac{x^3}{6} + \frac{1}{2x}$, $\frac{1}{2} \le x \le 1$
2. Find the area of the surface obtained by rotating about y-axis.
   i. $y = \sqrt[3]{x}$, $1 \le y \le 2$ ii. $x = \sqrt{a^2 - y^2}$, $0 \le y \le \frac{a}{2}$
3. The curve $y = \sqrt{4 - x^2}$, $-1 \le x \le 1$ is an arc of the circle $x^2 + y^2 = 4$. Find the area of the surface obtained by rotating this arc about x-axis.
4. If $a > 0$, find the area of the surface generated by rotating the loop of the curve $3ay^2 = x(a - x)^2$ about y-axis.

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5. Find the area of the sphere generated by revolution of the circle $x^2 + y^2 = r^2$ about x-axis.
6. Find the area of the surface generated by rotating the curve $y = \frac{x^2}{2} + \frac{1}{2}$ about y-axis for $0 \le x \le 1$.
7. Find the area of the surface generated by rotating the portion of the curve $y = \frac{1}{3}(x^2 + 2)^{3/2}$ between $x = 0$ and $x = 3$ and y-axis.
8. Find the area of the surface generated by rotating curve $x = \cos t$, $y = 1 + \sin t$, $0 \le t \le 2\pi$ about x-axis.
9. Find the area of the surface generated by rotating the hypocycloid $x = a\cos^3\theta$, $y = a\sin^3\theta$ about x-axis.
10. The loop of the curve $9x^2 = y(3 - y)^2$ is rotated about x-axis. Find the surface area generated.
11. Find the area of the surface obtained by rotating the curve $y = \cos x$, $0 \le x \le \frac{\pi}{3}$ about x-axis.
12. Find the surface area of solid generated by revolving the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ about x-axis. (**Hint:** *The parametric equation for asteroid is* $x = a\cos^3\theta$, $y = a\sin^3\theta$)
13. Show that the area of the curved surface generated when one loop of the curve $x^2(a^2 - x^2) = 8a^2y^2$ is revolved about x-axis is $\frac{\pi a^2}{4}$.

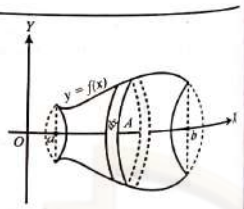
Answer

1. i. $\frac{\pi}{27}(145\sqrt{145} - 1)$ ii. $\frac{\pi}{6}(37\sqrt{37} - 17\sqrt{17})$ iii. $\pi\left[1 + \frac{1}{4}(e^2 - e^{-2})\right]$ iv. $\frac{21\pi}{2}$
2. i. $\frac{\pi}{27}(145\sqrt{145} - 10\sqrt{10})$ ii. πa^2 3. 8π
4. $\frac{56\sqrt{3}a^2\pi}{45}$ 5. $4\pi r^2$ 6. $\frac{2\pi}{3}(2\sqrt{2} - 1)$
7. $\frac{99\pi}{2}$ 8. $4\pi^2$ 9. $\frac{12\pi a^2}{5}$ 10. $\frac{56}{5}\sqrt{3}\,\pi$
11. $\pi\left(\frac{\sqrt{21}}{4} + \log\frac{\sqrt{7} + \sqrt{3}}{2}\right)$ 12. $\frac{12}{5}\pi a^2$

We finally discuss the estimation of volume of solid of revolution.

4.4 Volume of solid of revolution

Let S be a solid with volume V and solid lies between $x = a$ and $x = b$. Suppose $A(x)$ is the area of cross-section obtained by slicing the solid through x-axis with thickness of slice Δx. Then $A(x)\,\Delta x$ estimates the volume of a slice of width Δx. Then adding the volumes of all such slices from a to b gives the total volume. Hence, the volume of solid is



$$V = \int_a^b A(x)\,dx.$$

4.4.1 Solid of revolution

If a plane region is revolved about x-axis then a solid will be generated, for example: when a semicircle is revolved about x-axis then a sphere is formed. The solid so generated is called the solid of revolution. To find the volume of solid of revolution, two cases for solids are common. They are:

i. **Disk formula:** This method is applied for the solid of revolution of region bounded by single curve.
   a. The volume of solid revolution about x-axis is given by
   $$V = \int_a^b \pi[R(x)]^2\,dx, \text{ where } R(x) \text{ is the radius.}$$
   b. The volume of solid revolution about y-axis is given by
   $$V = \int_c^d \pi[R(y)]^2\,dy, \text{ where, } R(y) \text{ is the radius.}$$

ii. **Washer formula:** This method is applied for the solid of revolution of the region bounded by two curves.
   a. The volume of the solid when rotated about x-axis is given by
   $$V = \int_a^b \pi[[R(x)]^2 - [r(x)]^2]\,dx$$
   when $R(x)$ is outer radius and $r(x)$ is inner radius.
   b. The volume of the solid when rotated about y-axis is given by
   $$V = \int_c^d \pi[[R(y)]^2 - [r(y)]^2]\,dy$$
   when $R(y)$ is outer radius and $r(y)$ is inner radius.

We follow the following outlines to estimate the volume.
i. Sketch the curves of the given functions.
ii. Identify the radius, both $R(x)$ and $r(x)$ or $R(y)$ and $r(y)$.
iii. Find the limits of integration.
iv. Evaluate the volume by the suitable formula.

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Example 20. The circle $x^2 + y^2 = a^2$ revolves round the x- axis; show that the volume of the whole sphere generated is $\frac{4}{3}\pi a^3$.

Solution:

The given equation of circle is

$x^2 + y^2 = a^2$

or, $y^2 = a^2 - x^2$ .... (i)

Let O be the centre and a be the radius of the circle $ABA'B'$.

The sphere is formed by the revolution of the area of circle (i) about x-axis.

The volume of sphere (V)

= volume of the solid formed by the revolution of the area of circle (i) about x-axis

= volume of the solid formed by the revolution of the area of the semi-circle ABA'

= 2 × volume of the solid formed by the revolution of the area of the quadrant $ABOA$ about x-axis

$$= 2\int_0^a \pi y^2 dx$$

$$= 2\pi\int_0^a (a^2 - x^2)\, dx \qquad [\because \text{Using (i)}]$$

$$= 2\pi\left[a^2x - \frac{x^3}{3}\right]_0^a$$

$$= 2\pi\left[a^3 - \frac{a^3}{3}\right]$$

$$= 2\pi\left[\frac{2a^3}{3}\right]$$

$$= \frac{4\pi a^3}{3} \text{ cubic unit.}$$

$\therefore$ Volume $= \frac{4\pi a^3}{3}$ cubic unit.

Example 21. Prove that the volume of the ellipsoid formed by the revolution of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$ about the x-axis is $\frac{4\pi ab^2}{3}$.

Solution:

The given equation of ellipse is

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

or, $\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$

or, $y^2 = b^2\left(1 - \frac{x^2}{a^2}\right)$

Here, $O(0, 0)$ is the centre of the ellipse.

The volume of the solid formed by the revolution of the ellipse about the x-axis

= volume formed by the revolution of the portion ABA'

= 2 (volume formed by the revolution of the portion AOB) about x-axis.

$$= 2\pi\int_0^a y^2 dx$$

$$= 2\pi\int_0^a b^2\left(1 - \frac{x^2}{a^2}\right) dx$$

$$= 2\pi b^2\left[x - \frac{x^3}{3a^2}\right]_0^a$$

$$= 2\pi b^2\left(a - \frac{a}{3}\right)$$

$$= \frac{4\pi ab^2}{3}$$

$\therefore$ Volume $= \frac{4\pi ab^2}{3}$ cubic unit.

Example 22. The area bounded by the parabola $y^2 = 4px$ and the line $x = a$ revolves about the x-axis. Find the volume generated.

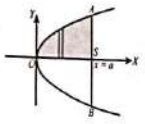
Solution:

The given equation of parabola is

$y^2 = 4px$ .... (i)

Let the line $x = a$ meets the parabola at A and B and the x-axis at S.

Then the volume generated by the area $AOBSA$ is



$$V = \int_0^a \pi y^2 dx$$

$$= \pi\int_0^a y^2 dx$$

$$= \pi\int_0^a 4px\, dx$$

$$= 4p\pi\int_0^a x\, dx$$

$$= 4p\pi\left[\frac{x^2}{2}\right]_0^a$$

$$= 4p\pi\left[\frac{a^2}{2} - \frac{0^2}{2}\right]$$

$$= 4\pi p\frac{a^2}{2}$$

$$= 2\pi pa^2 \text{ cubic units.}$$

$\therefore$ Volume $= 2\pi pa^2$ cubic unit

Example 23. Find the volume of the solid in the region in the first quadrant evolved by the parabola $y = x^2$, y-axis and the line $y = 1$ revolving about the line $x = \frac{3}{2}$.

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Solution:

The given parabola is $y = x^2$.

x	-2	-1	0	1	2
y	4	1	0	1	4

Now from figure, limit of integration is $y = 0$ to $y = 1$.

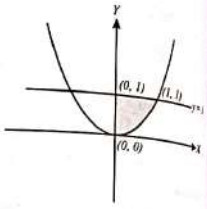
$$\therefore \quad V = \pi \int_0^1 \left[\left\{ R(y) - \frac{3}{2} \right\}^2 - \left\{ r(y) - \frac{3}{2} \right\}^2 \right] dy$$

$$= \pi \int_0^1 \left[\left(0 - \frac{3}{2}\right)^2 - \left(\sqrt{y} - \frac{3}{2}\right)^2 \right]$$

$$= \pi \int_0^1 \left(\frac{9}{4} - y + 3\sqrt{y} - \frac{9}{4} \right)$$

$$= \pi \left[-\frac{y^2}{2} + \frac{3y^{3/2}}{3/2} \right]_0^1$$

$$= \pi \left[-\frac{1}{2} + 2 \right] = \frac{3\pi}{2}$$



$\therefore$ volume $V = \frac{3\pi}{2}$ cubic units.

Example 24. Find the volume of the solid generated by revolution of the region bounded by $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$.

Solution:

The given curve is $y = \sqrt{x}$ and passes through the following points.

x	0	1	4
y	0	1	± 2

The revolution is about $y = 1$.

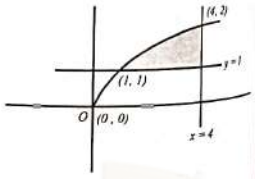
$\therefore$ Radius $R(x) = \sqrt{x} - 1$

$$\therefore \quad V = \int_a^b \pi [R(x)]^2 \, dx$$

$$= \int_1^4 \pi (\sqrt{x} - 1)^2 \, dx$$

$$= \pi \int_1^4 (x - 2\sqrt{x} + 1) \, dx$$

$$= \pi \left[\frac{x^2}{2} - \frac{4}{3} x^{3/2} + x \right]_1^4 = \frac{7\pi}{6}$$



$\therefore$ Volume is $\frac{7\pi}{6}$ cubic units.

Example 25. Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2 + 1$ and the line $y = x + 3$ about x-axis.

Solution:

The given equation of parabola is $y = x^2 + 1$.

The points on its locus are

x	-2	-1	0	1	2
y	5	2	1	2	5

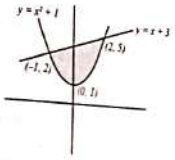
and the line is $y = x + 3$

x	-2	-1	0	1	2
y	1	2	3	4	5

The points of intersection are $(-1, 2)$ and $(2, 5)$.

Now the outer radius is $R(x) = x + 3$

and inner radius is $r(x) = x^2 + 1$



$$\therefore \quad V = \pi \int_{-1}^{2} [\{R(x)\}^2 - \{r(x)\}^2] \, dx$$

$$= \pi \int_{-1}^{2} [(x + 3)^2 - (x^2 + 1)^2] \, dx$$

$$= \pi \int_{-1}^{2} (x^2 + 6x + 9 - x^4 - 2x^2 - 1) \, dx$$

$$= \pi \int_{-1}^{2} (-x^4 - x^2 + 6x + 8) \, dx$$

$$= \pi \left[\frac{-x^5}{5} - \frac{x^3}{3} + 3x^2 + 8x \right]_{-1}^{2}$$

$$= \frac{117\pi}{5}$$

$\therefore$ Volume $= \frac{117\pi}{5}$ cubic units.

Exercise 4.4

1. Find the volume of solids generated by revolving the regions bounded by the lines and the curves about x-axis.

   i. $y = x^3$, $y = 0$, $x = 2$

   ii. $y = \sqrt{\cos x}$, $0 \le x \le \frac{\pi}{2}$, $y = 0$, $x = 0$

   iii. $y = 2\sqrt{x}$, $y = 2$, $x = 0$

   iv. $y = \sec x$, $y = \sqrt{2}$, $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$

   v. $y = 5x - x^2$, $x = 0$, $x = 5$

2. Find the volume of solid generated by revolving the regions bounded by the lines and the curves about y-axis.

   i. y-axis and curve $x = \frac{2}{y}$, $1 \le y \le 4$

   ii. $x = y^{3/2}$, $x = 0$, $y = 2$

   iii. $x = \sqrt{2 \sin 2y}$, $0 \le y \le \frac{\pi}{2}$, $x = 0$

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3. i. Find the volume of the solid in the region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ revolved about x-axis.

ii. Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2$ and the line $y = 1$ about the line $y = -1$.

iii. Find the volume of the solid generated by revolving the region bounded between $x^2 = 4y$ and $y = |x|$ about $y = -2$.

iv. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 2$, $x = 0$ about the line $y = 2$.

4. i. Find the volume of the solid generated by revolving the region enclosed by triangle with vertices (1, 0), (2, 1) and (1, 1) about y-axis.

ii. The region in the first quadrant bounded by the parabola $y = x^2$ and the line $y = x$ is revolved about y-axis. Find the volume of the solid thus generated.

iii. Find the volume of the solid generated by revolving the region in the first quadrant bounded on the left by the circle $x^2 + y^2 = 3$ and on the right by the line $x = \sqrt{3}$ and above by the line $y = \sqrt{3}$ about y-axis.

iv. Find the volume of the solid generated by revolving the region bounded on the left by the line $x = 4$ and on the right by the circle $x^2 + y^2 = 25$.

v. Find the volume of the solid in the region in the first quadrant bounded by the curve $y = x^2$ below by the x-axis and on the right by the line $x = 1$ about the line $x = -1$.

vi. Find the volume of the solid generated by revolving the region by the parabola $y = 3x^2$ and the line $y = 3x$ in first quadrant about y-axis.

5. Show that the volume of the solid generated by revolving the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ about x-axis is $\frac{32}{105}\pi a^3$.

Answer

1. i. $\frac{128\pi}{7}$ ii. π iii. 2π iv. $\pi(\pi - 2)$ v. $\frac{625\pi}{6}$
2. i. 3π ii. 4π iii. 2π
3. i. $\frac{117\pi}{5}$ ii. $\frac{64\pi}{15}$ iii. $\frac{192\pi}{5}$ iv. $\frac{8\pi}{3}$
4. i. $\frac{4\pi}{3}$ ii. $\frac{\pi}{6}$ iii. $\pi\sqrt{3}$ iv. 36π v. $\frac{7\pi}{6}$ vi. $\frac{\pi}{2}$

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Multiple Choice Questions

1. The area of quadrant of the ellipse $16x^2 + 9y^2 = 144$ is
   a. 3π b. 20π
   c. 10π d. $\frac{20}{\pi}$
2. The area between the curve $y = x^2$ and the line $y = 2x$ is
   a. 4 b. 2
   c. 4/3 d. 3/5
3. The area of a hyperbola $xy = c^2$ bounded by the x– axis and the ordinates $x = a$ and $x = b$ is given by
   a. $c \log(ab)$ b. $c^2 \log \frac{a}{b}$
   c. $\frac{c^2 a}{b}$ d. $c^2 \log \frac{b}{a}$

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4. The area bounded by the curve $y = x^2$, two ordinates $x = 0$ and $x = 2$ and x-axis is equal to
   a. 8/3 b. 16/3
   c. 2 d. 4

5. The area bounded by the curve $r = f(\theta)$ and the radii vectors $\theta = \alpha$ and $\theta = \beta$ is given by the definite integral
   a. $2\int_\alpha^\beta r^2 d\theta$ b. $\frac{1}{2}\int_\alpha^\beta r^2 d\theta$
   c. $\int_\alpha^\beta r d\theta$ d. $\frac{1}{2}\int_\alpha^\beta \frac{d\theta}{r^2}$

6. The area bounded by the parabola $y^2 = x$, straight line $y = 4$ and y-axis is
   a. $\frac{64}{3}$ b. $\frac{16}{3}$
   c. $7\sqrt{3}$ d. none of these

7. The area bounded by the parabola $y^2 = 4x$ and latus ratum is
   a. 3 b. $\frac{3}{4}$
   c. $\frac{8}{3}$ d. none of these

8. The area of the region bounded by the function $f(x) = x^3$, the x-axis and the lines $x = -1$ and $x = 1$ is
   a. $\frac{1}{4}$ b. $\frac{1}{3}$
   c. $\frac{1}{8}$ d. $\frac{1}{2}$

9. The area enclosed by $y = x$, $y = 1$, $y = 3$ and the y-axis is
   a. 2 b. $\frac{9}{2}$
   c. 4 d. $\frac{3}{2}$

10. The area enclosed by $y = \cos x$, $\frac{-\pi}{2} \le x \le \frac{\pi}{2}$ and the x-axis is
   a. 1 b. 4
   c. 2 d. π

11. The area bounded by the parabola $y^2 = 4ax$ and its latus rectum is
   a. $\frac{4}{3}a^2$ b. $\frac{8}{3}a^2$
   c. $\frac{16}{3}a^2$ d. $\frac{32}{3}a^2$

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12. Area bounded by $y^2 = 4x$, $x^2 = 4y$ is
   a. $\frac{4}{3}$ b. $\frac{8}{3}$
   c. $\frac{16}{3}$ d. $\frac{32}{3}$

13. Smaller of the area bounded by $x^2 + y^2 = 4$, $x + y = 2$ is
   a. $\pi - 2$ b. $4\pi - 2$
   c. $\pi - 4$ d. $2\pi - 4$

14. Which of the following gives the length of the curve $r = f(\theta)$ between two points for which $\theta = \theta_1$ and $\theta = \theta_2$?
   a. $\int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$ b. $\int_{r_1}^{r_2} \sqrt{1 + \left(r\frac{d\theta}{dr}\right)^2}\, dr$
   c. $\int_{\theta_1}^{\theta_2} r^2 \sin^2\theta \frac{d}{d\theta}(r\sin\theta)$ d. $\pi\int_{\theta_1}^{\theta_2} r^3 \sin\theta\, d\theta$

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15. The arc length of a smooth Cartesian curve $y = f(x)$ from $x = a$ to $x = b$ in

a. $\int_a^b f(x)\,dx$

b. $(b-a)^2 + (f(b) - f(a))^2$

c. $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

d. $\int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy$

16. The area of the surface generated by the revolution of the curve $y = f(x)$ about x-axis between the abscissa $x = x_1$ and $x = x_2$ is

a. $2\pi \int_{x_1}^{x_2} y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

b. $2\pi \int_{x_1}^{x_2} y \sqrt{a + \left(\frac{dx}{dy}\right)^2}\,dx$

c. $2\pi \int_{x_1}^{x_2} x \left[1 + \left(\frac{dx}{dy}\right)^2\right] dx$

d. $2\pi \int_{x_1}^{x_2} y \left[1 + \left(\frac{dy}{dx}\right)^2\right] dx$

17. The area of the surface generated by the revolution of the curve $x = f(y)$ about y-axis bounded by $y = y_1$ and $y = y_2$ is

a. $2\pi \int_{y_1}^{y_2} x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy$

b. $2\pi \int_{y_1}^{y_2} x \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dy$

c. $2\pi \int_{y_1}^{y_2} x \left[1 + \left(\frac{dy}{dx}\right)^2\right] dy$

d. $2\pi \int_{y_1}^{y_2} x \left[1 + \left(\frac{dx}{dy}\right)^2\right] dy$

18. The volume of the solid of revolution formed by revolving the curve $y = f(x)$ about the x-axis and bounded by the sections $x = x_1$ and $x = x_2$ is given by

a. $2\pi \int_{x_1}^{x_2} y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

b. $\pi \int_{x_1}^{x_2} y^2 x$

c. $\pi \int_{x_1}^{x_2} y^3 d\theta$

d. $\pi \int_{x_1}^{x_2} y^{3/2}\,d\theta$

19. The volume of the solid generated by the revolution of the parabola $y^2 = 8x$ about the x-axis between the ordinates $x = 0$ and $x = 5$ is

a. 100π

b. 200π

c. 300π

d. 400π

20. What is the volume generated when the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is revolved about its minor axis?

a. $4ab$ cubic units

b. $\frac{4}{3} a^2 b$ cubic units

c. $\frac{4}{3} ab$ cubic units

d. 4 cubic units

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21. What is the volume generated when the region surrounded by $y = \sqrt{x}$, $y = 2$ and $y = 0$ is revolved about y-axis?

a. 32 cubic units

b. $\frac{32}{5}$ cubic units

c. $\frac{32}{\pi}$ cubic units

d. $\frac{5\pi}{32}$ cubic units

Answer

1. a.	2. c.	3. d.	4. a.	5. b.	6. a.	7. c.
8. d.	9. c.	10. c.	11. b.	12. c.	13. a.	14. a.
15. c.	16. a.	17. a.	18. b.	19. a.	20. b	21. c

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Unit 5

Partial Derivative

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- knowledge of limit, continuity and derivatives of functions of single variable.

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- functions of two and three variables and their graphs.
- limit and continuity of functions of two and three variables.
- partial derivatives of functions of three variables with their geometrical interpretations.
- Euler's theorem on homogeneous functions for two and three variables.
- total derivatives and derivative of implicit function.

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5.0 Introduction

Marquis de Condorcet was the first person to use the concept of partial derivatives from 1770, and he used it for partial differences. The symbol was originally introduced in 1770 by Nicolas de Condorcet, who used it for a partial differential, and adopted for the partial derivative by Adrien-Marie Legendre in 1786. The modern partial derivative notation was developed by Adrien-Marie Legendre (1786), although he later abandoned it; Carl Gustav Jacob Jacobi reintroduced the symbol in 1841.

Adrien-Marie Legendre (1752-1833)

Before starting partial derivatives, we first discuss the limit and continuity of function of and three variables. To better understand these concepts, the concept of two or more variable functions and their geometry are unavoidable. So we begin with functions of two and three variables.

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5.1 Functions of two and three variables

There are various physical quantities in real world which depends on more than one variable. For examples, temperature at a point on earth surface depends on the longitude and latitude of the point, rate of propagation of heat on a metallic rod depends on the position and time, volume of cone and cylinders depend on radius and height and there are various other examples as well. Dependency of these type physical quantities can be mathematically explained using the concept of two or more variables. We now give the formal definition of two and three variables functions.

Function of two variables: A function f of two variables is a rule that gives a unique real number at each ordered pair of real numbers (x, y) in a set D. The unique real number associated to the ordered pair (x, y) is denoted by $z = f(x, y)$. Here the set D is the domain of the function f and the range of the function f is the set of values given by f i.e., $\{f(x, y): (x, y) \in D\}$. Thus function of two variables is a function whose domain is a subset of $\mathbb{R}^2$ and range is a subset of real number $\mathbb{R}$.

Example 1. Find the domain and range of $f(x, y) = \sqrt{1 - \frac{x^2}{4} - \frac{y^2}{9}}$.

Solution:

Here the domain of the function is the set of all (x, y) for which we have

$$1 - \frac{x^2}{4} - \frac{y^2}{9} \geq 0.$$

i.e., $\frac{x^2}{4} + \frac{y^2}{9} \leq 1$

which is the interior of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Since its square root function is always non-negative, we have $f(x, y) \geq 0$. Moreover, the largest value of f occurs when both $x = 0$ and $y = 0$ and the value is $\sqrt{1 - \frac{0}{4} - \frac{0}{9}} = 1$. Thus the range of the function $f(x, y)$ is $[0, 1]$.

Example 2. Find the domain and range of $f(x, y) = \sin(xy)$.

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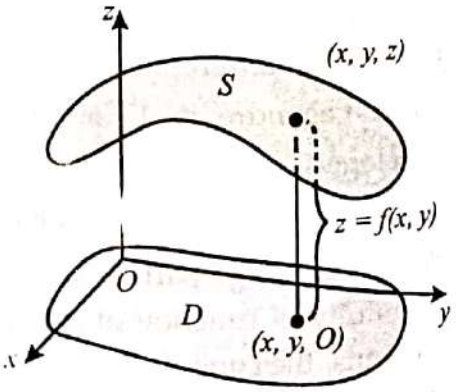
Solution:

Here the domain of the function is the set of all ordered pair (x, y) which is the entire xy-plane and the range of the function is $[-1, 1]$.

Definition

Let $z = f(x, y)$ be a function of two variables. Then by graph of the function $f(x, y)$ we mean a set of all points (x, y, z), where the point (x, y) is in the domain of the function f. Thus the graph of function of two variables represents a surface in three-dimensional space. The set of (x, y, z) in space such that $f(x, y, z) = c$ is level surface of $z = f(x, y)$.

Figure aside shows the graph of two variables function and its domain.



The surface of $z = f(x, y)$ can also be written as a set $S = \{(x, y, z): \quad x, y)$ a $\quad (x, y) \in D\}$.

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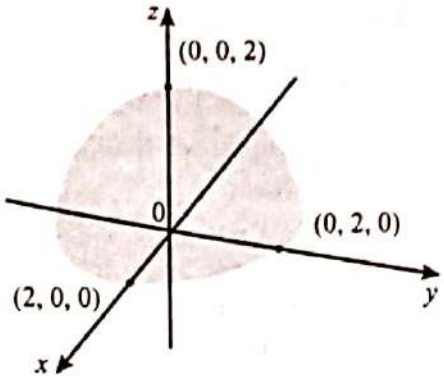
Example 3. Draw the graph of the function $f(x, y) = \sqrt{4 - x^2 - y^2}$.

Solution:

We note that the graph is given by $z = \sqrt{4 - x^2 - y^2}$. Squaring both sides we get $z^2 = 4 - x^2 - y^2$. This gives $x^2 + y^2 + z^2 = 4$ which is the sphere of radius 2. Hence the graph of given function is the interior of the sphere of radius 2 as shown in adjoining figure.



We now define the functions of three variables.

Definition

By a function f of three variables we mean a rule that assigns to each ordered triplet (x, y, z) in the domain $D \subset \mathbb{R}^3$, a unique real number denoted by $f(x, y, z)$.

Example 4. Find the domain and range of the function $u = f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$.

Solution:

Here domain is the set of all ordered triplet $S(x, y, z)$. So the domain is the entire space and the range is the set $[0, \infty)$.

Next we discuss the limit and continuity of two variables functions.

5.1.1 Limit of function of two variables

The concept of limit of function of two variables is analogous (similar) to the concept of limit of function of single variable $y = f(x)$ as discussed in unit one.

Definition

Let $z = f(x, y)$ be a function of two variables. Let (a, b) be any point in the domain D of $f(x, y)$. A number L is said to be limit of $f(x, y)$ as (x, y) approaches to (a, b) if $f(x, y)$ can be brought very close to L [(i.e., $|f(x, y) - L| < \varepsilon, \varepsilon > 0$ a very small number)] by bringing (x, y) sufficiently close to (a, b) [(i.e., $0 < |(x, y) - (a, b)| < \delta, \delta > 0$ a very small number)] which is written as

$$\lim_{(x, y) \to (a, b)} f(x, y) = L.$$

Example 5. Evaluate the limit $\lim_{(x, y) \to (0, 0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2}$.

Solution:

We note that $\sin\theta \to \theta$ for very small values as $\sin 0 = 0$.

$$\therefore \lim_{(x, y) \to (0, 0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \frac{x^2 + y^2}{x^2 + y^2} = 1.$$

Remark: In xy-plane, a point (a, b) can be approached through infinite paths. Due to this, in the definition of limit, we see that the point (x, y) approach to (a, b) through infinite possible paths. So for the limit to exists, value of limits in all possible paths are equal. If value of limit for any two paths are not same, then limit in such case does not exist. This will be explained in examples.

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Example 6. Show that $\lim_{(x, y) \to (0, 0)} \frac{x - y}{x + y}$ does not exist.

Solution:

Let $f(x, y) = \frac{x - y}{x + y}$.

We first approach to (0, 0) through x-axis. Along x-axis, $y = 0$.

So, we have

$$f(x, y) = \frac{x - y}{x + y}$$

$$= \frac{x - 0}{x + 0}$$

$$= \frac{x}{x} = 1, \text{ for } x \neq 0.$$

This gives that $f(x, y) \to 1$ as $(x, y) \to (0, 0)$ through x-axis.

Next we approach to (0, 0) through y-axis. Along y-axis, $x = 0$.

So, we have

$$f(x, y) = \frac{x - y}{x + y}$$

$$= \frac{0 - y}{0 + y} = 1, \text{ for } y \neq 0.$$

This gives that $f(x, y) \to -1$ as $(x, y) \to (0, 0)$ through y-axis.

Since along two different directions, limit of $f(x, y)$ are different. This concludes that $\lim_{(x, y) \to (0, 0)} \frac{x - y}{x + y}$ does not exist.

5.1.2 Continuity of function of two variables

A function f of two variable x and y is said to be continuous at a point (a, b) if

$$\lim_{(x, y) \to (a, b)} f(x, y) = f(a, b).$$

If f is continuous at every point of a set $D \subseteq \mathbb{R}^2$, then we say f is continuous on D.

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Geometrical meaning of continuity of function of two variables says that if (x, y) changes by a small amount then the value of $f(x, y)$ also changes by a small amount. Consequently, there is a continuous surface of functions of two variable and has no hole or break on it.

Remark: If $f(x, y)$ is defined (finite) in a domain D, then it is continuous in D.

Example 7. Check the continuity of $f(x, y) = x^2y^3 + x^3y^2 + x + y$ at (1, 1).

Solution:

We have, $f(x, y) = x^2y^3 + x^3y^2 + x + y$.

Now,

$f(1, 1) = 1 + 1 + 1 + 1 = 4$.

Again,

$$\lim_{(x, y) \to (1, 1)} f(x, y) = \lim_{(x, y) \to (1, 1)} (x^2y^3 + x^3y^2 + x + y)$$

$$= 1 \cdot 1 + 1 \cdot 1 + 1 + 1 = 4$$

i.e., $\lim_{(x, y) \to (1, 1)} f(x, y) = f(1, 1)$.

So, $f(x, y)$ is continuous at (1, 1).

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Exercise 5.1

1. Find the limit, if it exists.

a. $\lim_{(x,y)\to(1,2)} (5x^3 - x^2y^2)$

b. $\lim_{(x,y)\to(2,1)} \left(\frac{4-xy}{x^2+3y^2}\right)$

c. $\lim_{(x,y)\to(1,-1)} e^{-xy}\cos x + y$

d. $\lim_{(x,y)\to(0,0)} \left(\frac{xy\cos y}{3x^2+y^2}\right)$

e. $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}}$

f. $\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

g. $\lim_{(x,y,z)\to\left(\pi,0,\frac{1}{3}\right)} [e^{y^2}\tan(xz)]$

2. Find the set of the points where the function is continuous.

a. $f(x, y) = \frac{y}{1+e^{x-y}}$

b. $f(x, y) = \sin(\sqrt{1+x^2-y^2})$

c. $f(x, y) = \frac{1+x^2+y^2}{1-x^2-y^2}$

d. $f(x, y) = \log(x^2+y^2-4)$

Answer

1. a. 1 b. $\left(\frac{2}{7}\right)$ c. e

d. does not exist [Hint: choose the path $y = mx$ for $(x, y) \to (0, 0)$] show the limit depends on m and hence does not exist.

e. 0 f. 2 g. $\sqrt{3}$

2. a. $D = \{(x, y): x \in R, y \in R\} = R^2$ b. $D = \{(x, y): x \in R, y \in R\} = R^2$

c. $D = \{(x, y): x^2 + y^2 < 1\}$ d. $\{(x, y): x^2 + y^2 > 4\}$

5.2 Partial derivative

When a physical quantity depends only on a single variable, then its rate of change is measured with the help of ordinary derivatives. But in real world, a physical quantity is normally dependent on two or more independent variables. In such case, we can express rate of change of the quantity with respect to one of the independent variable using the partial derivatives. We note that partial derivatives are the ordinary derivatives obtained by assuming the other variables as a constant. So, we need to use partial derivatives to obtain the rate of change of function or quantity of two or more variables. Thus, we need partial derivatives in various situations in daily life. For example, partial derivatives are used in obtaining equation of tangent plane, determining the maximum, minimum values of functions of two or more variables, in partial differential equations, to explain various laws in physics, in the marginal analysis in economics, in image processing in engineering to name just a few cases. We now give the definition.

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Definition

Let $u = f(x, y)$ be the function of two independent variables.

Then, the limit $\lim_{\Delta x\to 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$, if exists, is called partial derivative of f with respect to x and is denoted by $\frac{\partial f}{\partial x}$.

Similarly, $\frac{\partial f}{\partial y} = \lim_{\Delta y\to 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y}$, represents partial derivative of f with respect to y.

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5.2.1 Evaluation of partial derivative

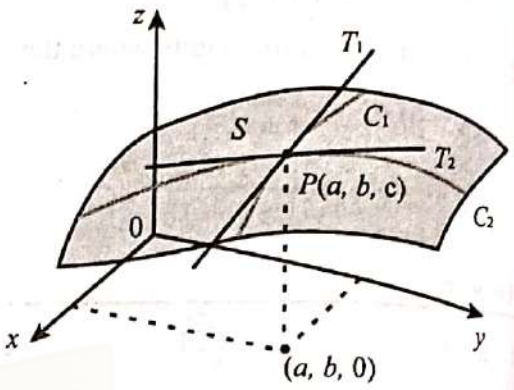
The partial derivative of f with respect to x is obtained by differentiating $u = f(x, y)$ with respect to x treating y as a constant. Similarly partial derivative of f with respect to y is obtained by differentiating $u = f(x, y)$ with respect to y treating x as a constant.

The partial derivative of u with respect to x is denoted by u_x or f_x or $\frac{\partial u}{\partial x}$.

Similarly, we can define partial derivative of function $u = f(x, y, z)$ of three variables.

5.2.2 Geometrical interpretation of the partial derivatives

Let $z = f(x, y)$ be a given function, we recall that $z = f(x, y)$ represents a surface S. Let us take a point $(x, y) = (a, b)$. Then the point $P(a, b, f(a, b))$ is a point on the surface S as shown in figure. We now fix $x = a$ and $y = b$. Then the intersection of plane $x = a$ and surface S is a curve say C_1 and that of $y = b$ and S is another curve say C_2 which is shown in figure. In terms of function C_1 is the curve given by $g(y) = f(a, y)$ and C_2 is the curve given by $h(x) = f(x, b)$. Now the slope of tangent T_1 to the C_1 is the $g'(b) = f_y(a, b)$ and slope of tangent T_2 to C is $h'(a) = f_x(a, b)$. Thus, the partial derivative measures the slope of tangent to a curve formed by intersection of surface and plane.



5.2.3 Higher order partial derivatives

If $u = f(x, y)$ be the function of two independent variables x and y, then $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ are the partial derivatives with respect to x and y respectively. If these partial derivatives are again differentiated partially either with respect to x and y, then the resulting derivatives are called the partial derivatives of second order,

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i.e., $u_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = f_{xx}$, $\quad u_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = f_{xy}$,

$u_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = f_{yy}$ and $\quad u_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = f_{yx}$

Similarly, the partial derivatives of third order and fourth order can be defined.

We note that notation f_{xy} or $\frac{\partial^2 f}{\partial y \partial x}$ means that we first differentiate with respect to x and then with respect y. In the case of f_{yx} order is reversed. In general f_{xy} and f_{yx} are not equal. They are equal when both are continuous. This is given by following theorem.

Theorem 1. (Clairaut's Theorem)

If both f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$.

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Example 8. If $u = x^3 + 2x^2y + 3y^2x + y^3$, find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$.

Solution:

We have, $u = x^3 + 2x^2y + 3y^2x + y^3$ .... (i)

Then, differentiating (i) partially with respect to x, we get

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(x^3 + 2x^2y + 3y^2x + y^3) = 3x^2 + 4xy + 3y^2$$

Similarly, $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^3 + 2x^2y + 3y^2x + y^3) = 2x^2 + 6xy + 3y^2$

Again, differentiating $\frac{\partial u}{\partial x}$ partially with respect to x, we get

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}(3x^2 + 4xy + 3y^2) = 6x + 4y$$

Differentiating $\frac{\partial u}{\partial y}$ partially with respect to y, we get

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}(2x^2 + 6yx + 3y^2) = 6x + 6y.$$

Example 9. If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

Solution:

We have, $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$

Then, $\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(\frac{y}{z} + \frac{z}{x} + \frac{x}{y}\right) = \left(-\frac{z}{x^2} + \frac{1}{y}\right)$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\frac{y}{z} + \frac{z}{x} + \frac{x}{y}\right) = \frac{1}{z} - \frac{x}{y^2}$$

$$\frac{\partial u}{\partial z} = \frac{\partial}{\partial z}\left(\frac{y}{z} + \frac{z}{x} + \frac{x}{y}\right) = -\frac{y}{z^2} + \frac{1}{x}$$

$$\therefore \quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = x\left(-\frac{z}{x^2} + \frac{1}{y}\right) + y\left(\frac{1}{z} - \frac{x}{y^2}\right) + z\left(-\frac{y}{z^2} + \frac{1}{x}\right)$$

$$= -\frac{z}{x} + \frac{x}{y} + \frac{y}{z} - \frac{x}{y} - \frac{y}{z} + \frac{z}{x} = 0$$

Hence, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

Example 10. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x}{y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$.

Solution:

We have, $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x}{y}\right)$

Now, $\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1 - \left(\frac{x}{y}\right)^2}} \times \frac{1}{y} + \frac{1}{1 + \frac{x^2}{y^2}} \times \frac{1}{y}$

$$\therefore \quad \frac{\partial u}{\partial x} = \frac{1}{\sqrt{y^2 - x^2}} + \frac{y}{x^2 + y^2}$$

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Multiplying by x, we get

$$x\frac{\partial u}{\partial x} = \frac{x}{\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2} \quad \ldots\ldots (i)$$

Again, $\frac{\partial u}{\partial y} = \frac{1}{\sqrt{1-\frac{x^2}{y^2}}}\left(-\frac{x}{y^2}\right) + \frac{1}{1+\left(\frac{x}{y}\right)^2}\left(-\frac{x}{y^2}\right)$

$$\therefore \quad \frac{\partial u}{\partial x} = -\left[\frac{x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2}\right]$$

Multiplying by y, we get

$$y\frac{\partial u}{\partial y} = -y\left[\frac{x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2}\right] \quad \ldots\ldots (ii)$$

Adding equations (i) and (ii), we get

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{-xy}{y\sqrt{y^2-x^2}} - \frac{xy}{x^2+y^2} + \frac{xy}{\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2} = 0$$

Hence, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$.

Example 11. If $z(x+y) = x^2+y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.

Solution:

We have, $z(x+y) = x^2+y^2$ $\ldots\ldots (i)$

Differentiating partially with respect to x, we get

$$(x+y)\frac{\partial z}{\partial x} + z = 2x$$

$$\therefore \quad \frac{\partial z}{\partial x} = \frac{2x-z}{x+y}$$

Again, differentiating partially with respect to y, we get

$$(x+y)\frac{\partial z}{\partial y} + z = 2y$$

or, $\frac{\partial z}{\partial y} = \frac{2y-z}{x+y}$

Now, $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = \left(\frac{2x-z}{x+y} - \frac{2y-z}{x+y}\right)^2 = 4\left(\frac{x-y}{x+y}\right)^2$ $\ldots\ldots (ii)$

Also, $4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = 4\left(1 - \frac{2x-z}{x+y} - \frac{2y-z}{x+y}\right)$

$$= 4\left(\frac{(x+y) - 2x + z - 2y + z}{x+y}\right)$$

$$= 4\left(\frac{-(x+y)+2z}{x+y}\right)$$

$$= 4\,\frac{-(x+y)+2\frac{(x^2+y^2)}{x+y}}{x+y} \qquad [\because z = \frac{x^2+y^2}{x+y}]$$

$$= 4\,\frac{-(x+y)^2 + 2(x^2+y^2)}{(x+y)^2}$$

$$= 4\left(\frac{x-y}{x+y}\right)^2$$

$$= \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 \qquad [\because \text{from (ii)}]$$

Hence, $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.

Example 12. If $u = \log(x^3+y^3+z^3-3xyz)$, then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{-3}{(x+y+z)^2}$.

Solution:

We have, $u = \log(x^3+y^3+z^3-3xyz)$

Then, $\frac{\partial u}{\partial x} = \frac{1}{x^3+y^3+z^3-3xyz} \times (3x^2-3yz)$

$$= \frac{3(x^2-yz)}{x^3+y^3+z^3-3xyz}$$

Similarly, $\frac{\partial u}{\partial y} = \frac{3(y^2-zx)}{x^3+y^3+z^3-3xyz}$

and $\frac{\partial u}{\partial z} = \frac{3(z^2-xy)}{x^3+y^3+z^3-3xyz}$

Again, $\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{3(x^2-yz)}{x^3+y^3+z^3-3xyz}\right)$

$$= \frac{6x(x^3+y^3+z^3-3xyz) - (3x^2-3yz)(3x^2-3yz)}{(x^3+y^3+z^3-3xyz)^2}$$

$$= \frac{-3(x^4-2xy^3-2xz^3+3y^2z^2)}{(x^3+y^3+z^3-3xyz)^2}$$

Similarly, $\frac{\partial^2 u}{\partial y^2} = \frac{-3(y^4-2yz^3-2yx^3+3z^2x^2)}{(x^3+y^3+z^3-3xyz)^2}$

and $\frac{\partial^2 u}{\partial z^2} = \frac{-3(z^4-2zx^3-2zy^3+3x^2y^2)}{(x^3+y^3+z^3-3xyz)^2}$

On adding, we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{-3x^4+6xy^3+6xz^3-9y^2z^2-3y^4+6yz^3+6yx^3-9z^2x^2-3z^4+6zx^3+6zy^3-9x^2y^2}{(x^3+y^3+z^3-3xyz)^2}$$

$$= \frac{-3(x^4+y^4+z^4)+6(xy^3+xz^3+x^3y+yz^3+x^3z+y^3z)-9(x^2y^2+y^2z^2+x^2z^2)}{(x^3+y^3+z^3-3xyz)^2}$$

$$= \frac{-3[(x^4+y^4+z^4)-2(xy^3-xz^3+x^3y+yz^3+x^3z+y^3z)+3(x^2y^2+y^2z^2+x^2z^2)]}{(x^3+y^3+z^3-3xyz)^2}$$

$$= \frac{-3(x^2+y^2+z^2-xy-yz-zx)^2}{[(x+y+z(x^2+y^2+z^2-xy-yz-zx))]^2} = \frac{-3}{(x+y+z)^2}$$

$[\because a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)]$

Exercise 5.2

1. Find the first order partial derivatives of the following functions:
   a. $u = 3x^2+6xy+8$
   b. $u = 7x^5+39xy^2-7y^3$
   c. $u = \cos x + \sin y$
   d. $u = xe^y + y\sin x$
2. Verify the $u_{xy} = u_{yx}$ for the following functions.
   a. $u = 4xy^3 - 3x^2y^2 + 6x + 9y$
   b. $u = 3x^2y^3 - 4^3y^2 + 6x^2y^2$
   c. $u = \sin^{-1}\frac{x}{y}$
   d. $u = \tan^{-1}\frac{x}{y}$
   e. $u = \sqrt{x^2+y^2}$
   f. $u = e^{ax}\cos by$
   g. $u = \log(y\sin x + x\sin y)$

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3. If $u = e^{xyz}$ find $\frac{\partial^3 u}{\partial x \partial y \partial z}$

4. If $u = x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, show that $\frac{\partial^2 u}{\partial y \partial z} = \frac{x^2 - y^2}{x^2 + y^2}$.

5. If $u = e^{\frac{x}{y}} + e^{\frac{y}{z}} + e^{\frac{z}{x}}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

6. If $u = \log(x^2 + y^2 + z^2)^{\frac{1}{2}}$, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{x^2 + y^2 + z^2}$.

7. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that

   i. $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$.

   ii. $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x + y + z)^2}$.

8. If $u = (x^2 + y^2 + z^2)^{\frac{-1}{2}}$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

9. If $u = \log(x^3 + y^3 - x^2y - y^2x)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 u = \frac{-4}{(x + y)^2}$

10. If $u = (1 - 2xy - y^2)^{\frac{-1}{2}}$, show that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = -y^2u^3$

11. If $u = e^{xyz} f\left(\frac{xz}{y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2xyzu$

12. If $u = \tan^{-1}\frac{xy}{\sqrt{1 + x^2 + y^2}}$, show that $\frac{\partial^2 u}{\partial x \partial y} = (1 + x^2 + y^2)^{-\frac{3}{2}}$

13. If $u = (x^2 + y^2 + z^2)^{m/2}$, find the value of $m \neq 0$ that will make $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.

Answers

1. a. $u_x = 6x + 6y$, $u_y = 6x$
   b. $u_x = 42x^5 + 39y^2$, $u_y = 78xy - 21y^2$
   c. $u_x = -\sin x$, $u_y = \cos y$
   d. $u_x = e^y + y\cos x$, $u_y = xe^y + \sin x$
3. $(1 + 3xyz + x^2y^2z^2)e^{xyz}$
13. -1

5.3 Homogeneous function

A function $f(x, y)$ of two independent variables x and y is said to be homogeneous of degree n if it satisfies the property $f(tx, ty) = t^n f(x, y)$ for all values of t, t being independent of x and y.

In such case, the function $f(x, y)$ can be written as $f(x, y) = x^n\phi\left(\frac{y}{x}\right)$ or $f(x, y) = y^n\psi\left(\frac{x}{y}\right)$.

For example, if $u = f(x, y) = x^4 + x^3y + x^2y^2 + xy^3 + y^4$, then

$$\begin{aligned} f(tx, ty) &= (tx)^4 + (tx)^3(ty) + (tx)^2(ty)^2 + (tx)(ty)^3 + (ty)^4 \\ &= t^4x^4 + t^4x^3y + t^4x^2y^2 + t^4xy^3 + t^4y^4 \\ &= t^4(x^4 + x^3y + x^2y^2 + xy^3 + y^4) \\ &= t^4 f(x, y) \end{aligned}$$

Hence, $f(x, y) = x^4 + x^3y + x^2y^2 + xy^3 + y^4$ is a homogenous function of degree 4.

Alternatively, $u = f(x, y) = x^4 + x^3y + x^2y^2 + xy^3 + y^4$

$$= x^4\left[1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2 + \left(\frac{y}{x}\right)^3 + \left(\frac{y}{x}\right)^4\right] = x^4\phi\left(\frac{y}{x}\right)$$

Thus, u is a homogeneous function of degree 4.

It is to be noted that the degree of homogeneous function may be a fraction and negative.

Consider a function $u = f(x, y) = \frac{\sqrt{x} + \sqrt{y}}{x^4 + y^4}$

$$= \frac{x^{\frac{1}{2}}\left(1 + \sqrt{\frac{y}{x}}\right)}{x^4\left[1 + \left(\frac{y}{x}\right)^4\right]} = x^{\left(\frac{1}{2} - 4\right)}\frac{\left(1 + \sqrt{\frac{y}{x}}\right)}{\left[1 + \left(\frac{y}{x}\right)^4\right]}$$

$$= x^{-\frac{7}{2}}\left[\frac{x + \sqrt{\frac{y}{x}}}{1 + \left(\frac{y}{x}\right)^4}\right] = x^{-\frac{7}{2}}\phi\left(\frac{y}{x}\right)$$

$\therefore$ $u = f(x, y)$ is a homogeneous function of degree $n = -\frac{7}{2}$.

5.4 Total differential

Let $u = f(x, y)$ be a homogenous function of degree n with f_x and f_y be the partial derivatives with respect to x and y respectively.

Then, the total differential of u denoted by du, is defined as:

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

i.e., $du = u_x\,dx + u_y\,dy$

For example,

let $u = ax^2 + 2lxy + by^2$ is a homogeneous function of degree 2.

Then, $u_x = 2ax + 2hy$

and $u_y = 2lx + 2by$.

The total differential of u is given by

$$\begin{aligned} du &= u_x\,dx + u_y\,dy \\ &= (2ax + 2hy)\,dx + (2lx + 2by)\,dy \end{aligned}$$

$\therefore$ $du = 2(ax + hy)\,dx + 2(lx + ly)\,dy$ is the required total differential of u.

5.4.1 Composite function and its derivative

Let $u = f(x, y)$ be a function of two variables x and y. Further, let x and y be both functions of single variable t, i.e., $x = \phi(t)$ and $y = \psi(t)$.

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Then, u becomes the function of t and is known as composite function of t. Then

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$$

is the total derivative of u with respect to t.

For example, the total derivative of

$u = x^3 + 3x^2y + 3xy^2 + y^3$ if $x = t^2 - 1$, $y = 2t + 1$, is given by

Now, $\frac{\partial u}{\partial x} = 3x^2 + 6xy + 3y^2$ and $\frac{\partial u}{\partial y} = 3x^2 + 6xy + 3y^2$

Also, $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 2$

Now the total derivative of u is

$$\begin{aligned}\frac{du}{dt} &= \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}\\ &= (3x^2 + 6xy + 3y^2)2t + (3x^2 + 6xy + 3y^2)2\\ &= 6t(x^2 + 2xy + y^2) + 6(x^2 + 2xy + y^2)\\ &= (6t + 6)(x^2 + 2xy + y^2)\\ &= 6(t + 1)[(t^2 - 1)^2 + 2(t^2 - 1)(2t + 1) + (2t + 1)^2]\\ &= 6(t + 1)[t^4 - 2t^2 + 1 + 4t^3 + 2t^2 - 4t - 2 + 4t^2 + 4t + 1]\\ &= 6(t + 1)(t^4 + 4t^3 + 4t^2)\\ &= 6t^2(t + 1)(t^2 + 4t + 4)\end{aligned}$$

5.5 Euler's theorem

5.5.1 Euler's theorem on homogeneous function of two variables

Let $u(x, y)$ be a homogeneous function of two variables x, and y of degree n. Then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu.$$

Proof:

Here $u(x, y)$ is a homogeneous function of x and y of degree n, we can write

$$u = x^n f\left(\frac{y}{x}\right) \quad \ldots\ (i)$$

Differentiating (i) partially with respect to x, we have

$$\begin{aligned}\frac{\partial u}{\partial x} &= nx^{n-1} f\left(\frac{y}{x}\right) + x^n f'\left(\frac{y}{x}\right)\left(\frac{-y}{x^2}\right)\\ &= nx^{n-1} f\left(\frac{y}{x}\right) - x^{n-2} y f'\left(\frac{y}{x}\right) \quad \ldots (i)\end{aligned}$$

Again, differentiating (i) partially with respect to y, we get

$$\frac{\partial u}{\partial y} = x^n f'\left(\frac{y}{x}\right)\frac{1}{x} \quad \ldots (ii)$$

Now, $x\frac{\partial u}{\partial y} + y\frac{\partial u}{\partial y} = x\left[nx^{n-1} f\left(\frac{y}{x}\right) - x^{n-2} y f'\left(\frac{y}{x}\right)\right] + y\left[x^n f'\left(\frac{y}{x}\right)\frac{1}{x}\right]$

$$= nx^n f\left(\frac{y}{x}\right) - x^{n-1} y f'\left(\frac{y}{x}\right) + x^{n-1} y f'\left(\frac{y}{x}\right)$$

$$= nx^n f\left(\frac{y}{x}\right)$$

$\therefore x\frac{\partial u}{\partial y} + y\frac{\partial u}{\partial y} = nu$, which completes the proof.

5.5.2 Euler's theorem on homogeneous function of three variables

Let $u = f(x\ y\ z)$ be a homogeneous function of three independent variables x, y and z of degree n.

Then, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$.

Proof:

Since $u = f(x, y, z)$ be a homogeneous function of x, y and z of degree n. Then

$$u = x^n g\left(\frac{y}{x}, \frac{z}{x}\right) \quad \ldots\ (i)$$

Let $p = \frac{y}{x}$ and $q = \frac{z}{x}$

Then, $\frac{\partial p}{\partial x} = \frac{-y}{x^2}$, $\frac{\partial p}{\partial y} = \frac{1}{x}$, $\frac{\partial p}{\partial z} = 0$

and $\frac{\partial q}{\partial x} = \frac{-z}{x^2}$, $\frac{\partial q}{\partial y} = 0$, $\frac{\partial q}{\partial z} = \frac{1}{x}$

Adding (ii), (iii) and (iv)

Differentiating (i) partially with respect to x, y and z, we get

$$\begin{aligned}\frac{\partial u}{\partial x} &= nx^{n-1} g(p, q) + x^n\left[\frac{\partial g}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial g}{\partial q}\frac{\partial q}{\partial x}\right]\\ &= nx^{n-1} g(p, q) + x^n\left[\frac{\partial g}{\partial p}\left(\frac{-y}{x^2}\right) + \frac{\partial g}{\partial q}\left(\frac{-z}{x^2}\right)\right] \quad \ldots (ii)\end{aligned}$$

and $\frac{\partial u}{\partial y} = x^n\left[\frac{\partial g}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial g}{\partial q}\frac{\partial q}{\partial y}\right]$

$$= x^n\left[\frac{\partial g}{\partial p}\left(\frac{1}{x}\right) + \frac{\partial g}{\partial q}\,0\right]$$

$\therefore \frac{\partial u}{\partial y} = x^{n-1}\frac{\partial g}{\partial p} \quad \ldots (iii)$

Also, $\frac{\partial u}{\partial z} = x^n\left[\frac{\partial g}{\partial p}\frac{\partial p}{\partial z} + \frac{\partial g}{\partial q}\frac{\partial q}{\partial z}\right]$

$$= x^n\left[\frac{\partial g}{\partial p}.0 + \frac{\partial g}{\partial q}\left(\frac{1}{x}\right)\right]$$

$\therefore \frac{\partial u}{\partial z} = x^{n-1}\frac{\partial g}{\partial q} \quad \ldots (iv)$

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From (ii), (iii) and (iv),

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z} = nx^n g(p,q) - yx^{n-1}\frac{\partial g}{\partial p} - zx^{n-1}\frac{\partial g}{\partial q} + yx^{n-1}\frac{\partial g}{\partial p} + zx^{n-1}\frac{\partial g}{\partial q}$$
$$= nx^n g(p,q)$$

$\therefore\ x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z} = nu$, which completes the proof.

5.6 Total derivative of implicit function

If y is a function of x but the variable y can not be expressed explicitly in terms of x, then we say that y is implicit function of x. In such case the implicit function can be expressed as

$f(x, y) = c$, where, $y = y(x)$.

Then the total derivative of y with respect to x can be expressed as

$$\frac{df}{dx}=\frac{\partial f}{\partial x}\frac{dx}{dx}+\frac{\partial f}{\partial y}\frac{dy}{dx}.$$

But, $\frac{df}{dx}=\frac{dc}{dx}=0$

or, $\frac{\partial f}{\partial x}\frac{dx}{dx}+\frac{\partial f}{\partial y}\frac{dy}{dx}=0$

$\therefore\ \frac{dy}{dx}=\frac{-\partial f/\partial x}{\partial f/\partial y}$.

Example 13. Verify Euler's theorem for the function $u = 2x^2 + 3xy - 4y^2$.

Solution

We have, $u = f(x, y) = 2x^2 + 3xy - 4y^2$ .... (i)

Now, $f(tx, ty) = 2(tx)^2 + 3(tx)(ty) - 4(ty)^2$
$= 2t^2x^2 + 3t^2xy - 4t^2y^2$
$= t^2(2x^2 + 3xy - 4y^2) = t^2 f(x, y)$

Thus, $f(x, y)$ is a homogeneous function of degree 2. So, we have to show $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=2u$.

Again, differentiating equation (i) partially with respect to x, we have

$\frac{\partial u}{\partial x} = 4x + 3y$ .... (ii)

Differentiating equation (i) partially with respect to y, we have

$\frac{\partial u}{\partial y} = 3x - 8y$ .... (iii)

From (ii) and (iii)

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x(4x + 3y) + y(3x - 8y)$
$= 4x^2 + 3xy + 3xy - 8y^2$
$= 4x^2 + 6xy - 8y^2$
$= 2(2x^2 + 3xy - 4y^2) = 2u$

$\therefore\ x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = 2u.$

Example 14. Verify the Euler's theorem for the function $u = x^n \tan^{-1}\left(\frac{y}{x}\right)$.

Solution:

We have, $u = x^n \tan^{-1}\left(\frac{y}{x}\right) = x^n\phi\left(\frac{y}{x}\right)$

Hence, u is the homogenous function of degree n.

So, we have to show $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = nu$

Now, $\frac{\partial u}{\partial x}=\frac{\partial}{\partial x}\left(x^n \tan^{-1}\frac{y}{x}\right) = nx^{n-1}\tan^{-1}\frac{y}{x} + x^n\frac{1}{1+(y/x)^2}\left(\frac{-y}{x^2}\right)$
$= nx^{n-1}\tan^{-1}\frac{y}{x} - \frac{x^n y}{x^2+y^2}$

Again, $\frac{\partial u}{\partial y} = x^n\frac{1}{1+(y/x)^2}\cdot\frac{1}{x}$
$= \frac{x^{n+1}}{x^2+y^2}$

Now, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x\left[nx^{n-1}\tan^{-1}\frac{y}{x} - \frac{x^n y}{x^2+y^2}\right] + y\cdot\frac{x^{n+1}}{x^2+y^2}$
$= nx^n\tan^{-1}\frac{y}{x} - \frac{x^{n+1}y}{x^2+y^2} + \frac{x^{n+1}y}{x^2+y^2}$
$= nx^n\tan^{-1}\frac{y}{x}$
$= nu$

$\therefore\ x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = nu.$

Thus, Euler's theorem is verified.

Example 15. Verify Euler's theorem for $u = \frac{x^7 + y^7 + x^2y^5}{x^2 + y^2}$

Solution:

We have, $u = f(x, y) = \frac{x^7 + y^7 + x^2y^5}{x^2 + y^2}$
$= \frac{x^7\left[1+\left(\frac{y}{x}\right)^7+\left(\frac{y}{x}\right)^5\right]}{x^2\left[1+\left(\frac{y}{x}\right)^2\right]}$
$= \frac{x^5\left[1+\left(\frac{y}{x}\right)^7+\left(\frac{y}{x}\right)^5\right]}{1+\left(\frac{y}{x}\right)^2}$
$= x^5\phi\left(\frac{y}{x}\right)$

Hence, it is a homogeneous function of degree 5.

Now, $\frac{\partial u}{\partial x} = \frac{(x^2+y^2)(7x^6+2xy^5)-(x^7+y^7+x^2y^5)2x}{(x^2+y^2)^2}$

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and $\frac{\partial u}{\partial y} = \frac{(x^2+y^2)(7y^6+5x^2y^4)-(x^7+y^7+x^2y^5)2y}{(x^2+y^2)^2}$

Now, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x\left\{\frac{(x^2+y^2)(7x^6+2xy^5)-(x^7+y^7+x^2y^5)2x}{(x^2+y^2)^2}\right\}+y\left\{\frac{(x^2+y^2)(7y^6+5x^2y^4)-(x^7+y^7+x^2y^5)2y}{(x^2+y^2)^2}\right\}$

$= \frac{5x^9+5y^9+5x^7y^2+10x^2y^7+5x^4y^5}{(x^2+y^2)^2}$

$= \frac{5(x^9+y^9+x^7y^2+2x^2y^7+x^4y^5)}{(x^2+y^2)^2}$

$= \frac{5(x^9+2x^2y^7+x^4y^5+x^7y^2+y^9)}{(x^2+y^2)^2}$

$= \frac{5(x^9+x^2y^7+x^4y^5+y^2x^7+y^9+x^2y^7)}{(x^2+y^2)^2}$

$= \frac{5\{x^2(x^7+y^7+x^2y^5)+y^2(x^7+y^7+x^2y^5)\}}{(x^2+y^2)^2}$

$= \frac{5(x^2+y^2)(x^7+y^7+x^2y^5)}{(x^2+y^2)^2}$

$= 5\left(\frac{x^7+y^7+x^2y^5}{x^2+y^2}\right)$

$= 5u$

$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = 5u.$

Hence, Euler's theorem is verified.

Example 16. Verify Euler's theorem for

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \frac{1}{20}u, \text{ where } u = \frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}.$$

Solution:

We have, $u = \frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}$

$= \frac{x^{1/4}\left(1+\left(\frac{y}{x}\right)^{1/4}\right)}{x^{1/5}\left(1+\left(\frac{y}{x}\right)^{1/5}\right)}$

$= x^{1/20}\frac{\left[1+\left(\frac{y}{x}\right)^{1/4}\right]}{\left[1+\left(\frac{y}{x}\right)^{1/5}\right]}$

$\therefore \quad u = x^{1/20}\phi\left(\frac{y}{x}\right)$

Thus, u is a homogeneous function of degree $\frac{1}{20}$

Then, $\frac{\partial u}{\partial x} = \frac{1}{20}x^{-\frac{19}{20}}\phi\left(\frac{y}{x}\right)+\frac{1}{20}x^{\frac{1}{20}}\phi'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right)$

and, $\frac{\partial u}{\partial x} = \frac{1}{20}x^{\frac{1}{20}}\phi'\left(\frac{y}{x}\right)\cdot\left(\frac{1}{x}\right)$

Then, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x\left[\frac{1}{20}x^{-\frac{19}{20}}\phi\left(\frac{y}{x}\right)+\frac{1}{20}x^{\frac{1}{20}}\phi'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right)\right]+y\left[\frac{1}{20}x^{\frac{1}{20}}\phi'\left(\frac{y}{x}\right)\cdot\left(\frac{1}{x}\right)\right]$

$= \frac{1}{20}x^{\frac{1}{20}}\phi\left(\frac{y}{x}\right)-\frac{1}{20}yx^{-\frac{19}{20}}\phi'\left(\frac{y}{x}\right)+\frac{1}{20}yx^{-\frac{19}{20}}\phi'\left(\frac{y}{x}\right)$

$= \frac{1}{20}x^{\frac{1}{20}}\phi\left(\frac{y}{x}\right)$

$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \frac{1}{20}u.$

Hence, Euler's theorem is verified.

Example 17. If $u = \tan^{-1}(x^2+2y^2)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \sin 2u$.

Solution:

We have, $u = \tan^{-1}(x^2+2y^2)$

or, $\tan u = x^2+2y^2 = t$ (say)

Here t is a homogeneous function of degree 2.

$\therefore \quad x\frac{\partial t}{\partial x}+y\frac{\partial t}{\partial y} = 2t \qquad \ldots\ldots(i)$

or, $x\frac{\partial \tan u}{\partial x}+y\frac{\partial \tan u}{\partial y} = 2\tan u$

or, $x\sec^2 u\frac{\partial u}{\partial x}+y\sec^2 u\frac{\partial u}{\partial y} = 2\tan u$

or, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \frac{2\tan u}{\sec^2 u}$

$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \sin 2u.$

Example 18. If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2} = \frac{-\sin u\cos 2u}{4\cos^3 u}.$$

Solution:

We have, $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$

or, $f(x,y) = \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$

Let $f = \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$

$= \frac{x\left(1+\frac{y}{x}\right)}{\sqrt{x}\left(1+\sqrt{\frac{y}{x}}\right)}$

$= x^{\frac{1}{2}}\frac{\left(1+\frac{y}{x}\right)}{\left(1+\sqrt{\frac{y}{x}}\right)} = x^{\frac{1}{2}}\phi\left(\frac{y}{x}\right)$

$\therefore$ f is a homogeneous function of degree $\frac{1}{2}$.

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Now by Euler's theorem, we have

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf$$

or, $x\frac{\partial \sin u}{\partial x}+y\frac{\partial \sin u}{\partial y}=\frac{1}{2}\sin u$

or, $x\cos u\frac{\partial u}{\partial x}+y\cos u\frac{\partial u}{\partial y}=\frac{1}{2}\sin u$

$\therefore\quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u$ .... (i)

Differentiating equation (i) with respect to x, we get

$$x\frac{\partial^2 u}{\partial x^2}+\frac{\partial u}{\partial x}+y\frac{\partial^2 u}{\partial x\partial y}=\frac{1}{2}sec^2u\frac{\partial u}{\partial x}$$

Multiplying this equation by x, we get

$$x\frac{\partial^2 u}{\partial x^2}+x\frac{\partial u}{\partial x}+xy\frac{\partial^2 u}{\partial x\partial y}=\frac{1}{2}xsec^2u\frac{\partial u}{\partial x}$$ .... (ii)

Again, differentiating (i) with respect to y and multiplying by y

$$xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}+y\frac{\partial u}{\partial y}=\frac{1}{2}ysec^2u\frac{\partial u}{\partial y}$$ .... (iii)

Adding equations (ii) and (iii), we get

$$x\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}+\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)=\frac{1}{2}sec^2u\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)$$

$$\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}+\frac{1}{2}\tan u=\frac{1}{2}\sec^2 u\frac{1}{2}\tan u$$

or, $x\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}=\frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u-1\right)$

or, $x\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}=\frac{1}{2}\tan u\left(\frac{1}{2\cos^2 u}-1\right)$

or, $x\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}=\frac{1}{2}\frac{\sin u}{\cos u}\left(\frac{1-2\cos^2 u}{2\cos^2 u}\right)$

$\therefore\quad x\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}=-\frac{\sin u\cos 2u}{4\cos^3 u}$

Example 19. If $\cos u=\frac{x+y}{\sqrt{x}+\sqrt{y}}$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{1}{2}\cot u=0$.

Solution:

Let $v=\cos u=\frac{x+y}{\sqrt{x}+\sqrt{y}}$.

Then, $v(tx, ty)=\frac{tx+ty}{\sqrt{tx}+\sqrt{ty}}=t^{1/2}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$

$\therefore$ v is a homogeneous function of degree $n=\frac{1}{2}$. So, by Euler's theorem

$$x\frac{\partial v}{\partial x}+y\frac{\partial v}{\partial y}=\frac{1}{2}v$$ ....(i)

Now, $v=\cos u$, $\frac{\partial v}{\partial x}=-\sin u\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial y}=-\sin u\frac{\partial u}{\partial y}$

$\therefore$ From (i)

$$x\left(-\sin u\frac{\partial u}{\partial x}\right)+y\left(-\sin u\frac{\partial u}{\partial y}\right)=\frac{1}{2}\cos u$$

or, $\left(-x\frac{\partial u}{\partial x}-y\frac{\partial u}{\partial y}\right)\sin v=\frac{1}{2}\cos v$

or, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{1}{2}\cot u=0$.

Example 20. If $z=f(x, y)$ such that $x=e^u+e^{-v}$ and $y=e^{-u}-e^v$, then prove that $\frac{\partial z}{\partial u}-\frac{\partial z}{\partial v}=x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}$.

Solution:

We have, $\frac{\partial x}{\partial u}=e^u$, $\left(\frac{\partial x}{\partial v}\right)=-e^{-v}$, $\left(\frac{\partial y}{\partial u}\right)=-e^{-u}$, $\left(\frac{\partial y}{\partial v}\right)=-e^v$

L.H.S. $=\frac{\partial z}{\partial u}-\frac{\partial z}{\partial v}$

$$=\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)-\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}\right)$$

$$=\left(\frac{\partial z}{\partial x}e^u+\frac{\partial z}{\partial y}(-e^{-u})\right)-\left(\frac{\partial z}{\partial x}(-e^{-v})+\frac{\partial z}{\partial y}(-e^v)\right)$$

$$=\frac{\partial z}{\partial x}(e^u+e^{-v})-\frac{\partial z}{\partial y}(e^{-u}-e^v)$$

$$=\frac{\partial z}{\partial x}x-\frac{\partial z}{\partial y}y=x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=\text{R.H.S.}$$

Example 21. If $u=\log\left(\frac{x^5+y^5+z^5}{x^2+y^2+z^2}\right)$, prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=3$.

Solution:

We have, $u=\log\left(\frac{x^5+y^5+z^5}{x^2+y^2+z^2}\right)$.

or, $e^u=\frac{x^5+y^5+z^5}{x^2+y^2+z^2}$.

Suppose $v=e^u=\frac{x^5+y^5+z^5}{x^2+y^2+z^2}$.

Then, $v(tx, ty, tz)=\frac{(tx)^5+(ty)^5+(tz)^5}{(tx)^2+(ty)^2+(tz)^2}=t^3\frac{x^5+y^5+z^5}{x^2+y^2+z^2}$

$\therefore$ v is a homogeneous function of degree 3.

Hence, by Euler's theorem

$$x\frac{\partial v}{\partial x}+y\frac{\partial v}{\partial y}+z\frac{\partial v}{\partial z}=3v$$

or, $x\frac{\partial e^u}{\partial x}+y\frac{\partial e^u}{\partial y}+z\frac{\partial e^u}{\partial z}=3e^u$

or, $xe^u\frac{\partial u}{\partial x}+ye^u\frac{\partial u}{\partial y}+ze^u\frac{\partial u}{\partial z}=3e^u$

$\therefore$ $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=3$.

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Example 22. Find $\frac{dy}{dx}$ by implicit differentiation of the curve $xy^2 = \sin y^3$.

Solution:

We have, the curve $xy^2 = \sin y^3$.

Differentiating both sides with respect to x, we get

$$\frac{d}{dx}(xy^2) = \frac{d}{dx}(\sin y^3)$$

or, $y^2 + 2xy\frac{dy}{dx} = \cos y^3 \cdot 3y^2 \cdot \frac{dy}{dx}$

or, $2xy\frac{dy}{dx} - \cos y^3 \cdot 3y^2 \cdot \frac{dy}{dx} = -y^2$

or, $\frac{dy}{dx} = \frac{-y^2}{2xy - 3y^2\cos y^3}$

$\therefore \frac{dy}{dx} = \frac{-y^2}{-y(3y\cos y^3 - 2x)} = \frac{y}{3y\cos y^3 - 2x}$.

Exercise 5.3

1. Verify Euler's theorem for the give functions.
   a. $u = xy^2 - x^2y - y^3$ b. $u = \sqrt{x^2 + y^2}$
   c. $u = 3xz^2 - 2xyz + y^2z$ d. $u = x^n \tan^{-1}\left(\frac{y}{x}\right)$
   $u = \log\left(\frac{x^2 + y^2}{xy}\right)$ f. $u = \sin\frac{x^2 + y^2}{xy}$
2. If $u = \tan^{-1}\left(\frac{x^2}{x}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{\sin 2u}{2}$.
3. If $u = \sin^{-1}\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$, show that $\frac{\partial u}{\partial x} = -\frac{y}{x}\frac{\partial u}{\partial y}$.
4. If $u = \log\frac{x^2 + y^2}{x + y}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1$.
5. If $\sin u = \frac{x^2 + y^2}{x + y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$.
6. If $u = \log\left(\frac{x^4 + y^4}{x + y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$.
7. If $\sin u = \frac{x + 9y + 10z}{\sqrt{x^3 + y^3 + z^3}}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + 3\tan u = 0$.
8. If $\cos u = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$.
9. If $u = \sqrt{x^2 + y^2}\sin^{-1}\left(\frac{y}{x}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u$.
10. If $u = \sin\left(\frac{xy + yz + zx}{x^2 + y^2 + z^2}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.
11. If $v = \frac{x^3y^3}{x^3 + y^3}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3v$.

12. If $u = \frac{x^2 + y^2}{\sqrt{x + y}}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}u$.
13. If $u = \log\frac{x^2 + y^2}{\sqrt{x} + \sqrt{y}}$, prove that $x^2\frac{\partial^2 u}{dx^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{3}{2}$.
14. If $u = \sin^{-1}\left[\frac{x + y}{\sqrt{x} + \sqrt{y}}\right]$, prove that
   a. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$. b. $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{-\sin u\cos 2u}{4\cos^3 u}$.
15. a. If $u = (x^3 + y^3)^{1/3}$, show that $x^2\frac{\partial^2 u}{dx^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{2}{9}(x^2 + y^2)^{1/3}$.
   b. If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, show that
      i. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.
      ii. $x^2\frac{\partial^2 u}{dx^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u$.
16. Find $\frac{dz}{dt}$ in following functions
   a. $z = 2x^2 - 3xy + 7y^2$, $x = \sin t$, $y = \cos t$ b. $z = \log(x^2 + y^2)$, $x = e^{-t}$, $y = e^t$
   c. $z = x^3 - y^3$, $x = \frac{1}{t + 1}$, $y = \frac{t}{t + 1}$
17. If $u = e^{-at}\cos(x - at)$, show that $\frac{\partial^2 u}{dt^2} = a^2\frac{\partial^2 u}{dx^2}$.
18. If $u = x^2 + y^2 + z^2$, $x = e^{2t}$, $y = e^{2t}\cos 3t$, $z = e^{2t}\sin 3t$, show that $\frac{du}{dt} = 8e^{4t}$.
19. Using implicit differentiation, find $\frac{dy}{dx}$
   a. $(\tan x)^y + (y)^{\tan x} = 0$ b. $x^y + y^x = a$
   c. $x^y y^x = (x + y)^{x+y}$
20. If $u = f(x, y)$ and $x = e^t\cos\theta$, $y = e^t\sin\theta$, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-2t}\left[\frac{\partial^2 u}{dt^2} + \frac{\partial^2 u}{\partial\theta^2}\right]$.

Answer

15. a. $3\sin^2 t - 10\sin t\cos t - 3\cos^2 t$ b. $2\tanh 2t$ c. $\frac{-3(1 + t^2)}{(t + 1)^4}$
19. a. $\frac{-\sec^2 x\,[y(\tan x)^{y-1} + \log y\, y^{\tan x}]}{(\tan x)^y\log(\tan x) + \tan x\cdot y^{\tan x - 1}}$ b. $-\frac{y\,x^{y-1} + y^x\log y}{x^y\log x + xy^{x-1}}$ c. $\frac{y}{x}$

Multiple Choice Questions

1. If $u = \frac{x^2y + y^3}{x + y}$, then value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is
   a. $3u$ b. $2u$
   c. u d. 0

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2. If $u = \sin^{-1}\left(\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}}\right)$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

a. $\sqrt{2}u$ b. $\frac{1}{\sqrt{2}}u$

c. $2u$ d. 0

3. If $u = \log\left(\frac{x^3 + x^2y - y^3}{x - y}\right)$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

a. $2e^u$ b. e^{2u}

c. 2 d. $\frac{1}{2}$

4. If $f(x,y) = \sin x + \cos y$, $x = \cos t$, $y = \sin t$, then at $t = 0$, $\frac{df}{dt}$ is

a. 2 b. -2

c. 1 d. 0

5. If $u = \tan^{-1}\frac{y}{x}$, then by Euler's theorem, the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

a. $\tan u$ b. $\sin u$

c. 0 d. $\cos 2u$

6. If $r^2 = x^2 + y^2 + z^2$, th

a. $\frac{y}{r}$ b. $2x$

c. $2y$ d. $\frac{xz}{r}$

7. If $f(x, y) = y\sin(xy)$, then the value of $f_x(\pi, 1)$ is

a. 0 b. 1

c. -1 d. 2.5

8. The value of $\lim_{(x,y)\to(0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ is

a. 1 b. 0

c. -1 d. does not exist

9. The value of $\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4 + y^2}$ is

a. 0 b. 1

c. 2 d. does not exist

10. If $u = f(x + ay) + g(x - ay)$, then $\frac{\partial^2 u}{\partial y^2}$ is

a. $\frac{\partial^2 u}{\partial x^2}$ b. $a\frac{\partial^2 u}{\partial x^2}$

c. $a^2\frac{\partial^2 u}{\partial x^2}$ d. $\frac{\partial^2 u}{\partial x\,\partial y}$

Answer

1. b. 2. d 3. c. 4. d. 5. c.
6. a. 7. c. 8. b. 9. d. 10. c.

✧✧✧

Unit 6

Application of Partial Derivative

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- concept of extreme values of functions of single variable.
- evaluation of partial derivatives of various orders for functions of two and three variables.

Expected learning outcomes

After completion of this unit, students will develop sufficient knowledge and evaluation skills on

- equation of tangent plane to a surface.
- determination of various stationary, saddle and inflection points of functions of two or three variables.
- application of partial derivatives in evaluating extreme values of functions of two and three variables.
- computation of stationary points using Lagrange's method of multiplier up to two constraints.

6.0 Introduction

We recall that ordinary derivatives $\left[\frac{dy}{dx} \text{ for } y = f(x)\right]$ of a function of single variable are applicable to find the equation of tangent line at some point (x_0, y_0) on the curve of $y = f(x)$. We can generalize this idea of determining equation of tangent line for a function of a single variable to determine equation of tangent plane at a given point on the surface of function of two variables.

Moreover, we are familiar with the method of finding maximum and minimum value of a function $y = f(x)$ of single variable x using ordinary derivative. We will generalize this concept and evaluate maximum and minimum values of function of two or three variables.

6.1 Determination of tangent plane to the surface

We know the function $z = f(x, y)$ represents a surface in three dimensional space, which is locus of all (x, y, z) in the space such that $z = f(x, y)$.

Also, the equation of the plane is of the form

$ax + by + cz + d = 0$

i.e., a first degree (linear) equation in x, y and z represents a plane.

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Equation of plane to $z = f(x, y)$ at (x_0, y_0)

If (x_0, y_0, z_0) is any point on the surface of $z = f(x, y)$ and $f(x, y)$ has continuous partial derivatives $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$, then equation of plane at (x_0, y_0, z_0) on the surface of $z = f(x, y)$ is given by

$$z - z_0 = (x - x_0) f_x(x_0, y_0) + (y - y_0) f_y(x_0, y_0).$$

Example 1. Find the equation of tangent plane to the surface of $z = 3y^2 - 2x^2 + x$ at $(2, -1, -3)$.

Solution:

We have given equation of surface

$z = f(x, y) = 3y^2 - 2x^2 + x$ .... (i)

and given point on the surface $(x_0, y_0, z_0) = (2, -1, -3)$.

Differentiating (i) partially with respect to x and y

$$\frac{\partial f}{\partial x} = f_x = -4x + 1 \text{ and } \frac{\partial f}{\partial y} = f_y = 6y$$

At $(x_0, y_0) = (2, -1)$,

$f_x(x_0, y_0) = -4 \times 2 + 1 = -7$ and $f_y(x_0, y_0) = -6$

$\therefore$ Equation of tangent plane at (x_0, y_0, z_0)

$z - z_0 = (x - x_0) f_x(x_0, y_0) + (y - y_0) f_y(x_0, y_0)$

or, $z - (-3) = (x - 2)(-7) + (y + 1)(-5)$

or, $z + 3 = -7x + 14 - 5y - 5$

$\therefore$ $7x + 5y + z - 5 = 0$ is the required equation of tangent plane.

Exercise 6.1

Find an equation of tangent plane to the given surface $z = f(x, y)$ at the specified point.

1. $z = 2x^2 + y^2$ at $(1, 1, 3)$
2. $z = \sqrt{xy}$ at $(1, 1, 1)$
3. $z = x \sin(x + y)$ at $(-1, 1, 0)$
4. $z = 3(x - 1)^2 + 2(y + 3)^2 + 7$ at $(2, 2, 12)$
5. $z = x e^{xy}$ at $(2, 0, 2)$
6. $z = \log(x - 2y)$ at $(3, 1, 0)$

Answer

1. $4x + 2y - z - 3 = 0$
2. $x + y - 2z = 0$
3. $x + y + z = 0$
4. $6x + 20y - z - 40 = 0$
5. $z = x + 4y - z = 0$
6. $x - 2y - z - 1 = 0$

6.2 Extrema of function of two variables

The maximum or minimum value of a function of two variables is known as extreme value of the function.

6.2.1 Maximum value

A function $z = f(x, y)$ is said to have maximum value at (a, b) if $f(a, b)$ is the greatest of all values of $f(x, y)$ in the neighbourhood of $f(a, b)$,

i.e., $f(a, b) > f(a + h, b + k)$ for all $h > 0, k > 0$

Equivalently, $f(a, b) - f(a + h, b + k) > 0$.

Since $f(a, b)$ is greatest of all value in the neighbourhood of (a, b), so it is called local maximum value of $f(x, y)$.

If $z_{ab} = f(a, b)$ is maximum value of the function $z = f(x, y)$ then the point (a, b, z_{ab}) will be highest point (peak) on the surface compared to the points in the neighbourhood of (a, b, z_{ab})

6.2.2 Minimum value

A function $z = f(x, y)$ is said to have minimum value at (a, b) if $f(a, b)$ is the smallest of all values of $f(x, y)$ in the neighbourhood of (a, b),

i.e., $f(a, b) < f(a + h, b + k)$ for all $h > 0, k > 0$

Equivalently, $f(a, b) - f(a + h, b + k) < 0$.

Since $f(a, b)$ is smallest of all values in the neighbourhood of (a, b), so it is called local minimum value of $f(x, y)$.

If $z_{ab} = f(a, b)$ is the minimum value of the function $z = f(x, y)$, then the point (a, b, z_{ab}) will be at the lowest point (bottom) on the surface compared to the points in the neighbourhood of (a, b, z_{ab}).

6.2.3 Critical points (stationary points)

The critical points on the surface of $z = f(x, y)$ are the points where the surface changes its direction.

Note that the change in direction of surface is measured by the change in direction of tangent plane. The maximum and minimum value (extrema) of a function occurs at critical (stationary) points only. The stationary points on the surface of $z = f(x, y)$ are given by $f_x = 0$ and $f_y = 0$.

6.2.4 Determination of extrema

A. The necessary condition for extrema

If $f(x, y)$ has a local maximum or minimum value at (a, b) and first order partial derivatives f_x, f_y exist, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Note: If f_{xx}, f_{xy}, f_{yy} are second order derivatives of $f(x, y)$ then the determinant

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

$$= f_{xx} f_{yy} - (f_{xy})^2 \qquad [\because f_{xy} = f_{yx}]$$

is called discriminant of second order partial derivative. The discriminant D is also known as *Hessian* determinant.

B. Sufficient condition

If $f(x, y)$ has local maximum or minimum value at (a, b) and the second order partial derivatives f_{xx}, f_{xy} $(= f_{yx})$, f_{yy} exists, then

i. $f(x, y)$ has maximum value at (a, b) if $f_{xx} < 0, D > 0$.

ii. $f(x, y)$ has minimum value *at* (a, b) if $f_{xx} > 0, D > 0$.

Note: The maximum or minimum value is $f(a, b)$ which is the functional value of $f(x, y)$ at stationary point (a, b).

Remark 1: If $D < 0$, even if $f_x = 0$ and $f_y = 0$, then the function $f(x, y)$ has neither maximum nor minimum value at stationary point (a, b). Such a point is called saddle point.

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Example 2. Find stationary points and examine for maxima and minima for the function $x^3y + xy^3 + 4x^2 - xy + 4y^2 - 4$.

Solution:

Let $f(x, y) = x^3y + xy^3 + 4x^2 - xy + 4y^2 - 4$

Differentiating partially, we get

$f_x = 3x^2y + y^3 + 8x - y$, $f_y = x^3 + 3xy^2 - x + 8y$

$f_{xx} = 6xy + 8$, $f_{xy} = 3x^2 + 3y^2 - 1$, $f_{yy} = 6xy + 8$

For stationary points,

$f_x = 0$ .... (i)

or, $3x^2y + y^3 + 8x - y = 0$

and $f_y = 0$ .... (ii)

$x^3 + 3xy^2 - x + 8y = 0$

Adding (i) and (ii), we get

$3x^2y + y^3 + 8x - y + x^3 + 3xy^2 - x + 8y = 0$

or, $7x + 7y + x^3 + y^3 + 3xy(x + y) = 0$

or, $7(x + y) + (x + y)^3 = 0$

or, $(x + y)(x^2 + 2xy + y^2 + 7) = 0$

or, $(x + y)\{(x + y)^2 + 7\} = 0$

Either $(x + y)^2 + 7 = 0$ or, $x + y = 0$

which gives

$x = -y$ .... (iii)

Now, from equation (ii)

$(-y)^3 + 3(-y)y^2 - (-y) + 8y + = 0$

or, $-y^3 - 3y^3 + y + 8y = 0$

or, $-4y^3 + 9y = 0$

or, $y(9 - 4y^2) = 0$

Either $y = 0$ or $9 - 4y^2 = 0$

or, $y = \pm\frac{3}{2}$

$\therefore$ The values of y are $0, \frac{3}{2}, -\frac{3}{2}$ and the values of x are $0, -\frac{3}{2}, \frac{3}{2}$.

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Hence, the function has three stationary points $(0, 0)$, $\left(\frac{3}{2}, \frac{-3}{2}\right)$ and $\left(\frac{-3}{2}, \frac{3}{2}\right)$.

At $(0, 0)$, $f_{xx} = 8 > 0$, $f_{yy} = 8$, $f_{xy} = -1$

Now, $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 8 & -1 \\ -1 & 8 \end{vmatrix} = 64 - 1 = 63 > 0.$

$\therefore$ The function is minimum at $(0, 0)$.

At $\left(\frac{3}{2}, -\frac{3}{2}\right)$, $f_{xx} = -\frac{27}{2} + 8 = \frac{-11}{2} < 0$,

$f_{yy} = -\frac{27}{2} + 8 = \frac{-11}{2}$ and

$f_{xy} = 3\left(\frac{9}{4} + \frac{9}{4}\right) - 1 = \frac{25}{2}$

Now, $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} \frac{-11}{2} & \frac{25}{2} \\ \frac{25}{2} & \frac{-11}{2} \end{vmatrix} = \frac{121}{4} - \frac{625}{4} = -126 < 0.$

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Hence, the function is neither maximum nor minimum at $\left(\frac{3}{2}, -\frac{3}{2}\right)$.

At $\left(-\frac{3}{2}, \frac{3}{2}\right)$, $f_{xx} = \frac{-11}{2}, f_{yy} = \frac{-11}{2}, f_{xy} = \frac{-25}{2}$.

Now, $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} \frac{-11}{2} & \frac{25}{2} \\ \frac{25}{2} & \frac{-11}{2} \end{vmatrix} = -126 < 0$

The function is neither minimum nor maximum at $\left(-\frac{3}{2}, \frac{3}{2}\right)$.

Thus, the saddle points are $\left(-\frac{3}{2}, \frac{3}{2}\right)$ and $\left(\frac{3}{2}, -\frac{3}{2}\right)$.

Remark 2: If $D = 0$, we can not conclude whether $f(x, y)$ has maximum or minimum value. We need further investigation for determining maximum or minimum value of the function.

Example 3. Show that the function $f(x, y) = y^2 + x^2y + x^4$ has minimum at (0, 0).

Solution:

Here, $f(x, y) = y^2 + x^2y + x^4$

Differentiating partially, we get

So, $f_x = 2xy + 4x^3$, $f_y = 2y + x^2$

$f_{xx} = 2y + 12x^2$, $f_{yy} = 2$, $f_{xy} = 2x = f_{yx}$

At (0, 0), $f_x = 0$, $f_y = 0, f_{xx} = 0$, $f_{yy} = 2$ and $f_{xy} = 0$.

Thus, (0, 0) is the stationary point of $f(x, y)$.

Now, $f_{xx}f_{yy} - (f_{xy})^2 = 0.\ 2 - (0)^2 = 0$.

So there is no information and hence, further analysis is necessary.

Also, $f(0, 0) = 0$

Now, $f(0 + h, 0 + k) - f(0, 0) = f(h, k) - 0$

$= k^2 + h^2k + h^4$

$= k^2 + 2k\frac{h^2}{2} + \frac{h^4}{4} + \frac{3h^4}{4}$

$= \left(k + \frac{h^2}{2}\right)^2 + \frac{3h^4}{4} > 0.$

$\therefore$ $f(0 + h,\ 0 + k) - f(0, 0)$ is always positive for any small value of h and k.

By the definition of local minimum value, $f(x, y)$ has local minimum value at (0, 0) and minimum value is $f(0, 0) = 0$.

Remark 3: The table given below gives a comprehensive picture of the conditions for maxima and minima of the function $f(x, y)$ of two variables at a point.

	For function $z = f(x, y)$ to be			
Conditions	**Maximum**	**Minimum**	**Saddle point (Neither maxima nor minima)**	**Undecided information (further analysis is necessary)**
First order partial derivatives	$f_x = f_y = 0$	$f_x = f_y = 0$	$f_x = f_y = 0$	$f_x = f_y = 0$
Second order partial derivatives	i. $f_{xx} < 0$ ii. $f_{xx}f_{yy} - (f_{xy})^2 > 0$	i. $f_{xx} > 0$ ii. $f_{xx}f_{yy} - (f_{xy})^2 > 0$	i. $f_{xx}f_{yy} - (f_{xy})^2 < 0$	i. $f_{xx}.f_{yy} - (f_{xy})^2 = 0$

Next, we discuss the case of three variables.

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6.2.5 Conditions for extreme values in case of three variables

If $f(x, y, z)$ be a function of three independent variables x, y and z then the necessary and sufficient conditions for the functions of three independent variables x, y and z to have an extreme values are as follows:

Recall that a determinant whose elements are the partial derivatives of the function called **Hessian determinant**

i.e., the Hessian determinant $= \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix}$.

Case I: For maximum value of $f(x, y, z)$ at (a, b, c)

i. $f_x = 0, f_y = 0$ and $f_z = 0$

ii. $f_{xx} < 0, \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0$ and $\begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} < 0$

i.e., principal minors are alternatively negative and positive.

Case II: For minimum value of $f(x, y, z)$ at (a, b, c)

i. $f_x = 0, f_y = 0$ and $f_z = 0$

ii. $f_{xx} > 0, \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0$ and $\begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} > 0$

i.e., principal minors are all positive.

We now apply above criteria to find local maximum and minimum values in various functions as given by following examples.

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Example 4. Determine the maximum or minimum values of the function $f(x, y) = x^3 + y^3 - 3axy$.

Solution:

Here, $f(x, y) = x^3 + y^3 - 3axy$ .... (i)

Differentiating partially, we get

$f_x = 3x^2 - 3ay$, $f_y = 3y^2 - 3ax$, $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3a = f_{yx}$

For extreme value, $f_x = 0$, $f_y = 0$

Then, $3x^2 - 3ay = 0$ .... (ii)

and $3y^2 - 3ax = 0$ .... (iii)

Adding equations (ii) and (iii), we get

$x^2 - y^2 + (x - y)\, a = 0$

or, $(x - y)(x + y + a) = 0$

Since, $x + y + a \neq 0$. Therefore, $x = y$

Thus, from (ii), we get

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$x^2 - ax = 0$

or, $x(x - a) = 0$

$\therefore$ $x = 0, a$

which gives, $y = 0, y = a$

Therefore, the points (0, 0) and (a, a) are stationary points.

At (0, 0), $f_{xx} = 0$. So the function $f(x, y)$ has neither maxima nor minima at (0, 0).

At (a, a), $f_{xx} = 6a > 0$, $f_{yy} = 6a$, $f_{xy} = -3a$

Now, $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$

$= 6a \times 6a - (-3a)^2$

$= 36a^2 - 9a^2$

$= 27a^2 > 0.$

Thus, we see that f satisfies $f_{xx}f_{yy} - (f_{xy})^2 > 0$ and $f_{xx} > 0$ at $(-a, a)$.

So, $f(x, y)$ at (a, a) has minimum value.

Minimum value $(f_{min}) = (a)^3 + (a)^3 - 3a \times a \times a$

$= a^3 + a^3 - 3a^3$

$= -a^3.$

Example 5. Find the local maximum and minimum of the function $U(x, y) = 2x^2 - 3xy + 4y^2 + 5$.

Solution:

Here, $U(x, y) = 2x^2 - 3xy + 4y^2 + 5$

Differentiating partially, we get

$U_x = 4x - 3y$, $U_y = -3x + 8y$

$U_{xx} = 4 > 0$, $U_{yy} = 8$

$U_{xy} = -3$, $U_{yx} = -3$

For maxima and minima, we have $U_x = 0$ and $U_y = 0$

So, $4x - 3y = 0$ .... (i)

and $-3x + 8y = 0$ .... (ii)

Solving equations (i) and (ii), we get

$x = 0$ and $y = 0$

Hence, the stationary point is (0, 0).

At (0, 0), $U_{xx} = 4 > 0$, $U_{yy} = 8$, $U_{xy} = -3$, $U_{yx} = -3$.

$\therefore$ $U_{xx}\,U_{yy} - (U_{xy})^2 = 4 \times 8 - (-3)^2 = 23 > 0$

Hence, the function $U(x, y)$ has minimum value at (0, 0).

The minimum value is $U_{min} = 2 \times 0 - 3 \times 0 + 4 \times 0 + 5 = 5.$

Example 6. Investigate for the extreme values for the function $F(x, y) = x^3 + 3x^2 + 4xy + y^2$.

Solution:

Here, $F(x, y) = x^3 + 3x^2 + 4xy + y^2$

Differentiating partially, we get

$F_x = 3x^2 + 6x + 4y$, $F_y = 4x + 2y$

$F_{xx} = 6x + 6$, $F_{yy} = 2$ and $F_{xy} = F_{yx} = 4.$

For extreme values, we have

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$F_x = 0 \Rightarrow 3x^2 + 6x + 4y = 0$ .... (i)

and $F_y = 0 \Rightarrow 4x + 2y = 0$

which gives $2y = -4x \Rightarrow y = -2x$ .... (ii)

Substituting $y = -2x$ in (i), we get

$3x^2 + 6x + 4y = 0$

or, $3x^2 + 6x + 4(-2x) = 0$

or, $3x^2 - 2x = 0$

or, $x(3x - 2) = 0$

$\therefore \quad x = 0 \text{ or } \frac{2}{3}$.

Then, when $x = 0$, $y = 0$ and when $x = \frac{2}{3}$, $y = -\frac{4}{3}$

Thus, the stationary points are $(0, 0)$ and $\left(\frac{2}{3}, \frac{-4}{3}\right)$.

At $(0, 0)$, $F_{xx} = 6 > 0$, $F_{xy} = 2$, $F_{xy} = F_{yx} = 4$

Now, $\begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 6 & 4 \\ 4 & 2 \end{vmatrix} = 12 - 16 = -4 < 0$.

Thus, there is neither maxima nor minima at $(0, 0)$. Hence, $(0, 0)$ is a saddle point.

At $\left(\frac{2}{3}, -\frac{4}{3}\right)$, $F_{xx} = 6 + 6 \times \frac{2}{3} = 10 > 0$, $F_{xy} = 2$ and $F_{yx} = F_{xy} = 4$.

Now, $\begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 10 & 4 \\ 4 & 2 \end{vmatrix} = 20 - 16 = 4 > 0$.

Therefore, $\left(\frac{2}{3}, -\frac{4}{3}\right)$ is a point at which the function has minimum value.

The minimum value is

$$F\left(\frac{2}{3}, -\frac{4}{3}\right) = \left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2 + 4\left(\frac{2}{3}\right)\left(\frac{-4}{3}\right) + \left(\frac{-4}{3}\right)^2$$

$$= \frac{8}{27} + \frac{4}{3} - \frac{32}{9} + \frac{16}{9} = \frac{-4}{27}.$$

Example 7. Investigate maxima or minima of the function $u(x, y) = 16 - (x + 2)^2 - (y - 2)^2$.

Solution:

Here, $u(x, y) = 16 - (x + 2)^2 - (y - 2)^2$.

Differentiating partially, we get

$u_x = -2(x + 2)$, $u_y = -2(y - 2)$

$u_{xx} = -2$, $u_{yy} = -2$, $u_{xy} = 0$.

Now for extreme value, $u_x = 0$, $u_y = 0$. Therefore,

$-2(x + 2) = 0$, $-2(y - 2) = 0$

or, $x = -2$ and $y = 2$.

Thus, $(-2, 2)$ is the stationary point.

At $(-2, 2)$, $u_{xx}u_{yy} - (u_{xy})^2 = (-2)(-2) - (0)^2 = 4 > 0$

and $u_{xx} = -2 < 0$.

Thus we see that u satisfies $u_{xx}u_{yy} - (u_{xy})^2 > 0$ and $u_{xx} < 0$ at $(-2, 2)$.

So u has a maximum value at $(-2, 2)$ and its maximum value is

$u_{max} = 16 - (-2 + 2)^2 - (2 - 2)^2 = 16$.

Example 8. Examine for the maximum and minimum values of the function $x^3 + y^3 - 3x - 12y + 10$.

Solution:

Let $f(x, y) = x^3 + y^3 - 3x - 12y + 10$.

Differentiating partially, we get

$f_x = 3x^2 - 3$, $f_y = 3y^2 - 12$

$f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = f_{yx} = 0$

For maximum and minimum values,

$f_x = 0$ and $f_y = 0$

or, $3x^2 - 3 = 0$ and $3y^2 - 12 = 0$

or, $x^2 - 1 = 0$ and $y^2 - 4 = 0$

$\therefore \quad x = \pm 1$ $\quad y = \pm 2$

The stationary points are $(1, 2)$, $(-1, 2)$, $(1, -2)$, $(-1, -2)$.

At $(1, 2)$, $f_{xx}f_{yy} - (f_{xy})^2 = (6 \times 1) \times (6 \times 2) - 0 = 72 > 0$

and $f_{xx} = 6 \times 1 = 6 > 0$

So f has minimum value at $(1, 2)$ and minimum value is

$f_{min} = 1^3 + 2^3 - 3 \times 1 - 12 \times 2 + 10$

$= 1 + 8 - 3 - 24 + 10$

$= -8$

At $(-1, 2)$, $f_{xx}f_{yy} - (f_{xy})^2 = 6 \times (-1) \times (6 \times 2) - 0 = -72 < 0$

and $f_{xx} = 6 \times (-1) = -6 < 0$.

$\therefore$ $f(x, y)$ has neither maximum nor minimum value at $(-1, 2)$.

At $(1, -2)$, $f_{xx}f_{yy} - (f_{xy})^2 = 6 \times (1) \times 6 \times (-2) - 0 = -72 < 0$

and $f_{xx} = 6 \times (1) = 6 > 0$.

$\therefore$ $f(x, y)$ has neither maximum nor minimum value at $(1, -2)$.

At $(-1, -2)$, $f_{xx}f_{yy} - (f_{xy})^2 = 6 \times (-1) \times 6 \times (-2) - 0$

$= 72 > 0$

and $f_{xx} = 6 \times (-1) = -6 < 0$.

So, $f(x, y)$ has maximum value at $(-1, -2)$ and maximum value is

$f(-1, -2) = (-1)^3 + (-2)^3 - 3 \times (-1) - 12 \times (-2) + 10$

$= -1 - 8 + 3 + 24 + 10 = 28$.

Example 9. Discuss the maxima or minima of the function $U(x, y) = -x^2 - 2y^2 + 6x - 4y$ and obtain these values.

Solution:

Here, $U(x, y) = -x^2 - 2y^2 + 6x - 4y$

Differentiating partially, we get

$U_x = -2x + 6$, $U_y = -4y - 4$, $U_{xx} = -2$, $U_{yy} = -4$, $U_{xy} = U_{yx} = 0$.

For maxima and minima, $U_x = 0$ and $U_y = 0$

or, $-2x + 6 = 0$, $-4y - 4 = 0$

$\therefore$ $x = 3$, $y = -1$.

Hence, $(3, -1)$ is stationary point.

At $(3, -1)$,

$U_{xx} = -2 < 0$ and $U_{yy} = -4$ and $U_{xx} \cdot U_{yy} - (U_{xy})^2 = -2 \times -4 - (0)^2 = 8 > 0$.

Hence, the function has maximum value at $(3, -1)$ and the maximum value is

$U(3, -1) = -(3)^2 - 2 \times (-1)^2 + 6 \times 3 - 4 \times -1 = -9 - 2 + 18 + 4 = 11$.

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Example 10. Investigate the extreme values of the function
$u(x, y, z) = -(2x+3)^2 - (y-4)^2 - (z+1)^2 + 100$.

Solution:

Here, $u(x, y, z) = -(2x+3)^2 - (y-4)^2 - (z+1)^2 + 100$

Differentiating partially with respect to x, y and z, we get

$u_x = -4(2x+3)$, $u_y = -2(y-4)$ and $u_z = -2(z+1)$.

For extreme values, we have

$u_x = 0 \Rightarrow -4(2x+3) = 0 \Rightarrow x = \frac{-3}{2}$

$u_y = 0 \Rightarrow -2(y-4) = 0 \Rightarrow y = 4$

$u_z = 0 \Rightarrow -2(z+1) = 0 \Rightarrow z = -1$.

$\therefore$ The stationary point is $(x, y, z) = \left(\frac{-3}{2}, 4, -1\right)$

Again, $u_{xx} = -8 < 0$, $u_{yy} = -2 < 0$ and $u_{xy} = 0 = u_{yx}$

Now, $\begin{vmatrix} u_{xx} & u_{xy} \\ u_{yx} & u_{yy} \end{vmatrix} = \begin{vmatrix} -8 & 0 \\ 0 & -2 \end{vmatrix} = 16 > 0$

and $\begin{vmatrix} u_{xx} & u_{xy} & u_{xz} \\ u_{yz} & u_{yy} & u_{yz} \\ u_{zx} & u_{zy} & u_{zz} \end{vmatrix} = \begin{vmatrix} -8 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{vmatrix} = -32 < 0.$

Therefore, the function has maximum value at $\left(\frac{-3}{2}, 4, -1\right)$. The maximum value is

$u\left(\frac{-3}{2}, 4, -1\right) = (-3+3)^2 - (4-4)^2 - (-1+1)^2 + 100 = 100.$

Example 11. Discuss the local minimum and maximum values of the function
$1 + (x+y+z)^3 - 3(x+y+z) - 24xyz$.

Solution:

Let $F(x, y, z) = 1 + (x+y+z)^3 - 3(x+y+z) - 24xyz$

Differentiating partially with respect to x, y and z, we get

$F_x = 3(x+y+z)^2 - 3 - 24yz$, $F_y = 3(x+y+z)^2 - 3 - 24xz$, $F_z = 3(x+y+z)^2 - 3 - 24xy$

For stationary points, we have

$F_x = 0 \Rightarrow 3(x+y+z)^2 - 3 - 24yz = 0$ .... (i)

$F_y = 0 \Rightarrow 3(x+y+z)^2 - 3 - 24xz = 0$ .... (ii)

$F_z = 0 \Rightarrow 3(x+y+z)^2 - 3 - 24xy = 0$ .... (iii)

Solving (i) and (ii), on subtracting, we get

$24z(x-y) = 0$

Either $z = 0$ or $x = y$ .... (iv)

Similarly on subtracting (i) and (iii), we get

$24y(x-z) = 0$

Either $y = 0$ or $x = z$ .... (v)

Similarly from (ii) and (iii) $x = 0$ or $y = z$ .... (vi)

$\therefore$ From (iv), (v) and (vi)

$(x, y, z) = (0, 0, 0)$ or $x = y = z$

If $x = y = z$, from (i)

$3(3x)^2 - 3 - 24x.x = 0$

or, $27x^2 - 3 - 24x^2 = 0$

or, $3x^2 - 3 = 0$

$\Rightarrow$ $x = \pm 1$, $\Rightarrow y = \pm 1$, and $z = \pm 1$.

Hence, $(x, y, z) = (0, 0, 0)$ or $(1, 1, 1)$ or $(-1, -1, -1)$

Again

$F_{xx} = 6(x+y+z)$ $\quad F_{yy} = 6(x+y+z)$

$F_{zz} = 6(x+y+z)$ $\quad F_{xy} = 6(x+y+z) - 24z$

$F_{yz} = 6(x+y+z) - 24x$ $\quad F_{xz} = 6(x+y+z) - 24y$

At $(0, 0, 0)$, $F_{xx} = 0$, $F_{yy} = 0$, $F_{zz} = 0$, $F_{xy} = 0$, $F_{yz} = 0$ and $F_{xz} = 0$

Now $\begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$

This does not give any information.

At $(1, 1, 1)$, $F_{xx} = 18$, $F_{yy} = 18$, $F_{zz} = 18$, $F_{xy} = -6$, $F_{yz} = -6$ and $F_{xz} = -6$

Now, $\begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 18 & -6 \\ -6 & 18 \end{vmatrix} = 324 - 36 = 288 > 0$

and $\begin{vmatrix} F_{xx} & F_{xy} & F_{xz} \\ F_{yx} & F_{yy} & F_{yz} \\ F_{zx} & F_{zy} & F_{zz} \end{vmatrix} = \begin{vmatrix} 18 & -6 & -6 \\ -6 & 18 & -6 \\ -6 & -6 & 18 \end{vmatrix} = 3456 > 0.$

$\therefore$ $F(x, y, z)$ is minimum at $(1, 1, 1)$. The minimum value is

$F_{min} = 1 + (1+1+1)^3 - 3(1+1+1) - 24 \cdot (1)(1)(1) = 1 + 27 - 9 - 24 = -5$

At $(-1, -1, -1)$, $F_{xx} = -18$, $F_{yy} = -18$, $F_{zz} = -18$, $F_{xy} = 6$, $F_{yz} = 6$ and $F_{zx} = 6$.

Now, $\begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} -18 & 6 \\ 6 & -18 \end{vmatrix} = 324 - 36 = 288 > 0$

and $\begin{vmatrix} F_{xx} & F_{xy} & F_{xz} \\ F_{yx} & F_{yy} & F_{yz} \\ F_{zx} & F_{zy} & F_{zz} \end{vmatrix} = \begin{vmatrix} -18 & 6 & 6 \\ 6 & -18 & 6 \\ 6 & 6 & -18 \end{vmatrix} = -3456 < 0$

$\therefore$ $F(x, y, z)$ has maximum value at $(-1, -1, -1)$ and the maximum value is

$F_{max} = 1 + (-1-1-1)^3 - 3(-1-1-1) - 24 \cdot (-1)(-1)(-1) = 1 - 27 + 9 + 24 = 7.$

Next we discuss some applied problems.

Example 12. Find the dimension of the rectangular box open at the top with volume 32 cc requiring least material for the construction.

Solution:

Let x, y, z be the required length, breadth and height of the box respectively. Then volume is $V = xyz$

or, $32 = xyz$ $\Rightarrow y = \frac{32}{xz}$

Again, surface area $S = 2(x+y)z + xy$ .... (i)

or, $S = 2xz + \frac{64}{x} + \frac{32}{z}$ .... (ii)

Differentiating (ii) partially with respect to x, we get

$S_x = 2z - \frac{64}{x^2}$ .... (iii)

and differentiating (ii) partially with respect to z, we get

$S_z = 2x - \frac{32}{z^2}$ .... (iv)

For maxima and minima, we have

$S_x = 0 \Rightarrow 2z - \frac{64}{x^2} = 0$ $\Rightarrow z = \frac{32}{x^2}$ .... (v)

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$\Rightarrow x = \frac{16}{z^2}$ .... (vi)

i.e. $S_z = 0 \Rightarrow 2x - \frac{32}{z^2} = 0$

Using equations (v) and (vi), we have

$$z = \frac{32}{\left(\frac{16}{z^2}\right)^2} = \frac{32 \times z^4}{16 \times 16} = \frac{z^4}{8}$$

or, $8z - z^4 = 0$

or, $z(8 - z^3) = 0$

Either $z = 0$ or $z = 2$ $\therefore z = 2$

$\Rightarrow x = 4, y = 4$

Again, $S_{xx} = \frac{128}{x^3} = \frac{128}{64} = 2 > 0$

and $S_{zz} = 2, S_{zz} = \frac{64}{z^3} = \frac{64}{8} = 8 > 0$

$\therefore S_{xx} S_{zz} - (S_{xz})^2 = 2 \times 8 - 2^2 = 16 - 4 = 12 > 0.$

Hence, surface is minimum for $x = 4$cm, $y = 4$cm, $z = 2$cm.

6.3 Lagrange's method of multiplier

Many practical problems in science, engineering and in daily life are formulated as constrained optimization problems and one has to solve these problems. For example, in a production company, the production of two different types of goods is to be maximized or minimized under the given budget constraints and this type of problems are formulated in constrained optimization problems. These type of constrained optimization problems can be solved using the method of Lagrange's method of multiplier. The Lagrange's method of multipliers is named after Joseph-Louis Lagrange, an Italian mathematician. This method is considered as an important technique for solving constrained optimization problems. We note that the Lagrange's method of multiplier is just to find the point of extreme values (stationary point), but not for obtaining maximum or minimum values of the function. We are required to use the Hessain determinant test for determining whether the function has maximum or minimum values at the stationary point.

We now discuss the method.

6.3.1 Lagrange's multipliers

The working process of Lagrange's method of multipliers is as follows:

i. Functions of two variables $f(x, y)$ under condition $\phi(x, y) = 0$

Let $f(x, y)$ be a function whose extreme values are to be determined under the constraint or condition $\phi(x, y) = 0$.

Assume that both f and ϕ have continuous first order partial derivatives.

$F(x, y, \lambda) = f(x, y) + \lambda\phi(x, y)$, where λ is called Lagrange's multiplier.

For extreme values, we solve $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial \lambda} = 0.$

Solving these, we get the stationary points $(a, b, c), \ldots$

At these stationary points $(a, b, c), \ldots$, the function $f(x, y)$ will have

maximum value if $\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} > 0$ and

minimum value if $\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} < 0.$

ii. Functions of three variables $f(x, y, z)$ under condition $\phi(x, y, z) = 0$

Let $f(x, y, z)$ be a function whose extreme values are to be determined under the condition $\phi(x, y, z) = 0$.

Assume that both f and ϕ have continuous first order partial derivatives. Let us introduce Lagrange multiplier λ and define Lagrange function F by

$$F(x, y, z, \lambda) = f(x, y, z) + \lambda\phi(x, y, z)$$

For extreme values, $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0, \frac{\partial F}{\partial \lambda} = 0$

Solving the above condition gives the stationary points $(a, b, c), \ldots$.

Then we find the Hessian determinants, say $|H_{B1}|$ and $|H_{B2}|$, as follows:

$$|H_{B1}| = \begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} \text{ and } |H_{B2}| = \begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & F_{xx} & F_{xy} & F_{xz} \\ \phi_y & F_{yx} & F_{yy} & F_{yz} \\ \phi_z & F_{zx} & F_{zy} & F_{zz} \end{vmatrix}.$$

We need to evaluate the value of $|H_{B1}|$ and $|H_{B2}|$ at the stationary points $(a, b, c), \ldots$.

The function $f(x, y, z)$ with constraint $\phi(x, y, z) = 0$ will be

i. maximum value when $|H_{B1}| > 0$ and $|H_{B2}| < 0$.

ii. minimum value when $|H_{B1}| < 0$ and $|H_{B2}| < 0$.

In case if the above conditions are not satisfied, then the function $f(x, y)$ will have neither a maximum nor a minimum.

iii. Function of three variables $f(x, y, z)$ under two constraints (conditions)

Lagrange's method of undetermined multipliers in the case of three variables function $f(x, y, z)$ connected by two constraints $\phi(x, y, z) = 0$ and $\psi(x, y, z) = 0$, can be explained stepwise as:

We first define $F = f + \lambda\phi + \mu\psi$. ....(i)

where constants λ and μ are called Lagrange's multipliers. Here, we note that the number of multipliers is equal to the number of constraints in the given problem.

Next we differentiate (i) partially with respect to x, y, z and λ, μ, we get

$F_x = f_x + \lambda\phi_x + \mu\psi_x$

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$F_y = f_y + \lambda\phi_y + \mu\psi_y$

$F_z = f_z + \lambda\phi_z + \mu\psi_z$

$F_\lambda = \phi$ and $F_\mu = \psi$.

For the stationary points, we have

$F_x = F_y = F_z = F_\lambda = F_\mu = 0$.

Solving these equations, we get the stationary points (a, b, c),.... Rule of deciding the local maximum and minimum value points is similar to the case of single constraint in which we find the Hessian determinants corresponding the two multipliers μ and λ. But in practice, the points of extreme values are determined by reducing into single variable depending on number of constraints. We now discuss some examples.

Example 13. Find the extreme value for the function $f(x, y) = x^2 + y^2$ under the condition $x + 4y = 2$.

Solution:

Here, $f(x, y) = x^2 + y^2$ under the condition $x + 4y = 2$.

Now, expressing f as a function of y alone, we get

$f(y) = (2 - 4y)^2 + y^2 = 4 - 16y + 17y^2$

and $\frac{df}{dy} = -16 + 34y$ and $\frac{d^2f}{dy^2} = 34$

For extreme values,

$\frac{df}{dy} = 0$

or, $-16 + 34y = 0$

or, $y = \frac{16}{34} = \frac{8}{17}$.

Substituting, $y = \frac{8}{17}$ in $x + 4y = 2$, we get

$x = 2 - 4 \times \frac{8}{17} = \frac{2}{17}$

$\therefore$ Stationary point is $\left(\frac{2}{17}, \frac{8}{17}\right)$.

Since $\frac{d^2f}{dy^2} = 34 > 0$, the given function has a minimum value at $\left(\frac{2}{17}, \frac{8}{17}\right)$.

$f_{min} = \left(\frac{2}{17}\right)^2 + \left(\frac{8}{17}\right)^2 = \frac{4}{17}$. $[\because f = x^2 + y^2]$

Example 14. Use Lagrange's multiplier to find the maximum and minimum values of the function $u(x, y) = ax + by$ subject to the constraint $x^2 + y^2 = 1$, where a and b are positive constants.

Solution:

Here, $u(x, y) = ax + by$ subject to the condition $\phi = x^2 + y^2 - 1 = 0$.

Let $F = u + \lambda\phi = ax + by + \lambda(x^2 + y^2 - 1)$

$= ax + by + \lambda(x^2 + y^2 - 1)$

where λ is the Lagrange's multiplier. Then, ... (i)

$F_x = a + 2\lambda x$, $F_y = b + 2\lambda y$, $F_\lambda = x^2 + y^2 - 1$

$F_{xx} = 2\lambda$, $F_{yy} = 2\lambda$, $F_{xy} = F_{yx} = 0$.

For extreme values, we have

$F_x = 0$, $F_y = 0$, $F_\lambda = 0$

i.e., $a + 2\lambda x = 0$, $b + 2\lambda y = 0$, $x^2 + y^2 = 1$

or, $x = \frac{-a}{2\lambda}$, $y = \frac{-b}{2\lambda}$

Using $F_\lambda = 0$,

$x^2 + y^2 = 1$

or, $\left(\frac{-a}{2\lambda}\right)^2 + \left(\frac{-b}{2\lambda}\right)^2 = 1$

Also, $a^2 + b^2 = 4\lambda^2$

$\therefore \lambda = \pm\frac{\sqrt{a^2 + b^2}}{2}$

When $\lambda = \frac{\sqrt{a^2 + b^2}}{2}$,

$x = \frac{-a \times 2}{2\sqrt{a^2 + b^2}} = \frac{-a}{\sqrt{a^2 + b^2}}$ and

$y = \frac{-b \times 2}{2\sqrt{a^2 + b^2}} = \frac{-b}{\sqrt{a^2 + b^2}}$

When $\lambda = -\frac{\sqrt{a^2 + b^2}}{2}$, $x = \frac{-a \times 2}{-2\sqrt{a^2 + b^2}} = \frac{a}{\sqrt{a^2 + b^2}}$, and $y = \frac{b}{\sqrt{a^2 + b^2}}$.

Thus, the stationary points are

$\left(\frac{-a}{\sqrt{a^2+b^2}}, \frac{-b}{\sqrt{a^2+b^2}}\right)$ and $\left(\frac{a}{\sqrt{a^2+b^2}}, \frac{b}{\sqrt{a^2+b^2}}\right)$.

At $\left(\frac{-a}{\sqrt{a^2+b^2}}, \frac{-b}{\sqrt{a^2+b^2}}\right)$, $F_{xx} = 2\lambda = 2 \times \frac{\sqrt{a^2+b^2}}{2} = \sqrt{a^2+b^2}$

$F_{yy} = 2\lambda = 2 \times \frac{\sqrt{a^2+b^2}}{2} = \sqrt{a^2+b^2}$

$\phi_x = 2x = 2 \times \frac{-a}{\sqrt{a^2+b^2}} = \frac{-2a}{\sqrt{a^2+b^2}}$

$\phi_y = 2y = 2 \times \frac{-b}{\sqrt{a^2+b^2}} = \frac{-2b}{\sqrt{a^2+b^2}}$

Now, $\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 0 & \frac{-2a}{\sqrt{a^2+b^2}} & \frac{-2b}{\sqrt{a^2+b^2}} \\ \frac{-2a}{\sqrt{a^2+b^2}} & \sqrt{a^2+b^2} & 0 \\ \frac{-2b}{\sqrt{a^2+b^2}} & 0 & \sqrt{a^2+b^2} \end{vmatrix}$

$= \frac{2a}{\sqrt{a^2+b^2}}(-2a - 0) - \frac{2b}{\sqrt{a^2+b^2}}(0 + 2b)$

$= \frac{-4}{\sqrt{a^2+b^2}}(a^2+b^2) = -4\sqrt{a^2+b^2} < 0$, a negative.

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$\therefore$ The function has minimum value at point $\left(\frac{-a}{\sqrt{a^2+b^2}}, \frac{-b}{\sqrt{a^2+b^2}}\right)$.

The Minimum value, $u_{min} = a\frac{-a}{\sqrt{a^2+b^2}} + b\frac{-b}{\sqrt{a^2+b^2}} = -\frac{(a^2+b^2)}{\sqrt{a^2+b^2}} = -\sqrt{a^2+b^2}$

At $\left(\frac{a}{\sqrt{a^2+b^2}}, \frac{b}{\sqrt{a^2+b^2}}\right)$, $F_{xx} = 2\lambda = 2\left(-\frac{\sqrt{a^2+b^2}}{2}\right) = -\sqrt{a^2+b^2} < 0$

$F_{yy} = 2\lambda = 2\left(-\frac{\sqrt{a^2+b^2}}{2}\right) = -\sqrt{a^2+b^2}$

$\phi_x = 2x = 2\frac{a}{\sqrt{a^2+b^2}} = \frac{2a}{\sqrt{a^2+b^2}}$

$\phi_y = 2y = 2\frac{b}{\sqrt{a^2+b^2}} = \frac{2b}{\sqrt{a^2+b^2}}$

Now, $\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 0 & \frac{2a}{\sqrt{a^2+b^2}} & \frac{2b}{\sqrt{a^2+b^2}} \\ \frac{2a}{\sqrt{a^2+b^2}} & -\sqrt{a^2+b^2} & 0 \\ \frac{2b}{\sqrt{a^2+b^2}} & 0 & -\sqrt{a^2+b^2} \end{vmatrix}$

$= \frac{\cdots}{\sqrt{a^2+b^2}}(0+2a) + \frac{2b}{\sqrt{a^2+b^2}}(0+2b)$

$= \frac{4}{\sqrt{a^2+b^2}}(a^2+b^2) \quad = 4\sqrt{a^2+b^2} > 0$, a positive.

$\therefore$ The function has maximum value at $\left(\frac{a}{\sqrt{a^2+b^2}}, \frac{b}{\sqrt{a^2+b^2}}\right)$

Maximum value $(u_{max}) = a\frac{a}{\sqrt{a^2+b^2}} + b\frac{b}{\sqrt{a^2+b^2}} = \frac{(a^2+b^2)}{\sqrt{a^2+b^2}} = \sqrt{a^2+b^2}$.

Example 15. Use Lagrange's multiplier to find the minimum value of $x^2 + y^2 + z^2$ subject to the constraint $ax + by + cz = p$.

Solution:

Let $f = x^2 + y^2 + z^2$ and $\phi = ax + by + cz - p$.

To solve this problem, by using Lagrange's multiplier method, let

$F = f + \lambda\phi = (x^2 + y^2 + z^2) + \lambda(ax + by + cz - p)$

where λ is Lagrange's multiplier.

Differentiating partially, we get

$F_x = 2x + \lambda a$, $F_y = 2y + \lambda b$,

$F_z = 2z + \lambda c$, $F_\lambda = ax + by + cz - p$.

For extreme values, we have

$F_x = 0$, $F_y = 0$, $F_z = 0$ and $F_\lambda = 0$

or, $2x + \lambda a = 0$, $2y + \lambda b = 0$, $2z + \lambda c = 0$, $ax + by + cz - p = 0$.

Then, $x = -\frac{\lambda a}{2}$, $y = -\frac{\lambda b}{2}$, $z = -\frac{\lambda c}{2}$.

Now, $ax + by + cz - p = 0$ gives

$a\left(-\frac{\lambda a}{2}\right) + b\left(-\frac{\lambda b}{2}\right) + c\left(-\frac{\lambda c}{2}\right) = p$

or, $-\lambda\frac{(a^2+b^2+c^2)}{2} = p$ $\quad \therefore \lambda = \frac{-2p}{a^2+b^2+c^2}$.

Therefore, $x = \frac{ap}{a^2+b^2+c^2}$, $y = \frac{bp}{a^2+b^2+c^2}$, $z = \frac{cp}{a^2+b^2+c^2}$.

So the stationary point $= \left(\frac{ap}{a^2+b^2+c^2}, \frac{bp}{a^2+b^2+c^2}, \frac{cp}{a^2+b^2+c^2}\right)$.

Again, $F_{xx} = 2$, $F_{yy} = 2$, $F_{zz} = 2$

$F_{xy} = F_{yz} = F_{zx} = 0$, $\phi_x = a$, $\phi_y = b$, $\phi_z = c$

Now, $|H_{B1}| = \begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix} = \begin{vmatrix} 0 & a & b \\ a & 2 & 0 \\ b & 0 & 2 \end{vmatrix} = -(2a^2 + 2b^2) < 0$,

and $|H_{B2}| = \begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & F_{xx} & F_{xy} & F_{xz} \\ f_y & F_{yx} & F_{yy} & F_{yz} \\ \phi_z & F_{zx} & F_{zy} & F_{zz} \end{vmatrix} = \begin{vmatrix} 0 & a & b & c \\ a & 2 & 0 & 0 \\ b & 0 & 2 & 0 \\ c & 0 & 0 & 2 \end{vmatrix}$

$= -a\begin{vmatrix} a & 0 & 0 \\ b & 2 & 0 \\ c & 0 & 2 \end{vmatrix} + b\begin{vmatrix} a & 2 & 0 \\ b & 0 & 0 \\ c & 0 & 2 \end{vmatrix} - c\begin{vmatrix} a & 2 & 0 \\ b & 0 & 2 \\ c & 0 & 0 \end{vmatrix}$

$= -a.4a + b(-4b) - c(4c)$

$= -(4a^2 + 4b^2 + 4c^2) < 0$.

At $\left(\frac{ap}{a^2+b^2+c^2}, \frac{bp}{a^2+b^2+c^2}, \frac{cp}{a^2+b^2+c^2}\right)$, $|H_{B1}| < 0$ and $|H_{B2}| < 0$.

Therefore, f has minimum value. The minimum value is

$f_{min} = \left(\frac{ap}{a^2+b^2+c^2}\right)^2 + \left(\frac{bp}{a^2+b^2+c^2}\right)^2 + \left(\frac{cp}{a^2+b^2+c^2}\right)^2 = \frac{p^2}{a^2+b^2+c^2}$.

Example 16. Use Lagrange's multiplier method to find the extreme point of the function $U(x, y) = x^2 + xy + y^2 + 3z^2$ under the constraint $x + 2y + 4z = 60$. Also find the extreme value.

Solution:

Here, $U(x, y) = x^2 + xy + y^2 + 3z^2$ such that $x + 2y + 4z - 60 = 0$.

Let $\phi(x, y, z) = x + 2y + 4z - 60$

We define $F = U + \lambda\phi$, where λ is Lagrange's multiplier.

$= x^2 + xy + y^2 + 3z^2 + \lambda(x + 2y + 4z - 60)$

Differentiating partially with respect to x, y, z, and λ, we get

$F_x = 2x + y + \lambda$ $\quad F_\lambda = x + 2y + 2\lambda$

$F_z = 6z + 4\lambda$ $\quad F_\lambda = x + 2y + 4z - 60$

For extreme values, we have

$F_x = 0 \Rightarrow 2x + y + \lambda = 0$ .... (i)

$F_y = 0 \Rightarrow x + 2y + 2\lambda = 0$ .... (ii)

$F_z = 0 \Rightarrow 6z + 4\lambda = 0$ .... (iii)

$F_\lambda = 0 \Rightarrow x + 2y + 4z - 60 = 0$ .... (iv)

Now, solving (i) and (ii), we have

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$$2x + y + \lambda = 0$$
$$\underline{-2x \pm 4y \pm 4\lambda = 0}$$
$$-3y - 3\lambda = 0$$

$\Rightarrow y = -\lambda$ .... (v)

Substituting $y = -\lambda$ in (i), we have

$2x - \lambda + \lambda = 0$

$\therefore x = 0$ .... (vi)

and from (iii), $6z = -4\lambda$

$\therefore z = -\frac{4}{6}\lambda = \frac{-2\lambda}{3}$ .... (vii)

Now substituting the value of (v), (vi) and (vii) in (iv), we have

$$0 + 2(-\lambda) + 4\left(\frac{-2}{3}\right)\lambda - 60 = 0$$

or, $-2\lambda - \frac{8\lambda}{3} - 60 = 0$

or, $-6\lambda - 8\lambda = 180$

or, $-14\lambda = 180$

$\therefore \lambda = \frac{-90}{7}$

Hence, $y = -\lambda = \frac{90}{7}$ and $z = \frac{-21}{3} = \frac{60}{7}$

Thus, the stationary point is $\left(0, \frac{90}{7}, \frac{60}{7}\right)$.

Again, $F_{xx} = 2$ $F_{yy} = 2$, $\phi_x = 1$
$F_{xy} = 1$ $F_{yz} = 0$ $\phi_y = 2$
$F_{zx} = 0$ $F_{zz} = 6$ $\phi_z = 4$

Now, $|H_{B1}| = \begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & F_{xx} & F_{xy} \\ \phi_y & F_{yx} & F_{yy} \end{vmatrix}$

$= \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix}$

$= 0\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$

$= 0 - 0 + 2(1 - 4)$

$= -6 < 0$

and $|H_{B2}| = \begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & F_{xx} & F_{xy} & F_{xz} \\ \phi_y & F_{yx} & F_{yy} & F_{yz} \\ \phi_z & F_{zx} & F_{zy} & F_{zz} \end{vmatrix} = \begin{vmatrix} 0 & 1 & 2 & 4 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 0 & 0 & 6 \end{vmatrix}$

$= -4\begin{vmatrix} 1 & 2 & 4 \\ 2 & 1 & 0 \\ 1 & 2 & 0 \end{vmatrix} - 0 - 0 + 6\begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix}$

$= -4 \times 4\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + 6\left\{-1\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}\right\}$

$= -16 \times 3 + 6(-6) = -48 - 36 = -84 < 0.$

Thus at $\left(0, \frac{90}{7}, \frac{60}{7}\right)$, $|H_{B1}| < 0$ and $|H_{B2}| < 0$. So f has minimum value at $\left(0, \frac{90}{7}, \frac{60}{7}\right)$. The minimum value is $U_{min} = 0 + 0 \times \frac{90}{7} + \left(\frac{90}{7}\right)^2 + 3\left(\frac{60}{7}\right)^2 = \frac{8100}{49} + \frac{10800}{49} = \frac{18900}{49}$

Example 17. Find the maximum and minimum distances of the point (3, 4, 12) from the sphere $x^2 + y^2 + z^2 = 1$.

Solution

The given equation of sphere is $x^2 + y^2 + z^2 = 1$ where (x, y, z) is any point in the sphere. The distance between (x, y, z) and (3, 4, 12) is

$d = \sqrt{(x-3)^2 + (y-4)^2 + (z-12)^2}$

Let $f(x, y, z) = (x-3)^2 + (y-4)^2 + (z-12)^2$

under the constraint $x^2 + y^2 + z^2 = 1$

i.e. $\phi(x, y, z) = x^2 + y^2 + z^2 - 1$.

Now, $F = f + \lambda\phi = (x-3)^2 + (y-4)^2 + (z-12)^2 + \lambda(x^2 + y^2 + z^2 - 1)$

$\therefore \frac{\partial F}{\partial x} = 2(x-3) + 2\lambda x$, $\frac{\partial F}{\partial y} = 2(y-4) + 2\lambda y$ and $\frac{\partial F}{\partial z} = 2(z-12) + 2\lambda z$

For the critical points, using $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$, we have

$2(x-3) + 2\lambda x = 0$ ....(i)

$2(y-4) + 2\lambda y = 0$ ....(ii)

$2(z-12) + 2\lambda z = 0$ ....(iii)

Multiplying equation (i) by x, equation (ii) by y and equation (iii) by z and adding, we get

$x^2 + y^2 + z^2 - 3x - 4y - 12z + \lambda(x^2 + y^2 + z^2) = 0$

or, $1 - 3x - 4y - 12z + \lambda = 0$ ....(iv)

Also, from equation (i), $x = \frac{3}{1+\lambda}$ ....(v)

From equation (ii), $y = \frac{4}{1+\lambda}$ ....(vi)

and, from equation (iii), $z = \frac{12}{1+\lambda}$ ....(vii)

Substituting these values in equation (iv), we get

$1 - 3x - 4y - 12z + \lambda = 0$

or, $1 + \lambda - \frac{9}{1+\lambda} - \frac{16}{1+\lambda} - \frac{144}{1+\lambda} = 0$

or, $(1+\lambda)^2 = 169 \Rightarrow 1 + \lambda = \pm 13$

Hence, from equation (v), (vi) and (vii)

$x = \pm\frac{3}{13}, y = \pm\frac{4}{13}, z = \pm\frac{12}{13}$

$\therefore$ The minimum distance at $\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ is $d_{min} = \sqrt{\left(3 - \frac{3}{13}\right)^2 + \left(4 - \frac{4}{13}\right)^2 + \left(12 - \frac{12}{13}\right)^2} = 12$

and maximum distance at $\left(-\frac{3}{13}, -\frac{4}{13}, -\frac{12}{13}\right)$ is $d_{max} = \sqrt{\left(3 + \frac{3}{13}\right)^2 + \left(4 + \frac{4}{13}\right)^2 + \left(12 + \frac{12}{13}\right)^2} = 14.$

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Example 18. Investigate the extreme values of $u(x, y, z) = 2x + 3y + z$ subject to the constraints $x + z = 1$ and $x^2 + y^2 = 5$ (apply Lagrange multiplier method).

Solution:

Here, $u(x, y, z) = 2x + 3y + z$, $\phi = x + z - 1$ and $\psi = x^2 + y^2 - 5$.

The Lagrange's function is

$F(x, y, z, \lambda, \mu) = 2x + 3y + z + \lambda(x + z - 1) + \mu(x^2 + y^2 - 5)$

where λ and μ are Lagrange multipliers.

Differentiating partially with respect to x, y, z, and λ, μ, we get

$F_x = 2 + \lambda + 2x\mu$, $F_y = 3 + 2y\mu$, $F_z = 1 + \lambda$,

$F_\lambda = x + z - 1$ and $F_\mu = x^2 + y^2 - 5$.

Using the first order condition for extreme values, we have

$F_x = 0 \Rightarrow 2 + \lambda + 2x\mu = 0$ .... (i)

$F_y = 0 \Rightarrow 3 + 2\mu y = 0$ .... (ii)

$F_z = 0 \Rightarrow 1 + \lambda = 0 \Rightarrow \lambda = -1$ .... (iii)

$F_\lambda = 0 \Rightarrow x + z - 1 = 0 \Rightarrow x + z = 1$ .... (iv)

$F_\mu = 0 \Rightarrow x^2 + y^2 - 5 = 0 \Rightarrow x^2 + y^2 = 5$ .... (v)

From (i) and (iii)

$2 - 1 + 2x\mu = 0$

or, $2x\mu + 1 = 0 \Rightarrow x = -\frac{1}{2\mu}$

and from (ii), $3 + 2\mu y = 0 \Rightarrow y = -\frac{3}{2\mu}$

Substituting the values of x and y in (v), we obtain

$\frac{1}{4\mu^2} + \frac{9}{4\mu^2} = 5$ or, $\frac{10}{4\mu^2} = 5$

$\therefore 2\mu^2 = 1 \Rightarrow \mu = \pm\frac{1}{\sqrt{2}}$

Then, $x = -\frac{1}{2\mu} = \pm\frac{1}{\sqrt{2}}$, $y = -\frac{3}{2\mu} = \pm\frac{3}{\sqrt{2}}$

and from (iv), $x + z = 1 \Rightarrow z = 1 \pm \frac{1}{\sqrt{2}}$.

If $(x, y, z) = \left(\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}}\right)$, then

$u(x, y, z) = 2x + 3y + z = \frac{2}{\sqrt{2}} + \frac{9}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} = 5\sqrt{2} + 1$

If $(x, y, z) = \left(\frac{-1}{\sqrt{2}}, \frac{-3}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}}\right)$, then

$u(x, y, z) = 2x + 3y + z = 2\left(\frac{-1}{\sqrt{2}}\right) + 3\left(\frac{-3}{\sqrt{2}}\right) + \left(1 + \frac{1}{\sqrt{2}}\right)$

$= \frac{-2}{\sqrt{2}} - \frac{9}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} = 1 - 5\sqrt{2}$

Thus, the extreme values of $u(x, y, z)$ are $5\sqrt{2} + 1$ and $1 - 5\sqrt{2}$.

6.4 Absolute maximum and minimum values

If we are given a function of two (or three variables) defined in some specified D of xy-plane then we can compare the maximum or minimum values at stationary points with the functional values of stationary points with the functional values of the function on the boundary of D.

In such case the largest of all maximum value and functional value on the boundary of D is called absolute maxima.

In order to examine absolute maxima we need to find functional value stationary points and functional value on boundaries of the domain D.

Similarly, the smallest of the all minimum values and functional value on the boundary of D is called absolute minima.

Example 19. Find absolute maxima and minima values of $f(x, y) = x^2 + y^2 - 2x$ in D, where D is closed triangular region with vertices (2, 0), (0, 2) and (0, – 2).

Solution:

Given $f(x, y) = x^2 + y^2 - 2x$ .... (i)

For stationary points, differentiating (i) partially with respect to x and y respectively.

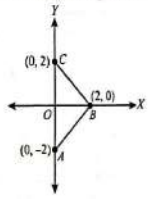
$f_x = 2x - 2$, $f_y = 2y$

Solving $f_x = 0$ and $f_y = 0$, we get the stationary point $(x, y) = (1, 0)$.

From (i) the functional value at stationary point is

$f(1, 0) = -1$.

Now, for functional value along the boundary of D. The domain D is a triangular region with vertices (2, 0), (0, 2) and (0, – 2) is shown in figure.



Along AB,

$\frac{x}{2} + \frac{y}{-2} = 1$

or, $\frac{y}{-2} = 1 - \frac{x}{2}$

or, $y = \frac{-2(2 - x)}{2}$

$\therefore y = x - 2$

From (i), we have

$f(x, y) = x^2 + (x - 2)^2 - 2x$

$= x^2 + x^2 - 4x + 4 - 2x$

$= 2x^2 - 6x + 4$

$f(x, y) = 2x^2 - 6x + 4$

At $A(0, -2)$, $f(0, -2) = 4$

At $B(2, 0)$, $f(2, 0) = 0$

Along BC,

$\frac{x}{2} + \frac{y}{2} = 1$

$\therefore y = (2 - x)$

From (i),

$f(x, y) = x^2 + (2 - x)^2 - 2x$

$= 2x^2 + 4 - 4x - x^2 - 2x$

$= x^2 - 6x + 4$

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And at $C(0, 2)$, $f(0, 2) = 4$.
At $B(2, 0)$, $f(2, 0) = -4$.
Thus, the values of function $f(x, y)$ at

a. Stationary point $(1, 0)$, $f(1, 0) = -1$
b. At vertex $A(0, -2)$, $f(0, -2) = 4$
c. At vertex $B(2, 0)$, $f(2, 0) = 0$
d. At vertex $C(0, 2)$, $f(0, 2) = 4$

Therefore, the absolute maxima is at $(0, \pm 2)$ and $f(0, \pm 2) = 4$ and absolute minima is at $(2, 0)$ and $f(2, 0) = 0$.

Example 20. Find absolute maximum and minimum values of the function

$f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y): 0 \le x \le 3, 0 \le y \le 2\}$

Solution:

We have given equation

$f(x, y) = x^2 - 2xy + 2y$ .... (i)

Differentiating (i) partially with respect to x and y respectively,

$f_x = 2x - 2y, f_y = -2x + 2$

For stationary points,

$f_y = 0$ i.e., $x - y = 0$ and $f_x = 0$ i.e., $x - 1 = 0$

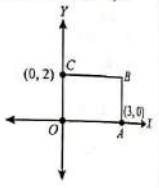
$\therefore$ The stationary point is $(x, y) = (1, 1)$.

The functional values of (i) at stationary point $(1, 1)$

$f(1, 1) = 1 - 2 + 2 = 1$.

Now, for functional value of $f(x, y)$ along the boundaries of $D = \{(x, y): 0 \le x \le 3, 0 \le y \le 2\}$.

The domain D is a rectangle as shown in figure.



Along OA, $y = 0, 0 \le x \le 3$
$f(x, 0) = x + 2$
At $O(0, 0)$, $f(0, 0) = 0$
At $A(3, 0)$, $f(3, 0) = 5$

Along AB, $x = 3, 0 \le y \le 2$ [$\because$ from (i)]
$f(3, y) = 9 - 6y + 2y = 9 - 4y$
At $A(3, 0)$, $f(3, 0) = 9$
At $B(3, 2)$, $f(3, 2) = 9 - 8 = 1$

Along BC, $y = 2, 0 \le x \le 3$, [$\because$ from (i)]
$f(x, 2) = x^2 - 4x + 4$
At $B(3, 2)$, $f(3, 2) = 1$
At $C(0, 2)$, $f(0, 2) = 4$

Thus, the functional value of $f(x, y)$ at

a. Stationary point $(1, 1)$, $f(1, 1) = 1$
b. At $O(0, 0)$, $f(0, 0) = 0$
c. At $A(3, 0)$, $f(3, 0) = 9$
d. At $B(3, 2)$, $f(3, 2) = 1$
e. At $C(0, 2)$, $f(0, 2) = 4$.

Therefore, the absolute maxima is at $B(3, 0)$ and $f(3, 0) = 9$, and absolute minima is at $O(0, 0)$ and $f(0, 0) = 0$.

Exercise 6.2

1. Find the stationary points and determine the extreme values of the following functions.
   a. $f = x^2 + xy + y^2$.
   b. $u = -x^2 - y^2 + 2x + 2y + 2$
   c. $f = 2x^2 + 3xy + 4y^2 - 5x + 2y$
   d. $\phi = x^2 + y^2 + (px + qy + r)^2$
   e. $f = x^3 + x^2 - xy + y^2 + 4$.
   f. $f = 6x^2 - 2x^3 + 3y^2 + 6xy$
   g. $f = x^3 + y^3 - 12x - 3y + 20$.
   h. $u = 6a\,xy - x^2y - xy^2$
   i. $u = \frac{a^3}{x} + \frac{a^3}{y} + xy$
   j. $f = x^4 + y^4 - 2x^2 + 4xy - 2y^2$
   k. $u = x^2y^2 - 5x^2 - 8xy - 5y^2$
   l. $\phi = 4xy - x^4 - y^4$
   m. $f = (x^2 + y^2)^2 - 8(x^2 - y^2)$
2. a. Show that the function $u(x, y) = x^3 + y^3 - 3xy$ has a saddle point at $(0, 0)$
   b. Show that the function $u(x, y) = 4x^2y - y^2 - 8x^4$ is maximum at $(0, 0)$.
3. Find the extreme values of the following functions.
   a. $u(x, y, z) = 2x^2 + xy + 4y^2 + xz + z^2 + 2$
   b. $f(x, y, z) = 20 - x^2 - y^2 - z^2$
   c. $u(x, y, z) = x^4(4x - 5) + y^2 + 2z^2$
   d. $g(x, y, z) = 6xz - 3x^2 + 4y - 2y^2 - 6z^2$
   e. $f(x, y, z) = -3x^2 + 6xz + 4y - 2y^2 - 6z^2$
   f. $f = (x + y + z)^3 - 3(x + y + z) - 24xyz + 1$.
   g. $\phi = xz + x^2 - y + yz + y^2 + 3z^2$.
4. a. If the product of two numbers x and y is 16, what is the minimum value of their sum?
   b. Find the extreme value of $u(x, y) = xy$ when $2x + y = 6$.
   c. Find the extreme values of $u = xy^2$ subject to the constraint $x + y = 1$.
   d. If $z(x, y) = x^2 + y^2$ subject to the constraint $x + 4y = 2$, find the point where z is maximum and also maximum value of z.
   e. Find the maximum and minimum values of the function $f(x, y) = x^2 + y^2$ subject to $x^2 + y^2 - 2x - 4y = 0$.
   f. Find the maximum value of $u(x, y) = 48 - (x - 5)^2 - 3(y - 4)^2$ subject to the constraint $x + 3y = 9$.
   g. Find the extreme point of the function $f(x, y) = x^2 + 3xy - 5y^2$ subject to the constraint $2x + 3y = 6$.
   h. Find the point on the ellipse $x^2 + 2y^2 = 1$ where $f(x, y) = xy$ has its extreme values.
5. Determine the values of x, y, z that maximizes the function $3x + 5y + z - x^2 - y^2 - z^2$ subject to the constraint $x + y + z = 6$.

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6. Find the extreme values of the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints
   a. $x + y + z = 3a$
   b. $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$
   c. $ax + by + cz = a + b + c$
   d. $xy + yz + zx = 3a^2$
   e. $xyz = a^3$
7. a. Find the extreme point of the function $P = xyz$ under the condition $2x + 3y + z = 10$. Find the maximum value of the product P.
   b. If the sum of three positive numbers x, y and z is 24, what is the maximum value of their product?
   c. Find the largest product of the numbers x, y and z subject to the condition $x^2 + y^2 + z^2 = 9$.
8. a. Divide 24 into three parts such that continued product of first, square of second and cube of third is maximum.
   b. Divide 120 into three parts so that the sum of their products taken two at a time shall be maximum.
   c. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on unit sphere $x^2 + y^2 + z^2 = 1$.
9. a. Determine the extreme value of $x^2 + y^2 + z^2$ subject to the conditions $x + z = 1$ and $2y + z = 2$.
   b. Find the minimum value of $u = x^2 + y^2 + z^2$ subject to the constraints $x + y + z = 1$ and $xyz + 1 = 0$.
10. a. Determine the dimensions of the rectangular box open at the top of maximum volume whose surface area is 432 sq. cm.
    b. If the sum of the dimensions of a rectangular swimming pool is a. Prove that the amount of water in the pool is maximum when it is a cube.
11. Find absolute maxima and minima of $f(x, y)$ in given domain.
    a. $f(x, y) = x^2 + y^2 + x^2y + 4$, $D = \{(x, y): |x| \le 1, |y| \le 1\}$
    b. $f(x, y) = x^4 + y^4 - 4xy + 2$, $D = \{(x, y): 0 \le x \le 3, 0 \le y \le 2\}$
12. a. Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.
    b. Find the point in the plane $x + 2y + 3z = 13$ nearest to the point $(1, 1, 1)$.

Answer

1. a. Minimum value = 0 at (0, 0). b. maximum value 4 at (1, 1)
   c. minimum at (2, −1), min value = −6
   d. minimum at $\left(\frac{-pr}{1+p^2+q^2}, \frac{-qr}{1+p^2+q^2}\right)$, minimum value = $\frac{r^2}{1+p^2+q^2}$
   e. minimum value = 4 at (0, 0), saddle point at $\left(-\frac{1}{2}, \frac{1}{4}\right)$.
   f. minimum at (0, 0), min value = 0, saddle point (1, −1)
   g. minimum value = 2 at (2, 1); maximum value = 38 at (-2, -1), saddle point at (-2, 1) and (2, -1).
   h. max at $(2a, 2a)$, max. value = $8a^2$ i. minimum value = $3a^2$ at point (a, a)
   j. minimum at $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$, minimum value = −16
   k. max at (0, 0), max value = 0, no minimum
   l. maximum at (1, 1) and (−1, −1), max value = 2, saddle point (0,0)
   m. no minimum or maximum at (0, 0), minimum value at −16 at (2, 0) and (−2, 0)

3. a. minimum value 2 at (0, 0, 0) b. maximum 20 at (0, 0, 0)
   c. minimum −1 at (1, 0, 0) d. maximum 2 at (0, 1, 0)
   e. maximum 2 at (0, 1, 0) f. minimum value = -5 at (1, 1, 1) & maximum value = 7 at (-1, -1, -1).
   g. minimum value = $\frac{-11}{40}$ at $\left(\frac{1}{20}, \frac{11}{20}, \frac{-1}{10}\right)$
4. a. minimum Value = −8 at (−4, −4) b. extreme value = $\frac{9}{2}$ at $x = \frac{3}{2}, y = 3$
   c. maximum value = $\frac{4}{27}$ at $\left(\frac{1}{3}, \frac{2}{3}\right)$ and minimum value = 0 at (1, 0).
   d. maximum value = $\frac{4}{17}$ at $\left(\frac{2}{17}, \frac{8}{17}\right)$ e. maximum of 20 at (2, 4), minimum = 0 at (0, 0)
   f. maximum value = 32 g. extreme value at (3, 0)
   h. maximum at $\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{2}\right)$, minimum at $\left(\frac{1}{\sqrt{2}}, \frac{-1}{2}\right)\left(\frac{-1}{\sqrt{2}}, \frac{1}{2}\right)$
5. $(x, y, z) = (2, 3, 1)$
6. a. minimum value= $3a^2$ at (a, a, a) b. minimum value 27 at (3, 3, 3)
   c. minimum value = $\frac{(a+b+c)^2}{a^2+b^2+c^2}$ d. minimum value = $3a^2$ at (a, a, a) and $(-a, -a, -a)$
   e. $3a^2$
7. a. maximum at $\left(\frac{5}{3}, \frac{10}{9}, \frac{10}{3}\right)$ b. maximum value = 512 at (8, 8, 8).
   c. $3\sqrt{3}$
8. a. (12, 8, 4) b. (40, 40, 40)
   c. maximum temperature = 50°
9. a. minimum value = 1 b. minimum value = 3.
10. $(l, b, h) = (6, 6, 3)$
11. a. maximum $f(\pm 1, 1) = 7$, minimum $f(0, 0) = 4$ b. maximum $f(3, 0) = 83$, minimum $f(1, 1) = 0$
12. a. $\sqrt{6}, 3\sqrt{6}$ b. $\left(\frac{3}{2}, 2, \frac{5}{2}\right)$

Multiple Choice Questions

1. What is the saddle point?
   a. Point where function has maximum value
   b. Point where function has minimum value
   c. Point where function has zero value
   d. Point where function neither have maximum value nor minimum value
2. The drawback of Lagrange's method of maxima and minima is
   a. Maxima or minima is not fixed
   b. Nature of stationary point can not be known
   c. Accuracy is not good
   d. Nature of stationary point is known but can not give maxima or minima
3. The maximum value of product of two numbers whose sum is 12 is
   a. 32 b. 35
   c. 36 d. 20

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4. For the function $f(x,y) = \sin x \sin y$, which of the following is a critical point ?
   a. $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ b. $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
   c. $\left(0, \frac{\pi}{2}\right)$ d. $(0, 0)$
5. The point $(0, 0)$ in the domain of $f(x, y) = \sin(xy)$ is a _____ point?
   a. saddle b. minima
   c. maxima d. constant
6. The minimum value of the function $x^2 + y^2 + 6x + 12$ is
   a. $\frac{1}{2}$ b. 2
   c. 1 d. 3
7. The point $(0, 0)$ for $f(x,y) = x^3 + y^3 - 3axy$ is
   a. a maximum point b. a minimum point
   c. a saddle point d. none of these
8. If $x + y = k$, $x > 0$, $y > 0$, then xy is maximum when
   a. $x = ky$ b. $kx = y$
   c. $x = y$ d. none of these
9. The sum of two positive numbers is 50. What are the numbers if their product is to be largest possible.
   a. 24 and 26 b. 28 and 22
   c. 25 and 25 d. 20 and 30

Answer

1. d. 2. b. 3. c. 4. c. 5. a.
6. d. 7. c. 8. b. 9. c.

✧✧✧

Unit 7

First Order Ordinary Differential Equation

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- the evaluation of various fundamental integrals.
- the concept of differential equations, their types, and order and degree of the equations.
- the solution techniques of first order and first degree ordinary differential equations.

Expected learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- the solutions of first order linear, Bernoulli and Riccati differential equations with their engineering applications.
- mathematical modelling of various engineering problems (mixing problem, motion of under gravity, growth and decay problems, heating system, electric circuits) using first order equations.

7.0 Introduction

A differential equation states how a rate of change (in terms of derivative or differential) in a function is related to independent variable.

The measurement of rate of change describes the dynamics of a system or problem in engineering, physics and all branches of knowledge of applied fields.

Fundamentally, antiderivative is the reverse process of differentiation so while solving differential equation the skill of antiderivative is basic tool. So we expect student's skill in evaluating basic integrals.

Differential equations are outsprings of differential calculus and was first time used and defined by Newton and Leibnitz.

Differential equations are broadly used in geometry, mechanics, isometric problems and optimization problems.

Let us consider a case of motion of a particle.

If y is the distance covered by a particle in motion that depends on time t then, $y = y(t)$, i.e., y is a function of time t.

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Let $y = y(t) = 2t^2 + 4t - 5$ then its velocity (rate of change of distance per unit time) is given by

$$v = \frac{dy}{dt} = \frac{d}{dt}(2t^2 + 4t - 5) = 4t + 4 = 4(t + 1)$$

$$\therefore \quad v = \frac{dy}{dt} = 4(t + 1).$$

which is a differential equation of a moving particle whose distance covered is a function of time.

Some physical or engineering systems can be described by higher order derivatives and hence such equations are called higher order differential equation.

7.1 Order and degree of a differential equation

The highest order derivative occurring in the differential equation is termed as the order of differential equation.

For example, $\frac{d^2y}{dt^2} = y$ is a second order differential equation.

The positive integral power or exponent of highest order derivative in the equation is termed as the **degree** of the differential equation. For determining degree, we should remove any radical powers (roots) of function or its derivatives.

For example, $\left(\frac{d^2y}{dt^2}\right)^4 + 7\frac{dy}{dt} + 9y = 6e^t$ is third order fourth degree differential equation.

7.2 Types of differential equation

A. Based on derivative

1. **Ordinary differential equation:** A differential equation that contains ordinary differential or derivatives is known as ordinary differential equation.

   $\frac{d^2y}{dt^2} = g$ (constant) is a differential equation of falling body under gravity which is an ordinary differential equation of second order.

2. **Partial differential equation:** A differential equation that contains partial derivatives of a function of two or more variables is known as partial differential equation.

   If $u = u(x, y)$ then, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is a partial differential equation of second order.

B. Based on order of derivative

1. **First order differential equation:** A differential equation that contains first order derivatives of a function, which may also contain function itself is known as first order differential equation.

   $\frac{dy}{dt} = yt$, i.e., $y' = yt$

$\frac{dy}{dt} = \frac{1}{y}$, i.e., $y' = \frac{1}{y}$

where, $y = y(t)$ is a function of independent variable t.

2. **Higher order differential equation:** A differential equation that contains second order derivative or higher derivative of a function, which may also contain function itself is known as higher order differential equation. For example,

   $\frac{d^2y}{dt^2} = -t$, i.e., $y'' = -t$ is a second order differential equation.

C. Based on degree (linear and non-linear)

1. **Linear differential equation:** A differential equation that does not contain product terms of function (y) and its derivatives and can be written as a polynomial in terms of the function y with respect to its derivative as degree of polynomial is called linear differential equation.

   For example, $y'' + 2y' + 3y = t$ where $y = y(t)$ is a second order linear differential equation as it is polynomial in y with respect to the order of derivative i.e., zero order, first order and second order derivatives as power form a polynomial in y.

   Alternatively, a differential equation is called linear if in the equation

   i. dependent variable and derivatives occur only in the first degree,

   ii. product of the derivatives do not occur,

   iii. product of the dependent variable do not occur.

   Otherwise it is said to be non-linear differential equation.

2. **Non-linear differential equation:** All differential equation which are not a linear differential equations are termed as non-linear differential equations.

   For example, the equations

   i. $y' = \frac{1}{y}$ ii. $y' = y^2$

   iii. $y'' + \sin y = 0$ iv. $yy'' = 1$

   are non-linear differential equation as non of the equations can be written as polynomial in y in terms of derivative.

D. Based on function of independent variable

1. **Homogeneous differential equation:** A differential equation that contains only the terms of function $y = y(t)$ and its derivatives and no any constant term or any other function of t is called homogeneous differential equation.

   For example, the differential equation

   i. $2y' + y = 0$ is a first order linear homogeneous differential equation

   ii. $y''' + 3x^2y'' + yy' = 0$ is a second order non-linear homogeneous differential equation

   Note that the equations may have variable coefficients of the function $y(t)$ and its derivatives.

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2. **Non-homogeneous differential equation:** A differential equation that contains some constant terms or any function of independent variable is called non-homogeneous differential equation. For examples,
   i. The differential equation $y' + y = t$ is a first order non-homogeneous differential equation.
   ii. The equation $y'' + 4y = 8x^2$ is a second order non-homogeneous differential equation.

7.3 Solution of a differential equation

By the solution of a differential equation, we mean a function $y = y(t)$ that satisfies the given differential equation i.e. the equation becomes an identity on substituting y and its derivatives. The solution methods for different types of equation will be discussed on subsequent sections and units. The idea of antiderivative will be used to eliminate the derivatives or differentials from the equation as a fundamental tool for solving a differential equation.

For example, for the equation $y' = ky$, the function $y = ce^{kt}$ is a solution for any value of k as $y' = cke^{kt}$ and from $y' = ky$, we have $cke^{kt} = cke^{kt}$ which is true, so $y = ce^{kt}$ is the general solution of the equation $y' = ky$.

7.3.1 Types of solutions of differential equations

1 **General solution:** The solution of a differential equation that contains the same number of arbitrary constants as the order of differential equation is called general solution. For example,
   i. The differential equation $y' = ky$ has general solution as $y = ce^{kt}$, where c is an arbitrary constant.
   ii. The differential equation $y'' - 4y = 0$, $y = y(t)$ has $y = c_1e^{2t} + c_2e^{-2t}$ as the general solution as it contains two constants c_1 and c_2 and equation is of order 2.

2. **Particular solution:** A solution of a differential equation that is obtained from general solution by imposing some initial or boundary conditions is known as particular solution.
   For example, the initial value problem $y' = 5y$, $y(0) = 2$ has general solution as $y = ce^{5t}$ and using $y(0) = 2$ we get, $y(0) = ce^0 = c = 2$ thus $y = 2e^{5t}$ is a particular solution.

3. **Singular solution:** For some differential equations, we can have solutions which are neither general solution (that does not contain arbitrary constant) nor can be obtained from general equation, such a solution is called singular solution.
   For example, the differential equation $(y')^2 = 4y$ has the general solution as $y = (x + c)^2$, which represents a family of parabolas. The function $y = 0$ is also a solution but it is not a parabola rather it is a straight line (x-axis) which is also a solution but can not be obtained from general solution $y = (x + c)^2$. The solution $y = 0$ is a singular solution of the equation.

7.3.2 Initial value problem (IVP) and boundary value problem (BVP)

Initial value problem: A differential equation together with some initial condition is known as initial value problem (IVP). The initial condition, i.e., value of function and its derivative when the independent variable has value at single point normally at x is denoted by $y(0) = k$ (some constant) i.e., $y = k$ when $x = 0$.

Examples:

i. $y' + 2y = 0$, $y(0) = 1$, is a first order initial value problem.

ii. $y'' + 4y = x$, $y(0) = 1$, $y'(0) = 2$ is a second order initial value problem.

Boundary value problem: A differential equation together with some boundary condition is known as boundary value problem. The boundary value, i.e., value of the function and its derivative when the independent variable has values at different points of its boundaries is denoted by $y(0) = a$, $y'(1) = b$, where $x = 0$ and $x = 1$ are boundaries of independent variable.

Example:

i. $y'' = ky^{\frac{1}{2}}$, $y(0) = a$, $y'(2) = b$ is a second order boundary value problem.

7.4 Methods of solution of first order differential equation

The method for solving first order differential equation depends on its forms. A first order differential equation can in general be written as

$$y' = f(x, y), \quad y = y(x) \qquad \ldots.(i)$$

or, $$F(x, y, y') = 0 \qquad \ldots.(ii)$$

We prefer representation (i) for first order differential equation.

Equation (i) can also be written as

$$M(x, y)\,dx + N(x, y)\,dy = 0 \qquad \ldots.(iii)$$

7.4.1 Variable separable form

The first order differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ is said to be in separable form if it can be reduced to $f_1(x)\,dx + f_2(y)\,dy = 0$.

Integration of each term gives the solution of such differential equation. We now illustrate the solution process for such equations.

Example 1. Solve: $ydx - xdy = xydx$

Solution:

We separate the variables as

$$\frac{ydx - xdy}{xy} = dx$$

or, $$\frac{dx}{x} - \frac{dy}{y} = dx$$

Integrating, we get

$$\log x - \log y = x + c$$

or, $$\log\left(\frac{x}{y}\right) = x + c$$

or, $$\log\frac{x}{y} = x + c$$

or, $$\log\frac{x}{y} = x + c$$

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or, $\frac{x}{y} = e^{(x+c)}$

or, $y = x\,e^{-(x+c)}$

$= ke^{-x}$, where $k = e^{-c}$

$\therefore$ $y = kxe^{-x}$ is the required solution.

Example 2. Solve: $(x^2 + 1)\frac{dy}{dx} = 1$.

Solution:

The given differential equation is $(x^2 + 1)\frac{dy}{dx} = 1$

or, $dy = \frac{dx}{x^2 + 1}$

Integrating, $\int dy = \int \frac{1}{x^2 + 1}\,dx$

$\therefore$ $y = \tan^{-1} x + c$, is the required solution.

Example 3. Solve: $\log\left(\frac{dy}{dx}\right) = ax + by$.

Solution:

The given differential equation is $\log\left(\frac{dy}{dx}\right) = ax + by$

or, $\frac{dy}{dx} = e^{ax + by} = e^{ax}\,e^{by}$

or, $\frac{dy}{e^{by}} = e^{ax}\,dx$

or, $e^{-by}\,dy = e^{ax}\,dx$

$\frac{e^{by}}{-b} = \frac{e^{ax}}{a} + C'$

$\therefore$ $be^{ax} + ae^{by} = (ab)C'$

$\therefore$ $be^{ax} + ae^{by} = C$ is required solution, where $C = (ab)C'$

Example 4. Solve: $(\sin x + \cos x)\,dy = (\cos x - \sin x)\,dx$.

Solution:

The given differential equation is $(\sin x + \cos x)\,dy = (\cos x - \sin x)\,dx$

or, $dy = \frac{\cos x - \sin x}{\sin x + \cos x}\,dx$

Integrating, we get

$\int dy = \int \frac{(\cos x - \sin x)}{(\sin x + \cos x)}\,dx = \int \frac{d(\sin x + \cos x)}{(\sin x + \cos x)}$

or, $y = \log(\sin x + \cos x) + \log c \quad \left(\because \int \frac{d[f(x)]}{f(x)} = \log f(x) + c\right)$

or, $y = \log[c(\sin x + \cos x)]$

$\therefore$ $e^y = c(\sin x + \cos x)$, is the required solution.

Example 5. Solve: $(xy^2 + x)\,dx + (yx^2 + y)\,dy = 0$.

Solution:

The given differential equation is

$(xy^2 + x)\,dx + (yx^2 + y)\,dy = 0$

or, $x(y^2 + 1)\,dx + y(x^2 + 1)\,dy = 0$

Dividing by $(x^2 + 1)(y^2 + 1)$, $\frac{x\,dx}{x^2 + 1} + \frac{y\,dy}{y^2 + 1} = 0$

or, $\frac{2x\,dx}{x^2 + 1} + \frac{2y\,dy}{y^2 + 1} = 0$

or, $\frac{d(x^2 + 1)}{x^2 + 1} + \frac{d(y^2 + 1)}{(y^2 + 1)} = 0$, $[\because\ d(x^2 + 1) = 2x\,dx$ and $d(y^2 + 1) = 2y\,dy]$

Integrating, $\log(x^2 + 1) + \log(y^2 + 1) = \log c$

or, $\log[(x^2 + 1)\,.\,(y^2 + 1)] = \log c$

$\therefore$ $(x^2 + 1)(y^2 + 1) = c$ is the required solution.

Example 6. Solve the initial value problem $y' = 2y^2 + xy^2$, $y(0) = 1$.

Solution:

The given differential equation is $y' = 2y^2 + xy^2$

or, $\frac{dy}{dx} = y^2(2 + x)$

or, $\frac{dy}{y^2} = (2 + x)\,dx$

Integrating, $-\frac{1}{y} = \frac{x^2}{2} + 2x + c$ .... (i)

Using, $y(0) = 1$, we get $c = -1$

Substituting the value of c in equation (i), we get $-\frac{1}{y} = \frac{x^2}{2} + 2x - 1$

or, $-\frac{2}{y} = x^2 + 4x - 2$

$\therefore$ $y = \frac{-2}{x^2 + 4x - 2}$ is the required solution.

Example 7. Solve the initial value problem $2xy' = 3y$ given $y = 4$, when $x = 1$.

Solution:

The given differential equation is $2x\frac{dy}{dx} = 3y$

or, $\frac{2}{y}\,dy = \frac{3}{x}\,dx$

Integrating, $2\log y = 3\log x + \log c$

or, $\log y^2 = \log x^3 + \log c$

or, $\log y^2 = \log(cx^3)$ $\quad\therefore$ $y^2 = cx^3$ .... (i)

Now, when $x = 1$, $y = 4$

$4^2 = c\,.\,1^3$

$\therefore$ $c = 16$.

Substituting the value of c in (i), $y^2 = 16x^3$

$\therefore$ $y = 4x^{3/2}$, which is the required solution.

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Example 8. Solve the initial value problem

$y(1-x^2)\frac{dy}{dx} + x(1-y^2) = 0$ given that $y = 1$, when $x = 0$.

Solution:

The given differential equation is

$$y(1-x^2)\frac{dy}{dx} + x(1-y^2) = 0$$

or, $y(1-x^2)\,dy + x(1-y^2)\,dx = 0$

Dividing both sides by $(1-x^2)(1-y^2)$, $\frac{y}{(1-y^2)}dy + \frac{x}{(1-x^2)}dx = 0$

or, $\frac{-2y}{1-y^2}dy + \frac{-2x}{1-x^2}dx = 0$

Integrating, $\log(1-y^2) + \log(1-x^2) = \log c$

or, $\log[(1-x^2)(1-y^2)] = \log c$

$\therefore\ (1-x^2)(1-y^2) = c$ .... (i)

When $x = 0, y = 1, (1-0)(1-1) = c \quad \therefore c = 0$.

Substituting the value of c in equation (i), we get

$(1-x^2)(1-y^2) = 0$, which is the required particular solution.

7.4.2 Reduction into variable separable form by change of variables

Sometimes, we can solve differential equations by making suitable substitution so that the variables of the given differential equations are easily separated.

For example, if the differential equation is expressed in the form

$$\frac{dy}{dx} = f(ax + by + c) \quad \ldots\ (i)$$

Put $ax + by + c = v$.

Then, $a + b\frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{1}{b}\left(\frac{dv}{dx} - a\right)$

Hence, equation (i) becomes,

$$\frac{1}{b}\left(\frac{dv}{dx} - a\right) = f(v)$$

or, $\frac{dv}{dx} = a + b\,f(v)$

$\therefore\ \frac{dv}{a + b\,f(v)} = dx$

which is in variable separable form.

Example 9. Solve: $\frac{dy}{dx} + 1 = e^{x+y}$

Solution:

The given differential equation is

$$\frac{dy}{dx} + 1 = e^{x+y} \quad \ldots\ (i)$$

Let $x + y = v$. Then, $1 + \frac{dy}{dx} = \frac{dv}{dx}$

Then, equation (i) becomes $\frac{dv}{dx} = e^v$

or, $\frac{dv}{e^v} = dx$

or, $e^{-v}\,dv = dx$

Integrating, $\frac{e^{-v}}{-1} = x + c$

or, $-e^{-v} = x + c$ .... (ii)

Substituting the $v = x + y$ in (ii), we get $-e^{-(x+y)} = x + c$

or, $-1 = \frac{x + c}{e^{(x+y)}}$

or, $-1 = (x + c)\,e^{x+y}$

$\therefore\ (x + c)\,e^{x+y} + 1 = 0$, which is the required solution.

Example 10. Solve: $(x + y + 1)\frac{dy}{dx} = 1$.

Solution:

The given differential equation is

$$(x + y + 1)\frac{dy}{dx} = 1 \quad \ldots\ (i)$$

Let $x + y + 1 = v$. Then,

$1 + \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{dv}{dx} - 1$

Then, equation (i) becomes

$v\left(\frac{dv}{dx} - 1\right) = 1$

or, $\frac{dv}{dx} - 1 = \frac{1}{v}$

or, $\frac{dv}{dx} = 1 + \frac{1}{v}$

or, $\frac{dv}{dx} = \frac{v + 1}{v}$

or, $\left(\frac{v}{v + 1}\right)dv = dx$

or, $\left[\frac{v + 1}{v + 1} - \frac{1}{v + 1}\right]dv = dx$

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or, $\left[1-\frac{1}{v+1}\right]dv = dx$

Integrating, $v - \log(v+1) = x + c$ .... (ii)

Substituting the value of $v = x + y + 1$ in (ii), we get

$x + y + 1 - \log(x + y + 2) = x + c$

$\therefore$ $y - \log(x + y + 2) = c$ is the required solution.

Example 11. Solve: $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$.

Solution:

The given differential equation is

$\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ .... (i)

Let $x + y = v$. Then, $1 + \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{dv}{dx} - 1$

Then, equation (i) becomes $\frac{dv}{dx} - 1 = \sin v + \cos v$

or, $\frac{dv}{dx} = (1 + \cos v) + \sin v$.

or, $\frac{dv}{dx} = 2\cos^2\frac{v}{2} + 2\sin\frac{v}{2}\cos\frac{v}{2}$

or, $\frac{dv}{2\cos^2\frac{v}{2} + 2\sin\frac{v}{2}.\cos\frac{v}{2}} = dx$

or, $\frac{1}{2}\frac{\sec^2\frac{v}{2}}{1+\tan\frac{v}{2}}dv = dx$

or, $\frac{d\left(1+\tan\frac{v}{2}\right)}{1+\tan\frac{v}{2}} = dx$

Integrating, $\log\left(1+\tan\frac{v}{2}\right) = x + c$

$\therefore$ $\log\left[1+\tan\frac{(x+y)}{2}\right] = x + c$, which is the required solution.

Example 12. Solve: $\left(\frac{x+y-a}{x+y-b}\right)\frac{dy}{dx} = \frac{x+y+a}{x+y+b}$.

Solution:

The given differential equation is $\left(\frac{x+y-a}{x+y-b}\right)\frac{dy}{dx} = \frac{x+y+a}{x+y+b}$.

or, $\frac{dy}{dx} = \frac{(x+y+a)}{(x+y+b)} \times \frac{(x+y-b)}{(x+y-a)}$ .... (i)

Let $x + y = v$. Then, $1 + \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{dv}{dx} - 1$

Then, equation (i) becomes $\frac{dv}{dx} - 1 = \frac{(v+a)}{(v+b)} \times \frac{(v-b)}{(v-a)}$

or, $\frac{dy}{dx} = 1 + \frac{(v+a)(v-b)}{(v+b)(v-a)}$

or, $\frac{dy}{dx} = \frac{(v+b)(v-a)+(v+a)(v-b)}{(v+b)(v-a)}$

or, $\frac{dy}{dx} = \frac{v^2 - va + bv - ab + v^2 - vb + av - ab}{(v+b)(v-a)}$

or, $\frac{dy}{dx} = \frac{2v^2 - 2ab}{(v+b)(v-a)}$

or, $\frac{dy}{dx} = \frac{2(v^2-ab)}{v^2 + (b-a)v - ab}$

or, $\frac{v^2 + (b-a)v - ab}{v^2 - ab}dv = 2dx$

or, $\frac{(v^2-ab) + (b-a)v}{v^2-ab}dv = 2dx$

or, $\left[1 + \frac{b-a}{2}.\left(\frac{2v}{v^2-ab}\right)\right]dv = 2dx$

Integrating, $v + \frac{b-a}{2}\log(v^2 - ab) = 2x + k$

or, $\frac{b-a}{2}\log(v^2 - ab) = 2x - v + k$ .... (ii)

Substituting the value of $v = x + y$ in (ii), we get

$\frac{b-a}{2}\log[(x+y)^2 - ab] = 2x - x - y + k$

$\therefore$ $(b-a)\log[(x+y)^2 - ab] = 2(x-y) + c$, where $c = 2k$, which is the required solution.

Example 13. Solve: $(x^2 + y^2 + 2xy + 1)\,dy = (x+y)\,dx$.

Solution:

The given differential equation is $(x^2 + y^2 + 2xy + 1)\,dy = (x+y)\,dx$.

or, $[(x+y)^2 + 1]\,dy = (x+y)\,dx$

or, $\frac{dy}{dx} = \frac{x+y}{(x+y)^2+1}$ .... (i)

Let $x + y = v$. Then, $1 + \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{dv}{dx} - 1$

Then, equation (i) becomes $\frac{dv}{dx} - 1 = \frac{v}{v^2+1}$

or, $\frac{dv}{dx} = 1 + \frac{v}{v^2+1} = \frac{v^2+1+v}{v^2+1}$

or, $\frac{v^2+1}{v^2+1+v}dv = dx$

or, $\left[\frac{v^2+v+1-v}{v^2+v+1}\right]dv = dx$

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or, $\left[1-\frac{v}{v^2+v+1}\right]dv = dx$

or, $dv - \frac{v}{v^2+v+1}dv = dx$

or, $dv - \frac{\frac{1}{2}(2v+1)dv - \frac{1}{2}dv}{v^2+v+1} = dx$

or, $dv - \frac{1}{2}\frac{d(v^2+v+1)}{v^2+v+1} + \frac{1}{2}\frac{dv}{v^2+v+1} = dx$

or, $dv - \frac{1}{2}\frac{d(v^2+v+1)}{v^2+v+1} + \frac{1}{2}\frac{dv}{\left(v+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = dx$

Integrating, we get

$$v - \frac{1}{2}\log(v^2+v+1) + \frac{1}{2}\cdot\frac{1}{\frac{\sqrt{3}}{2}}\tan^{-1}\left(\frac{v+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = x + c \quad \ldots\ldots \text{(ii)}$$

Substituting the value of $v = x + y$ in (ii), we get

$$x + y - \frac{1}{2}\log[(x+y)^2 + (x+y) + 1] + \frac{1}{\sqrt{3}}\tan^{-1}\left[\frac{2(x+y)+1}{\sqrt{3}}\right] = x + c$$

$\therefore$ $y - \frac{1}{2}\log[(x+y)^2 + (x+y) + 1] + \frac{1}{\sqrt{3}}\tan^{-1}\left[\frac{2(x+y)+1}{\sqrt{3}}\right] = c$, which is the required solution.

7.5 Homogeneous differential equation

A differential equation of the form $Mdx + Ndy = 0$ is said to be ***homogeneous differential equation*** of first order if both M and N are homogeneous functions of x and y. In such case $Mdx + Ndy$ can be expressed in the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$. In homogeneous equation there are no constant or function of x other than $y = y(x)$ and its derivatives.

For examples:

i. The differential equation $(x^2 + y^2)\,dx - xy\,dy = 0$ is homogeneous, since

$$\frac{dy}{dx} = \frac{x^2+y^2}{xy} = \frac{x}{y} + \frac{y}{x} = \frac{1}{\left(\frac{y}{x}\right)} + \left(\frac{y}{x}\right) = f\left(\frac{y}{x}\right).$$

ii. The differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$ is homogeneous, since

$$\frac{dy}{dx} = \left(\frac{y}{x}\right)^2 - \left(\frac{y}{x}\right) = f\left(\frac{y}{x}\right).$$

The process of solving such homogeneous differential equations involve following steps:

Step 1 : Express the given differential equation into the homogenous differential equation of the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right) \quad \ldots\ldots \text{(i)}$$

Step 2 : Substitute $y = vx$, which reduces homogeneous equation into variable separable form.

Note that the differentiation of $y = vx$ gives, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

Step 3 : Substitute $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in (i)

to obtain $v + x\frac{dv}{dx} = f\left(\frac{vx}{x}\right)$

or, $v + x\frac{dv}{dx} = f(v)$

or, $x\frac{dv}{dx} = f(v) - v$

$\therefore$ $\frac{dv}{f(v)-v} = \frac{dx}{x}$, which is a variable separable form.

Step 4 : Integrate each term and substitute back $v = \frac{y}{x}$ to get the required solution.

Example 14. Solve: $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$.

Solution:

The given differential equation is $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

or, $\frac{dy}{dx} = \frac{y^2}{x^2} - \frac{y}{x}$ $\quad \ldots\ldots \text{(i)}$

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$ and $\frac{y}{x} = v$.

and equation (i) becomes $v + x\frac{dv}{dx} = v^2 - v$

or, $x\frac{dv}{dx} = v^2 - v - v$

or, $x\frac{dv}{dx} = v(v-2)$

or, $\frac{dv}{v(v-2)} = \frac{dx}{x}$

or, $\frac{1}{2}\left[\frac{1}{v-2} - \frac{1}{v}\right]dv = \frac{dx}{x}$

Integrating, $\frac{1}{2}[\log(v-2) - \log v] = \log x + \log k$

or, $\log\frac{v-2}{v} = 2\log(kx)$

or, $\log\frac{v-2}{v} = \log(kx)^2$

or, $\frac{v-2}{v} = (kx)^2$

or, $1 - \frac{2}{v} = k^2x^2$ $\quad \ldots\ldots \text{(ii)}$

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Substituting $v = \frac{y}{x}$ in (ii), we get

$1 - \frac{2x}{y} = k^2x^2$

or, $\frac{y - 2x}{y} = k^2x^2$

or, $y - 2x = k^2x^2y$

or, $2x - y = (-k^2)x^2y$

$\therefore$ $2x - y = cx^2y$, where $c = -k^2$ is the required solution.

Example 15. Solve: $(x^2 + y^2)dx + 2xy\,dy = 0$.

Solution:

The given differential equation is $(x^2 + y^2)dx + 2xy\,dy = 0$.

or, $\frac{dy}{dx} = -\frac{(x^2 + y^2)}{2xy}$ .... (i)

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

and, equation (i) becomes $v + x\frac{dv}{dx} = -\frac{(x^2 + v^2x^2)}{2x \cdot vx}$

or, $v + x\frac{dv}{dx} = -\frac{(1 + v^2)}{2v}$

or, $x\frac{dv}{dx} = -\frac{(1 + 3v^2)}{2v}$

or, $\frac{2v}{1 + 3v^2}dv + \frac{dx}{x} = 0$

or, $\frac{1}{3}\frac{d(1 + 3v^2)}{(1 + 3v^2)} + \frac{dx}{x} = 0$

or, $\frac{d(1 + 3v^2)}{(1 + 3v^2)} + \frac{3}{x}dx = 0$

Integrating, $\log(1 + 3v^2) + 3\log x = \log c$

or, $\log(1 + 3v^2) + \log x^3 = \log c$

or, $\log[x^3(1 + 3v^2)] = \log c$

or, $x^3(1 + 3v^2) = c$ .... (ii)

Substituting $v = \frac{y}{x}$ in (ii), we get $x^3\left(1 + 3\frac{y^2}{x^2}\right) = c$

$\therefore$ $x(x^2 + 3y^2) = c$ is the required solution.

Example 16. Solve: $y^2dx + (xy + x^2)\,dy = 0$.

Solution:

The given differential equation is $y^2dx + (xy + x^2)\,dy = 0$

or, $\frac{dy}{dx} = \frac{-y^2}{x^2 + xy}$ .... (i)

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

and equation (i) becomes $v + x\frac{dv}{dx} = \frac{-v^2x^2}{x^2 + x.vx}$

or, $v + x\frac{dv}{dx} = \frac{-v^2}{1 + v}$

or, $x\frac{dv}{dx} = \frac{-v^2}{1 + v} - v$

or, $x\frac{dv}{dx} = \frac{-2v^2 - v}{1 + v}$

or, $\frac{1 + v}{2v^2 + v}dv = \frac{-dx}{x}$

or, $\left[\frac{1}{4}\left(\frac{4v + 1}{2v^2 + v}\right) + \frac{3}{4.(2v^2 + v)}\right]dv = \frac{-dx}{x}$

or, $\left[\frac{1}{4}\left(\frac{4v + 1}{2v^2 + v}\right) + \frac{3}{8}\frac{1}{\left(v^2 + \frac{v}{2}\right)}\right]dv = \frac{-dx}{x}$

or, $\left[\frac{1}{4}\left(\frac{4v + 1}{2v^2 + v}\right) + \frac{3}{8}\frac{1}{\left(v + \frac{1}{4}\right)^2 - \frac{1}{16}}\right]dv = \frac{-dx}{x}$

Integrating, $\frac{1}{4}\log(2v^2 + v) + \frac{3}{8}\cdot\frac{1}{2\cdot\frac{1}{4}}\log\frac{v + \frac{1}{4} - \frac{1}{4}}{v + \frac{1}{4} + \frac{1}{4}} = -\log x + \log k$ $\left[\because \int\frac{1}{x^2 - a^2}dx = \frac{1}{2a}\log\frac{x - a}{x + a}\right]$

or, $\frac{1}{4}\log(2v^2 + v) + \frac{3}{4}\log\left(\frac{v}{v + \frac{1}{2}}\right) = \log\frac{k}{x}$

or, $\log(2v^2 + v) + 3\log\left(\frac{2v}{2v + 1}\right) = 4\log\frac{k}{x}$

or, $\log(2v^2 + v) + \log\left(\frac{2v}{2v + 1}\right)^3 = \log\frac{k^4}{x^4}$

or, $\log\left\{(2v^2 + v)\cdot\left(\frac{2v}{2v + 1}\right)^3\right\} = \log\frac{k^4}{x^4}$

or, $(2v^2 + v)\cdot\frac{(2v)^3}{(2v + 1)^3} = \frac{k^4}{x^4}$

or, $\frac{v(2v + 1)\cdot 8v^3}{(2v + 1)^3} = \frac{k^4}{x^4}$

or, $\frac{8v^4}{(2v + 1)^2} = \frac{k^4}{x^4}$

$\therefore$ $v^4 = \frac{k^4}{8x^4}(2v + 1)^2$ ....(ii)

Substituting $v = \frac{y}{x}$ in (ii), we get $\frac{y^4}{x^4} = \frac{k^4}{8x^4}\left(\frac{2y}{x} + 1\right)^2$

or, $y^4 = \frac{k^4}{8}\frac{(2y + x)^2}{x^2}$

Taking square root on both sides, $y^2 = \frac{k^2}{\sqrt{8}}\frac{(2y + x)}{x}$

or, $xy^2 = \frac{k^2}{\sqrt{8}}(x + 2y)$

$\therefore$ $xy^2 = c(x + 2y)$, where $c = \frac{k^2}{\sqrt{8}}$ is constant, is the required solution.

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Example 17. Solve: $x(x-y)dy + y^2\,dx = 0$.

Solution:

The given differential equation is $x(x-y)dy + y^2\,dx = 0$

or, $\frac{dy}{dx} = \frac{-y^2}{x(x-y)}$ .... (i)

or, $\frac{dy}{dx} = \frac{y^2}{x(y-x)}$

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

and equation (i) becomes $v + x\frac{dv}{dx} = \frac{(vx)^2}{x(vx-x)}$

or, $v + x\frac{dv}{dx} = \frac{v^2}{v-1}$

or, $x\frac{dv}{dx} = \frac{v^2}{v-1} - v$

or, $x\frac{dv}{dx} = \frac{v^2 - v^2 + v}{v-1}$

or, $x\frac{dv}{dx} = \frac{v}{v-1}$

or, $\frac{(v-1)}{v}dv = \frac{dx}{x}$

or, $\left(1 - \frac{1}{v}\right)dv = \frac{dx}{x}$

Integrating, $v - \log v = \log x + \log k$

or, $v = \log(kx) + \log v$

or, $v = \log(kxv)$ .... (ii)

Substituting $v = \frac{y}{x}$, we get $\frac{y}{x} = \log\left(kx.\frac{y}{x}\right)$

or, $\frac{y}{x} = \log(ky)$

or, $ky = e^{y/x}$

or, $y = \frac{1}{k}e^{y/x}$

$\therefore$ $y = ce^{y/x}$ is the required solution, where $c = \frac{1}{k}$.

Example 18. Solve: $x\frac{dy}{dx} = y - \sqrt{x^2 + y^2}$.

Solution:

The given differential equation is $x\frac{dy}{dx} = y - \sqrt{x^2 + y^2}$

or, $\frac{dy}{dx} = \frac{y}{x} - \frac{\sqrt{x^2 + y^2}}{x}$ .... (i)

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

and equation (i) becomes $v + x\frac{dv}{dx} = \frac{vx}{x} - \frac{\sqrt{x^2 + v^2x^2}}{x}$

or, $v + x\frac{dv}{dx} = v - \sqrt{1 + v^2}$

or, $x\frac{dv}{dx} = -\sqrt{1 + v^2}$

or, $\frac{dv}{\sqrt{1 + v^2}} + \frac{dx}{x} = 0$

Integrating, $\log\left(v + \sqrt{v^2 + 1}\right) + \log x = \log c$

or, $\log\left[\left(v + \sqrt{v^2 + 1}\right)x\right] = \log c$

or, $\left(v + \sqrt{v^2 + 1}\right)x = c$ .... (ii)

Substituting $v = \frac{y}{x}$ in (ii), we get $\left(\frac{y}{x} + \sqrt{\frac{y^2}{x^2} + 1}\right)x = c$

or, $y + \sqrt{y^2 + x^2} = c$

$\therefore$ $c - y = \sqrt{x^2 + y^2}$ is the required solution.

Example 19. Solve: $x\sin\frac{y}{x}dy = \left(y\sin\frac{y}{x} - x\right)dx$.

Solution:

The given differential equation is $x\sin\frac{y}{x}dy = \left(y\sin\frac{y}{x} - x\right)dx$

or, $\frac{dy}{dx} = \frac{y\sin\frac{y}{x} - x}{x\sin\frac{y}{x}}$ .... (i)

which is the homogeneous differential equation of first order.

Let $y = vx$. Then, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

and equation (i) becomes $v + x\frac{dv}{dx} = \frac{vx\sin\frac{vx}{x} - x}{x\sin\frac{vx}{x}}$

or, $v + x\frac{dv}{dx} = \frac{vx\sin v - x}{x\sin v}$

or, $v + x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin v}$

or, $x\frac{dv}{dx} = \frac{v\sin v - 1}{\sin v} - v$

or, $x\frac{dv}{dx} = \frac{v\sin v - 1 - v\sin v}{\sin v}$

or, $x\frac{dv}{dx} = -\frac{1}{\sin v}$

or, $\sin v\,dv + \frac{dx}{x} = 0$

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Integrating, $\cos y + \log x = c$ .... (ii)

or, log[illegible] $\cos y + c$

Substituting $y = \frac{y}{x}$ in (ii), we get

$\therefore \log x = \cos\frac{y}{x} + c$, which is the required solution.

Example 20. Solve: $(1 + e^{x/y})\,dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$.

Solution:

The given differential equation is $(1 + e^{x/y})\,dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$

or, $\frac{dx}{dy} = -\frac{e^{x/y}}{1 + e^{x/y}}\left(1 - \frac{x}{y}\right)$ .... (i)

which is the homogeneous differential equation of first order.

Let $x = vy$. Then, $\frac{dx}{dy} = v.1 + y\frac{dv}{dy}$

and, equation (i) becomes $v + y\frac{dv}{dy} = -\frac{e^v}{1 + e^v}\left(1 - \frac{vy}{y}\right) = \frac{-e^v(1 - v)}{1 + e^v}$

or, $y\frac{dv}{dy} = \frac{-e^v(1 - v) - v(1 + e^v)}{1 + e^v}$

$\frac{dv}{dy} = \frac{-e^v - v}{1 + e^v}$

or, $\left(\frac{1 + e^v}{v + e^v}\right)dv + \frac{dy}{y} = 0$

Integrating, $\log(v + e^v) + \log y = \log c$

or, $\log[(v + e^v)y] = \log c$

or, $(v + e^v)y = c$ .... (ii)

Substituting $v = \frac{x}{y}$ in (ii), we get $\left(\frac{x}{y} + e^{x/y}\right)y = c$

$\therefore x + ye^{x/y} = c$ is the required solution.

Exercise 7.1

1. Solve the following differential equations

a. $\sqrt{1-x^2}\,dy + \sqrt{1-y^2}\,dx = 0$

b. $(x^2 + 1)\frac{dy}{dx} = xy$

c. $(1 + x)y\,dx + (1 + y)x\,dy = 0$

d. $\tan y\,dx + \tan x\,dy = 0$

e. $\frac{dy}{dx} = \frac{x^2 + 1}{y^2 + 1}$

f. $\sec^2 x \tan y\,dx + \sec^2 y \tan x\,dy = 0$

g. $\frac{dy}{dx} + \frac{1 + \cos 2y}{1 - \cos 2x} = 0$

h. $e^{x-y}\,dx + e^{y-x}\,dy = 0$

i. $\log\left(\frac{dy}{dx}\right) = 3x + 4y$

j. $(a^2 + y^2)x\,dx + (x^2 - a^2)\,dy = 0$

k. $\frac{dy}{dx} = \sin^{-1}x$

2. Solve the following initial value problems

a. $xy' + y = 0,\ y(1) = 1$

b. $y' = \frac{x^2}{y},\ y(1) = 0$

c. $y' = -2xy,\ y(0) = 1$

d. $y'\cos^2 x - \sin^2 y = 0,\ y(0) = \frac{\pi}{2}$

e. $xyy' + 2,\ y(2) = 0$

f. $y' = 2(1 + x)(1 + y^2),\ y(0) = 0$

g. $\cos y\,dx + (1+2e^x)\sin y\,dy = 0,\ y(0) = \frac{\pi}{4}$

h. $y' = \sec y,\ y(0) = 0$

i. $e^x y' = 2(x + 1)y^2,\ y(0) = \frac{1}{6}$

j. $y' = \frac{(3x^2 - e^x)}{(2y - 5)},\ y(0) = 1$

k. $y'(x^2 + 1) = -2xy, y(1) = 2.$

3. a. Find the curve in the xy-plane that passes through $(0, 3)$ and whose tangent line at a point (x, y) has slope $\frac{2x}{y^2}$.

b. Find an equation of a curve with x-intercept 2 whose tangent line at any point (x, y) has slope xe^y.

4. Solve the following differential equations.

a. $(x + y + 1)\frac{dy}{dx} = 1$

b. $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$

c. $\frac{dy}{dx} = \frac{y + 1}{y + 1 + x}$

d. $(x + y)^2\frac{dy}{dx} = a^2$

e. $\frac{dy}{dx} = (4x + y + 1)^2$

f. $\frac{dy}{dx} = \sqrt{y - x}$

g. $y' = (y + 4x)^2$

5. Solve the following initial value problems.

a. $xyy' = 2y^2 + 4x^2,\ y(2) = 4$

b. $xy' = y + 3x^4\cos^2\left(\frac{y}{x}\right),\ y(1) = 0$

c. $y' = \frac{y - x}{y - x - 1},\ y(-5) = 5$

d. $y' = \frac{1 - 2y - 4x}{1 + y + 2x},\ y(0) = 0$

6. Solve the following differential equations

a. $\frac{dy}{dx} = \frac{3xy + y^2}{3x^2}$

b. $2x + x\frac{dy}{dx} = y$

c. $\frac{dy}{dx} = \frac{-x^2y}{x^3 + y^3}$

d. $(x^2 - y^2)\,dy = xy\,dx$

e. $(x + y)\,dx + (y - x)\,dy = 0$

f. $\frac{dy}{dx} = \frac{y^3}{x^3}$

g. $\frac{dy}{dx} + \frac{3xy + y^2}{x^2 + xy} = 0$

h. $x\frac{dy}{dx} = y + \sqrt{x^2 + y^2}$

i. $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$

7. Solve the following initial value problems:

a. $y^3y' + x^3 = 0,\ y(0) = 1$

b. $(x + y)y' = x - y,\ y(2) = 0$

c. $(x^3 + y^3)y' = x^2y,\ y(0) = 1$

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Answer

1. a. $\sin^{-1}x + \sin^{-1}y = c$
   b. $y = c(x^2+1)^{1/2}$
   c. $\log(x.y) + (x+y) = c$
   d. $\sin x \cdot \sin y = c$
   e. $\frac{y^4}{4} + y = \frac{x^4}{4} + x + c$
   f. $\tan x \cdot \tan y = c$
   g. $\tan y - \cot x = c$
   h. $e^{2x} + 2^{2y} = c$
   i. $4e^{3x} + 3e^{-4y} = c$
   j. $(x^2-a^2)(y^2+a^2) = c$
   k. $y = (\sin^{-1}x)x + \sqrt{1-x^2} + c$
2. a. $xy = 1$
   b. $2x^3 - 3y^2 = 2$
   c. $y = e^{-x^2}$
   d. $\cot y = -\tan x$
   e. $y - 2\log(y+2) = \log\frac{x}{8}$
   f. $y = \tan(x^2+2x)$
   g. $(e^x+2)\sec y = 3\sqrt{2}$
   h. $\sin y = x$
   i. $y = \frac{1}{2+(4+2x)e^x}$
   j. $y = \frac{5}{2} - \sqrt{x^3 - e^x + \frac{13}{4}}$
   k. $(x^2+1)y = 4$
3. a. $\frac{1}{3}y^3 = x^2 + 9$
   b. $e^y = \frac{x^2}{2} - 4$
4. a. $y - \log(x+y+2) = c$
   b. $\tan(x+y) - \sec(x+y) = x + c$
   c. $y + 1 = ce^{\left(\frac{c}{y+1}\right)}$
   d. $y = a\tan^{-1}\left(\frac{x+y}{a}\right) + c$
   e. $\frac{1}{2}\tan^{-1}\left(\frac{(4x+y+1)}{2}\right) = x + c$
   f. $\sqrt{y-x} + \log\left(\sqrt{y-x}-1\right) = \frac{x}{2} + c$
   g. $y = 2\tan(2x+c) - 4x$
5. a. $y = (2x^4 - 4x^3)^{1/2}$
   b. $y = x\tan^{-1}(x^3 - 1)$
   c. $(y-x)^2 - 2y = 90$
   d. $2y - 2x + (y+2x)^2 = 0$
   b. $3x + y\log x = cy$
   b. $y + 2x\log cx = 0$
   c. $\frac{x^3}{y^3} = 3\log y + c$
   d. $x^2 + 2y^2\log(cy) = 0$
   e. $\tan^{-1}\frac{y}{x} - \frac{1}{2}\log(x^2+y^2) = c$
   f. $y^2 - x^2 = cx^2y^2$
   g. $x^2y(y+2x) = c$
   h. $y + \sqrt{x^2+y^2} = cx^2$
   i. $x = c\sin\frac{y}{x}$
7. a. $x^4 + y^4 = 1$
   b. $x^2 - 2xy - y^2 = 4$
   c. $x^3 = 3y^3\log y$

7.6 Exact differential equations

A differential equation of the form $Mdx + Ndy = 0$, where both M and N are functions of x and y is said to be an ***exact differential equation*** if there exists a function $u(x, y)$ such that $Mdx + Ndy = du$.

In other words, $Mdx + Ndy$ should be perfect differential.

For such differential equation, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

For Examples:

i. $xdy + ydx = 0$ is exact, since $xdy + ydx = d(xy)$.

ii. $xdx + ydy = 0$ is exact. Since $xdx + ydy$ can be written as

$$xdx + ydy = d\left(\frac{x^2+y^2}{2}\right)$$

iii. $\sin x\,dy + y\cos x\,dx = 0$ is exact, since

$$\sin x\,dy + y\cos x\,dx = d(y\sin x).$$

7.6.1 Necessary and sufficient condition for exactness

Theorem 1.

The necessary and sufficient condition for the differential equation of the form $Mdx + Ndy = 0$, where M and N are the functions of x and y, to be exact is that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Proof:

Condition is necessary

Suppose that $Mdx + Ndy = 0$ .... (i)

is exact. We need to show that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Since (i) is exact, let $U(x, y) = c$ .... (ii)

be the primitive of (i).

Taking total derivative on (ii), we get

$$\frac{\partial W}{\partial x}dx + \frac{\partial U}{\partial y}dy = 0. \quad \text{.... (iii)}$$

By assumption, (i) and (iii) are both derivatives of primitive (ii). Hence comparing (i) and (iii), we get

$$M = \frac{\partial U}{\partial x} \text{ and } N = \frac{\partial U}{\partial y}.$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial^2 U}{\partial y \partial x} \text{ and } \frac{\partial N}{\partial x} = \frac{\partial^2 U}{\partial y \partial x}.$$

Thus, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence the condition is necessary.

Condition is Sufficient

Suppose that the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ holds.

We need to show that $Mdx + Ndy = 0$ is exact.

For this, let $M = \frac{\partial U}{\partial x}$

or, $\frac{\partial M}{\partial y} = \frac{\partial^2 U}{\partial y \partial x}$

So, $\frac{\partial^2 U}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\therefore \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial U}{\partial y}\right)$$

Integrating, $N = \frac{\partial U}{\partial y} + f(y)$

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Thus, $Mdx + Ndy = \frac{\partial U}{\partial x}dx + \left(\frac{\partial U}{\partial y} + f(y)\right)dy$

$= \frac{\partial U}{\partial x}dx + \frac{\partial U}{\partial y}dy + f(y)\,dy$

$= d[U + \int f(y)\,dy] = d[U + F(y)]$

Let $U(x, y) + F(y) = u(x, y)$. Then, $Mdx + Ndy = d[u(x, y)]$.

It shows that the expression $Mdx + Ndy$ is perfect differential and hence the differential equation $Mdx + Ndy = 0$ is exact.

Steps of solving exact differential equations

The steps of solving such an equation $Mdx + Ndy = 0$ are as follows:

Step 1. Check whether the given equation is exact or not.

Step 2. Integrate M with respect to x taking y as constant.

Step 3. Integrate N with respect to y taking x as constant.

Step 4. Reject the terms in step 3 that are already obtained in step 2.

Step 5. Equate the sum of these integrals to a constant to get the required solution.

Note: The rules of applying the differential operator d is same as the rules of derivative (i.e., $\frac{d}{dx}$)

7.6.2 Solution by inspection

There are few exact differentials which help us to find an integrating factor by inspection. The differential rules exactly follow the differentiation rules. We list some standard differentials.

1. $xdy + ydx = d(xy)$
2. $xdx + ydy = \frac{1}{2}d(x^2 + y^2)$
3. $\frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right)$
4. $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$
5. $\frac{xdy - ydx}{xy} = d\left(\log\frac{y}{x}\right)$
6. $\frac{2xy\,dy - y^2\,dx}{x^2} = d\left(\frac{y^2}{x}\right)$
7. $\frac{2xy\,dx - x^2\,dy}{y^2} = d\left(\frac{x^2}{y}\right)$
8. $\frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$
9. $\frac{ydx - xdy}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$
10. $\frac{dx + dy}{x + y} = \log(x + y)$
11. $\frac{xdy + ydx}{\sqrt{1 - (xy)^2}} = d(\sin^{-1}xy)$

7.6.3 Reduction into exact differential equation

If a differential equation is not exact, then it can be made exact by multiplying a suitable function of x or y or both. Such a function is called an ***integrating factor*** and written as I.F.

For example:

Let $xdy - ydx = 0$ is a differential equation. Here, $M = -y$ and $N = x$.

Then it is not exact, since $\frac{\partial M}{\partial y} = -1$, $\frac{\partial N}{\partial x} = 1$

$\therefore \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$.

To make it exact, multiplying by $\frac{1}{x^2}$, we get

$$\frac{1}{x^2}(xdy - ydx) = 0 \Rightarrow d\left(\frac{y}{x}\right) = 0,$$

which is an exact equation. Hence, $\frac{1}{x^2}$ is an integrating factor.

Remarks: A differential equation may have more than one integrating factors.

Consider a differential equation

$ydx - xdy = 0$. .... (i)

Comparing (i) with $Mdx + Ndy = 0$ .... (ii)

we get $M = y$ and $N = -x$.

Now, $\frac{\partial M}{\partial y} = 1$, $\frac{\partial N}{\partial x} = -1$

$\therefore \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$.

So the equation (i) is not exact.

Hence, to make equation (i) exact, we need an integrating factor.

Case I: Multiplying the given equation (i) by $\frac{1}{y^2}$, we get

$$\frac{ydx - xdy}{y^2} = 0$$

$\Rightarrow \quad d\left(\frac{x}{y}\right) = 0$, which is an exact.

Integrating, $\frac{x}{y} = c$

$\therefore \quad x = cy$ .... (iii)

So, $\frac{1}{y^2}$ is an integrating factor of (i).

Case II: Multiplying the equation (i) by $\frac{1}{x^2}$, we get

$$\frac{ydx - xdy}{x^2} = 0$$

$\Rightarrow \quad \frac{xdy - ydx}{x^2} = 0$

i.e. $d\left(\frac{y}{x}\right) = 0$, which is an exact.

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Integrating, $\frac{y}{x} = c$

$\therefore \quad y = cx \ldots$ (iv)

$\therefore \quad \frac{1}{x^2}$ is also an integrating factor of (i).

Case III: Again, multiplying the equation (i) by $\frac{1}{xy}$, we get

$$\frac{ydx - xdy}{xy} = 0$$

or, $\frac{dx}{x} - \frac{dy}{y} = 0$

Integrating, we get

$\log x - \log y = \log c$

or, $\log \frac{x}{y} = \log c$

or, $\frac{x}{y} = c$

or, $x = cy \ldots$ (v)

$\therefore \quad \frac{1}{xy}$ is also an integrating factor of (i)

From case I, case II and case III, we see that the number of integrating factors of a differential equation may be more than one and hence sometimes the solutions of differential equations can be in different form.

7.6.4 Rules of determining integrating factors

Consider a differential equation $Mdx + Ndy = 0 \quad \ldots$ (i)

Now, the condition that (i) to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Then, we have following rules:

Rule 1: $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x alone say $f(x)$.

Then, $e^{\int f(x)\,dx}$ will be an integrating factor of (i).

Rule 2: If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M}$ is a function of y alone, say $f(y)$

Then, $e^{\int f(y)\,dy}$ will be an I.F. of (i).

Rule 3: If $Mx + Ny \neq 0$ and the equation $Mdx + Ndy = 0$ is a homogeneous equation then the integrating factor is $\frac{1}{Mx + Ny}$.

Rule 4: If $Mx - Ny \neq 0$ and the equation $Mdx + Ndy = 0$ can be expressed in the form $yf(x, y)dx + xg(x, y)dy = 0$, then the integrating factor is $\frac{1}{Mx - Ny}$.

Example 21. Solve: $(x^2 + 2xy^2)\,dx + (2x^2y + y^2)\,dy = 0$.

Solution:

The given differential equation is

$(x^2 + 2xy^2)\,dx + (2x^2y + y^2)\,dy = 0$

or, $x^2dx + y^2dy + (2xy^2\,dx + 2x^2y\,dy) = 0$

or, $x^2dx + y^2dy + y^2 \cdot 2x\,dx + x^2 \cdot 2ydy = 0$

or, $x^2dx + y^2dy + d(x^2 \cdot y^2) = 0$

Integrating, $\frac{x^3}{3} + \frac{y^3}{3} + x^2y^2 = \frac{c}{3}$

$\therefore \quad x^3 + y^3 + 3x^2y^2 = c$ is the required solution.

Example 22. Solve: $xdy + (x + 1)\,ydx = 0$.

Solution:

The given differential equation is

$xdy + (x + 1)\,ydx = 0$

or, $xdy + xydx + ydx = 0$

Dividing by xy, $\frac{1}{y}dy + dx + \frac{1}{x}dx = 0$

Integrating, $\log y + x + \log x = c$

or, $(\log x + \log y) + x = c$

$\therefore \quad \log(xy) + x = c$ is the required solution.

Example 23. Solve: $x\frac{dy}{dx} + y = y^2 \log x$.

Solution:

The given differential equation is

$x\frac{dy}{dx} + y = y^2 \log x$

or, $xdy + ydx = y^2 \log x\,dx$

or, $y^2 \log x\,dx - xdy - ydx = 0$.

Dividing both sides by x^2y^2,

$\frac{\log x}{x^2}dx - \frac{1}{xy^2}dy - \frac{1}{x^2y}dx = 0$

or, $\frac{\log x}{x^2}dx + \left(\frac{1}{x}\right)\left(-\frac{1}{y^2}\right)dy + \left(\frac{1}{y}\right)\left(-\frac{1}{x^2}\right)dx = 0$

or, $\frac{\log x}{x^2}dx + d\left(\frac{1}{x} \cdot \frac{1}{y}\right) = 0.$

Integrating, $\int \frac{\log x}{x^2}dx + \int d\left(\frac{1}{xy}\right) = 0 \quad \ldots$ (i)

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Now, $\int \frac{\log x}{x^2}\,dx = \log x \frac{x^{-2+1}}{-2+1} - \int \frac{x^{-2+1}}{-2+1} \cdot \frac{1}{x}\,dx$

$= \log x \cdot \frac{x^{-1}}{-1} + \int \frac{1}{x^2}\,dx = -\frac{1}{x}\log x - \frac{1}{x}$

Then, equation (i) becomes $-\frac{1}{x}\log x - \frac{1}{x} + \frac{1}{xy} + c = 0$

or, $\frac{-y\log x - y + 1 + cxy}{xy} = 0$

or, $-(y\log x + y) + 1 + cxy = 0$

∴ $1 + cxy = y(1 + \log x)$ is the required solution.

Example 24. Solve: $(x^2 + y^2 + 2x)\,dx + xy\,dy = 0$.

Solution:

The given differential equation is

$(x^2 + y^2 + 2x)\,dx + xy\,dy = 0$

Multiplying by $2x$ on both sides

$(2x^3 + 2xy^2 + 4x^2)dx + 2x^2y\,dy = 0$

or, $2x^3\,dx + 4x^2\,dx + (x^2 \cdot 2ydy + y^2 \cdot 2xdx) = 0$

or, $2x^3\,dx + 4x^2dx + d(x^2y^2) = 0$

Integrating, $\frac{2x^4}{4} + \frac{4x^3}{3} + x^2y^2 = \frac{c}{6}$

∴ $3x^4 + 8x^3 + 6x^2y^2 = c$ is the required solution.

Example 25. Solve: $(x^3y^2 - y)\,dx - (x^2y^3 + x)\,dy = 0$.

Solution:

The given differential equation is

$(x^3y^2 - y)\,dx - (x^2y^3 + x)\,dy = 0$

or, $x^3y^2\,dx - x^2y^3dy - ydx - xdy = 0$

Dividing by x^2y^2,

$xdx - ydy + \left[\frac{-ydx - xdy}{x^2y^2}\right] = 0$

or, $xdx - ydy + \left[-\frac{dx}{x^2y} - \frac{dy}{xy^2}\right] = 0$

or, $xdx - ydy + \left[\frac{1}{y} \cdot d\left(\frac{1}{x}\right) + \frac{1}{x} \cdot d\left(\frac{1}{y}\right)\right]$

or, $xdx - ydy + d\left(\frac{1}{x} \cdot \frac{1}{y}\right) = 0$

Integrating, $\frac{x^2}{2} - \frac{y^2}{2} + \frac{1}{xy} = k$

or, $x^3y - xy^3 + 2 = 2k\,xy$.

∴ $x^3y - xy^3 + 2 = cxy$, where $c = 2k$ is the required solution.

Example 26. Solve: $xdx + ydy + (x^2 + y^2)\,dy = 0$.

Solution:

The given differential equation is $xdx + ydy + (x^2 + y^2)\,dy = 0$

Dividing both sides by $x^2 + y^2$, $\frac{xdx + ydy}{x^2 + y^2} + dy = 0$

or, $\frac{1}{2}\frac{d(x^2 + y^2)}{x^2 + y^2} + dy = 0$

Integrating, $\frac{1}{2}\log(x^2 + y^2) + y = k$

or, $\log(x^2 + y^2) = 2k - 2y$

or, $x^2 + y^2 = e^{2k - 2y}$

or, $x^2 + y^2 = e^{2k} \cdot e^{-2y}$

∴ $x^2 + y^2 = ce^{-2y}$ is the required solution, where $c = e^{2k}$.

Example 27. Solve: $\sin x\,dy - y\cos x\,dx + y^2\,dx = 0$.

Solution:

The given differential equation is $\sin x\,dy - y\cos x\,dx + y^2\,dx = 0$

Dividing by $-y^2$, $-\frac{1}{y^2}dy.\sin x + \frac{1}{y}\cos x\,dx - dx = 0$

or, $d\left(\frac{1}{y}\right).\sin x + \frac{1}{y}.d(\sin x) - dx = 0$

or, $d\left(\frac{1}{y}.\sin x\right) - dx = 0$

Integrating, $\frac{1}{y}\sin x - x = c$

or, $\sin x - xy = cy$

∴ $\sin x = y(x + c)$ is the required solution.

Example 28. Solve: $y\sin 2x\,dx - (y^2 + \cos^2 x)\,dy = 0$.

Solution:

The given differential equation is $y\sin 2x\,dx - (y^2 + \cos^2 x)\,dy = 0$

or, $y\sin 2x\,dx - \cos^2 x\,dy - y^2dy = 0$

or, $y\sin 2x\,dx - \frac{(1 + \cos 2x)}{2}dy - y^2dy = 0$

or, $y\sin 2x\,dx - \frac{\cos 2x}{2}dy - \frac{1}{2}dy - y^2\,dy = 0$

or, $2y\sin 2x\,dx - \cos 2x\,dy - dy - 2y^2dy = 0$

Multiplying by -1, $-2y\sin 2x\,dx + \cos 2x\,dy + dy + 2y^2\,dy = 0$

or, $[y.d(\cos 2x) + \cos 2x.dy] + dy + 2y^2\,dy = 0$

or, $d(y\cdot\cos 2x) + dy + 2y^2dy = 0$

Integrating,

$y\cos 2x + y + \frac{2y^3}{3} = k$

or, $3y\cos 2x + 2y^3 + 3y = 3k$

∴ $3y\cos 2x + 2y^3 + 3y = c$ which is the required solution, where $3k = c$.

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Exercise 7.2

1. Solve the following differential equations:

a. $2xydx + x^2dy = 0$
b. $3x^3y^2dx - 2x^3ydy = 0$
c. $(x^2 - ay)\,dx - (ax - y^2)\,dy = 0$
d. $x\frac{dy}{dx} = y + x^2\log x$
e. $(x + y)(dx - dy) = dx + dy$
f. $(x^2 + y^2 + 2x)\,dx + 2ydy = 0$
g. $(x^2 + y^2)\,dx = 2xy\,dy$
h. $\sin x\cos x\,dx + \sin y\cos y\,dy = 0$
i. $xdy - y\,dx - 2\sqrt{x^2 - y^2}\,dx = 0$
j. $\sec^2 x\tan y dx + \sec^2 y\tan x\,dy = 0$

Answer

1. a. $x^2y = c$ b. $x^3 = cy^2$ c. $x^3 + y^3 - 3axy = c$ d. $y = x^2\log x - x^2 + cx$
e. $x - y + c = \log(x + y)$ f. $x + \log(x^2 + y^2) = c$ g. $x^2 - y^2 = cx$ h. $\sin^2 x + \sin^2 y = c$
i. $y = x\sin[\log(cx^2)]$ j. $\tan x \cdot \tan y = c$

7.7 Linear differential equations of first order

A differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$ in which P and Q are functions of x alone or constant, is called a **linear differential equation of first order**. This equation is non-homogeneous and becomes homogeneous if $Q(x) = 0$.

7.7.1 First order linear homogeneous equation and its solution

If $Q(x) = 0$,

then, $\frac{dy}{dx} + P(x)y = Q(x)$ ....(i)

reduces to $\frac{dy}{dx} + P(x)y = 0$ ....(ii)

Equation (ii) is said to be homogeneous equation corresponding to (i). A general solution of (i) generally depends on the solution of (ii). Equation (ii) is solvable by separation of variables as,

$$\frac{dy}{dx} = -P(x)y$$

or, $\frac{dy}{y} = -P(x)dx$

On integrating

$$\log y = -\int P(x)dx + \log C$$

$\therefore\ y = Ce^{-\int P(x)dx}$

is the general solution of (ii). Thus, we see that the solution of a homogeneous differential equation is always an exponential function.

7.7.2 Solution of first order linear non-homogeneous differential equation

The equation $\frac{dy}{dx} + P(x)y = Q(x)$ ....(i)

is general first order linear non-homogeneous differential equation which can be solved by the method of exactness by finding integrating factor. Equation (i) can be written as

$$[P(x)y - Q(x)]\,dx + dy = 0 \quad \text{....(ii)}$$

For solving (ii) we need an integrating factor (I.F.) that makes (ii) exact differential equation. Comparing (ii) with $Mdx + Ndy = 0$, we get

Since, $N = 1$, so $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{N}\left(\frac{\partial M}{\partial y}\right) = P(x)$, a function of x.

$\therefore$ Integrating factor of (ii) is I.F. $= e^{\int P(x)dx}$

Multiplying (i) by I.F.,

$$e^{\int P(x)dx}\left[\frac{dy}{dx} + P(x)y\right] = \frac{d}{dx}(ye^{\int P(x)dx}) = e^{\int P(x)dx}Q(x)$$

or, $d(y\,e^{\int P(x)dx}) = e^{\int P(x)dx}Q(x)$

On integrating, we get

$$y\,e^{\int P(x)dx} = \int Q(x)\,e^{\int P(x)dx}\,dx + C$$

$\therefore\ y = e^{-\int P(x)dx}\left[\int Q(x)\,e^{\int P(x)dx}\,dx + C\right]$ gives the general solution of equation (i).

We summarize the above process of solving a linear non-homogeneous differential equation (i) in the following steps:

Step 1: Reduce the given differential equation in the form

$$\frac{dy}{dx} + Py = Q. \quad \text{.... (i)}$$

Step 2: Find integrating factor (I.F.) $= e^{\int Pdx}$

Step 3: Multiply both sides of (i) by I.F. $= e^{\int Pdx}$, we get

i.e. $e^{\int Pdx}\cdot\frac{dy}{dx} + Py\,e^{\int Pdx} = Qe^{\int Pdx}$

or, $\frac{d}{dx}(y.e^{\int Pdx}) = Qe^{\int Pdx}$

$\therefore\ d\left(y.e^{\int Pdx}\right) = Qe^{\int Pdx}dx$ .... (ii)

Step 4: Integrating both sides of (ii), we get

$$ye^{\int Pdx} = \int Q\,e^{\int Pdx}\,dx + c$$

$\therefore\ y = e^{-\int Pdx}\left[\int Qe^{\int Pdx}dx + c\right]$ is the required solution.

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Remark: If the linear differential equation is of the form $\frac{dx}{dy} + P(y)x = Q(y)$ then

I.F. $e^{\int P(y)dy}$ and solution will be $x = e^{-\int P(y)dy}[Q\, e^{\int P(y)dy}\, dy + c]$.

For illustration see example 35.

Example 29. Solve: $\frac{dy}{dx} + 2y = 4x$

Solution:

The given differential equation is $\frac{dy}{dx} + 2y = 4x$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = 2, Q = 4x$.

Then, I.F. $= e^{\int P dx} = e^{\int 2dx} = e^{2x}$.

Multiplying both sides of (i) by I.F. $= e^{2x}$, we get $e^{2x}\frac{dy}{dx} + 2y.\, e^{2x} = 4xe^{2x}$

or, $\frac{d}{dx}(e^{2x}.y) = 4xe^{2x}$

or, $d(e^{2x}.y) = 4x\, e^{2x}\, dx$

Integrating, $y.e^{2x} = 4\int x\, e^{2x}\, dx = 4.\left[x.\frac{e^{2x}}{2} - \int\frac{e^{2x}}{2}.1\, dx\right]$

or, $ye^{2x} = 2x\, e^{2x} - e^{2x} + c$

$\therefore\ y = 2x - 1 + ce^{-2x}$ is the required solution.

Example 30. Solve: $(1+x)\frac{dy}{dx} - xy = 1 - x$.

Solution:

The given differential equation is $(1+x)\frac{dy}{dx} - xy = 1 - x$

or, $\frac{dy}{dx} + \left(-\frac{x}{1+x}\right)y = \frac{1-x}{1+x}$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = \left(-\frac{x}{1+x}\right)$, $Q = \frac{1-x}{1+x}$

Then, I.F. $= e^{\int P dx}$

$= e^{\int\frac{-1}{1+x}dx}$

$= e^{\int\left[\frac{1+x-1}{1+x}\right]dx}$

$= e^{-\int\left[1-\frac{1}{1+x}\right]dx}$

$= e^{-[x - \log|(1+x)|]}$

$= e^{-x}.\, e^{\log|1+x|}$

$= (1+x)\, e^{-x}$.

Multiplying both sides of (i) by I.F. $= (1+x)\, e^{-x}$, we get

$(1+x)\, e^{-x}\frac{dy}{dx} - e^{-x}y = (1-x)\, e^{-x}$

or, $\frac{d}{dx}[(1+x)\, e^{-x}.y] = (1-x)\, e^{-x}$

or, $d[(1+x)\, e^{-x}\, y] = (1-x)\, e^{-x}\, dx$.

Integrating, $(1+x)\, e^{-x}\, y = \int(1-x)\, e^{-x}\, dx$

$= (1-x)\frac{e^{-x}}{-1} - \left[\int(-1).\frac{e^{-x}}{-1}\, dx\right]$

$= -(1-x)\, e^{-x} - \int e^{-x}\, dx$

$= -(1-x)\, e^{-x} + e^{-x} + c$

$= (-1+x+1)\, e^{-x} + c$

or, $(1+x)\, e^{-x}\, y = xe^{-x} + c$

$\therefore\ y(1+x) = x + ce^{x}$ is the required solution.

Example 31. Solve: $\frac{dy}{dx} + y\tan x = \sec x$

Solution:

The given differential equation is $\frac{dy}{dx} + y\tan x = \sec x$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = \tan x$, $Q = \sec x$

Then, I.F. $= e^{\int P dx} = e^{\int\tan x\, dx} = e^{\log\sec x} = \sec x$. $\left[\because \int\tan x dx = \log\sec x\right]$

Multiplying both sides of (i) by I.F. $= \sec x$, $\frac{dy}{dx}\sec x + y.\tan x.\sec x = \sec^2 x$

or, $\frac{d}{dx}(y.\sec x) = \sec^2 x$

or, $d(y\sec x) = \sec^2 x\, dx$.

Integrating, $y\sec x = \int\sec^2 x\, dx + c$

or, $y\sec x = \tan x + c$

or, $y = \frac{\tan x}{\sec x} + \frac{c}{\sec x}$

$\therefore\ y = \sin x + c\cos x$ is the required solution.

Example 32. Solve: $\sin 2x\frac{dy}{dx} - y = \tan x$.

Solution:

The given differential equation is $\sin 2x\frac{dy}{dx} - y = \tan x$

or, $\frac{dy}{dx} + \left(-\frac{1}{\sin 2x}\right)y = \frac{\tan x}{\sin 2x}$

or, $\frac{dy}{dx} + (-\operatorname{cosec} 2x)\, y = \frac{1}{2}\frac{\sin x}{\cos x.\sin x.\cos x}$

or, $\frac{dy}{dx} + (-\operatorname{cosec} 2x)\, y = \frac{1}{2}\sec^2 x$ .... (i)

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which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = -\operatorname{cosec} 2x$, $Q = \frac{1}{2}\sec^2 x$

Now, $\int P\,dx = -\int \operatorname{cosec} 2x\,dx$

$= -\frac{1}{2}\log\tan\frac{2x}{2}$

$= -\frac{1}{2}\log\tan x = \log(\tan x)^{1/2}$

Then, I.F. $= e^{\int P\,dx} = e^{\log(\tan x)^{-1/2}} = \frac{1}{\sqrt{\tan x}}$

Multiplying both sides of (i) by $\frac{1}{\sqrt{\tan x}}$, we get

$\frac{1}{\sqrt{\tan x}}\frac{dy}{dx} - \frac{1}{\sqrt{\tan x}}\cdot\operatorname{cosec} 2x \cdot y = \frac{1}{2\sqrt{\tan x}}\sec^2 x$

or, $\frac{d}{dx}\left[y\cdot\frac{1}{\sqrt{\tan x}}\right] = \frac{1}{2\sqrt{\tan x}}\sec^2 x$ $\left[\because \frac{d}{dx}(y \times \text{I.F.}) = Q \times \text{I.F.}\right]$

or, $d\left(\frac{y}{\sqrt{\tan x}}\right) = \frac{1}{2\sqrt{\tan x}}\sec^2 x\,dx$.

Integrating, $\frac{y}{\sqrt{\tan x}} = \int\frac{1}{2\sqrt{\tan x}}\sec^2 x\,dx$

Put, $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

We have, $\frac{y}{\sqrt{\tan x}} = \frac{1}{2}\int\frac{1}{\sqrt{t}}\,dt = \frac{1}{2}\frac{\sqrt{t}}{\frac{1}{2}} + c = \sqrt{t} + c = \sqrt{\tan x} + c$.

$\therefore$ $y = \tan x + c\sqrt{\tan x}$, is the required solution.

Example 33. Solve: $x\frac{dy}{dx} + 2y = x^2\log x$.

Solution:

The given differential equation is $x\frac{dy}{dx} + 2y = x^2\log x$

or, $\frac{dy}{dx} + \frac{2}{x}y = x\log x$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = \frac{2}{x}$, $Q = x\log x$

Then, I.F. $= e^{\int P\,dx} = e^{\int\frac{2}{x}dx} = e^{2\log x} = e^{\log x^2} = x^2$.

Multiplying both sides of (i) by its I.F. $= x^2$, $x^2\frac{dy}{dx} + 2xy = x^3\log x$

or, $\frac{d}{dx}(x^2\cdot y) = x^3\log x$

or, $d(x^2 y) = x^3\log x\,dx$.

Integrating, $x^2 y = \int\log x\cdot x^3\,dx$

or, $x^2 y = \log x\cdot\frac{x^4}{4} - \int\frac{x^4}{4}\cdot\frac{1}{x}\,dx$

or, $x^2 y = \frac{1}{4}x^4\log x - \frac{1}{4}\int x^3\,dx$

or, $x^2 y = \frac{1}{4}x^4\log x - \frac{1}{16}x^4 + k$

$\therefore$ $16x^2 y = 4x^4\log x - x^4 + c$ which is a required solution, where $c = 16k$.

Example 34. Solve: $(x^3 + 1)\frac{dy}{dx} + 3x^2 y = \sin^2 x$.

Solution:

The given differential equation is $(x^3 + 1)\frac{dy}{dx} + 3x^2 y = \sin^2 x$

or, $\frac{dy}{dx} + \frac{3x^2}{x^3 + 1}y = \frac{\sin^2 x}{x^3 + 1}$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = \frac{3x^2}{x^3 + 1}$, $Q = \frac{\sin^2 x}{x^3 + 1}$

Then, I.F. $= e^{\int P\,dx} = e^{\int\frac{3x^2}{x^3+1}dx} = e^{\log(x^3+1)} = x^3 + 1$.

Multiplying both sides of (i) by I.F. $= x^3 + 1$, we get $(x^3 + 1)\frac{dy}{dx} + 3x^2 y = \sin^2 x$

or, $\frac{d}{dx}[(x^3 + 1)y] = \sin^2 x$

or, $d[(x^3 + 1)y] = \sin^2 x\,dx$.

Integrating, $(x^3 + 1)y = \int\sin^2 x\,dx = \int\left(\frac{1 - \cos 2x}{2}\right)dx$

or, $(x^3 + 1)y = \frac{x}{2} - \frac{\sin 2x}{4} + k$

or, $4y(x^3 + 1) = 2x - \sin 2x + 4k$

$\therefore$ $4y(x^3 + 1) = 2x - \sin 2x + c$ is the required solution where $4k = c$.

Example 35. Solve: $(1 + x + xy^2)\,dy + (y + y^3)\,dx = 0$.

Solution:

The given differential equation is $(1 + x + xy^2) + (y + y^3)\frac{dx}{dy} = 0$

or, $\frac{dx}{dy} + \frac{1 + x + xy^2}{y + y^3} = 0$

or, $\frac{dx}{dy} + \frac{x(1 + y^2)}{y + y^3} + \frac{1}{y + y^3} = 0$

or, $\frac{dx}{dy} + \frac{x}{y} = -\frac{1}{y(1 + y^2)}$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dx}{dy} + Px = Q$, $P = \frac{1}{y}$, $Q = -\frac{1}{y(1 + y^2)}$

Then, I.F. $= e^{\int P\,dy} = e^{\int(1/y)\,dy} = e^{\log y} = y$.

Multiplying both sides of (i) by I.F. $= y$, $y\cdot\frac{dx}{dy} + x = -\frac{1}{1 + y^2}$

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or, $\frac{d}{dx}(xy) = -\frac{1}{1+y^2}$

or, $d(xy) = -\frac{1}{1+y^2}dy$.

Integrating, $xy = -\int \frac{1}{1+y^2}dy$

.... (ii)

or, $xy = -\tan^{-1} y + c$

$\therefore$ $xy + \tan^{-1} y = c$ is the required solution.

Example 36. Solve: $\frac{dy}{dx} + \frac{y}{x} = x^2$, if $y = 1$ when $x = 1$

Solution:

The given differential equation is $\frac{dy}{dx} + \frac{y}{x} = x^2$

or, $\frac{dy}{dx} + \left(\frac{1}{x}\right)y = x^2$ .... (i)

which is the linear differential equation of first order.

Comparing equation (i) with $\frac{dy}{dx} + Py = Q$, we get $P = \frac{1}{x}$, $Q = x^2$

Then, I.F. $= e^{\int (1/x)\,dx} = e^{\log x} = x$.

Multiplying both sides of equation (i) by I.F. $= x$, $x\frac{dy}{dx} + x.\frac{1}{x}y = x^3$

or, $\frac{d}{dx}(xy) = x^3$

or, $d(xy) = x^3\,dx$

Integrating, $xy = \int x^3\,dx$

$\therefore$ $xy = \frac{x^4}{4} + c$ .... (ii)

When $x = 1$, $y = 1$, so that (ii) gives $1.1 = \frac{1}{4} + c$

$\therefore$ $c = 1 - \frac{1}{4} = \frac{3}{4}$.

Substituting the value of c in equation (ii), $xy = \frac{x^4}{4} + \frac{3}{4}$

$\therefore$ $4xy = x^4 + 3$ is the required solution.

7.8 Bernoulli's equations

An equation of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$ is called *Bernoulli's equation*. The Bernoulli's equation is a first order non-homogeneous and non-linear differential equation. Such type of equation can be solved by reducing into linear differential equation.

The method of solving a Bernoulli's equation can be summarized as follows:

Step 1: Dividing both sides of (i) by y^n, we get

$y^{-n}\frac{dy}{dx} + Py^{-n+1} = Q$ .... (ii)

Step 2: Putting $y^{-n+1} = v$, then differentiating with respect to, x

$(-n \quad 1)\, y^{-n}\frac{dy}{dx} = \frac{dv}{dx}$

$\therefore$ $y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}$

The equation (ii) then becomes $\frac{1}{1-n}\frac{dv}{dx} + Pv = Q$

$\therefore$ $\frac{dv}{dx} + [(1-n)P]v = (1-n)Q$ .... (iii)

Step 3: The equation (iii) is linear differential equation in v and x. It can be solved by the method of linear equation.

Remark: We may have different forms in place of y and y^n. (See examples 41 a..d 42).

Example 37. Solve: $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Solution:

The given differential equation is $\frac{dy}{dx} + \left(\frac{1}{x}\right)y = \left(\frac{1}{x^2}\right)\cdot y^2$

which is the Bernoulli's differential equation.

Dividing both sides by y^2, $y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x^2}$ .... (i)

Let $y^{-1} = v$, then

$-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$

or, $y^{-2}\frac{dy}{dx} = -\frac{dv}{dx}$

Then, equation (i) becomes $-\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x^2}$

or, $\frac{dv}{dx} + \left(-\frac{1}{x}\right)v = -\frac{1}{x^2}$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{-\int \frac{1}{x}dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$.

Multiplying both sides of (ii) by I.F. $= \frac{1}{x}$, $\frac{1}{x}\frac{dv}{dx} + \left(-\frac{1}{x^2}\right)v = -\frac{1}{x^3}$

or, $\frac{d}{dx}\left(\frac{1}{x}.v\right) = -\frac{1}{x^3}$

or, $d\left(\frac{v}{x}\right) = -\frac{1}{x^3}dx$.

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Integrating, $\frac{v}{x} = -\int \frac{1}{x^3}\,dx$

or, $\frac{v}{x} = \frac{-x^{-3+1}}{-3+1} + k$

or, $\frac{v}{x} = \frac{1}{2x^2} + k$

Substituting the value of $v = \frac{1}{y}$,

$\frac{1}{xy} = \frac{1}{2x^2} + k$

or, $\frac{1}{xy} = \frac{1 + 2kx^2}{2x^2}$

or, $2x = y(1 + 2kx^2)$

or, $2x = 2kx^2y + y$

$\therefore$ $2x = cx^2y + y$ is the required solution, where $c = 2k$.

Example 38. Solve: $x\frac{dy}{dx} + y = y^2 \log x$.

Solution:

The given differential equation is $x\frac{dy}{dx} + y = y^2 \log x$

$\frac{dy}{dx} + \left(\frac{1}{x}\right).y = \frac{\log x}{x}.y^2$

which is the Bernoulli's differential equation.

Dividing by y^2, $y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{\log x}{x}$ .... (i)

Let $y^{-1} = v$. Then, $-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$

or, $y^{-2}\frac{dy}{dx} = -\frac{dv}{dx}$

Then, equation (i) becomes $-\frac{dv}{dx} + \frac{1}{x}.v = \frac{\log x}{x}$

or, $\frac{dv}{dx} + \left(-\frac{1}{x}\right)v = -\frac{\log x}{x}$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{\int P\,dx} = e^{-\int \frac{1}{x}dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$.

Multiplying both sides of (ii) by I.F. $= \frac{1}{x}$, we get

$\therefore$ $\frac{1}{x}\frac{dv}{dx} - \frac{1}{x^2}v = -\frac{\log x}{x^2}$

or, $\frac{d}{dx}\left(\frac{1}{x}.v\right) = -\frac{\log x}{x^2}$

or, $d\left(\frac{v}{x}\right) = -\frac{\log x}{x^2}\,dx$

Integrating, $\frac{v}{x} = -\int \frac{\log x}{x^2}\,dx = -\int \frac{1}{x^2}\log x\,dx$

or, $\frac{v}{x} = -\left[\log x.\frac{x^{-1}}{(-1)} - \int \frac{x^{-1}}{-1} \times \frac{1}{x}\,dx\right]$

or, $\frac{v}{x} = \frac{\log x}{x} - \int \frac{1}{x^2}\,dx$

or, $\frac{v}{x} = \left(\frac{\log x}{x} + \frac{1}{x}\right) + c$

or, $\frac{v}{x} = \frac{\log x}{x} + \frac{1}{x} + c$.

Substituting the value of $v = \frac{1}{y}$,

$\frac{1}{xy} = \frac{\log x}{x} + \frac{1}{x} + c$

or, $1 = y\log x + y + cxy$

$\therefore$ $cxy + y(\log x + 1) = 1$ is the required solution.

Example 39. Solve: $\frac{dy}{dx} - y\tan x = -y^2 \sec x$.

Solution:

The given differential equation is $\frac{dy}{dx} + (-\tan x)y = -\sec x.y^2$

which is the Bernoulli's differential equation.

Dividing by y^2, $y^{-2}\frac{dy}{dx} - y^{-1}\tan x = -\sec x$ .... (i)

Let $y^{-1} = v$. Then,

$-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$

or, $y^{-2}\frac{dy}{dx} = -\frac{dv}{dx}$

Then, equation (i) becomes $-\frac{dv}{dx} - v\tan x = -\sec x$

or, $\frac{dv}{dx} + \tan x.v = \sec x$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{\int P\,dx} = e^{\int \tan x\,dx} = e^{\log \sec x} = \sec x$.

Multiplying both sides of equation (ii) by $\sec x$, we get

$\sec x.\frac{dv}{dx} + \sec x.\tan x.v = \sec^2 x$

or, $\frac{d}{dx}(\sec x.v) = \sec^2 x$

or, $d(\sec x\,v) = \sec^2 x\,dx$.

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Integrating, $\sec x \cdot v = \int \sec^2 x \, dx$

or, $\sec x \cdot v = \tan x + c$

Substituting the value of $v = \frac{1}{y}$, we get

$\frac{\sec x}{y} = \tan x + c$

$\therefore$ $\sec x = y(\tan x + c)$ is the required solution.

Example 40. Solve: $\cos x \, dy = y(\sin x - y) \, dx$.

Solution:

The given differential equation is $\frac{dy}{dx} = \frac{y(\sin x - y)}{\cos x}$

or, $\frac{dy}{dx} - y\frac{\sin x}{\cos x} = -\frac{y^2}{\cos x}$

or, $\frac{dy}{dx} - y\tan x = -\sec x \cdot y^2$

which is the Bernoulli's differential equation.

Dividing by y^2, $y^{-2}\frac{dy}{dx} - y^{-1}\tan x = -\sec x$ .... (i)

Let $y^{-1} = v$. Then,

$-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$

or, $y^{-2}\frac{dy}{dx} = -\frac{dv}{dx}$

Then, equation (i) becomes $-\frac{dv}{dx} - v\tan x = -\sec x$

or, $\frac{dv}{dx} + (\tan x)\, v = \sec x$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{\int P dx} = e^{\int \tan x \, dx} = e^{\log \sec x} = \sec x$

Multiplying both sides of (ii) by I.F. = $\sec x$, we get

$\sec x \cdot \frac{dv}{dx} + \sec x \cdot \tan x \cdot v = \sec^2 x$

or, $\frac{d}{dx}(\sec x \cdot v) = \sec^2 x$

or, $d(\sec x \cdot v) = \sec^2 x \, dx$.

Integrating, $\sec x \cdot v = \int \sec^2 x \, dx$

or, $\sec x \cdot v = \tan x + c$

Substituting the value of $v = \frac{1}{y}$, we get

$\frac{\sec x}{y} = \tan x + c$

$\therefore$ $\sec x = y(\tan x + c)$ is the required solution.

Example 41. Solve: $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$.

Solution:

The given differential equation is $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$

which is the Bernoulli's differential equation.

Dividing by e^y, $e^{-y}\frac{dy}{dx} + e^{-y}\frac{1}{x} = \frac{1}{x^2}$ .... (i)

Let $e^{-y} = v$. Then,

$-e^{-y}\frac{dy}{dx} = \frac{dv}{dx}$

or, $e^{-y}\frac{dy}{dx} = -\frac{dv}{dx}$

Then, equation (i) becomes, $-\frac{dv}{dx} + v \cdot \frac{1}{x} = \frac{1}{x^2}$

or, $\frac{dv}{dx} + \left(-\frac{1}{x}\right)v = -\frac{1}{x^2}$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{\int P dx} = e^{-\int \frac{1}{x} dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$.

Multiplying both sides of (ii) by I.F. $= \frac{1}{x}$, we get

$\frac{1}{x}\frac{dv}{dx} + \left(-\frac{1}{x^2}\right)v = -\frac{1}{x^3}$

or, $\frac{d}{dx}\left(\frac{1}{x} \cdot v\right) = -\frac{1}{x^3}$

or, $d\left(\frac{v}{x}\right) = -\frac{1}{x^3}dx$.

Integrating, $\frac{v}{x} = -\int \frac{1}{x^3} dx = -\frac{x^{-2}}{-2} + k$

$\therefore$ $\frac{v}{x} = \frac{1}{2x^2} + k$

Substituting the value of $v = e^{-y}$, we get

$\frac{1}{e^y \cdot x} = \frac{1}{2x^2} + k$

or, $\frac{1}{xe^y} = \frac{1 + 2kx^2}{2x^2}$

or, $2x = e^y(1 + 2kx^2)$

$\therefore$ $2x = (1 + cx^2)e^y$ is a required solution, where $c = 2k$.

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Example 42. Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)\,e^x \sec y$.

Solution:

The given differential equation is $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)\,e^x \sec y$

which is the Bernoulli's differential equation.

Dividing both sides by $\sec y$, $\cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x)\,e^x$

or, $\cos y \frac{dy}{dx} + \left(-\frac{1}{1+x}\right)\sin y = (1+x)\,e^x$ .... (i)

Let $\sin y = v$. Then, $\cos y \frac{dy}{dx} = \frac{dv}{dx}$

Then, equation (i) becomes $\frac{dv}{dx} + \left(-\frac{1}{1+x}\right)v = (1+x)\,e^x$ .... (ii)

The equation (ii) is a linear differential equation of first order.

Then, I.F. $= e^{\int P\,dx} = e^{-\int \frac{1}{1+x}dx} = e^{-\log(1+x)} = e^{\log(1+x)^{-1}} = (1+x)^{-1} = \frac{1}{1+x}$

Multiplying both sides of (ii) by $\frac{1}{1+x}$, we get

$\frac{1}{1+x}\frac{dv}{dx} - \frac{1}{(1+x)^2}v = e^x$

or, $\frac{d}{dx}\left(\frac{1}{1+x}v\right) = e^x$

or, $d\left(\frac{v}{1+x}\right) = e^x\,dx$.

Integrating, $\frac{v}{1+x} = \int e^x\,dx$

or, $\frac{v}{1+x} = e^x + c$

Substituting the value of $v = \sin y$, we get

$\frac{\sin y}{1+x} = e^x + c$

$\therefore$ $\sin y = (1+x)(e^x + c)$ is the required solution.

Exercise 7.3

1. Solve the following differential equations

a. $\frac{dx}{dt} + \frac{x}{t} = \frac{1}{t^2}$

b. $\frac{dy}{dx} + 3y = e^{-x}$

c. $(1-x^2)\frac{dy}{dx} - xy = 1$

d. $\cos x \frac{dy}{dx} + y\sin x = \sec^2 x$

e. $x\log x \frac{dy}{dx} + y = 2\log x$

f. $\sec x \frac{dy}{dx} + y\sec x = 1$

g. $(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$

h. $\sec x \frac{dy}{dx} + y\,\mathrm{cosec}\,x = 1$

i. $(x+y+1)\frac{dy}{dx} = 1$

2. Solve the following initial value problems

a. $xy' + 2y = x^2 - x + 1$, $y(1) = \frac{1}{2}$

b. $y' + 2y = 4x$, $y(0) = -1$

c. $x^2y' + 2xy - x + 1 = 0$, $y(1) = 0$

d. $y' + 2y\tan x = \sin x$, $y(0) = 0$

e. $y' + 2y = 6e^x$, $y(0) = 5$

3. **Solve the following differential equations**

a. $\frac{dy}{dx} + \frac{y}{x} = y^2$

b. $\frac{dy}{dx} + \frac{y}{x} = x^2y^6$

c. $\frac{dy}{dx} + y\tan x = y^3\sec x$

d. $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

e. $\frac{dy}{dx} + \frac{y}{x}\log y = \frac{y}{x^2}(\log y)^2$

4. **Solve the following initial value problems**

a. $y' - y\tan x = -y^3\sec x$, $y(0) = 1$

b. $y' + \frac{1}{x} = \frac{e^y}{x^2}$, $y(1) = 0$

c. $y' + \frac{1}{x}\tan y = \frac{1}{x^2}\tan y \sin y$, $y(1) = \frac{\pi}{2}$

Answer

1. a. $x = 1 + ce^{1/t}$ b. $y = \frac{1}{2}e^{-x} + ce^{-3x}$ c. $y\sqrt{1-x^2} = \sin^{-1}x + c$
   d. $3y\sec x = c + 3\tan x + \tan^3 x$ e. $y\log x = (\log x)^2 + c$ f. $2v = (\sin x + \cos x) + ce^{-x}$
   g. $2ye^{\tan^{-1}x} = e^{2\tan^{-1}x} + c$ h. $4y\sin x + \cos 2x = c$ i. $x + y + 2 = ce^x$
2. a. $y = \frac{x^2}{4} - \frac{x}{3} + \frac{1}{2} + \frac{1}{12x^2}$ b. $y = 2x - 1$ c. $y = \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}$
   d. $y\sec^2 x - \sec x + 1 = 0$ e. $y = 3e^{-2x} + 2e^x$
3. a. $1 = xy(c - \log x)$ b. $y^3(5x^3 + 2cx^5) = 2$ c. $(c - 2\sin x)y^2 = \cos^2 x$
   d. $e^{y^2}\tan y = \frac{e^{x^2}}{2}(x^2 - 1) + c$ e. $x = \log y\left(\frac{1}{2} + cx^2\right)$
4. a. $\sec x = y(\tan x + 1)$ b. $2x = (x^2 + 1)e^y$ c. $2x = (1 + x^2)\sin y$

7.9 Riccati's equation and its solution

An equation of the form

$$\frac{dy}{dx} + P(x)y + R(x)y^2 = Q(x) \quad \ldots.(i)$$

which is a non-linear equation is called the Riccati equation. The above form is considered as a standard form of Riccati equation.

If we take $R(x) = 0$, then the Riccati equation reduces to a first order linear equation. Again, if we take $Q(x) = 0$, then it reduces to Bernoulli's equation. The Riccati equation is needed in various applications of Mathematics that involve non-linear problems.

If we can find a particular solution say y_1, then we can reduce the Riccati equation into a standard form as discussed below:

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Case I: The substitution $y = y_1 + \frac{1}{u}$ reduces the Riccati equation into first order linear equation.

Case II: The substitution $y = y_1 + u$ reduces the Riccati equation into a Bernoulli's equation.

Case III: Moreover, a general substitution $y = \frac{1}{R(x)}\frac{dz}{dx}$ reduces the Riccati equation into a second order linear homogeneous equation given by

$$\frac{d^2z}{dx^2} + \left[P(x) - \frac{R'(x)}{R(x)}\right]\frac{dz}{dx} - R(x)\,Q(x)z = 0$$

Out of above cases, the case (I) is the most convenient way to solve the Riccati equation. We also use this substitution method to solve the Riccati equation. We consider the Riccati equation:

$$\frac{dy}{dx} + x^2y - xy^2 = 1 \qquad \ldots.(ii)$$

By hit and trial, we see that $y = x$ is a particular solution of equation (ii) since

$$\frac{dx}{dx} + x^2 \cdot x - x \cdot x^2 = 1$$

$$1 + x^3 - x^3 = 1$$

$1 = 1$ (true).

Let us substitute $y = x + \frac{1}{u}$ $\qquad \ldots.(iii)$

Differentiating both sides with respect to x,

$$\frac{dy}{dx} = 1 - \frac{1}{u^2}\frac{du}{dx}$$

Then (ii) becomes

$$1 - \frac{1}{u^2}\frac{du}{dx} + x^2\left(x + \frac{1}{u}\right) - x\left(x + \frac{1}{u}\right)^2 = 1$$

or, $$-\frac{1}{u^2}\frac{du}{dx} + x^3 + \frac{x^2}{u} - x\left(x^2 + 2\frac{x}{u} + \frac{1}{u^2}\right) = 0$$

or, $$-\frac{1}{u^2}\frac{du}{dx} + x^3 + \frac{x^2}{u} - x^3 - 2\frac{x^2}{u} - \frac{x}{u^2} = 0$$

or, $$-\frac{1}{u^2}\frac{du}{dx} - \frac{x^2}{u} - \frac{x}{u^2} = 0$$

$$\therefore \quad \frac{du}{dx} + x^2u = -x \qquad \ldots.(iii)$$

which is linear first order equation. We now find its solution.

Here, I.F. $= e^{\int P dx} = e^{\int x^2 dx} = e^{\frac{x^3}{3}}$

Multiplying (iii) by I.F., we get

$$e^{\frac{x^3}{3}}\frac{du}{dx} + x^2ue^{\frac{x^3}{3}} = -xe^{\frac{x^3}{3}}$$

$$\frac{d\left(ue^{\frac{x^3}{3}}\right)}{dx} = -xe^{\frac{x^3}{3}}$$

$$\int d\left(ue^{\frac{x^3}{3}}\right) = -\int xe^{\frac{x^3}{3}}\,dx + C$$

$$ue^{\frac{x^3}{3}} = -\int xe^{\frac{x^3}{3}}\,dx + C$$

We note that integral on the right side can not be obtained using elementary functions. So,

$$\therefore \quad u = e^{\frac{-x^3}{3}}\left(C - \int xe^{\frac{x^3}{3}}\,dx\right)$$

Substituting the value of u in $y = x + \frac{1}{u}$, we get

$$\therefore \quad y(x) = x + \frac{e^{\frac{x^3}{3}}}{C - \int x\,e^{\frac{x^3}{3}}\,dx}$$ is the solution of given equation.

Example 43. Solve: $y' = y^2 + 2y + 5$.

Solution:

It is a simple version of Riccati equation and can be solved easily.

$$\frac{dy}{dx} = y^2 + 2y + 5$$

$$\frac{dy}{y^2 + 2y + 5} = dx$$

$$\int \frac{dy}{(y+1)^2 + 4} = \int dx$$

$$\int \frac{dy}{(y+1)^2 + 2^2} = \int dx$$

$$\frac{1}{2}\tan^{-1}\left(\frac{y+1}{2}\right) = x + C$$

$$\therefore \quad \tan^{-1}\left(\frac{y+1}{2}\right) = 2x + C$$

Example 44. Show that the equation $\frac{dy}{dx} = x^3(y-x)^2 + x^{-1}y$ is a Riccati equation and hence solve it completely.

Solution:

Here, $\frac{dy}{dx} = x^3(y-x)^2 + x^{-1}y$ $\qquad \ldots.(i)$

or, $$\frac{dy}{dx} = x^3y^2 - 2x^4y + x^3 + x^{-1}y$$

or, $$\frac{dy}{dx} = -(2x^4 - x^{-1})y + x^3y^2 + x^3$$

$$\therefore \quad \frac{dy}{dx} + (2x^4 - x^{-1})y - x^3y^2 = x^3$$

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This shows that the given equation is Riccati equation. By inspection, $y = x$ is a solution of (i). As discussed in theory, let $y = x + u$ which gives

$\frac{dy}{dx} = 1 + \frac{du}{dx}$.

Then (i) becomes

$1 + \frac{du}{dx} = x^3u^2 + x^{-1}(x + u)$

or, $1 + \frac{du}{dx} = x^3u^2 + 1 + x^{-1}u$

$\therefore \quad \frac{du}{dx} = x^3u^2 + x^{-1}u$

Hence, $\frac{du}{dx} - x^{-1}u = x^3u^2$, which is a Bernoulli's equation.

Dividing both sides by u^2, we get

$\frac{1}{u^2}\frac{du}{dx} - \frac{1}{u}\frac{1}{x} = x^3$ .... (ii)

To solve this, let $\frac{1}{u} = v$.

Then, $\frac{dv}{dx} = \frac{d\left(\frac{1}{u}\right)}{du} \cdot \frac{du}{dx}$

So, $-\frac{dv}{dx} = \frac{1}{u^2}\frac{du}{dx}$

Then (ii) becomes.

$-\frac{dv}{dx} - v\frac{1}{x} = x^3$

$\frac{dv}{dx} + \frac{1}{x}v = -x^3$ .... (iii)

Finally (iii) is a linear differential equation which we now solve by finding integrating factor (I.F.). So,

I.F. $= e^{\int p dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$

Multiplying (iii) by x, we get

$x\frac{dv}{dx} + v = -x^4$

So, $\frac{d(v \cdot x)}{dx} = -x^4$

or, $d(v \cdot x) = -x^4\,dx$

$\int d(v \cdot x) = \int -x^4\,dx$

or, $v \cdot x = \frac{-x^5}{5} + C$

$\therefore \quad v = -\frac{x^4}{5} + \frac{C}{x}$

Again, using $v = \frac{1}{u} = \frac{1}{(y - x)}$, we get

$\frac{1}{y - x} = \left(\frac{C}{x} - \frac{x^4}{5}\right)$

$y - x = \left(\frac{C}{x} - \frac{x^4}{5}\right)^{-1}$

$\therefore \quad y = x + \left(Cx^{-1} - \frac{x^4}{5}\right)^{-1}$

Exercise 7.4

1. Show that the substitution $y = y_1 + u$ where y_1 is a solution of Riccati equation, reduces the Riccati equation to a Bernoulli's equation.
2. Show that the substitution $y = y_1 + \frac{1}{u}$ where y_1 is a solution of Riccati equation reduces the Riccati equation to first order linear equation.
3. Solve the Riccati equation $2\cos x\frac{dy}{dx} = 2\cos^2 x - \sin^2 x + y^2$.

[Take $y = \sin x$ as trial solution and then substitute $y = \sin x + \frac{1}{u}$]

4. Solve the Riccati equation $\frac{dy}{dx} = y^2 - \frac{y}{x} - \frac{1}{x^2}$, $x > 0$, $y(1) = 2$.

[Take $y = \frac{1}{x}$ is a trial solution and then substitute $y = \frac{1}{x} + u$.]

Answer

3. $y = \sin x + \frac{1}{C\cos x - \frac{1}{2}\sin x}$, where C is a constant

4. $y = \frac{2x}{C - x^2} + \frac{1}{x}$ (General solution) (Using $y(1) = 2$, $y = \frac{2x}{3 - x^2} + \frac{1}{x}$)

7.10 Mathematical modeling of engineering problems using first order equation

As we are observing that the natural phenomenon are changing everyday, these phenomenon are governed by fundamental laws of physics. Not limited to natural and technological changes, different social phenomenon may be population, finance, economic activities all are changing nature with time and position. As the derivative gives rate of change of one quantity (function), with respect to another quantity (variable), so it is natural to describe such rate of changes by differential equations.

In this section we shall deal with some of the fundamental processes that are modeled (described) by differential equations.

1. Mixing problem

In a mixing problem, a tank is filled with liquid to a specific level with a solution that contains a known amount of some soluble substance (say salt). The solution is made uniform by stirring and allowed to drain from the tank at some rate. At the same time a solution with known concentration of the soluble substance is added to the tank at a known rate. As the time progress, the concentration of the solution in the tank will change. We will determine the concentration of the solution in the tank at some specified point of time.

This kind of model can be used to solve different kind of problems such as discharge and filtration of pollutants in a river, injection and absorption of medicine in the bloodstream and migration of specifies into and out of an ecological system.

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We discuss some examples:

Example 45. A tank initially contains 4 kg of salt dissolved into 100 liter of water. Suppose that brine (salt solution) 2 kg of salt per litre is allowed to enter the tank at a rate of 5 litre/min and the uniform solution is drained from the tank at the same rate. Find the amount of salt in the tank after 10 minutes.

Solution

Let $y(t)$ be the amount of salt (in kgs) after t minutes. We note that:

$y(0) = 4$ and we are required to find $y(10)$.

We describe the problem by a differential equation whose solution is $y(t)$.

We recall that the rate of change of salt quantity is $\frac{dy}{dt}$ which is determined by the difference of the in and rate out.

Thus, the rate of change of salt in the solution in tank,

$\frac{dy}{dt}$ = rate in − rate out .... (i)

Now, rate in = (Concentration) (rate of flow in)

= (2 kg/litre) (5 litre/min)

$\therefore$ rate in = 10 kg/min.

Also, rate out = $\left(\frac{y(t)}{100}\text{ kg/liter}\right)(5\text{ kg/min}) = \frac{y(t)}{20}$ (kg/min)

Therefore from equation (i), $\frac{dy}{dt} = 10 - \frac{y}{20}$

$\therefore$ $\frac{dy}{dt} + \frac{y}{20} = 10$ ....(ii)

which is a first order linear differential equation. Also, we are given $y(0) = 4$.

So, the entire mixing problem can be expressed as initial value problem as

$\frac{dy}{dt} + \frac{y}{20} = 10, y(0) = 4$ ....(iii)

To solve it,

I.F. $= e^{\int \frac{1}{20} dt} = e^{\frac{t}{20}}$

On multiplying $\frac{dy}{dt} + \frac{y}{20} = 10$ by I.F. $= e^{\frac{t}{20}}$, we get

$\frac{dy}{dt} \cdot e^{\frac{t}{20}} + \frac{y}{20} \cdot e^{\frac{t}{20}} = 10\, e^{\frac{t}{20}}$

or, $\frac{d}{dt}\left(ye^{\frac{t}{20}}\right) = 10\, e^{\frac{t}{20}}$

or, $d\left(ye^{\frac{t}{20}}\right) = 10\, e^{\frac{t}{20}} dt$

Integrating, $e^{\frac{t}{20}} y = 10 \int e^{\frac{t}{20}} dt = 200\, e^{\frac{t}{20}} + C$

$\therefore$ $y(t) = 200 + C e^{\frac{-t}{20}}$

Using $y(0) = 4$, i.e., $y(t) = 4$ when $t = 0$. We get, ... (iv)

$4 = 200 + Ce^0$

$\therefore$ $C = -196$

$\therefore$ $y(t) = 200 - 196\, e^{\frac{-t}{20}}$ ....(v)

Which gives the amount of salt at any time t.

For $y(t)$ when $t = 10$

$y(10) = 200 - 196e^{-0.5} = 81.1$ kg

Example 46. A glucose solution is administered intravenously into the bloodstream at a constant rate r. As the glucose is added, it is converted into other substance and removed from the bloodstream at a rate that is proportional to the concentrate $C = C(t)$ of the glucose solution in the bloodstream and is given by, $\frac{dC}{dt} = r - KC$, where K is a positive constant.

a. Suppose that the concentrate at time $t = 0$ is C_0. Determine the concentrate at any time t by solving the above differential equation.

b. Assuming that $C_0 < \frac{r}{K}$, find $\lim_{t \to \infty} C(t)$ and interpret your answer.

Solution:

We are given the model equation of glucose concentration in the bloodstream

$\frac{dC}{dt} = r - KC$ ....(i)

or, $\frac{dC}{r - KC} = dt$

Integrating,

$\frac{\log(r - KC)}{-K} = t - \frac{\log a}{K}$, ($a$ is constant)

or, $\log(r - KC) = -Kt + \log a$

$\therefore$ $r - KC = ae^{-Kt}$

$C(t) = \left(\frac{r - ae^{-Kt}}{K}\right)$ ....(ii)

a. Given, $C(0) = C_0$

From (ii), we get

$C_0 = \frac{r - a}{K}$

$\therefore$ $KC_0 = r - a$

$\therefore$ $a = r - KC_0$

Substituting the value of a in (ii), we get

$C(t) = \frac{1}{K}\left[r - (r - KC_0)\, e^{-Kt}\right]$

$C(t) = \frac{r}{K} - \left(\frac{r - KC_0}{K}\right) e^{-Kt}$

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b. $\lim_{t\to\infty} C(t) = \lim_{t\to\infty}\left[\frac{r}{K} - \left(\frac{r - KC_0}{K}\right)e^{-Kt}\right]$

$= \frac{r}{K} - \left(\frac{r - KC_0}{K}\right)\lim_{t\to\infty} e^{-Kt}$

$= \frac{r}{K}$, as $\lim_{t\to\infty} e^{-Kt} = 0$.

$\therefore \lim_{t\to\infty} C(t) = \frac{r}{K}$.

Thus, the concentration over time approaches $\frac{r}{K}$ (constant) and is independent of C_0 which is shown in graph.

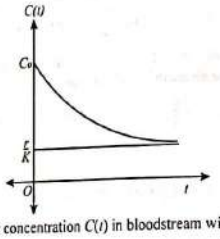

Fig: Showing sugar concentration $C(t)$ in bloodstream with respect to time t.

Example 47. A Jar with 2000 liter (L) of beer contains 4% alcohol (by Volume). Beer with 6% alcohol is pumped into the jar at a rate of 20 L/min and the mixture is pumped out at the same rate. What is the percentage of alcohol after one hour.

Solution:

Let $y(t)$ be the quantity of alcohol (in liters) in the jar at any instant t.

We are given,

$y(0)$ = 4% of 2000 L

$= \frac{4}{100} \times 2000\ L = 80\ L$

The rate of change of alcohol concentration in the Jar is given by the differential equation

$\frac{dy}{dt}$ = rate in − rate out .... (i)

Now,

Rate in = (6%) (20 L/m)

$= \frac{6}{100} \times 20(L/m) = \frac{6}{5}\ (L/m)$

Again,

rate out $= \left(\frac{y(t)}{2000}\right)$(20 L/min)

$= \frac{y(t)}{100}\ (L/m)$

$\therefore$ From (i), we have

$\frac{dy}{dt} = \frac{6}{5} - \frac{y}{100}$ .... (ii)

Solving (ii), by separation of variables

$\frac{dy}{dt} = \frac{120 - y}{1}$

or, $\frac{dy}{120 - y} = \frac{1}{100}\,dt$

Integrating,

$-\log(120 - y) = \frac{t}{100} - \log C$

or, $\log\left(\frac{120 - y}{C}\right) = \frac{-t}{100}$

or, $120 - y = C\,e^{\frac{-t}{100}}$

$\therefore\ y(t) = 120 - C\,e^{\frac{-t}{100}}$ .... (iii)

Using $y(0) = 80$ in (iii), we get

$80 = 120 - C$

$\therefore\ C = 40$.

$\therefore$ The solution (iii) becomes,

$y(t) = 120 - 40\,e^{\frac{-t}{100}}$ .... (iv)

For the quantity of alcohol in the solution in the jar after one hour (60 minute)

$y(60) = 120 - 40\left(e^{\frac{-60}{100}}\right)$

$= 120 - 40\,(e^{-0.6})$

$= 120 - 40\,(0.55)$

$= 120 - 21.95$

$= 120 - 22\ (21.95 \equiv 22) = 98$ liter.

$\therefore$ Percentage of alcohol concentration in the Jar $= \frac{98}{2000} \times 100\% = 4.9\%$.

Example 48. A 1500 gallon tank initially contains 600 gallons of water with 5 lbs of salt dissolved in it. Water enters a tank at the rate of 9 gal/hrs and the water entering the tank has a salt concentration of (1 + cos$t$) lbs/gal. If a well mixed solution leaves the tank at the rate of 6 gal/hr, how much salt is in the tank when it overflows?

Solution:

Let us assume that the instant the water enters the tank and somehow it disperses throughout the tank to give uniform concentration of the salt.

Now, we need to determine the concentration of the salt in the water existing in the tank.

Let $Q(t)$ be the amount of salt at time t.

Initially amount of water = 600 gallons with 5 lbs of salt dissolved in it

Also in every 1 hour, 9 gallons enters and 6 gallons flow away. So each hour 3 gallons add up.

$\therefore$ At any time t, there is $600 + 3t$ gallons of water.

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Hence, the initial value problem is

$$Q'(t) = 9\left(\frac{1}{3}(1+\cos t)\right) - 6\frac{Q(t)}{600+3t},\ Q(0) = 5$$

or, $Q'(t) = \frac{9}{5}(1+\cos t) - \frac{2Q(t)}{200+t},\ Q(0) = 5$

or, $Q'(t) + \frac{2Q(t)}{200+t} = \frac{9}{5}(1+\cos t)$

or, $\frac{dQ}{dt} + \frac{2Q(t)}{200+t} = \frac{9}{5}(1+\cos t)$ .... (i)

The equation (i) is linear in Q.

Hence, I.F. $= e^{\int \frac{2}{200+t}dt} = e^{2\log(200+t)} = (200+t)^2$

Multiplying both sides of (i) by I.F. $= (200+t)^2$, we get

$\frac{d}{dt}((200+t)^2 \cdot Q) = \frac{9}{5}(200+t)^2 \cdot (1+\cos t)$.

Integrating, the general solution is

$$(200+t)^2 Q = \int \frac{9}{5}(200+t)^2(1+\cos t)\,dt$$

or, $(200+t)^2 Q = \frac{9}{5}\left(\frac{1}{3}(200+t)^3 + (200+t)^2 \sin t + 2(200+t)\cos t - 2\sin t\right) + C$

$= \frac{9}{5}\left(\frac{1}{3}(200+t) + \sin t + \frac{2\cos t}{200+t} - \frac{2\sin t}{(200+t)^2}\right) + \frac{C}{(200+t)^2}$ .... (ii)

Now, using the initial value condition, put $t = 0, Q = 5$

$\therefore\ 5 = Q(0) = \frac{9}{5}\left(\frac{1}{3}(200) + \frac{2}{200}\right) + \frac{C}{(200)^2}$

or, $5 = \frac{9}{5}\left(\frac{200}{3} + \frac{1}{100}\right) + \frac{C}{40000}$

or, $C = -4600720$

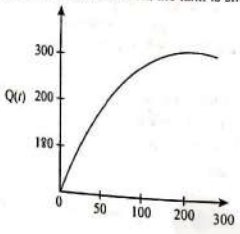
$\therefore$ The amount of salt in the tank at time t is

$$Q(t) = \frac{9}{5}\left(\frac{(200+t)}{3} + \sin t + \frac{2\cos t}{200+t} - \frac{2\sin t}{(2000+t)^2}\right) - \frac{4600720}{(200+t)^2}$$

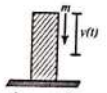
Now, the tank will overflow at $t = 300$ hrs.

The amount of salt in tank at that time is $Q(300) = 279.797$ lbs.

The graph of above solution before it overflows from the tank is shown in following figure.



2. Motion of free fall under gravity with air resistance



Suppose a body of mass m is falling freely from some height from earth surface. Let $y(t)$ be the vertical distance travelled by the body at any time t.

The mass m experiences two kind of forces: gravitational pull by earth (mg) and the air resistance (F_R) which is a drag force against the gravitational pull. So the net force experienced by the body of mass m is ($mg - F_R$), air resistance taken negative to indicate it is opposite to gravitational pull.

By Newton's second law of motion ($F = ma$), we have

$$mg - F_R = ma = m\frac{d^2y}{dt^2} = m\frac{d}{dt}\left(\frac{dy}{dt}\right)$$

$\therefore\ mg - F_R = m\frac{dv}{dt}$, where $v = \frac{dy}{dt}$ .... (i)

The drag force F_R is proportional to velocity i.e., $F_R = -Cv$, for some positive constant C depending upon shape of m and properties of air.

$\therefore\ mg + Cv = m\frac{dv}{dt}$

or, $\frac{dv}{dt} = \frac{C}{m}v + g$

$\therefore\ \frac{dv}{dt} - \frac{C}{m}v = g$

which is first order linear differential equation with $v = v(t)$ and known velocity $v(0)$ at $t = 0$.

Thus the free fall motion is described by the initial value problem

$$\frac{dv}{dt} + \frac{c}{m}v = g,\ v(0) = v_0$$

which is the differential equation of free fall and is a linear equation.

By using velocity function $v = v(t)$, the position at any instant $y = y(t)$ can be obtained by solving initial value problem.

$$\frac{dy}{dt} = v(t), \text{ with } y(0) = 0.$$

3. Exponential growth and decay model

A quantity $y = y(t)$ is said to have an exponential growth model if it increases at a rate which is proportional to the amount of the quantity present and it is said to have an exponential decay model if it decreases at a rate which is proportional to the amount of the quantity present.

Thus, for an exponential growth model the quantity $y = y(t)$ satisfies an equation of the form

$$\frac{dy}{dt} = ky\ (k > 0) \quad \text{.... (i)}$$

and for an exponential decay model, the quantity $y(t)$ satisfies an equation of the form,

$$\frac{dy}{dt} = -ky\ (k > 0) \quad \text{.... (ii)}$$

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where the constant k in (i) and (ii) is called growth and decay constant respectively.

Equation (i) and (ii) can also be written as

$$\frac{dy}{dt} - ky = 0 \text{ and } \frac{dy}{dt} + ky = 0$$

which are first order homogeneous linear equations and can be solved by separation of variables method.

Population growth model

The growth of population (of people, plants, bacteria, fruit, flies, etc.) are not constrained by environmental limitations. The growth of population is proportional to the current size of population.

Let $y = y(t)$ be the population at any instant t of certain species in a closed system. At each point of time the rate of growth of population is proportional to current population which is described by the equation

$$\frac{dy}{dt} = ky, k > 0$$

If $y(0) = y_0$ then the initial value problem

$$\frac{dy}{dt} = ky, k > 0, y(0) = y_0$$

pletely describes the population at any time t.

Radioactive decay

The decomposition of radioactive substance is proportional to the quantity $y(t)$ present at that time.

Thus the rate of decay of radioactive substance is given by

$$\frac{dy}{dt} = -ky, k > 0.$$

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The value of proportionality constant k is different for different substances and are determined by experiments.

If $y(0) = y_0$

then radioactive decay of a substance can be described by initial value problem as

$$\frac{dy}{dt} = ky, k > 0, y(0) = y_0$$

which is a linear homogeneous equation and can be solved by separation of variables.

Remark: For both exponential growth (population growth) and ential decay (radioactive decay). We can write,

$$k = \frac{dy/dt}{y}, (k > 0 \text{ for growth}, k < 0 \text{ for decay}).$$

The constant k, the growth (decay) rate and given by a fractio of change and the entire population (or mass of radioactive substance) known as growth de ate and is constant over time.

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Example 49. United Nations Organizations (UNO) report the world population in 1998 was approximately 5.9 billion and growing rate of about 13.33% per year. Formulate a differential equation for world population and estimate the world population at the end of the year 2025. When will the population be double of 1998?

Solution:

Let $y(t)$ be the total population (in millions) at any time t (in years).

Then the rate of growth of population is given by the initial value problem

$\frac{dy}{dt} = ky, y(t = 0) = 5.9$, taking 1998 as $t = 0$.

We need to find $y(t = 27)$ from 1998 to 2025.

Solving $\frac{dy}{dt} = ky$

or, $\frac{dy}{y} = kdt$

or, $\log y = kt + \log c$

$\therefore$ $y = ce^{kt}$, where c is integrating constant.

We have, $y(t) = 5.9$ when $t = 0$

i.e., $5.9 = c$

$\therefore$ The solution is $y(t) = 5.9\ e^{kt}$ ....(i)

Now, we are given $k = 1.33\% = 0.0133$ and for $t = 27$ years, we have

$$\begin{aligned} y(27) &= (5.9)e^{(0.0133)27} \\ &= (5.9)(e^{0.3591}) \\ &= (5.9)\ (1.432) \\ &= 8.45 \end{aligned}$$

$\therefore$ $y(27) = 8.45$ billions i.e., in 2025 the world population will be 8.45 billions.

For the time for the population to be doubled of the population in 1998, from (i)

$y(t) = 5.9\ e^{kt}$

$\therefore$ $2y(t) = 2\ (5.9) = 5.9e^{kt}$ as $y(t)$ is 5.9 in 1998.

or, $2 = e^{kt}$

$\therefore$ $kt = \log 2$

or, $$t = \frac{\log 2}{k}$$

$$= \frac{0.6931}{0.0133}$$

$$= 52.1 \text{ years}$$

Thus, the world population will be doubled by the year (1998 + 52) = 2050.

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4. Modeling of heating system using Newton's law of cooling

Newton's law of cooling says that "The time rate of change of temperature T of a body is proportional to the difference between T and the temperature of the surrounding medium (atmosphere)."

If $T(t)$ be the temperature of a body and T_A temperature of atmosphere, then

$$\frac{dT(t)}{dt} = k(T(t) - T_A)$$

where, k is proportionality constant, depends on the nature of body.

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Example 50. (Heating of a building): Suppose that in winter the daytime temperature in a certain office building is maintained at 21°C. The heating is shut off at 10 PM and turned on again at 6 AM. The outside temperature is 7°C. Determine the temperature of the building before turned on at 6 AM. Given that the temperature of building at 2 AM was 18°C.

Solution:

Let $T(t)$ be the temperature of the building at any time t. The atmospheric temperature outside the building is $T_A = 7°C$.

Using Newton's law of cooling,

$$\frac{d(T(t))}{dt} = k(T(t) - T_A), \text{ where } T = T(t) \quad \ldots.(i)$$

or, $$\frac{dT}{T - T_A} = kdt$$

Integrating both sides,

$$\log(T - T_A) = kt + \log C$$

or, $$T - T_A = Ce^{kt}$$

$\therefore$ $$T(t) = 7 + Ce^{kt} \quad \ldots.(ii)$$

We are given the initial temperature, when the heating is turned off at 10 PM, $T(0) = 21°C$.

From (ii), we have

$$21 = 7 + Ce^0$$

$\therefore$ $C = 14$

Hence, solution (ii) becomes,

$$T(t) = 7 + 14e^{kt} \quad \ldots.(iii)$$

To find k, we are given $T(t) = 18$ when $t = 4$ (from $t = 0$ at 10 PM, we have $t = 4$ at 2 AM)

$\therefore$ $$18 = 7 + 14e^{4k}$$

or, $$e^{4k} = \frac{11}{14}$$

$\therefore$ $$4k = \log\left(\frac{11}{14}\right)$$

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$\therefore$ $$k = \frac{1}{4}[\log(11) - \log(14)]$$

$$= \frac{1}{4}[2.398 - 2.640]$$

$$= \frac{1}{4}(-0.242) = -0.0605$$

$\therefore$ From equation (iii)

$$T(t) = 7 + 14\, e^{(-0.0605)t} \quad \ldots.(iv)$$

For temperature at 6 AM, put $t = 8$ (from 10 PM) in (iv), we get

$$T(8) = 7 + 14\, e^{(-0.0605)8}$$

$$= 7 + 14(e^{-0.484})$$

$$= 7 + 14\,(0.62)$$

$$= 7 + 8.68$$

$$T(8) = 15.68°C$$

Thus, the temperature of the building at 6 AM, $T(8) = 15.68°C$.

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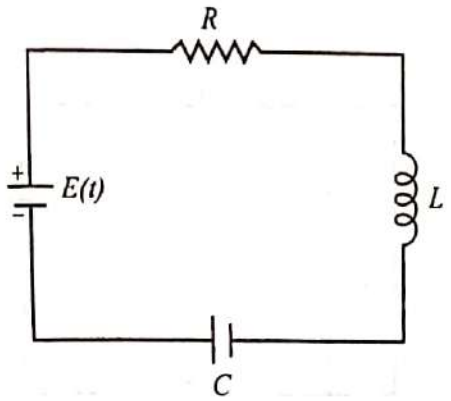
5. Modeling of electric circuits

An electric circuit is a series circuit in which we have a source of electric energy (electromotive force) such as a generator or a battery and a resistor, which uses energy, for example, an electric light bulb. If we close the switch, a current I will flow the resistor and this will cause a voltage drop, that is, the electric potential at two ends of the resistor will be different; this potential difference or voltage drop can be measured by voltmeter. Experiments show that the following laws hold:



A general RLC circuit

1. The voltage drop across a resistor is proportional to the instantaneous current I, say $E_R = RI$ (Ohm's law), where, the constant of proportionality R is called resistance of the resistor, the current I is measured in amperes, the resistance R in Ohm's and the voltage E_R in volts.

2. An inductor L in a circuit opposes a change in current i.e., inertial effect in electricity.

   The voltage drop E_L across an inductor is proportional to the instantaneous time rate of change of the current I, i.e.,

   $$E_L = L\frac{dI}{dt}$$

   where constant of proportionality L is called the inductance of the inductor and is measured in Henrys; time in seconds.

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3. A capacitor is an element that store electric energy.

   The voltage drop E_C across a capacitor is proportional to the instantaneous electric charge Q on the capacitor, i.e.,

   $$E_C = \frac{1}{C}Q$$

   where C is called capacitance and is measured in Farads, the charge Q is measured in Coulombs.

Note that, the current (I) is defined as rate of flow of charge (Q) per unit time, i.e., $I(t) = \frac{dQ}{dt}$.

Also note that

$$E_c = \frac{1}{C}\int_{t_0}^{t} I(x)\,dx.$$

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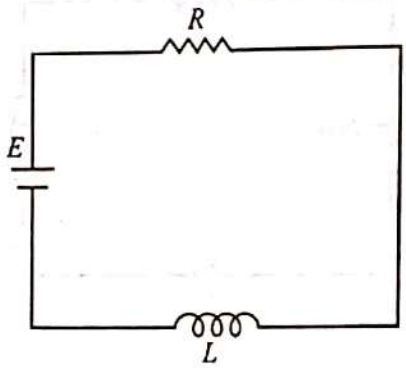
4. Kirchoff's voltage law: The algebraic sum of all the instantaneous voltage drops around any closed loop is zero. Equivalently, the voltage impressed on a closed loop is equal to the sum of the voltage drops in the rest of the loop.

Modeling of RL-Circuit

Consider the *RL*-circuit as shown in figure and derive a differential equation for the given figure.



This circuit contains three components electromotive force $E(t)$, Resistance R and inductor L, so by using Kirchoff's law, the equation for the given circuit is

$$E(t) = L\frac{dI}{dt} + RI$$

$$\therefore \quad \frac{dI}{dt} + \frac{R}{L} I = \frac{E}{L},\ E = E(t) \qquad \ldots.(i)$$

which is linear differential equation.

Its integrating factor $I.F. = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L}t}$

Multiplying (i) by I.F. we get

$$\frac{d}{dt}\left(I \times e^{\frac{R}{L}t}\right) = \frac{E}{L} e^{\frac{R}{L}t}$$

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Integrating, we get

$$Ie^{\frac{R}{L}t} = \int \frac{E}{L} e^{\frac{R}{L}t} dt + C$$

Thus the solution of (i) is given by

$$I(t) = e^{-\frac{R}{L}t}\left[\int e^{\frac{R}{L}t} \frac{E}{L} dt + C\right] \qquad \ldots.(ii)$$

The result of (ii) depends upon the form of electromotive force $E = E(t)$.

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Case I: If $E = E(t) = E_0$, (constant electromotive force). The solution (ii) becomes

$$I(t) = \frac{E_0}{L} e^{-\frac{R}{L}t}\left[\int e^{\frac{R}{L}t}\, dt + C\right]$$

$$= \frac{E_0}{L} e^{-\frac{R}{L}t}\left[\frac{L}{R}\, e^{\frac{R}{L}t} + C\right]$$

$$\therefore \quad I(t) = \frac{E_0}{R} + C e^{-\frac{R}{L}t} \qquad \ldots.(iii)$$

Case II: If $E = E(t)$, varying electromotive force

The solution (ii) becomes, $I(t) = e^{-\frac{R}{L}t}\left[\int e^{\frac{R}{L}t}\frac{E(t)}{L}\, dt + C\right]$

On integrating, we get the required solution.

For example, if $E(t) = E_0 \sin\omega t$, then above solution becomes

$$I(t) = e^{-\frac{R}{L}t}\left[\int e^{\frac{R}{L}t}\frac{E_0}{L}\sin\omega t\, dt + C\right]$$

$$= e^{-\frac{R}{L}t}\left[\frac{E_0}{L}\frac{e^{\frac{R}{L}t}}{\left(\frac{R}{L}\right)^2 + \omega^2}\left(\frac{R}{L}\sin\omega t - \omega\cos\omega t\right) + C\right]$$

$$\therefore \quad I(t) = \frac{E_0}{R^2 + \omega^2 L^2}(R\sin\omega t - \omega L\cos\omega t) + C e^{-\frac{R}{L}t}$$

which is the required general solution of the differential equation $\frac{dI}{dt} + \frac{R}{L}I = \frac{E}{L}$.

Modeling involving two quantities that change with respect to same variable

When we have physical system where two quantities change with respect to same variable, we model the system by a pair of differential equations. We illustrate this by the rainfall phenomena.

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Example 51. When a raindrop fall the size and velocity both increase with time. The initial velocity is always zero and initial mass $m(0) = m_0$. The growth rate of mass $m(t)$ is proportional to the current mass $m(t)$. Find the velocity $v(t)$ and mass $m(t)$ at any instant of time. Also find its terminal velocity when the drop hits the ground. (Note that, terminal velocity is given by $\lim_{t \to \infty} v(t)$)

Solution:

Let $m = m(t)$ and $v = v(t)$ be the mass and velocity of the raindrop at any instant of time t.
Since the growth rate of mass $m(t)$ is proportional to the existing mass,

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.... (i)

$\therefore \quad \frac{dm}{dt} = km$, for $k > 0$

or, $\frac{dm}{m} = kdt$

Integrating, we get

$\log m = kt - \log C$ .... (ii)

$\therefore \quad m = Ce^{kt}$

But we are given $m(0) = m_0$. Then (ii) gives

$m_0 = m(0) = Ce^0$

$\therefore \quad C = m_0$

From (ii) the mass at instant t is given by

$m(t) = m_0 e^{kt}$ .... (iii)

Again, using Newton's second law of motion in terms of momentum, is given by

$\frac{d}{dt}(mv) = mg$ .... (iv)

where, mv is the momentum

i.e., "the rate of change of momentum with respect to time is equal is equal to the force" ($w = mg$ is force)

$\therefore \quad m\frac{dv}{dt} + \frac{dm}{dt}v = mg$

Using (i), $\frac{dm}{dt} = km$

$m\frac{dv}{dt} + kmv = mg$

$\therefore \quad \frac{dv}{dt} = (g - kv)$ .... (v)

Solving (v), $\frac{dv}{g - kv} = dt$

Integrating, $\frac{\log(g - kv)}{-k} = t - \frac{\log C_1}{k}$

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or, $\log(g - kv) - \log C_1 = -kt$

or, $\frac{g - kv}{C_1} = e^{-kt}$

or, $g - kv = C_1 e^{-kt}$

or, $v = \frac{1}{k}(g - C_1 e^{-kt})$

$\therefore \quad$ Velocity $v = v(t) = \frac{1}{k}(g - C_1 e^{-kt})$ .... (vi)

Using $v(0) = 0$ in (vi), we get

$0 = \frac{1}{k}(g - C_1)$

$\therefore \quad C_1 = g$

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From (vi)

$$v(t)=\frac{g}{k}(1-e^{kt}) \quad \ldots\ldots \text{(vii)}$$

For terminal velocity $\lim_{t\to\infty} v(t)$

$$\lim_{t\to\infty} v(t) = \frac{g}{k}\lim_{t\to\infty}(1-e^{kt})$$

$$= \frac{g}{k}(1-0), \text{ as } \lim_{t\to\infty} e^{kt} = 0$$

$\therefore$ Terminal velocity $\lim_{t\to\infty} v(t)=\frac{g}{k}$.

Thus the complete solution of the rainfall problem are

Mass: $m(t) = m_0 e^{kt}$

Velocity: $v(t)=\frac{g}{k}(1-e^{-kt})$

Terminal velocity: $\lim_{t\to\infty} v(t)=\frac{g}{k}$.

Exercise 7.5

1. A tank initially contains 40 kg of salt dissolved into 200 liter of water. A solution of 2 kg of salt per liter is allowed to enter the tank at the rate of 5 liter per minute and uniform solution is drained from the tank at the same rate. Find the amount of salt at any time. Also determine the salt in tank in 15 minutes.
2. Suppose that you turn off the heater in your home at night 2 hours before you go to the bed. If temperature of the room is 66°F when you turn off the heater. The temperature of the room fall to 63°F at the time you go to bed. What will the temperature of the room in the morning after 8 hours you go to bed ? Assume that the outside atmospheric temperature is 32°F
3. The growth rate of a culture of bacteria is proportional to the number of bacteria present. After one day it is (1.5) times of original number. Find after how many days it will be (a) double (b) triple
4. A metal bar whose temperature is 20°C is placed in boiling water. How long does it take to heat the bar practically 100°C (say 99. 99° C) if the temperature of the bar after one minute is 51. 5° C.

Answer

1. 52. 6 kg 2. 53. 4° F 3. (a)1. 7 days, (b) 2.7 days 4. 13.4 min

Multiple Choice Questions

1. A differential equation is considered to be ordinary if it has
   a. one dependent variable b. more than one dependant variable
   c. one independent variable d. more than one independent variable
2. The number of arbitrary constants in general and particular solution of differential equation of second order are respectively
   a. 1, 4 b. 0, 2
   c. 2, 0 d. 4, 0

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3. The order and degree of differential equation $\sqrt{\left(\frac{dy}{dx}\right)^4} + 4 = \left(\frac{d^2y}{dx^2}\right)^2$ are respectively

a. 2, 6 b. 2, 3
c. 1, 4 d. 2, 4

4. The order of a differential equation whose general solution is $y = A \sin x + B \cos x$ is

a. 4 b. 2
c. 0 d. 3

5. Which of the following equation is linear?

a. $y' + xy^2 = 1$ b. $x^2y' + y^2 = e^x$
c. $y' + 3y = xy^2$ d. none

6. The solution of the differential equation $2x\frac{dy}{dx} - y = 0, y(1) = 2$ represents

a. straight line b. parabola
c. circle d. ellipse

7. $Pdx + x \sin y\, dy = 0$ is exact, then P is

a. $\sin y + \cos y$ b. $-\sin y$
c. $\cos y$ d. $x^2 - \cos y$

8. The solution of the differential equation $x\frac{dy}{dx} + y = 0$ is

a. $e^{xy} = c$ b. $y = cx$
c. $x = cy$ d. $e^x y = c$

9. The equation of curve passing through (2, 1) with slope $\frac{x^2 + y^2}{2xy}$ of tangent at (x,y) is

a. $2(x^2 - y^2) = 3x$ b. $2(x^2 - y^2) = 6y$
c. $x(x^2 - y^2) = 6$ d. $x(x^2 + y^2) = 10$

10. The solution of differential equations $xdy - ydx = 0$ represents

a. rectangular hyperbola b. lines passing through the origin
c. parabolas whose vertex in the origin d. circle with origin as centre

11. The curve $y = f(x)$ passes through (1, 2) and satisfies the differential equation $y(1 + xy)dx - xdy = 0$ then

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a. $f(x) = \frac{2x}{2 - x^2}$ b. $f(x) = \frac{x + 1}{x^2 + 1}$
c. $f(x) = \frac{x - 1}{4 - x^2}$ d. $f(x) = \frac{4x}{1 - 2x^2}$

12. The differential equation of the family of circles $x^2 + y^2 = a^2$ is

a. $xy\frac{dy}{dx} + y = 0$ b. $x\frac{dy}{dx} + y = 0$
c. $x + y\frac{dy}{dx} = 0$ d. none

13. The solution of differential equation $\frac{dx}{dy} + \frac{x}{3y} = 0$ with $y(1) = 1$ is given by

a. $x^3y = 1$ b. $xy^3 = 1$
c. $x^3 + y^3 = 2$ d. $xy = 1$

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14. The solution of the differential equation $2x\frac{dx}{dy} - y = 0$ represents

a. parabola
b. ellipse
c. rectangular hyperbola
d. hyperbola

15. The differential equation of family of lines passing through origin is

a. $x\frac{dy}{dx} + y = 0$
b. $\frac{dy}{dx} = y$
c. $\frac{dy}{dx} = x$
d. $x\frac{dy}{dx} - y = 0$

16. The differential equation corresponding to the primitive $y = cx^2$ is given by

a. $\frac{dy}{dx} = 2cx$
b. $\frac{d^2y}{dx^2} = 2c$
c. $\frac{dy}{dx} = 2\frac{y}{x}$
d. $\frac{d^3y}{dx^3} = 0$

17. The integral curves corresponding to the differential equation $\frac{dy}{dx} = x$ is a family of

a. circles
b. parabolas
c. hyperbolas
d. straight lines passing through the origin.

18. The integral curves corresponding to the differential equation $xdx + ydy = 0$, is a set of

a. parabolas
b. ellipses
c. circles
d. hyperbolas

19. What type of equation is $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x alone or constant?

a. linear equation of first order
b. linear equation of second order
c. Bernoulli's equation
d. linear equation of the second degree

20. The differential equation of all straight lines passing through origin is

a. $y\,dx = x\,dy$
b. $y = mx$
c. $x\,dx + y\,dy = 0$
d. $x\,dy + y\,dx = 0$

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21. Integrating factor of the differential equation $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \sin x$ is

a. $(1 + x^2)$
b. $2x$
c. $\frac{1}{1+x^2}$
d. $\frac{1}{1+x}$

22. Which of the following is a homogeneous differential equation?

a. $(3x - 5)dx = (4y - 1)dy$
b. $xy\,dx - (x^3 + y^3)\,dy = 0$
c. $y^2dx + (x^2 - xy - y^2)\,dy = 0$
d. $(x^2 + y)dx = (y^2 + x)\,dy$

Answer

1. c.	2. c.	3. d.	4. b.	5. d.	6. b.	7. d.
8. a.	9. a.	10. b.	11. a.	12. c.	13. d.	14. d.
15. d.	16. c.	17. b.	18. c.	19. a.	20. a.	21. a.
22. c.						

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Unit 8

Second Order Ordinary Differential Equation

Pre-requisite knowledge

Before starting this unit, students are expected to have fundamental concepts and evaluation skills on

- the techniques of solving the first order and first degree equations.

Expected Learning outcomes

After completion of this unit, student will develop sufficient knowledge and evaluation skills on

- the solution of second order homogeneous ODE with constant and variable coefficients, Euler-Cauchy equation.
- existence and uniqueness of solutions, Wronskian and general solutions for solving ODE.
- non-homogeneous second order ODE and solution by undetermined variation of parameters.
- engineering applications of second order differential equations (vibrating mass-spring, RLC circuits).

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8.0 Introduction

Due to the wide range of applications in engineering and the natural sciences, second order differential equations have received the great attention of researchers and scholars. In the previous chapter we studied about the types and the methods of solution of the first order differential equations. This chapter is the continuation of the differential equations but for the second order. Here, we will study homogeneous and non-homogeneous linear differential equation of second order and also the various methods to solve such equations using auxiliary equation, method of undetermined coefficients and variation of parameters. These problem-solving methods have been utilized to resolve issues with mechanical and electrical vibrations, particularly in the context of spring-mass problems and RLC circuit.

We first begin with definition of second order equation.

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8.1 Second order linear differential equation

A differential equation of the form

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x) \quad \ldots . (i)$$

or, $D^2y + P(x)Dy + Q(x)y = R(x)$, where $D = \frac{d}{dx}$ and $D^2 = \frac{d^2}{dx^2}$

where, $P(x)$, $Q(x)$ and $R(x)$ are functions of x or constants is called general second order differential equation.

We have following cases depending on $P(x)$, $Q(x)$ and $R(x)$:

1. If $P(x)$, $Q(x)$ are constant and $R(x) = 0$, then (i) is written as

   $$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$$

   which is called second order linear homogeneous equation with constant coefficients. For example, $y'' + 2y' + 3y = 0$.

2. If $P(x)$, $Q(x)$ are constants but $R(x)$ is a function of x, then (i) can be written as

   $$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R(x)$$

   which is called second order linear non-homogeneous equation with constant coefficients. For example, $y'' + 4y = e^{-x}\sin x$.

3. If $P(x)$, $Q(x)$ are functions of x and $R(x) = 0$, then (i) can be written as

   $$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

   which is called second order linear homogeneous equation with variable coefficients. For example, $y'' - 2x^2y' + 6xy = 0$.

4. If $P(x)$, $Q(x)$ are functions of x and $R(x) \neq 0$, then (i) can be written as

   $$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x)$$

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   which is called second order linear non- homogeneous equation with variable coefficients. For example, $y'' + x^2y' + xy = e^x \sin x$.

Note: The equation $x^2 y''y + 2yy' = 0$ is a non-linear second order differential equation.

Solution of second order differential equation

A solution of a second order differential equation (linear or non-linear) is a function $y = y(x)$ such that $y' = y'(x)$ and $y'' = y''(x)$ satisfy the equation, i.e., the differential equation becomes an identity when $y = y(x)$, $y' = y'(x)$ and $y'' = y''(x)$ are substituted in it.

General solution

If $y = y_1(x)$ and $y = y_2(x)$ are two solutions that are linearly independent (i.e., one solution is not proportional to other) then $y = C_1y_1(x) + C_2y_2(x)$ is the **general solution** for the second order homogeneous equation, by superposition principle (linearity principle).

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The two independent solutions $y = y_1(x)$, $y = y_2(x)$ that constitute the general solution $y = C_1y_1(x) + C_2y_2(x)$ of second order homogeneous differential equation form basis for the general solution, i.e., the set $\{y_1(x), y_2(x)\}$, where y_1 and y_2 are linearly independent is known as basis for the solution.

All linear differential equations with constant coefficients have solutions that are linear combinations of exponential and polynomial functions.

8.1.1 Solution method for second order linear homogeneous equation with constant coefficients

Consider a differential equation of second order $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$

where, P and Q are constants, which can be written as

$$(D^2 + PD + Q)y = 0 \quad \ldots\ldots \text{(i)}$$

As seen in the case of first order linear homogeneous differential equation, it is natural to expect solution in the exponential function.

Suppose $y = e^{mx}$ be a solution of the equation (i), then

$$D^2(e^{mx}) + PD(e^{mx}) + Q(e^{mx}) = 0$$

or, $m^2e^{mx} + Pme^{mx} + Qe^{mx} = 0$

or, $(m^2 + Pm + Q)e^{mx} = 0$

Since, $e^{mx} \neq 0$, we have

$$m^2 + Pm + Q = 0 \quad \ldots\ldots \text{(ii)}$$

Equation (ii) is called the **auxiliary equation** of (i) written as A.E. in short and is simply obtained by replacing D by m in $(D^2 + PD + Q)$ equation (i) and equating to zero. The equation (ii) is quadratic in m, so it gives two values of m. Let m_1 and m_2 be the two roots of equation (ii), then $y = e^{m_1x}$ and $y = e^{m_2x}$ are the solutions of (i). Based on the roots of the auxiliary equation, we have following cases:

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Case I : Both m_1 and m_2 are real and distinct

In this case, the solution of (i) in combined form can be written as

$$y = c_1e^{m_1x} + c_2e^{m_2x} \quad \ldots\ldots \text{(iii)}$$

Case II : Both m_1 and m_2 are real and equal

When the two roots of auxiliary equation are equal, say m,

then $m_1 = m_2 = m$. Then the solution of (i) becomes

$$y = c_1e^{mx} + c_2e^{mx} = (c_1 + c_2)e^{mx} = ce^{mx} \quad \ldots\ldots \text{(iv)}$$

Since the number of arbitrary constants in the general solution of a differential equation is equal to the order of the differential equation. The solution of the second order differential equation (i) must involve two arbitrary constants. Hence, the solution $y = ce^{mx}$ can not be the general solution of differential equation (i) because it contains only one arbitrary constant. So we shall deduce the correct general solution as follows:

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Since both roots of the auxiliary equation (ii) is m. So equation (i) can be written as

$(D - m)(D - m)\, y = 0$ .... (v)

Let us put $(D - m)\, y = v$.

Hence equation (v) will become

$(D - m)\, v = 0$.

or, $Dv = mv$

or, $\frac{dv}{dx} = mv$

$\therefore \quad \frac{dv}{v} = m dx$

Integrating, $\log v = mx + \log c_1$

or, $\log\left(\frac{v}{c_1}\right) = mx$

or, $\frac{v}{c_1} = e^{mx}$

or, $v = c_1 e^{mx}$

But $v = (D - m)\, y$.

$\therefore \quad (D - m)\, y = c_1 e^{mx}$

$\frac{y}{dx} - my = c_1 e^{mx}$ .... (vi)

The equation (vi) is linear equation of first order.

So, I.F. $= e^{\int P dx} = e^{\int -m dx} = e^{-mx}$

Multiplying both sides of (vi) by e^{-mx}, we get

$\frac{d}{dx}(y \,.\, e^{-mx}) = c_1$

or, $d(y \,.\, e^{-mx}) = c_1 dx$

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Integrating, $ye^{-mx} = \int c_1 dx = c_1 x + c_2$

$\therefore \quad y = (c_1 x + c_2)\, e^{mx}$, which is the required general solution.

Case III : If the roots are imaginary

The auxiliary equation has complex roots when its discriminant is negative. So, the roots will be $(\alpha + i\beta)$ and $(\alpha - i\beta)$ in which α can be zero but β cannot be zero. So the general solution in this case is

$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$

$y = e^{\alpha x}(c_1 e^{i\beta x} + c_2 e^{i\beta x})$

Using Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, we get

$y = e^{\alpha x}[c_1(\cos\beta x + i\sin\beta x) + c_2(\cos\beta x - i\sin\beta x)]$

$= (c_1 + c_2)\, e^{\alpha x} \cos\beta x + i(c_1 - c_2)\, e^{\alpha x} \sin\beta x$

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$\therefore y = Ae^{\alpha x}\cos\beta x + Be^{\alpha x}\sin\beta x$, where $A = c_1 + c_2$, $B = i(c_1 - c_2)$

$= e^{\alpha x}(A\cos\beta x + B\sin\beta x)$.

We now discuss some examples.

Example 1. Solve: $y'' - 7y' + 12y = 0$.

Solution:

The given differential equation is $y'' - 7y' + 12y = 0$ .... (i)

It's auxiliary equation is $m^2 - 7m + 12 = 0$

or, $m^2 - 3m - 4m + 12 = 0$

or, $(m-3)(m-4) = 0$

$\therefore m = 3, 4$.

So the general solution is $y = c_1 e^{3x} + c_2 e^{4x}$.

Example 2. Solve: $(D+3)^2 y = 0$.

Solution:

The given differential equation is $(D+3)^2 y = 0$ .... (i)

Then, its auxiliary equation is $(m+3)^2 = 0$

or, $(m+3)(m+3) = 0$

$\therefore m = -3, -3$.

So the roots are repeated. Hence, the general solution is $y = e^{-3x}(c_1 + c_2 x)$

Example 3. Solve: $y'' + 6y' + 25y = 0$.

Solution:

The given differential equation is $y'' + 6y' + 25y = 0$ .... (i)

It's auxiliary equation is $m^2 + 6m + 25 = 0$

$\therefore m = \dfrac{-6 \pm \sqrt{36 - 4.1.25}}{2.1} = \dfrac{-6 \pm \sqrt{-64}}{2} = \dfrac{-6 \pm 8i}{2} = -3 \pm i4.$

So, the general solution is $y = e^{-3x}(A\cos 4x + B\sin 4x)$.

Example 4. Solve: $6y'' - 5y' + y = 0$, $y(0) = 4$, $y'(0) = 0$.

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Solution:

The given differential equation is $6y'' - 5y' + y = 0$ .... (i)

It's auxiliary equation is $6m^2 - 5m + 1 = 0$

or, $6m^2 - 3m - 2m + 1 = 0$

or, $3m(2m-1) - 1(2m-1) = 0$

or, $(3m-1)(2m-1) = 0$

$\therefore m = \dfrac{1}{2}, \dfrac{1}{3}$

So, the general solution is $y = c_1 e^{\frac{1}{3}x} + c_2 e^{\frac{1}{2}x}$ .... (ii)

Using $y(0) = 4$ in (ii), $4 = c_1 + c_2$ .... (iii)

Now differentiating (ii), we have $y' = \dfrac{1}{3}c_1 e^{\frac{1}{3}x} + \dfrac{1}{2}c_2 e^{\frac{1}{2}x}$

Using $y'(0) = 0$, we have

$0 = \dfrac{1}{3}c_1 + \dfrac{1}{2}c_2$

$\therefore 2c_1 + 3c_2 = 0$ .... (iv)

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Solving (iii) and (iv), we get $c_1 = 12$ and $c_2 = -8$.

Substituting the value of c_1 and c_2 in (ii), we have

$y = 12e^{\frac{1}{3}x} - 8e^{\frac{1}{2}x}$ which is the required solution.

Example 5. Solve : $y'' - 4y' + 5y = 0$ given that $y(0) = 1, y'(0) = 2$.

Solution:

The given differential equation is $y'' - 4y' + 5y = 0$ .... (i)

Then, its auxiliary equation is $m^2 - 4m + 5 = 0$

$\therefore \quad m = \frac{-(-4) \pm \sqrt{16 - 4.1.5}}{2.1} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i$

So, the general solution is $y = e^{2x}(A \cos x + B \sin x)$ .... (ii)

If $x = 0, y = 1$, then $1 = e^0 (A \cos 0 + B \sin 0) \Rightarrow A = 1$ .... (iii)

Differentiating (ii) with respect to x, we get

$\frac{dy}{dx} = 2e^{2x}(A \cos x + B \sin x) + e^{2x}(-A \sin x + B \cos x)$

If $x = 0, \frac{dy}{dx} = 2$, then $2 = 2e^0 (A \cos 0 + B \sin 0) + e^0 (-A \sin 0 + B \cos 0)$

or, $2 = 2(A + 0) + 1(-0 + B)$

or, $2 = 2A + B$ .... (iv)

But from (iii), $A = 1$. So equation (iv) gives

$2 = 2.1 + B$ i.e., $B = 0$

$A = 1, B = 0$.

Substituting the values of A and B in equation (ii), we get

$y = e^{2x}(1. \cos x + 0. \sin x)$

or, $y = e^{2x}(\cos x + 0)$

$\therefore \quad y = e^{2x} \cos x$, which is the required solution.

Example 6. Solve : $\frac{d^2x}{dt^2} + \mu x = 0, \mu > 0$ given that $x = a$ and $\frac{dx}{dt} = 0$, when $t = \frac{\pi}{2\sqrt{\mu}}$.

Solution:

The given differential equation is $\frac{d^2x}{dt^2} + \mu x = 0$

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or, $(D^2 + \mu)x = 0$ .... (i)

Then, its auxiliary equation is $m^2 + \mu = 0$

$\therefore \quad m = \pm \sqrt{\mu}\, i$

Hence, the general solution is $x = c_1 \cos\sqrt{\mu}\, t + c_2 \sin\sqrt{\mu}\, t$ .... (ii)

If $x = a, t = \frac{\pi}{2\sqrt{\mu}}$, then equation (ii) gives

$a = c_1 \cos\left(\sqrt{\mu}\frac{\pi}{2\sqrt{\mu}}\right) + c_2 \sin\left(\sqrt{\mu}\frac{\pi}{2\sqrt{\mu}}\right)$

or, $a = c_1 \cos\frac{\pi}{2} + c_2 \sin\frac{\pi}{2}$

or, $a = c_2 . 1$

$\therefore \quad c_2 = a$

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Again, differentiating (ii), $\frac{dx}{dt} = -c_1\sqrt{\mu}\sin(\sqrt{\mu}\,t) + c_2\sqrt{\mu}\cos(\sqrt{\mu}\,t)$

If $\frac{dx}{dt} = 0, t = \frac{\pi}{2\sqrt{\mu}}$, then

$$0 = -c_1\sqrt{\mu}\sin\left(\sqrt{\mu}\frac{\pi}{2\sqrt{\mu}}\right) + c_2\sqrt{\mu}\cos\left(\sqrt{\mu}\frac{\pi}{2\sqrt{\mu}}\right)$$

$$= -c_1\sqrt{\mu}\sin\left(\frac{\pi}{2}\right) + c_2\sqrt{\mu}\cos\left(\frac{\pi}{2}\right)$$

or, $0 = -c_1\sqrt{\mu}$ i.e., $c_1 = 0$

$\therefore$ $c_1 = 0, c_2 = a.$

Hence equation (ii) gives $x = 0\cos\sqrt{\mu}\,t + a\sin\sqrt{\mu}\,t = a\sin\sqrt{\mu}\,t.$

Exercise 8.1

1. Solve the following differential equations:

a. $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0.$

b. $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0.$

c. $\frac{d^2y}{dx^2} - 9y = 0.$

d. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0.$

e. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0.$

f. $4\frac{d^2x}{dt^2} - 12\frac{dx}{dt} + 5x = 0.$

g. $(D^2 + 3aD - 4a^2)y = 0.$

h. $(D^2 - 4D + 4)y = 0.$

i. $(D^2 + 2D)y = 0.$

j. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0.$

k. $(8D^2 - 6D - 5)y = 0.$

l. $16\frac{d^2y}{dx^2} + 24\frac{dy}{dx} + 9y = 0.$

2. **Solve the following initial value problems:**

a. $y'' + 5y' + 6y = 0, y(0) = 2, y'(0) = 3$

b. $y'' - 16y = 0,\ y(0) = 1,\ y'(0) = 20$

c. $y'' - 3y' + 2y = 0,\ y(0) = 1, y'(0) = 1$

d. $y'' - 4y' + 4 = 0,\ y(0) = 3,\ y'(0) = 1$

e. $4y'' - 4y' - 3y = 0,\ y(-2) = e,\ y'(-2) = -\frac{e}{2}$

f. $2y'' + y' = 4y,\ y(0) = 0,\ y'(0) = 1$

g. $y'' + 4y' + 3y = 0,\ y(0) = 2,\ y'(0) = -1$

h. $y'' - 4y' + 5y = 0,\ y(0) = 1,\ y'(0) = 2$

i. $y'' + 8y' - 9y = 0,\ y(1) = 1,\ y'(1) = 0$

j. $y'' + 6y' + 9y = 0,\ y(0) = -4,\ y'(0) = 14$

k. $y'' + y' - 2y = 0, y(0) = 3, y'(0) = 0$

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Answer

1. a. $y = c_1e^{2x} + c_2e^{3x}$
   b. $y = c_1e^{x} + c_2e^{2x}$
   c. $y = c_1e^{3x} + c_2e^{-3x}$
   d. $y = e^{-x}(A\cos 2x + B\sin 2x)$
   e. $y = e^{-2x}(c_1 + c_2x)$
   f. $x = c_1e^{t/2} + c_2e^{5t/2}$
   g. $y = c_1e^{ax} + c_2e^{-4ax}$
   h. $y = (c_1 + c_2x)e^{2x}$
   i. $y = c_1 + c_2e^{-2x}$
   j. $y = e^{2x}(A\cos 3x + B\sin 3x)$
   k. $y = c_1e^{-x/2} + c_2e^{5x/4}$
   l. $y = (A + Bx)e^{-3x/4}$

2. a. $y = 9e^{-2x} - 7e^{-3x}$
   b. $y = 3e^{4x} - 2e^{-4x}$
   c. $y = e^{x}$
   d. $y = (3 - 5x)e^{2x}$
   e. $y = e^{-0.5x}$
   f. $y = \frac{1}{\sqrt{33}}e^{(-1+\sqrt{33})\frac{x}{4}} - \frac{2}{\sqrt{33}}e^{(-1+\sqrt{33})\frac{x}{4}}$
   g. $y = \frac{5}{2}e^{-x} - \frac{1}{2}e^{-3x}$
   h. $y = e^{2x}\cos x$
   i. $y = \frac{1}{10}e^{-9(x-1)} + \frac{9}{10}e^{x-1}$
   j. $y = (2x - 4)e^{-3x}$
   k. $y = 2e^{x} + e^{-2x}$

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8.2 Second order linear homogeneous equation with variable coefficient

The differential equation

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

where, $P(x)$ and $Q(x)$ are continuous functions of x is the general second order linear homogeneous equation with variable coefficients. In this section we solve such equations. One of the best kinds of such equation is Euler-Cauchy equation.

8.2.1 Euler-Cauchy equation and its solution

The second order homogeneous linear differential equation with variable coefficients of the form

$$x^2y'' + axy' + by = 0 \qquad \ldots\ldots(i)$$

where a and b are constants, is called Euler-Cauchy equation.

We now discuss its solution.

8.2.2 Solution of Euler-Cauchy equation

A solution to a differential equation $y = y(x)$ together with its derivative make the equation an identity. As the coefficients of y'' and y' are in power of x, so it is natural to expect the solution of equation (i) as $y = x^m$.

$\therefore \quad y' = m\,x^{m-1}$

and $\quad y'' = m\,(m-1)\,x^{m-2}$

On substituting in equation (i), we get

$$x^2\,m(m-1)\,x^{m-2} + axm\,x^{m-1} + bx^m = 0$$

or, $\quad x^m\,[m^2 - m + am + b] = 0$

$\therefore \quad m^2 + (a-1)m + b = 0 \qquad \ldots\ldots(ii)$

This equation is known as auxiliary equation of (i).

Since equation (ii) is quadratic in m, so m will have two values and hence we will get independent solutions for each values of m.

Depending on the values of m, we have the following cases:

Case I: Real and distinct (unequal) roots: If m_1, m_2 are distinct real roots of (ii), then

$y = C_1x^{m_1} + C_2\,x^{m_2}$ is the required general solution of (i).

Case II : Real and equal roots:

If the roots of auxiliary equation

$m^2 + (a-1)\,m + b = 0$ are real and equal, then its discriminant is zero. So,

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$$(a-1)^2 - 4b \times 1 = 0$$

or, $b = \frac{(a-1)^2}{4}$

$\therefore$ $m^2 + (a-1)m + \frac{(a-1)^2}{4} = 0$

or, $4m^2 - 4m(1-a) + (1-a)^2 = 0$

or, $[2m - (1-a)]^2 = 0$

$\therefore$ $m = \frac{1-a}{2}$

$\therefore$ $y_1 = x^{\left(\frac{1-a}{2}\right)}$ is a solution. The other linearly independent solution is given by, $y_2 = x^{\left(\frac{1-a}{2}\right)} \log x$.

$\therefore$ The general solution of (i) is, $y = C_1y_1 + C_2y_2 = C_1 x^{\left(\frac{1-a}{2}\right)} + C_2 x^{\left(\frac{1-a}{2}\right)} \log x$

i.e., $y = (C_1 + C_2 \log x)\, x^{\left(\frac{1-a}{2}\right)}$

Case III: Complex roots (unequal and imaginary roots): If roots of auxiliary equation

$$m^2 + (a-1)\, m + b = 0$$

are complex roots, say $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$,

then the general solution of (i) is given by

$$y = x^{\alpha} [A \cos(\beta \log x) + B \sin(\beta \log x)].$$

Example 7. Solve the equation $x^2 y'' - 2.5xy' - 2y = 0$.

Solution:

The given differential equation is $x^2y'' - \frac{5}{2}xy' - 2y = 0$.

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Its auxiliary equation is

$$m^2 + \left(\frac{-5}{2} - 1\right) m - 2 = 0$$

or, $2m^2 - 7m - 4 = 0$

or, $2m^2 - 8m + m - 4 = 0$

or, $2m\,(m-4) + (m-4) = 0$

or, $(2m+1)\,(m-4) = 0$

$\therefore$ $m = 4, m = -\frac{1}{2}$

$\therefore$ $y_1 = x^{\frac{-1}{2}}$ and $y_2 = x^4$

Thus, the general solution is $y = C_1y_1 + C_2y_2 = C_1 x^{\frac{-1}{2}} + C_2 x^4$.

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Example 8. Solve the differential equation $x^2y'' - 3xy' + 4y = 0$.

Solution:

The given differential equation is $x^2y'' - 3xy' + 4y = 0$.

Its auxiliary equation is

$m^2 + (a-1)m + b = 0$

or, $m^2 + (-3-1)m + 4 = 0$

or, $m^2 - 4m + 4 = 0$

or, $(m-2)^2 = 0$

or, $m_1 = m_2 = 2$, real and equal roots, so the general solution is

$y = (C_1 + C_2 \log x)x^2$.

Example 9. Solve the differential equation $x^2y'' + 7xy' + 13y = 0$

Solution:

The given differential equation is

$x^2y'' + 7xy' + 13y = 0$ ....(i)

It's auxiliary equation is

$m^2 + (a-1)m + b = 0$

or, $m^2 + (7-1)m + 13 = 0$

or, $m^2 + 6m + 13 = 0$

or, $m = \dfrac{-6 \pm \sqrt{36-52}}{2} = \dfrac{-6 \pm 4i}{2}$

$\therefore \quad m = -3 \pm 2i$

Thus, the general solution of (i) is $y = x^{-3}[A\cos(2\log x) + B\sin(2\log x)]$.

Exercise 8.2

1. Find the general solution of the differential equations

a. $x^2y'' - 4xy' + 6y = 0$ b. $x^2y'' - 20y = 0$

c. $x^2y'' - xy' + 2y = 0$ d. $x^2y'' + xy' + y = 0$

e. $(4x^2D^2 + 12xD + 3)y = 0$ f. $(x^2D^2 + 7xD + 9)y = 0$

2. Solve the following initial value problems:

a. $x^2y'' - 2xy' + 2y = 0, \quad y(1) = \dfrac{3}{2}, \quad y'(1) = 1$

b. $4x^2y'' + 24xy' + 25y = 0, \quad y(1) = 2, \quad y'(1) = -6$

c. $(x^2D^2 - 3xD + 4)y = 0, \quad y(1) = 0, \quad y'(1) = 3$

d. $(x^2D^2 + 3xD + 1)y = 0, \quad y(1) = 3, \quad y'(1) = -4$

Answers:

1. a. $y = C_1x^2 + C_2x^3$ b. $y = C_1x^{-4} + C_2x^5$

c. $y = x[A\cos(\log x) + B\sin(\log x)]$ d. $y = A\cos(\log x) + B\sin(\log x)$

e. $y = C_1x^{\frac{-1}{2}} + C_2x^{\frac{-3}{2}}$ f. $y = (C_1 + C_2\log x)x^{-3}$

2. a. $y = C_1x + C_2x^2, y = 2x - \dfrac{1}{2}x^2$ b. $y = (2 - \log x)x^{\frac{-5}{2}}$

c. $y = 3x^2\log x$ d. $y = \left(\dfrac{C_1 + C_2\log x}{x}\right), y = \left(\dfrac{3 - \log x}{x}\right)$

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8.3 Existence and uniqueness of solution

In previous two sections we discussed and solved the second order homogeneous differential equation of the form

$$y'' + P(x)y' + Q(x)y = 0 \quad \ldots . (i)$$

If $P(x)$ and $Q(x)$ are constant say p and q, we solved it by using auxiliary equation $m^2 + pm + q = 0$. Based on nature of roots of auxiliary equation and we have different cases of solution (see section 8.1.1).

If (i) can be written in the Euler-Cauchy equation

$$x^2y'' + axy + b = 0 \quad \ldots . (ii)$$

We solved equation by using auxiliary equation and we have different cases of solution (see section 8.2.2).

We now turn into the main theory of above solutions known as **existence and uniqueness of solution** of (i).

Following existence theorem guarantees the existence of solution of (i).

8.3.1 Existence theorem

The second order homogeneous differential equation

$$y'' + P(x)y' + Q(x)y = 0$$

where, $P(x)$ and $Q(x)$ are continuous functions of x (in some interval) has general solution $y = C_1y_1(x) + C_2y_2(x)$

where, $y_1(x)$ and $y_2(x)$ are linearly independent solutions obtained from roots of auxiliary equation.

8.3.2 Uniqueness theorem

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The initial value problem $y'' + P(x)y' + Q(x)y = 0$, $y(x_0) = k_0$ (constant), $y'(x_0) = k_1$ (constant) has unique solution obtained from

$$y = C_1y_1(x) + C_2y_2(x)$$

by determining C_1 and C_2 using the initial conditions $y(x_0) = k_0$ and $y'(x_0) = k_1$.

8.3.3 Linear independence and dependence solutions and use of Wronskian

The solutions of differential equation

$$y'' + P(x)y' + Q(x)y = 0,\ y = y_1(x) \text{ and } y = y_2(x)$$

obtained by using roots of auxiliary equation are said to be **linearly dependent** if one is proportional to other, i.e., either of $y_1(x)$ or $y_2(x)$ can be expressed in terms of other using some constants, i.e., $y_1(x) = ky_2(x)$.

The solutions $y = y_1(x)$ and $y = y_2(x)$ are **linearly independent** if one is not proportional to other.

The 2×2 determinant of solution and their derivatives given by

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$$W(y_1, y_2) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}$$

is known as **Wronskian determinant of the solutions.**

The Wronskian determinant provides testing tool for linear dependency and independency of solutions.

1. The solutions $y = y_1(x)$ and $y = y_2(x)$ are **linearly dependent if**

   $W(y_1, y_2) = 0.$

2. The solutions $y = y_1(x)$ and $y = y_2(x)$ are **linearly independent if**

   $W(y_1, y_2) \neq 0.$

8.4 Second order non-homogeneous equations with constant coefficients

The differential equation of the form

$$y'' + Py' + Qy = R(x) \quad \ldots\ldots \text{(i)}$$

where, P, Q are constants and $R(x)$ is a continuous functions of x or constant.

If $R(x) \neq 0$ equation (i) is called a *non-homogeneous linear differential equation of second order.*

8.4.1 Solution of second order non-homogeneous equation with constant coefficients

The general solution of a non-homogeneous linear equation (i) is of the form $y = y_h(x) + y_p(x)$, where $y_h(x)$ is the general solution of the corresponding homogeneous equation $y'' + Py' + Qy = 0$. The part y_h is called the ***complementary solution (CS)*** of the general solution.

And $y_p(x)$ is a *particular solution (PS)* of the non-homogeneous equation (i) and does not contain any arbitrary constant.

8.4.2 Method of determining the particular solution $y_p(x)$ by using method of undetermined coefficients

The function $R(x)$ in (i) is generally, exponential functions, polynomials, sines, cosines, or sums or products of such functions. These functions are their derivatives in the form similar to $R(x)$ itself. So we choose a particular solution $y_p(x)$ similar to that of $R(x)$ and involving unknown coefficients to be determined by substituting that choice for $y_p(x)$ in (i).

Rules for selection of y_p

i. **Basic Rule:** If $R(x)$ in (i) is one of the functions in the first column in table provided below, we choose the corresponding function $y_p(x)$ in the second column and determine its undetermined coefficients by substituting by $y_p(x)$ and its derivatives into equation (i).

ii. **Modification Rule:** If a term in a choice for $y_p(x)$ happens to be a solution of the homogeneous equation corresponding to (i), we then modify it by multiplying the choice of $y_p(x)$ by x (or by x^2 if this solution corresponds to a double root of the characteristic equation of the homogeneous equation).

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iii. **Sum Rule:** If $R(x)$ is a sum of functions in various rows in the first column of the table given below, then choose $y_p(x)$ as the sum of the functions in the corresponding rows of the second column.

Table for selecting y_p undetermined coefficients

Term in R(x)	*Choice for $y_p(x)$*
Ke^{ax}	Ae^{ax}
Kx^n $(n = 0, 1, \ldots.)$	$A_n x^n + A_{n-1}x^{n-1} + \ldots + A_1 x + A_0$
$K\cos ax$	$A\cos ax + B\sin ax$
$K\sin ax$	$A\cos ax + B\sin ax$
$Ke^{ax}\cos bx$	$e^{ax}(A\cos bx + B\sin bx)$
$Ke^{ax}\sin bx$	$e^{ax}(A\cos bx + B\sin bx)$

where, undetermined coefficients A, B, A_0, A_1, A_2, ..., A_n are to be determined.

We now apply the above method to find solution of some differential equations.

Example 10. Find the general solution of the following differential equation
$y'' - y' - 2y = 2e^{3x}$

Solution:

Given differential equation is $y'' - y' - 2y = 2e^{3x}$ .... (i)

The auxiliary equation of corresponding homogeneous equation is

$m^2 - m - 2 = 0$

or, $m^2 - 2m + m - 2 = 0$

or, $m(m-2) + 1(m-2) = 0$

or, $(m-2)(m+1) = 0$

$\therefore$ $m = 2, -1$

Thus, the complementary solution of (i) is $y_h(x) = c_1e^{2x} + c_2e^{-x}$.

For a particular solution, let $y_p(x) = Ae^{3x}$.

Then, $y_p'(x) = 3Ae^{3x}$, $y_p''(x) = 9Ae^{3x}$.

Then, (i) becomes

$9Ae^{3x} - 3Ae^{3x} - 2Ae^{3x} = 2e^{3x}$.

or, $4Ae^{3x} = 2e^{3x}$

or, $2Ae^{3x} = e^{3x}$

or, $2A = 1$

or, $A = \frac{1}{2}$

$\therefore$ $y_p(x) = \frac{1}{2}e^{3x}$.

Thus, the particular solution is $y_p(x) = \frac{1}{2}e^{3x}$.

Hence, the general solution is

$y = y_h(x) + y_p(x) = c_1e^{2x} + c_2e^{-x} + \frac{1}{2}e^{3x}$.

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Example 11. Find the general solution of $y'' - 3y' - 4y = 2e^{-x}$.

Solution:

The given differential equation is $y'' - 3y' - 4y = 2e^{-x}$ .... (i)

The auxiliary equation of corresponding homogeneous equation is

$m^2 - 3m - 4 = 0$

or, $m^2 - 4m + m - 4 = 0$

or, $m(m-4) + 1(m-4) = 0$

or, $(m-4)(m+1) = 0$

$\therefore$ $m = 4, -1$

Thus, the complementary solution is $y_h(x) = c_1e^{4x} + c_2e^{-x}$.

Since e^{x} is already in the homogeneous solution, we let $y_p(x) = Axe^{-x}$ as a particular solution of (i).

Then, $y_p'(x) = -Axe^{-x} + Ae^{-x}$

$y_p''(x) = Axe^{-x} - Ae^{-x} - Ae^{-x} = Axe^{-x} - 2Ae^{-x}$

Then, (i) becomes

$Axe^{-x} - 2Ae^{-x} - 3(-Axe^{-x} + Ae^{-x}) - 4Axe^{-x} = 2e^{-x}$

or, $(-2A - 3A)e^{-x} + (A + 3A - 4A)xe^{-x} = 2e^{-x}$

or, $-5Ae^{-x} = 2e^{-x}$

or, $-5A = 2$

$\therefore$ $A = -\frac{2}{5}$.

Thus, the particular solution is $y_p(x) = -\frac{2}{5}xe^{-x}$.

Hence, the general solution is $y = y_h(x) + y_p(x) = c_1e^{4x} + c_2e^{-x} - \frac{2}{5}xe^{-x}$.

Example 12. Find the general solution of $y'' + 4y = 8x^2$.

Solution:

The given differential equation is $y'' + 4y = 8x^2$ .... (i)

The auxiliary equation of corresponding homogeneous equation is

$m^2 + 4 = 0$

or, $m = 0 \pm 2i$

Thus, the complementary solution is $y_h(x) = c_1\cos 2x + c_2\sin 2x$.

For particular solution, let $y_p(x) = Ax^2 + Bx + C$ be a particular solution of (i).

Then, $y_p'(x) = 2Ax + B$, $y_p''(x) = 2A$

Then (i) becomes

$2A + 4(Ax^2 + Bx + C) = 8x^2$

or, $4Ax^2 + 4Bx + (2A + 4C) = 8x^2$

Equating the coefficients of x^2, x, x^0, we have

$4A = 8 \Rightarrow$ $A = 2$

$4B = 0 \Rightarrow$ $B = 0$

$2A + 4C = 0$ $\Rightarrow$ $C = -1$

$\therefore$ $y_p(x) = 2x^2 - 1$

Thus, the particular solution is $y_p(x) = 2x^2 - 1$.

Hence, the required solution is $y = y_h(x) + y_p(x) = c_1\cos 2x + c_2\sin 2x + 2x^2 - 1$.

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Example 13. Find the general solution of $y'' + 5y' = 15x^2$.

Solution:

The given differential equation is $y'' + 5y' = 15x^2$ .... (i)

It's auxiliary equation is $m^2 + 5m = 0$

or, $m(m + 5) = 0$

or, $m = 0, -5$

So, the complementary solution is $y_h(x) = c_1 + c_2e^{-5x}$.

For particular solution,

let $y_p(x) = x(Ax^2 + Bx + C) = Ax^3 + Bx^2 + Cx$ be a particular solution of (i).

Then, $y_p'(x) = 3Ax^2 + 2Bx + C$, $y_p''(x) = 6Ax + 2B$

Then, (i) becomes

$6Ax + 2B + 5(3Ax^2 + 2Bx + C) = 15x^2$

or, $6Ax + 2B + 15Ax^2 + 10Bx + 5C = 15x^2$

or, $15Ax^2 + (6A + 10B)x + (2B + 5C) = 15x^2$

Equating the coefficients of x^2, x, x^0, we get

$15A = 15 \Rightarrow A = 1$

$6A + 10B = 0 \Rightarrow B = -\frac{3}{5}$

$2B + 5C = 0 \Rightarrow C = \frac{6}{25}$

Thus, the particular solution is $y_p(x) = x^3 - \frac{3}{5}x^2 + \frac{6}{25}x$.

Hence, the general solution is $y = y_h(x) + y_p(x) = c_1 + c_2e^{-5x} + x^3 - \frac{3}{5}x^2 + \frac{6}{25}x$.

Example 14. Find the general solution of $y'' - y = \sin x$.

Solution:

The given differential equation is $y'' - y = \sin x$ .... (i)

It's auxiliary equation is $m^2 - 1 = 0$

or, $m = 1, -1$

So, the complementary solution is $y_h(x) = c_1e^x + c_2e^{-x}$.

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let $y_p(x) = A\sin x + B\cos x$ be the particular solution of (i).

Then $y_p'(x) = A\cos x - B\sin x$, $y_p''(x) = -A\sin x - B\cos x$

Then, (i) becomes

$-A\sin x - B\cos x - A\sin x - B\cos x = \sin x$

or, $-2A\sin x - 2B\cos x = \sin x$

Equating the coefficients, we get

$-2A = 1 \Rightarrow A = -\frac{1}{2}$

$-2B = 0 \Rightarrow B = 0$

Thus, the particular solution is $y_p(x) = -\frac{1}{2}\sin x$.

Hence, the general solution is

$y = y_h(x) + y_p(x) = c_1e^x + c_2e^{-x} - \frac{1}{2}\sin x$.

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Example 15. Find the general solution of $y'' + y = \cos x$.

Solution:

The given differential equation is $y'' + y = \cos x$ .... (i)

It's auxiliary equation is $m^2 + 1 = 0$

or, $m = 0 \pm i$

So, the complementary solution is $y_h(x) = c_1\cos x + c_2\sin x$

Let $y_p(x) = Ax\cos x + Bx\sin x$ be the particular solution of (i).

Then $y_p'(x) = A\cos x - Ax\sin x + B\sin x + Bx\cos x$

$y_p''(x) = -A\sin x - A(\sin x + x\cos x) + B\cos x + B(\cos x - x\sin x)$

$= -A\sin x - A\sin x - Ax\cos x + B\cos x + B\cos x - Bx\sin x$

$= -2A\sin x + 2B\cos x - Ax\cos x - Bx\sin x$

Then, (i) becomes

$-2A\sin x + 2B\cos x - Ax\cos x - Bx\sin x + Ax\cos x + Bx\sin x = \cos x$

or, $-2A\sin x + 2B\cos x = \cos x$

Equating the coefficients, we get

$-2A = 0 \Rightarrow A = 0$

$2B = 1 \Rightarrow B = \frac{1}{2}$

Thus, the particular solution of (i) is $y_p(x) = \frac{1}{2} x \sin x$.

Hence, the general solution is

$y = y_h(x) + y_p(x) = c_1\cos x + c_2\sin x + \frac{1}{2} x\sin x$.

Example 16. Find the general solution of $y'' + 4y' + 4y = 2\cosh x$.

Solution:

Given differential equation is $y'' + 4y' + 4y = 2\cosh x$

or, $y'' + 4y' + 4y = 2\left(\frac{e^x + e^{-x}}{2}\right)$

or, $y'' + 4y' + 4y = e^x + e^{-x}$

It's auxiliary equation is $m^2 + 4m + 4 = 0$

or, $(m + 2)^2 = 0$

or, $m = -2, -2$

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So, the complementary solution is $y_h(x) = (c_1 + c_2x)\, e^{-2x}$

For particular solution, let $y_p(x) = Ae^x + Be^{-x}$ be the particular solution of (i).

Then, $y_p'(x) = Ae^x - Be^{-x}$, $y_p''(x) = Ae^x + Be^{-x}$

Then, (i) becomes $Ae^x + Be^{-x} + 4(Ae^x - Be^{-x}) + 4(Ae^x + Be^{-x}) = e^x + e^{-x}$

or, $Ae^x + Be^{-x} + 4Ae^x - 4Be^{-x} + 4Ae^x + 4Be^{-x} = e^x + e^{-x}$

or, $9Ae^x + Be^{-x} = e^x + e^{-x}$

Equating the coefficients, we get

$9A = 1 \Rightarrow A = \frac{1}{9}$,

$B = 1$

Thus, the particular solution is $y_p(x) = \frac{1}{9} e^x + e^{-x}$.

Hence, the general solution is

$y = y_h(x) + y_p(x) = (c_1 + c_2x)\, e^{-2x} + \frac{1}{9} e^x + e^{-x}$.

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Example 17. Find the general solution of $y'' - 2y' = e^x \sin x$.

Solution:

The given differential equation is $y'' - 2y' = e^x \sin x$ .... (i)

It's auxiliary equation is $m^2 - 2m = 0$

or, $m(m-2) = 0$

or, $m = 0, 2$

So, the complementary solution is $y_h(x) = c_1 + c_2 e^{2x}$. For particular solution, let $y_p(x) = Ae^x \sin x + Be^x \cos x$ be the particular solution of (i).

Then, $y_p'(x) = Ae^x \sin x + Ae^x \cos x + Be^x \cos x - Be^x \sin x$ and

$y_p''(x) = Ae^x \sin x + Ae^x \cos x + Ae^x \cos x - Ae^x \sin x + Be^x \cos x - Be^x \sin x - Be^x \sin x - Be^x \cos x$

$= 2Ae^x \cos x - 2Be^x \sin x$

Then, (i) becomes

$2Ae^x \cos x - 2Be^x \sin x - 2(Ae^x \sin x + Ae^x \cos x + Be^x \cos x - Be^x \sin x) = e^x \sin x$

or, $2Ae^x \cos x - 2Be^x \sin x - 2Ae^x \sin x - 2Ae^x \cos x - 2Be^x \cos x + 2Be^x \sin x = e^x \sin x$

or, $-2Ae^x \sin x - 2Be^x \cos x = e^x \sin x$

Equality the coefficients, we have

$-2A = 1 \quad \Rightarrow \quad A = -\frac{1}{2}$

$-2B = 0 \quad \Rightarrow \quad B = 0$

Thus, the particular solution of (i) is $y_p(x) = -\frac{1}{2} e^x \sin x$.

Hence, the general solution is

$y = y_h(x) + y_p(x) = c_1 + c_2 e^{2x} - \frac{1}{2} e^x \sin x$.

Example 18. Find the general solution of the given initial value problem

$y'' - y' - 2y = 3e^x$, $y(0) = 0$, $y'(0) = -2$.

Solution:

The given differential equation is $y'' - y' - 2y = 3e^x$ .... (i)

It's auxiliary equation is $m^2 - m - 2 = 0$

or, $m^2 - 2m + m - 2 = 0$

or, $m(m-2) + 1(m-2) = 0$

or, $(m-2)(m+1) = 0$

or, $m = 2, -1$

So, the complementary solution is $y_h(x) = c_1 e^{2x} + c_2 e^{-x}$.

For particular solution, let $y_p(x) = Ae^x$ be the particular solution of (i).

Then, $y_p'(x) = Ae^x$, $y_p''(x) = Ae^x$. Then, (i) becomes

$Ae^x - Ae^x - 2Ae^x = 3e^x$

or, $-2Ae^x = 3e^x$

Equating the coefficients, we get

$-2A = 3 \quad \Rightarrow \quad A = -\frac{3}{2}$

So, the particular solution of (i) is $y_p(x) = -\frac{3}{2} e^x$.

Thus, the general solution of (i) is $y = y_h(x) + y_p(x)$.

or, $y = c_1 e^{2x} + c_2 e^{-x} - \frac{3}{2} e^x$ .... (ii)

Then, $y' = 2c_1 e^{2x} - c_2 e^{-x} - \frac{3}{2} e^x$ .... (iii)

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Using, $y(0) = 0$, we have $0 = c_1 + c_2 - \frac{3}{2}$ .... (iv)

Again, using $y'(0) = -2$, we have $-2 = 2c_1 - c_2 - \frac{3}{2}$ .... (v)

Solving (iv) and (v), we have $c_1 = \frac{1}{3}$ and $c_2 = \frac{7}{6}$

Substituting the value of c_1 and c_2 in (ii), we get $y = \frac{1}{3}e^{2x} + \frac{7}{6}e^{-x} - \frac{3}{2}e^{x}$

which is the required general solution of the given initial value problem.

Exercise 8.3

1. Find the particular solution of the following differential equations:
   a. $y'' + 2y' + y = x^2$
   b. $y'' + 9y = 6\cos 3x$
   c. $y'' - 6y' + 13y = 4e^{3x}$
   d. $y'' + y' - 2y = 14 + 2x - 2x^2$
   e. $y'' - y' - 2y = 3e^{2x}$
2. Find the general solution of the following differential equations:
   a. $y'' - 3y' - 4y = 3e^{2x}$
   b. $y'' - 3y' - 4y = 2e^{-x}$
   c. $y'' + 10y' + 25y = e^{-5x}$
   d. $y'' - y' - 2y = 10\cos x$
   e. $y'' + 4y = \sin 3x$
   f. $y'' + 4y = 2\sin 2x$
   g. $y'' + 2y' + 10y = 25x^2 + 3$
   h. $y'' + 4y' + 5y = 10$
   i. $y'' + 3y' = 28\cosh 4x$
   j. $y'' + y' = x$
   k. $y'' + 5y' = 15x^2$
   l. $y'' - y = 2e^{x} + 6e^{2x}$
   m. $y'' - y' - 6y = e^{-x} - 7\cos x$
   n. $y'' + 2y' - 35y = 12e^{5x} + 37\sin 5x$
   o. $y'' - 4y' - 5y = e^{x} + 4$
   p. $y'' + 2y' = 3x + 4\sin 2x$
   q. $y'' - 3y' - 4y = -8e^{x}\cos 2x$
   r. $y'' - 2y' - 3y = 3xe^{2x}$
   s. $y'' + 9y = x^2e^{3x} + 6$
3. Find the solution of the following initial value problems:
   a. $y'' - 6y' + 13y = 4e^{3x}$; $y(0) = 2$, $y'(0) = 4$
   b. $y'' + y' - 2y = 14 + 2x - 2x^2$; $y(0) = 0$, $y'(0) = 0$
   c. $y'' + y' - 2y = 2x$; $y(0) = 0$, $y'(0) = 1$
   d. $y'' + y' - 2y = -6\sin 2x - 18\cos 2x$; $y(0) = 2$, $y'(0) = 2$
   e. $y'' - 2y' - 3y = 3xe^{2x}$; $y(0) = 1$, $y'(0) = 0$
   f. $y'' + 2y' + y = e^{-x}$; $y(0) = -1$, $y'(0) = 1$
   g. $y'' + 2y' + 5y = 4e^{-x}\cos 2x$; $y(0) = 1$, $y'(0) = 0$

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Answer

1. a. $y_p(x) = x^2 - 4x + 6$
   b. $y_p(x) = x\sin 3x$
   c. $y_p(x) = e^{3x}$
   d. $y_p(x) = x^2 - 6$
   e. $y_p(x) = xe^{2x}$
2. a. $y = c_1e^{4x} + c_2e^{-x} - \frac{1}{2}e^{2x}$
   b. $y = c_1e^{4x} + c_2e^{-x} - \frac{2}{5}xe^{-x}$
   c. $y = (c_1 + c_2x)e^{-5x} + \frac{1}{2}x^2e^{-5x}$
   d. $y = c_1e^{-x} + c_2e^{2x} - 3\cos x - \sin x$
   e. $y = c_1\cos 2x + c_2\sin 2x - \frac{1}{5}\sin 3x$
   f. $y = c_1\cos x + c_2\sin x - \frac{1}{2}x\cos 2x$

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g. $y = e^{-x}(c_1 \cos 3x + c_2 \sin 3x) + \frac{5}{2}x^2 - x$

h. $y = e^{-2x}(c_1 \cos x + c_2 \sin x) + 2$

i. $y = c_1 + c_2 e^{-3x} + \frac{1}{2}e^{4x} + \frac{7}{2}e^{-4x}$

j. $y = c_1 + c_2 e^{x} + \frac{x^2}{2} - x$

k. $y = c_1 + c_2 e^{-5x} + x^3 + \frac{3x^2}{5} - \frac{6}{25}x$

l. $y = c_1 e^{x} + c_2 e^{-x} + xe^{x} + 2e^{2x}$

m. $y = c_1 e^{3x} + c_2 e^{-2x} - \frac{1}{4}e^{-x}$

n. $y = c_1 e^{-7x} + c_2 e^{5x} + xe^{5x} - 0.1\cos 5x - 0.6\sin 5x$

o. $y = c_1 e^{5x} + c_1 e^{-x} - \frac{e^{x}}{8} - \frac{4}{5}$

p. $y = c_1 + c_2 e^{-2x} + \frac{3}{2}x - \frac{1}{2}\sin 2x - \frac{1}{2}\cos 2x$

q. $y = c_1 e^{4x} + c_2 e^{-x} + \frac{10}{13}e^{x}\cos 2x + \frac{2}{13}e^{x}\sin 2x$

r. $y = e^{3x} + \frac{2}{3}e^{-x} - \frac{2}{3}e^{2x} - xe^{2x}$

s. $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{162}(9x^2 - 6x + 1)e^{3x} + \frac{2}{3}$

3. a. $y = e^{3x}(\cos 2x + 4\sin 2x) + e^{3x}$

b. $y = 4e^{x} + 2e^{-2x} + x^2 - 6$

c. $y = e^{x} - \frac{1}{2}e^{-2x} - x - \frac{1}{2}$

d. $y = -e^{-2x} + 3\cos 2x$

e. $y = e^{3x} + \frac{2}{3}e^{-x} - \frac{2}{3}e^{2x} - xe^{2x}$

f. $y = \left(-1 + \frac{x^2}{2}\right)e^{-x}$

g. $y = e^{-x}\cos 2x + \frac{1}{2}e^{-x}\sin 2x + xe^{-x}\sin 2x$

8.5 Second order non-homogeneous liner differential equation with variable coefficients

The equation $\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x)$

i.e., $y'' + P(x)y' + Q(x)y = R(x)$

where, $P(x)$, $Q(x)$, $R(x)$ are continuous functions of x is known as second order non-homogeneous differential equation with variable coefficients.

Since it is non-homogeneous equation so its solution $y(x)$ is given by

$$y(x) = y_h(x) + y_p(x)$$

We use method of variation of parameters for determining particular solution y_p.

The method of variation of parameter (due to Lagrange) is used to determine y_p.

This method is applicable for $P(x)$ and $Q(x)$ constant or function of x. Hence it is general method to solve the non-homogeneous equation with constant coefficients and variable coefficients.

We now describe the method in detail.

Consider the differential equation

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x) \quad \ldots.(i)$$

where $P(x)$, $Q(x)$ and $R(x)$ are continues functions of x (in some interval I).

Since it is non-homogeneous equation so its solution $y(x)$ is given by

$$y(x) = y_h(x) + y_p(x) \quad \ldots.(ii)$$

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where $y_h(x)$ is general solution of corresponding homogeneous equation and $y_p(x)$ is particular solution of (i).

For y_h:

The corresponding homogeneous equation of (i) is

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0 \quad \ldots.(iii)$$

The solution $y_h(x)$ of (iii) is given by

$$y_h(x) = C_1y_1(x) + C_2y_2(x) \quad \ldots.(iv)$$

where $y_1(x)$ and $y_2(x)$ are independent solutions of (iii) (see section 8.1.1 for constant $P(x)$ and $Q(x)$ and section 8.2 for variable coefficients $P(x)$ and $Q(x)$).

For determining particular solution $y_p(x)$

We use the idea of variation of parameter (the constants in the solution $y_h(x)$ of (iv) are known as parameter) is to replace C_1 and C_2 by some functions $u(x)$ and $v(x)$ of x.

i.e., we take $y_p = u(x)y_p + v(x)\, y_2(x)$ .... (v)

where $u(x)$ and $v(x)$ satisfy,

$u'(x)\, y_1(x) + v'(x)\, y_2(x) = 0$ .... (vi)

Since (v) is particular solution of (i), so y_p and y_p', y_p'' satisfy equation (i).

Differentiating (v) with respect to x, we get

$$y_p' = u'(x)\, y_1(x) + u(x)\, y_1'(x) + v'(x)y_2(x) + v(x)\, y_2'(x)$$

using (vi), $u'(x)\, y_1(x) + v'(x)y_2(x) = 0$, we get

$$y_p' = u(x)y_1'(x) + v(x)y_2'(x)$$

Again differentiating with respect to x, we get

$$y_p'' = u(x)y_1''(x) + u'(x)y_1'(x) + v(x)y_2''(x) + v'(x)y_2'(x)$$

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Substituting y_p, y_p', y_p'' in (i) and rearranging the terms, we get,

$$u(x)\,(y_1'' + P(x)y_1' + Q(x)y_1) + v(x)\,(y_2'' + P(x)y_2 + Q(x)y_2) + u'(x)\, y_1' + v'(x)y_2' = R(x)$$

Since $y_1(x)$ and $y_2(x)$ are solutions of (iii), so the first and second terms im above equations vanish.

$\therefore$ $u'(x)\, y_1'(x) + v'(x)\, y_2'(x) = R(x)$ .... (vii)

Also, from (vi), $u'(x)y_1 + v'(x)y_2 = 0$ ...(viii)

Equation (vii) and (viii) are system of linear equations, so solving by Cramer's rule, we get

$$u'(x)\,(y_1y_2' - y_2y_1') = -\,y_2R(x)$$

$$\therefore \quad u'(x) = -\frac{y_2R(x)}{y_1y_2' - y_2y_1'} = \frac{-\,y_2R(x)}{W(y_1, y_2)} \quad \ldots. (ix)$$

where $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is Wronskian determinant of solutions $y_1(x)$ and $y_2(x)$

Integrating (ix), we get

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$$u(x) = -\int \frac{y_2(x)\,R(x)}{W(y_1, y_2)} \quad \ldots\ldots (x)$$

Similarly, solving (vii) and (viii), we get

$$v'(x) = \frac{y_1 R(x)}{W(y_1, y_2)}$$

Integrating,

$$v(x) = \int \frac{y_2(x)\,R(x)}{W(y_1, y_2)} \quad \ldots\ldots (xi)$$

Thus the (complete) general solution of (i), from (ii)

$y(x) = y_h(x) + y_p(x)$ is given by

$y(x) = C_1y_1(x) + C_2y_2(x) + u(x)y_1(x) + v(x)\,y_2(x)$

where $u(x)$ and $v(x)$ are given by (x) and (xi)

Note: While applying method of variation of parameter to determine y_p for a differential equation with variable coefficients, we first divide the equation by coefficient function of y'' and rewrite the equation in the form $y'' + P(x)y' + Q(x) = R(x)$ (this will be illustrated in example 22)

Example 19. Find the general solution of the differential equations by the method of variation of parameters $y'' - 5y' + 6y = 2e^x$.

Solution:

The given differential equation is $y'' - 5y' + 6y = 2e^x$ .... (i)

Then, its auxiliary equation is

$m^2 - 5m + 6 = 0$

or, $m^2 - 2m - 3m + 6 = 0$

or, $m(m-2) - 3(m-2) = 0$

or, $(m-2)(m-3) = 0$

$\therefore \quad m = 2, 3$

Thus, $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the complementary solution of (i).

Here, $y_1(x) = e^{2x}$ and $y_2(x) = e^{3x}$ form a fundamental solutions of the corresponding homogeneous equation.

Then, $y_1'(x) = 2e^{2x}$, $y_2'(x) = 3e^{3x}$.

Now, $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} = 3e^{2x}.e^{3x} - 2e^{3x}.e^{2x} = e^{5x} \neq 0$

we have, $R(x) = 2e^x$

Then, $y_p(x) = -y_1(x)\int \frac{y_2(x)\,R(x)}{[W(y_1,y_2)]\,(x)}\,dx + y_2(x)\int \frac{y_1(x)\,R(x)}{[W(y_1,y_2)]\,(x)}\,dx$

$$= -e^{2x}\int \frac{2e^{3x}.e^x}{e^{5x}}\,dx + e^{3x}\int \frac{2e^{2x}.e^x}{e^{5x}}\,dx$$

$$= -2e^{2x}\int e^{-x}\,dx + 2e^{3x}\int e^{-2x}\,dx$$

$$= 2e^{2x}.e^{-x} - e^{3x}.e^{-2x} = 2e^x - e^x = e^x.$$

Hence, the general solution is $y = y_h(x) + y_p(x) = c_1e^{2x} + c_2e^{3x} + e^x$.

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Example 20. Find the general solution of $y'' + y = \sin x$ by method of variation of parameters.

Solution:

The given differential equation is $y'' + y = \sin x$ .... (i)

Then, its auxiliary equation is $m^2 + 1 = 0$

or, $m = \pm i$

Thus, $y_h(x) = c_1\cos x + c_2\sin x$ is a complementary solution of (i).

Here, $y_1(x) = \cos x$ and $y_2(x) = \sin x$ form a fundamental solutions of the corresponding homogenous equation.

In fact, $y_1'(x) = -\sin x$, $y_2'(x) = \cos x$

Then, $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0.$

we have, $R(x) = \sin x$

$$\text{Then, } y_p(x) = -y_1(x)\int \frac{y_2(x)\,R(x)}{[W(y_1,y_2)]\,(x)}\,dx + y_2(x)\int \frac{y_1(x)\,R(x)}{[W(y_1,y_2)]\,(x)}\,dx$$

$$= -\cos x \int \frac{\sin x.\sin x}{1}\,dx + \sin x \int \frac{\cos x.\sin x}{1}\,dx$$

$$= \cos x \int \sin^2 x\,dx + \sin x \int \sin x.\cos x\,dx$$

$$= -\cos x \int \left(\frac{1-\cos 2x}{2}\right)dx + \frac{1}{2}\sin x \int \sin 2x\,dx$$

$$= -\cos x \int \frac{1}{2}\,dx + \cos x \int \frac{\cos 2x}{2}\,dx + \frac{1}{2}\sin x \int \sin 2x\,dx$$

$$= -\frac{x}{2}\cos x + \frac{1}{2}\cos x.\frac{\sin 2x}{2} - \frac{1}{2}\sin x.\frac{\cos 2x}{2}$$

$$= -\frac{x}{2}\cos x + \frac{2}{4}\cos x\,(\sin x.\cos x) - \frac{1}{4}\sin x\,(2\cos^2 x - 1)$$

$$= -\frac{x}{2}\cos x + \frac{1}{2}\sin x.\cos^2 x - \frac{1}{2}\sin x.\cos^2 x + \frac{1}{4}\sin x$$

$$= -\frac{x}{2}\cos x + \frac{1}{4}\sin x.$$

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Here, the general solution is

$$y = y_h(x) + y_p(x)$$

$$= c_1\cos x + c_2\sin x - \frac{x}{2}\cos x + \frac{1}{4}\sin x$$

$$= c_1\cos x + c_2\sin x - \frac{x}{2}\cos x.$$

Example 21. Find the general solution of $y'' - 2y' + y = \frac{e^x}{x^3}$.

Solution:

The given differential equation is $y'' - 2y' + y = \frac{e^x}{x^3}$ .... (i)

Then, its auxiliary equation is $m^2 - 2m + 1 = 0$

or, $(m-1)^2 = 0$

or, $m = 1, 1$

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Thus, $y_h(x) = (c_1 + c_2x)\, e^x$ is a complementary solution of (i).

Here, $y_1(x) = e^x$, $y_2(x) = xe^x$ form a fundamental solutions of the corresponding homogeneous equation. From above, $y_1(x) = e^x$ and $y_2(x) = xe^x$, we have

$y_1'(x) = e^x, y_2'(x) = e^x + xe^x$

Then, $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} + xe^{2x} - xe^{2x} = e^{2x} \neq 0$

We have, $R(x) = \frac{e^x}{x^3}$

Then, $y_p(x) = -y_1(x) \int \frac{y_2(x)\, R(x)}{[W(y_1, y_2)]\,(x)}\, dx + y_2(x) \int \frac{y_1(x)\, R(x)}{[W(y_1, y_2)]\,(x)}\, dx$

$$= -e^x \int \frac{xe^x \cdot \frac{e^x}{x^3}}{e^{2x}}\, dx + xe^x \int \frac{e^x \cdot \frac{e^x}{x^3}}{e^{2x}}\, dx$$

$$= -e^x \int x^{-2}\, dx + xe^x \int x^{-3}\, dx$$

$$= -e^x \left[\frac{x^{-2+1}}{(-2+1)}\right] + xe^x \left[\frac{x^{-3+1}}{-3+1}\right]$$

$$= e^x . x^{-1} - \frac{xe^x . x^{-2}}{2}$$

$$= \frac{1}{x} . e^x - \frac{1}{2x} e^x = \frac{1}{2x} e^x$$

Hence, the general solution is $y = y_h(x) + y_p(x) = (c_1 + c_2x)\, e^x + \frac{1}{2x} . e^x$.

Example 22. Solve: $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 16\, x^3$; $y(1) = -1$, $y'(1) = 1$.

Solution:

The given equation is $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 16\, x^3$ .... (i)

which is a second order non-homogeneous linear differential equation with variable coefficients. Its solution is $y(x) = y_h(x) + y_p(x)$

To find y_h: Its corresponding homogeneous equation is

$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$ .... (ii)

which is Euler-Cauchy equation $x^2 \frac{d^2y}{dx^2} + ax \frac{dy}{dx} + by = 0$.

Its auxiliary equation is

$m^2 + (a - 1)m + b = 0$

or, $m^2 + (1 - 1)m - 1 = 0$

$\therefore$ $m = \pm 1$

$\therefore$ The linearly independent solutions of (ii) are

$y = y_1(x) = x$ and $y = y_2(x) = x^{-1} = \frac{1}{x}$

$\therefore$ The general solution of (ii) is

$y_h = C_1 y_1(x) + C_2 y_2(x)$

$y_h = C_1 x + \frac{C_2}{x}$ ...(iii)

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To find y_p: The Wronskian determinant of the linearly independent solution.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1 & y_2' \end{vmatrix} = \begin{vmatrix} x & \frac{1}{x} \\ 1 & -\frac{1}{x^2} \end{vmatrix} = -\frac{1}{x} - \frac{1}{x}$$

$$\therefore \quad W(y_1, y_2) = -\frac{2}{x}$$

For determining the particular solution y_p of equation (i)

We rewrite (i) as follows, dividing each term by coefficient x^2 of $\frac{d^2y}{dx^2}$,

$$\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 16x \qquad \ldots . (iv)$$

Comparing with $\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R$

We have $R = 16x$

Now, the particular solution is given by

$$y_p = -y_1 \int \frac{y_2 R}{W}\, dx + y_2 \int \frac{y_1 R}{W}\, dx$$

$$= -y_1 \int \frac{\left(\frac{1}{x}\right)16x}{\left(-\frac{2}{x}\right)}\, dx + y_2 \int \frac{x(16x)}{\left(-\frac{2}{x}\right)}\, dx$$

$$= 8y_1 \int x\,dx - 8y_2 \int x^3\,dx$$

$$= 8y_1\left(\frac{x^2}{2}\right) - 8y_2\left(\frac{x^4}{4}\right)$$

$$= 4.x.(x^2) - 2\left(\frac{1}{x}\right)x^4 = 4x^3 - 2x^3$$

$$\therefore \quad y_p = 2x^3 \qquad \ldots . (v)$$

The general solution of equation (i) is

$y(x) = y_h + y_p$, using (iii) and (v),

$$\therefore \quad y(x) = C_1 x + \frac{C_2}{x} + 2x^3 \qquad \ldots . (vi)$$

To determine C_1 and C_2, using initial condition

$y(1) = -1$ in (vi)

$-1 = C_1 + C_2 + 2$

$$\therefore \quad C_1 + C_2 = -3 \qquad \ldots . (a)$$

Also, differentiating (vi)

$$y'(x) = C_1 \frac{C_2}{x^2} + 6x^2$$

Using, $y'(1) = 1$

$1 = C_1 - C_2 + 6$

$$\therefore \quad C_1 - C_2 = -5 \qquad \ldots . (b)$$

Solving (a) and (b), we get $C_1 = -4$ and $C_2 = 1$

Thus, the solution of given initial value problem is, $y(x) = -4x + \frac{1}{x} + 2x^3$.

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Exercise 8.4

1. Find the general solution of the following differential equations by using the method of variation of parameters.

a. $y'' - y' - 2y = 2e^{-x}$ b. $y'' + 2y' + y = e^{-x}\cos x$

c. $y'' + 4y = 3\text{cosec}2x$ d. $y'' + 4y' + 4y = x^{-2}e^{-2x}$

e. $y'' - 2y' + y = 3x^{3/2}e^{x}$ f. $y'' + 4y' + 5y = 10$

g. $y'' + 2y' + y = 4e^{-x}\log x$ h. $y'' + 9y = \text{cosec}3x$

i. $y'' + 9y = \sec 3x$ j. $(D^2 + 2D + 2)y = 2e^{-x}\sec^2 x$

2. Find the solution of the following differential equations by using the method of variation of parameters.

a. $(x^2D^2 - 4xD + 6)y = 21\,x^{-4};\ D = \frac{d}{dx},\ D^2 = \frac{d^2y}{dx^2}$

b. $(x^2D^2 + xD - 9)y = 48x^5$

c. $(x^2D^2 + xD - 1)y = 16x^3, y(1) = -1, y'(1) = 11$

d. $(x^2D^2 - xD + 2)y = x\log x$

Answer

1. a. $y = c_1e^{-x} + c_2e^{2x} - \frac{2}{3}xe^{-x}$

b. $y = (c_1 + c_2x - \cos x)\,e^{-x}$

c. $y = c_1\cos 2x + c_2\sin 2x + \frac{3}{4}(\sin 2x)\ln\sin 2x - \frac{3}{2}x\cos 2x$

d. $y = (c_1 + c_2x)\,e^{-2x} - e^{-2x}\ln x$

e. $y = \left(c_1 + c_2x + \frac{12}{35}x^{7/2}\right)e^{x}$

f. $y = e^{-2x}(c_1\cos x + c_2\sin x) + 2$

g. $y = (c_1 + c_2x)\,e^{-x} + x^2e^{-x}(2\ln x - 3)$

h. $y = c_1\cos 3x + c_2\sin 3x - \frac{x}{3}\cos 3x + \frac{\sin 3x}{9}\log(\sin 3x)$

i. $y = c_1\cos 3x + c_2\sin 3x - \frac{1}{9}\cos 3x\log(\sec 3x) + \frac{x}{3}\sin 3x$

j. $y = e^{x}(C_1\cos x + C_2\sin x + \tan x\sin x)$

2. a. $y = C_1x^2 + C_2x^3 + \frac{1}{2x^4}$ b. $y = C_1x^{-3} + C_2x^3 + 3x^5$

c. $y = x - \frac{4}{x} + 2x^3$ d. $y = x[C_1(\log x) + C_2\sin(\log x)] + x\log x$

8.6 Engineering application of second-order differential equation

Second-order linear differential equations have various application in science and engineering. In this section, we discuss application in electric circuits and motion of vibrating springs.

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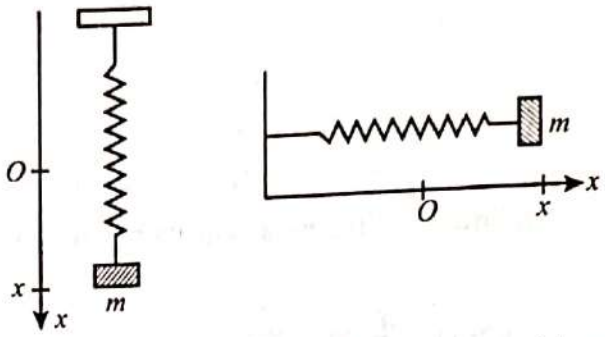
8.6.1 Vibrating mass-springs system

Consider the motion of an object with mass m at the end of a spring which in either vertical or horizontal position.



If the spring is stretched (or compressed) x units from its natural length then by Hooke's law is applicable. Hooke's law states that "The restoring force is proportional to the extension of the string."

$$\text{Restoring force} = -kx \qquad \ldots\ldots \text{(i)}$$

where, k is a positive constant known as the spring constant.

Also, by Newton's second law, $F = ma$, where m is mass and a is acceleration.

$$\text{Restoring force} = m\frac{d^2x}{dt^2} \qquad \ldots\ldots \text{(ii)}$$

From (i) and (ii),

$$m\frac{d^2x}{dt^2} = -kx$$

or, $$m\frac{d^2x}{dt^2} + kx = 0 \qquad \ldots\ldots \text{(iii)}$$

which is second-order linear differential equation.

AE of equation (iii) is

$$m\gamma^2 + k = 0$$

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or, $\gamma = \pm \omega i$, where $\omega = \sqrt{\frac{k}{m}}$

Then, the general solution is $x(t) = c_1 \cos\omega t + c_2 \sin\omega t$ which can be written as

$$x(t) = A\cos\delta\cos\omega t - A\sin\delta\sin\omega t, \text{ where } c_1 = A\cos\delta,\ c_2 = -A\sin\delta$$

Then,

$$x(t) = A\cos(\omega t + \delta)$$

where, frequency $= \omega = \sqrt{\frac{k}{m}}$

amplitude $= A = \sqrt{c_1^2 + c_2^2}$

and phase angle is δ.

This type of motion is known as simple harmonic motion.

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Example 23. A spring with a mass 2 kg has natural length 0.5 m. A force of 25.6 N is required to maintain if stretched to a length of 0.7 m. If the spring is stretched to a length of 0.7 m and then released with initial velocity 0, find the position of the mass at any point t.

Solution:

Using Hooke's law, the force required to stretch the spring is

$k \times 0.2 = 25.6$

or, $k = \frac{25.6}{0.2} = 128$

Also, $m = 2$

From equation (iii)

$$2\frac{d^2x}{dt^2} + 128x = 0 \quad \ldots.(iv)$$

It's AE is

$2\gamma^2 + 128 = 0$

or, $\gamma^2 = -64$

$\therefore \quad \gamma = \pm 8i$

Thus, general solution (iv) is

$x(t) = c_1 \cos 8t + c_2 \sin 8t \quad \ldots.(v)$

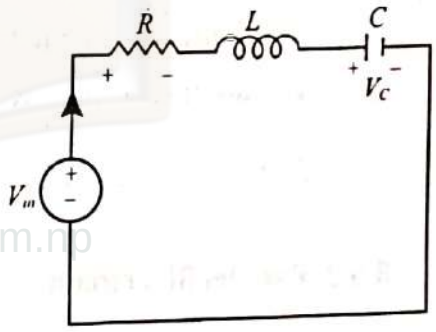
Then, $x'(t) = -8c_1 \cos 8t + 8c_2 \sin 8t \quad \ldots.(vi)$

Using $x(0) = 0.2$ in equation (v), we get $c_1 = 0.2$

Using $x'(0) = 0$ in (vi), we get $c_2 = 0$

Hence, required solution is $x(t) = \frac{1}{5}\cos 8t$.

8.6.2 Series RLC circuit



An RLC circuit constitutes three technical elements resistor (R), an inductor (I), and a capacitor (C).

In the above figure, the electrical elements resistor (R), inductor (L) and capacitor (C) are connected in series with the DC source (V_m), which is called the RLC series circuit.

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The Kirchoff's voltage law is applied around loop for $t < 0$ in which the voltage source of the circuit is calculated by the following

$$V_R + V_L + V_C = V_m \quad \ldots.(i)$$

The individual circuit elements (capacitance, inductance, resistance) are given by

Capacitance: $I = C\frac{dV_C}{dt}$

Resistance: $V_R = RI = RC\frac{dV_C}{dt}$

Inductance: $V_L = L\frac{dI}{dt} = L\frac{d}{dt}\left(C\frac{dV_C}{dt}\right) = LC\frac{d^2V_C}{dt^2}$

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From equation (i),

$$RC\frac{dV_C}{dt} + LC\frac{d^2V_C}{dt^2} + V_C = V_{in}$$

or, $$LC\frac{d^2V_C}{dt^2} + RC\frac{dV_C}{dt} + V_C = V_{in}$$

or, $$L\frac{d^2V_C}{dt^2} + R\frac{dV_C}{dt} + \frac{1}{C} + V_C = \frac{V_{in}}{C}$$

Damping factor

In RLC circuit, the damping factor is given by

$$\mu = \frac{\alpha}{\omega_0}$$

where, $\alpha = \frac{R}{2L}$ (damping coefficient)

and $$\mu = \frac{\frac{R}{2L}}{\frac{1}{\sqrt{LC}}} = \frac{R}{2}\sqrt{\frac{C}{L}}$$

RLC circuit is

i. overdamped when $\mu > 1$

ii. critically damped when $\mu = 1$ and

iii. underdamped when $\mu < 1$.

Moreover, another form of RLC circuit is

$$LQ'' + RQ'' + \frac{1}{C}Q = E(t)$$

8.6.3 Parallel RLC circuit

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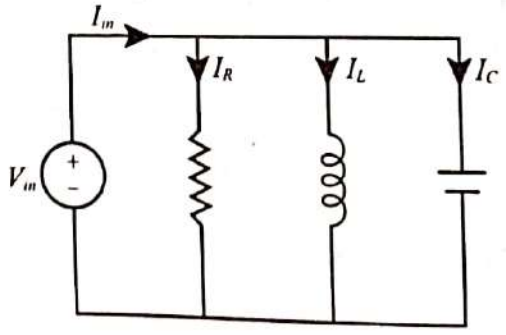

In the above figure, the electrical elements resistor (R), inductor (L) and capacitor (C) are connected in parallel with the DC source (V_s), which is called parallel RLC circuit.

The Kirchoff's current law,

$$I_{in} = I_R + I_L + I_C \qquad \ldots\ldots \text{(i)}$$

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Resistance: $I_R = \frac{V}{R}$

Capacitance: $I_C = C\frac{dV}{dt}$

Inductance: $I_L = \frac{1}{L}\int V dt$

From equation (i),

$$\frac{V}{R} + \frac{1}{L}\int V dt + C\frac{dV}{dt} = I_{in}$$

Differentiating with respect to t, we get

$$\frac{1}{R}\frac{dV}{dt} + \frac{1}{L}\frac{d}{dt}\int V dt + C\frac{d^2V}{dt^2} = \frac{dI_{in}}{dt}$$

or, $\frac{1}{R}\frac{dV}{dt} + \frac{1}{L}V + C\frac{d^2V}{dt^2} = 0$

or, $C\frac{d^2V}{dt^2} + \frac{1}{R}\frac{dV}{dt} + \frac{1}{L}V = 0$

or, $\frac{d^2V}{dt^2} + \frac{1}{RC}\frac{dV}{dt} + \frac{1}{LC}V = 0$

or, $LC\frac{d^2V}{dt^2} + L\frac{dV}{dt} + RV = 0.$

Damping factor

In parallel RLC circuit, the damping factor is given by

$$\mu = \frac{\alpha}{\omega_0}$$

where, $\alpha = \frac{1}{2Rc}$ (damping coefficient) and $\omega_0 = \frac{1}{\sqrt{LC}}$ (resonant frequency)

Then, $\mu = \frac{1}{2R}\sqrt{\frac{L}{C}}$

The circuit is overdamped when $\mu > 1$, critically damped when $\mu = 1$ and underdamped when $\mu > 1$.

Example 24. If a series has capacitor of 1.6×10^{-6} F and the inductor of $L = 0.4H$ then find the resistance R so that circuit is critically damped.

Solution:

We have, $L = 0.4H$, $C = 1.6 \times 10^{-6}$ F

For the critical damping,

$$\mu = \frac{R}{2}\sqrt{\frac{C}{L}} = 1$$

or, $\frac{R}{2} = \sqrt{\frac{1.6 \times 10^{-6}}{0.4}} = 1$

$\therefore$ $R = 10^3\ \Omega$

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Example 25. Solve second order differential equation of the series RLC circuit

$$L\frac{d^2V_C}{dt^2} + R\frac{dV_C}{dt} + \frac{1}{C}V_C = \frac{V_m}{C},$$

where $R = 10\,\Omega$, $L = 1H$, $C = 16 \times 10^{-4}\,F$, $V_m = 0$, $V_C(0) = 6V$, $V_C'(0) = 6A$.

Solution

The given differential equation is

$$1\frac{d^2V_C}{dt^2} + 40\frac{dV_C}{dt} + 625\,V_C = 0.$$

Its AE is

$$r^2 + 40r + 625 = 0$$

or, $$r = \frac{-40 \pm \sqrt{-900}}{2} = -20 \pm 15i$$

Then, general solution is $V_C = e^{-20t}(c_1 \cos 15t + c_2 \sin 15t)$.

Also, using $V_C(0) = 6V$

$$6 = c_1 \cos 0 + c_2 \sin 0$$

or, $$6 = c_1$$

$\therefore$ $$c_1 = 6$$

Also, $V_C' = -20e^{20t}(c_1 \cos 15t + c_2 \sin 15t) + e^{-20t}(-15c_1 \sin 15t + 15c_2 \cos 15t)$

Using $V_C'(0) = 6A$

or, $$6 = -20c_1 + 15c_2$$

or, $$6 = -20 \times 6 + 15c_2$$

or, $$15c_2 = 126$$

$\therefore$ $$c_2 = \frac{126}{15}$$

Hence, required solution is $V_C = e^{-200t}\left(6\cos 15t + \frac{126}{15}\sin 15t\right)$.

Exercise 8.5

1. A spring with a 4 kg mass has natural length 1 m and is maintained stretched to a length of 1.3 m by a force of 24.3 N. If the spring is compressed to a length of 0.8 m and then released with zero velocity, find the position of the mass at any point t.
2. If a parallel circuit has capacitor of 1.6×10^{-6} F and inductor of $L = 0.4H$ then find the resistance R so that circuit is critically damped.

   [Hint: For critically damped

   $$\mu = \frac{1}{2R}\sqrt{\frac{L}{C}} = 1$$

   or, $$R = \frac{1}{2}\sqrt{\frac{L}{C}} = \frac{1}{2}\sqrt{\frac{0.4}{1.6 \times 10^{-6}}} = \frac{1}{2}\sqrt{0.25 \times 10^{6}} = \frac{1}{2} \times 0.5 \times 10^{-3} = 0.25 \times 10^{-3}\,\Omega]$$

3. Solve second order differential equation of the parallel RLC circuit

   $$\frac{d^2V}{dt^2} + \frac{1}{RC}\frac{dV}{dt} + \frac{1}{LC}V = 0,\quad V(0) = 6,\quad \frac{dV(0)}{dt} = -12,\quad R = 20\Omega,\quad L = 50H,$$

   $$C = 6 \times 10^{-1}\,F$$

Answer

1. $x = -\frac{1}{5}e^{-6t} + \frac{6}{5}e^{-t}$
2. $0.25 \times 10^{-3}\,\Omega$
3. $6e^{-2t}$

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Multiple Choice Questions

1. The general solution of the differential equation $(D^2 - 2)y = 1$ is

a. $C_1e^{\sqrt{2}x} + C_2e^{-\sqrt{2}x}$ b. $(C_1 + C_2x)e^{\sqrt{2}x}$

c. $C_1e^{\sqrt{2}x} + C_2e^{-\sqrt{2}x} - \frac{1}{2}$ d. none of above

2. The non-zero particular integral can be found out for _____ differential equation

a. homogeneous b. non-homogeneous

c. second order homogeneous d. none of these

3. Which is the correct solution for the complementary function of $(D^2 + 2D + 2)y = 0$, where $D \equiv \frac{d}{dt}$?

a. $y = c_1e^{-it} + c_2e^{it}$ b. $y = ce^{-t}$

c. $(c_1t + c_2)\,e^{-t}$ d. $y = c_1e^{(-1-i)t} + c_2e^{(-1+i)t}$

4. The particular integral of $(D^2 + a^2)y = e^{iax}$, where $D \equiv \frac{d}{dt}$ is

a. $\frac{e^{iax}}{2ai}$ b. $\frac{x}{2a}\,e^{ax}$

c. $\frac{x}{2ai}e^{iax}$ d. $\frac{e^{iax}}{2}$

5. The Wronskian of 1, e^x is

a. e^x b. $1 + e^x$

c. 0 d. 1

6. The solution $y_1(x)$ and $y_2(x)$ are said to be linear independence if

a. $W(y_1, y_2) = 1$ b. $W(y_1, y_2) = 0$

c. $W(y_1, y_2) \neq 0$ d. none of the above

7. The particular integral of $(D + 5)(D - 4)\,y = 1000$ is

a. 50 b. -50

c. 100 d. -100

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8. Particular integral of $(D + 1)y = e^t$ where $D = \frac{d}{dt}$ is

a. te^t b. $\frac{1}{2}te^t$

c. $\frac{1}{2}e^t$ d. te^{2t}

9. Particular integral of $y'' - 5y' + 6y = e^{2x}$ is

a. $-xe^{2x}$ b. xe^{2x}

c. $\frac{-1}{2}xe^{2x}$ d. $-xe^{2x}$

Answer

1. c. 2. b. 3. d. 4. c. 5. a. 6. c. 7. b. 8. c. 9. d.

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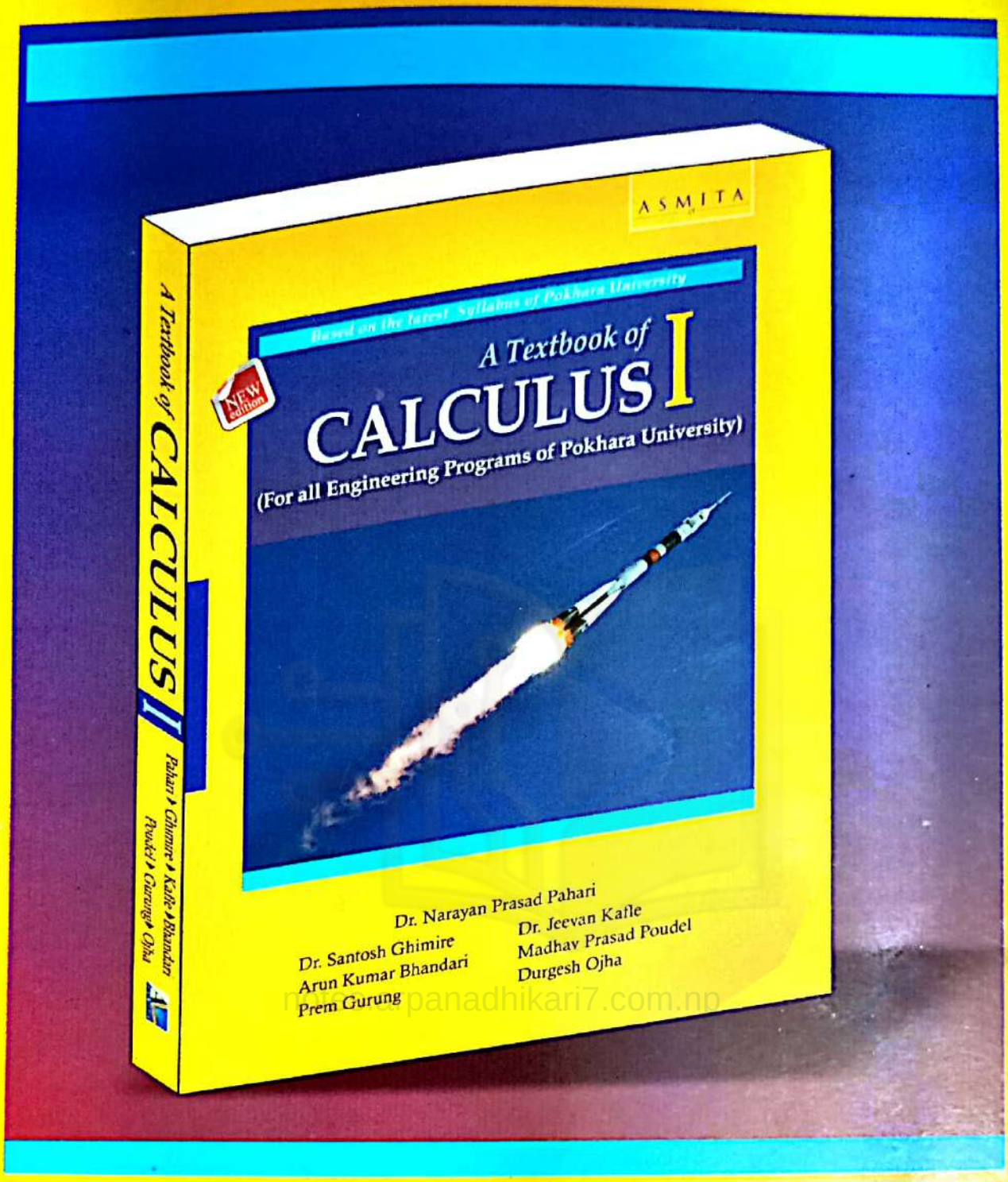

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